A few applications of GSP to machine learning and medicine

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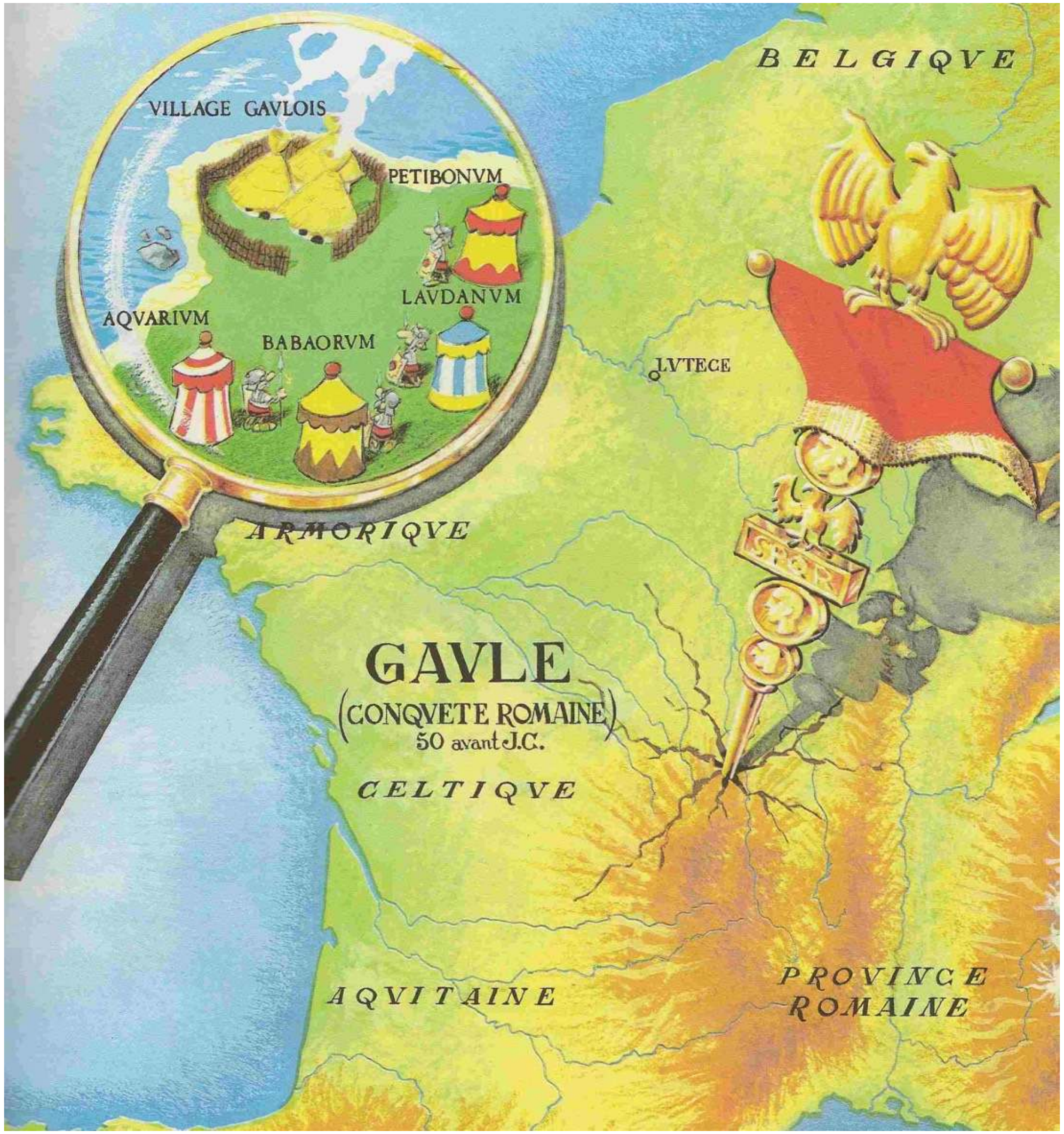
ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

Hello :)



BELGIQVE
VILLAGE GAVLOIS
PETIBONVM
LAVDANVM
AQVARIVM
BABAORVM
LVTECE
SPQR
ARMORIQVE
GAVLE
(CONQVETE ROMAINE)
50 avant J.C.
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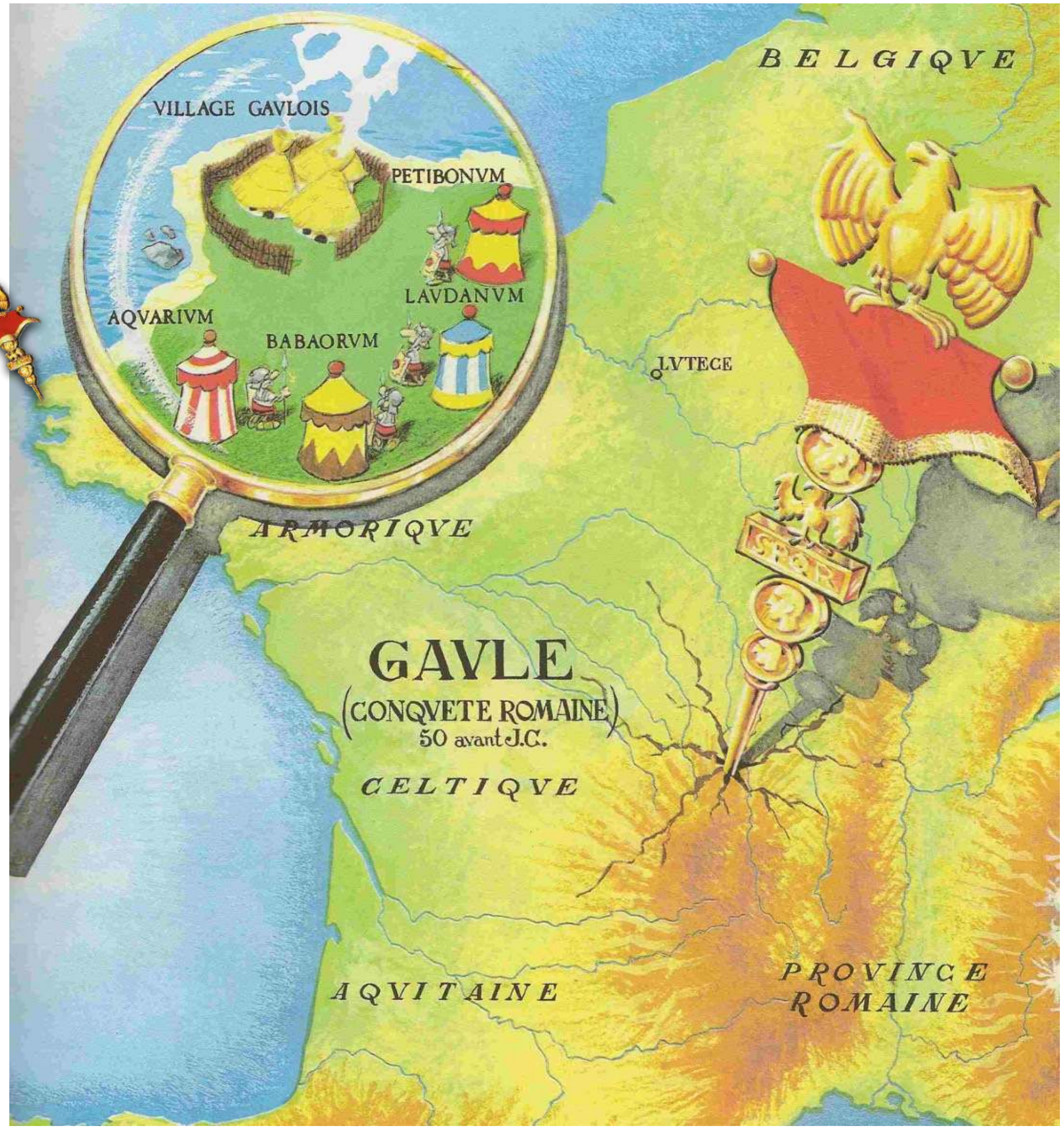
PROVINCE
ROMAINE

Hello :)

Brest
(random studies)



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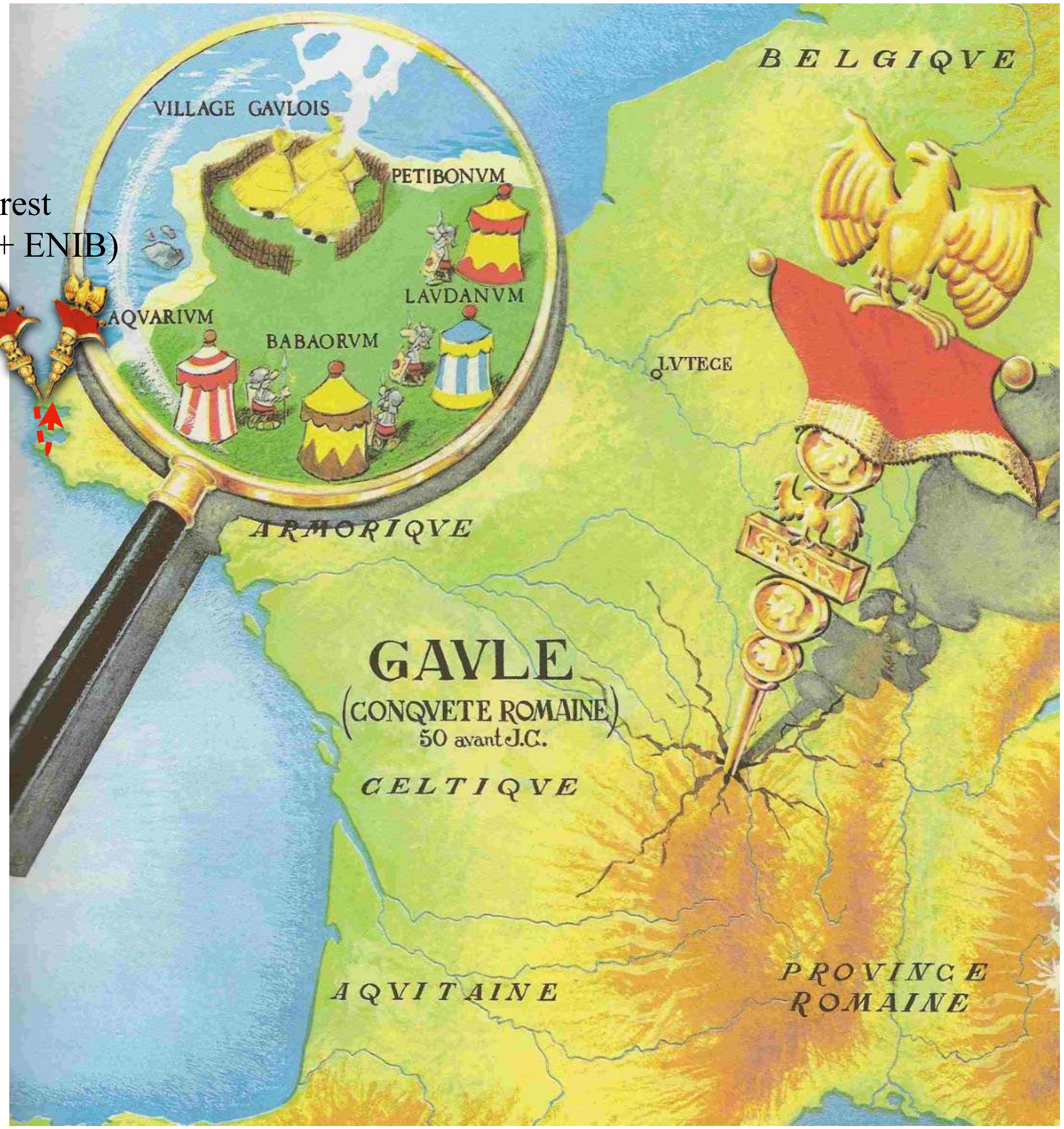

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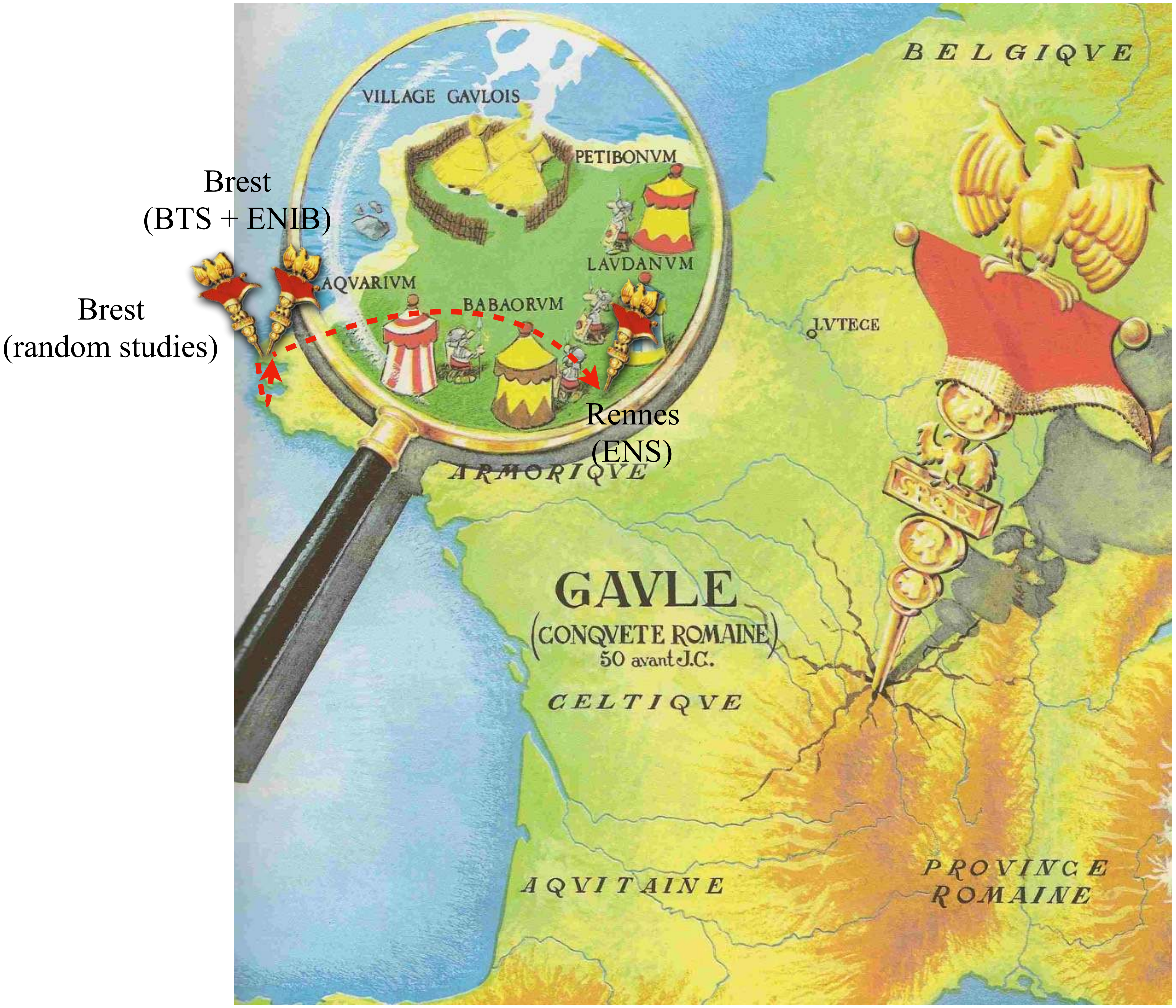
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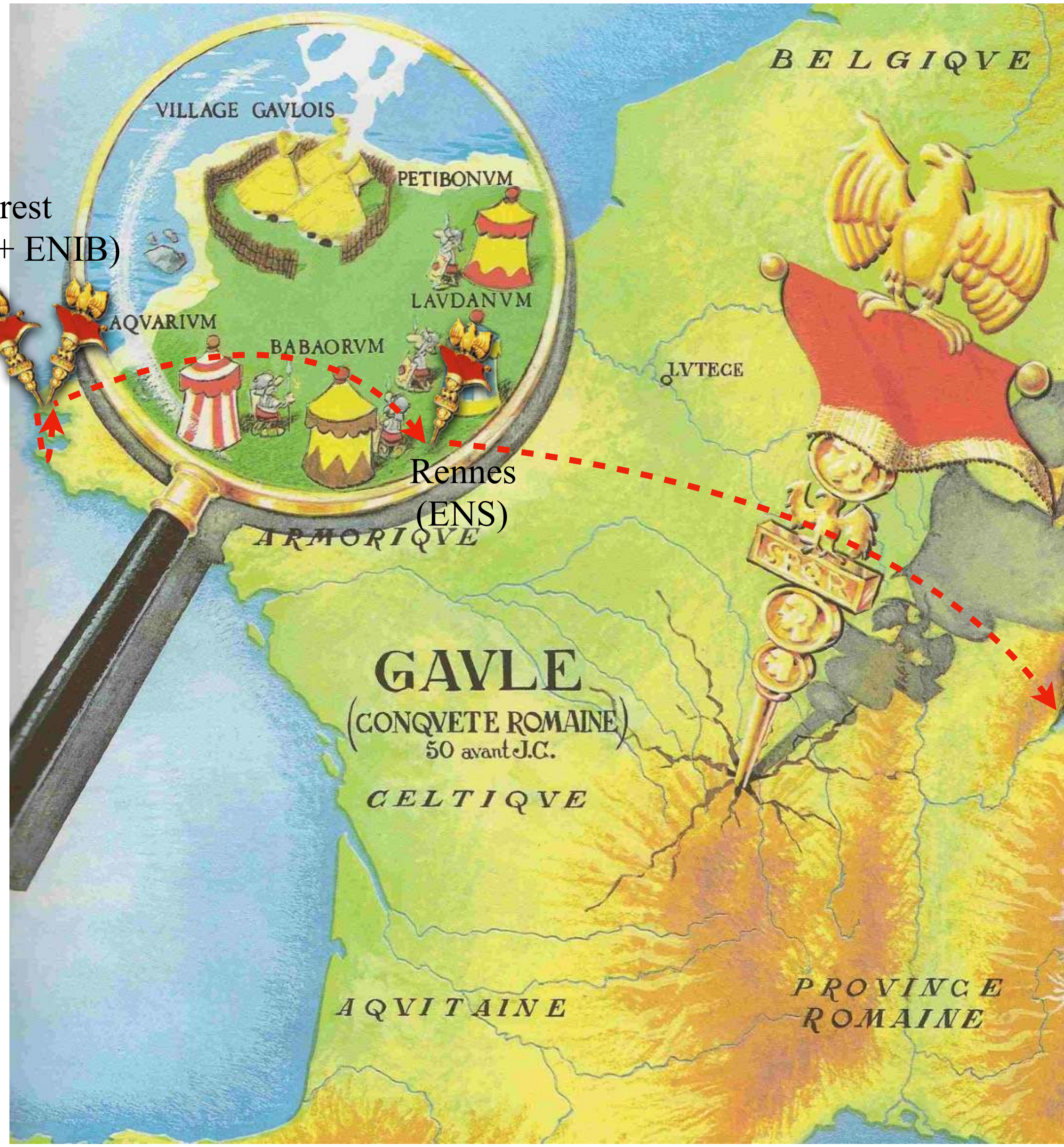

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Outline

- GSP for AI, *a.k.a.* "What I did during my Ph.D. and a bit after"
- GSP and personalized healthcare
- GSP and brain signals

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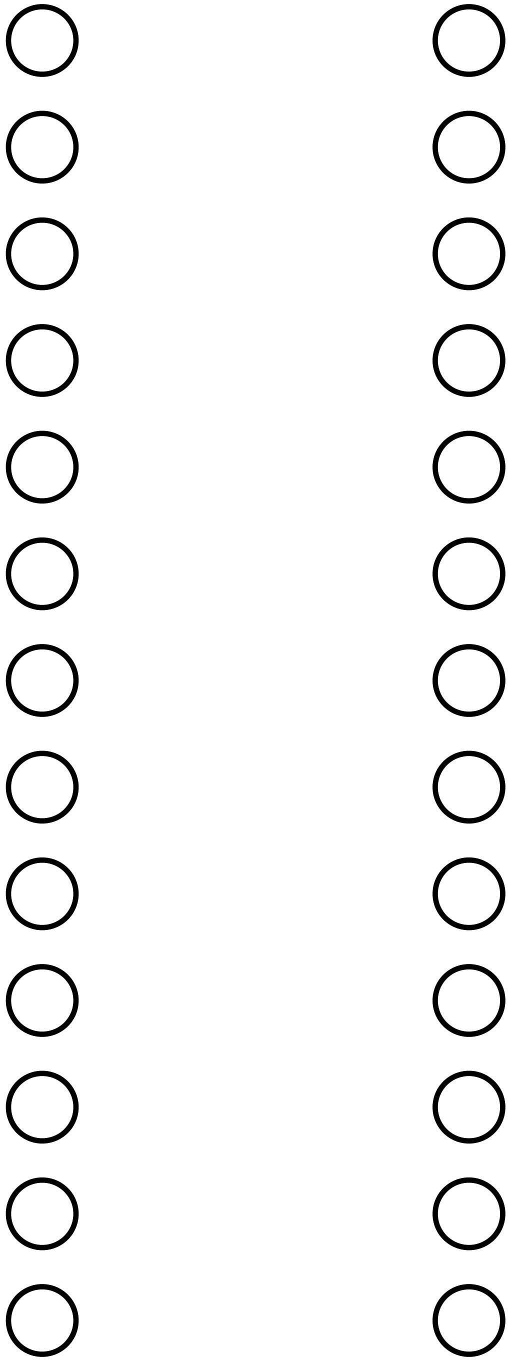
How to design a convolutional layer?



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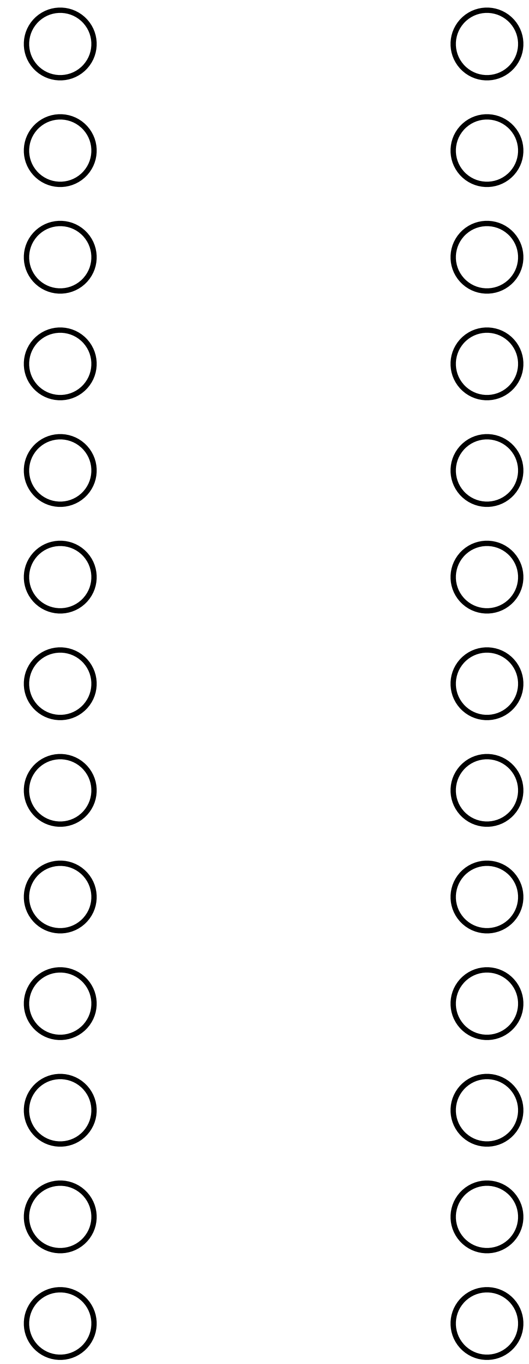

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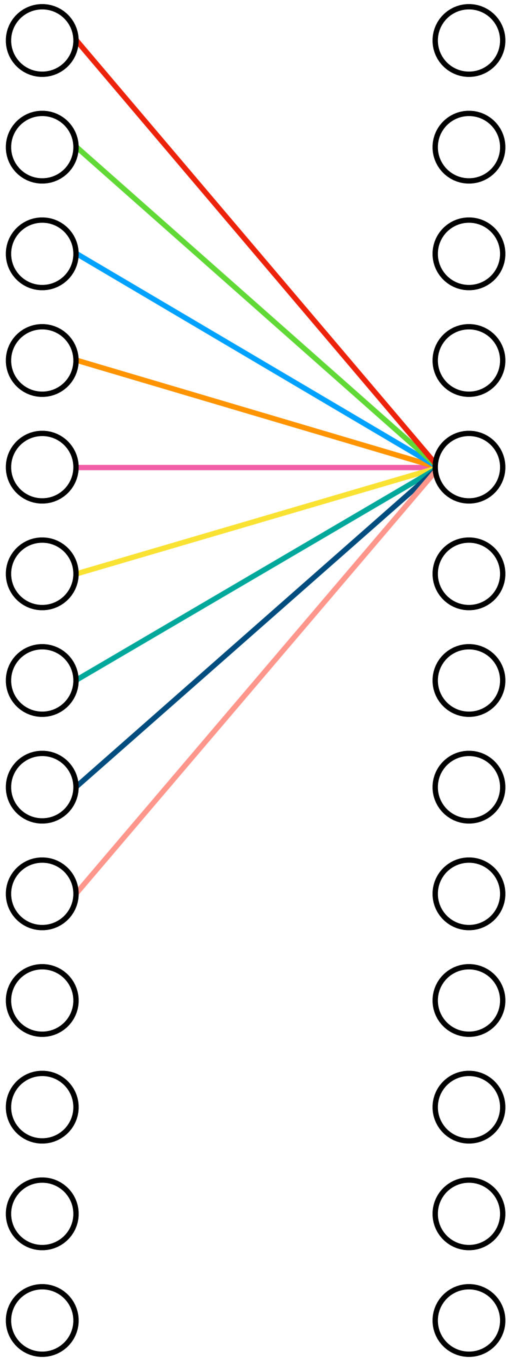

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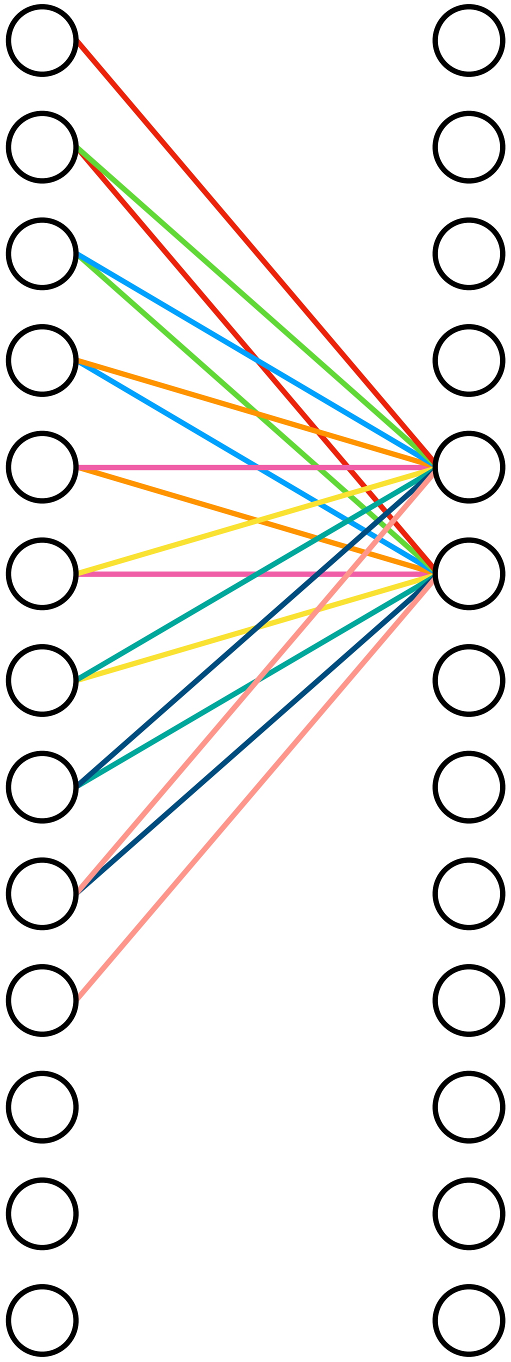

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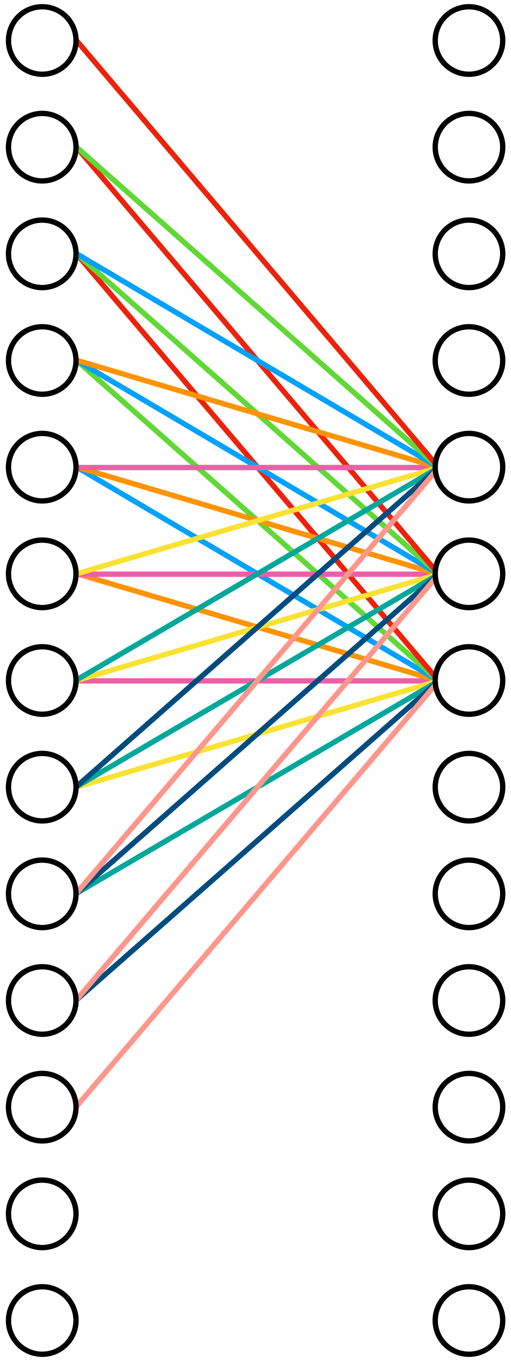

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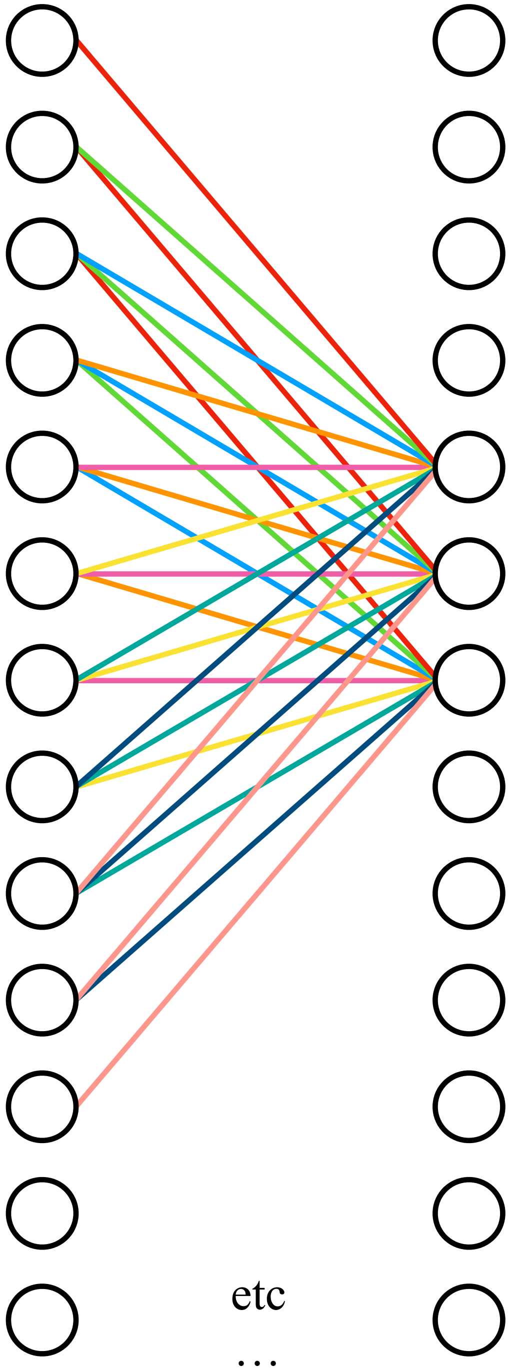


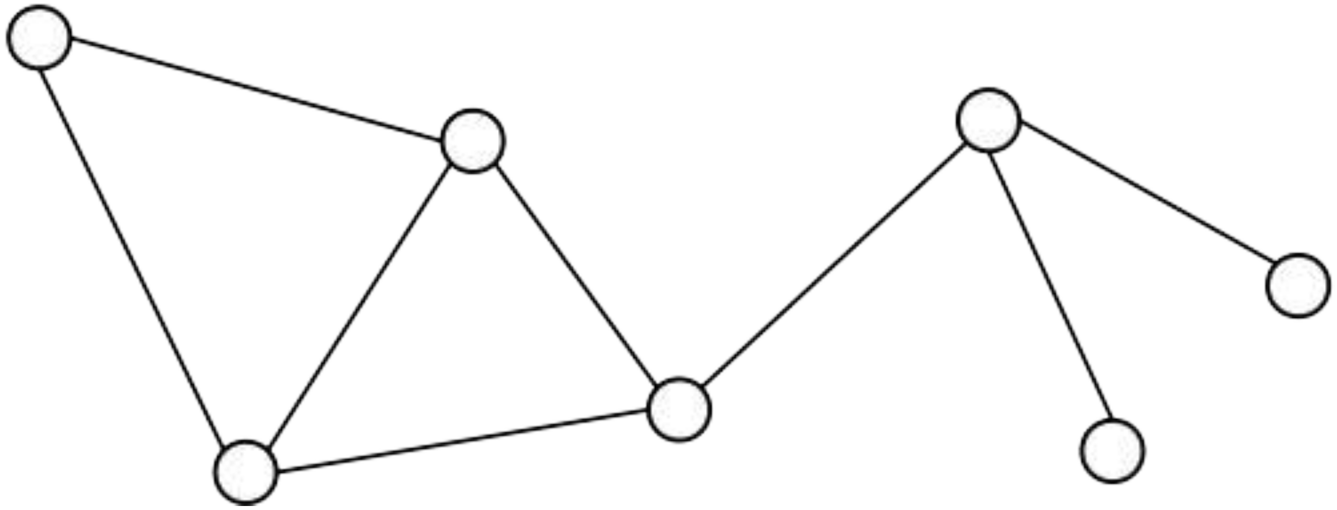
Why does it work?

Discriminating features are localized in the image (= locality)

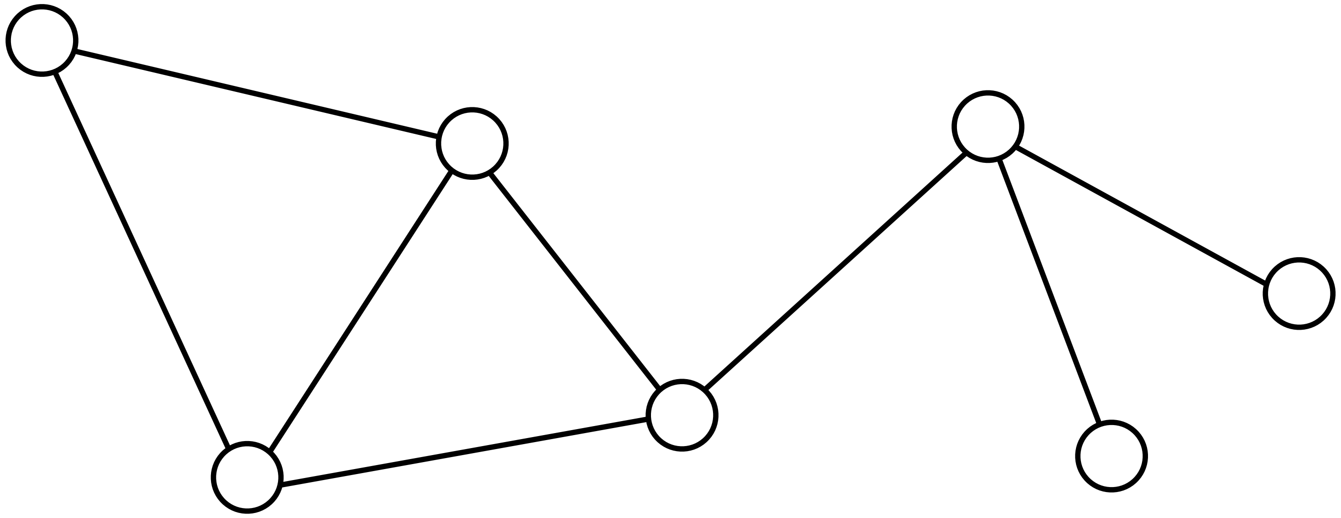
Statistical properties of discriminating features are invariant by translation (= stationarity)

A natural scene has an important low-frequency component (= smoothness)

Beyond images

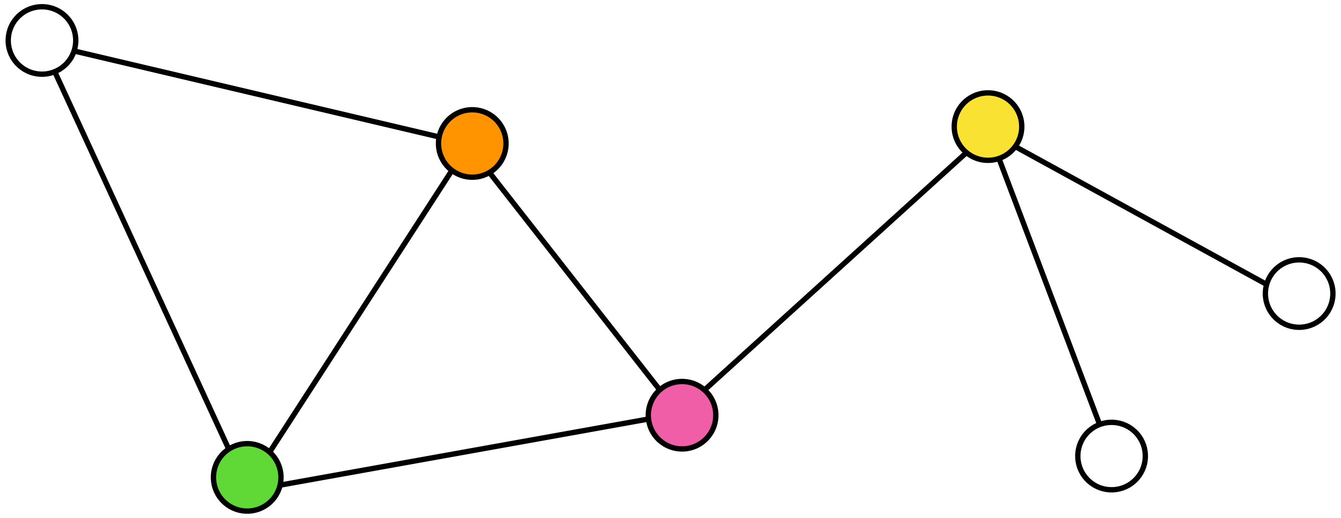


What do we need?



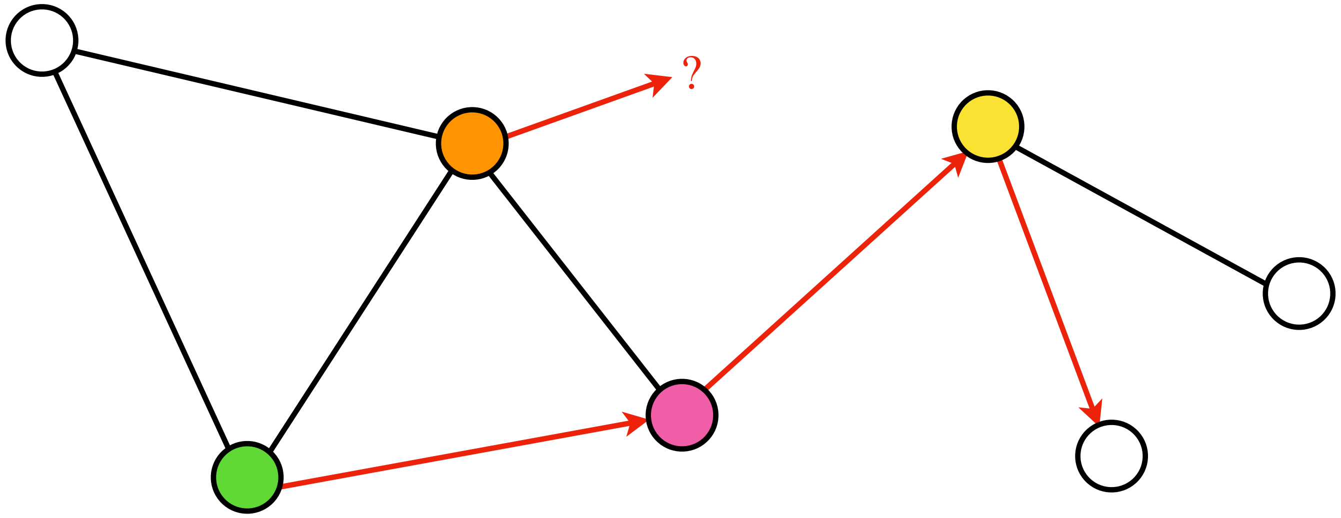
What do we need?

Assumption that discriminating features are localized



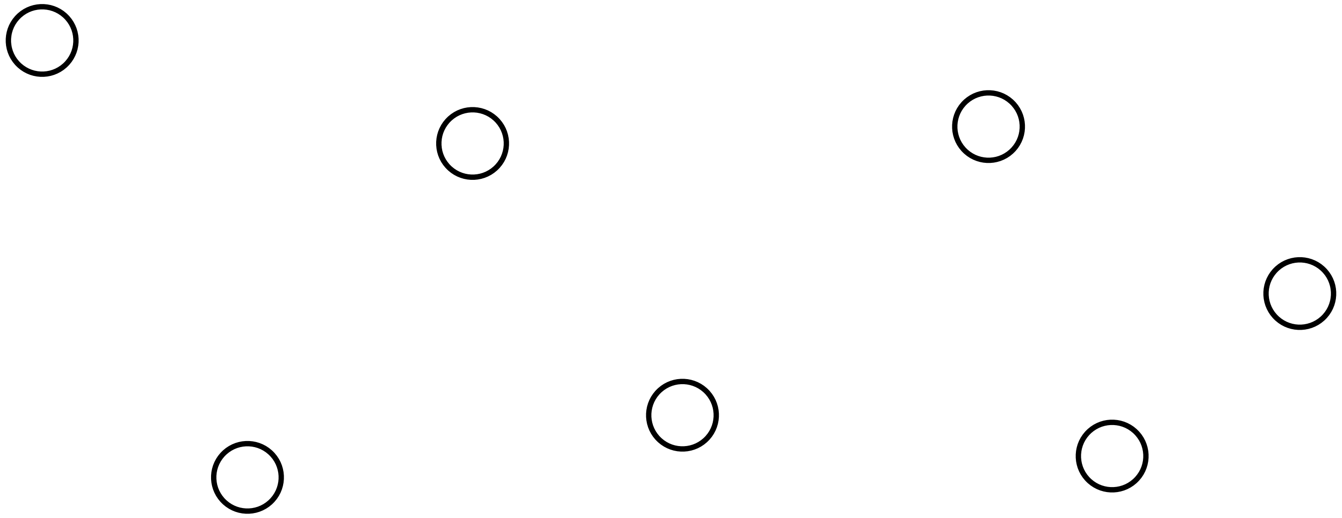
What do we need?

What is a translation on a graph domain?

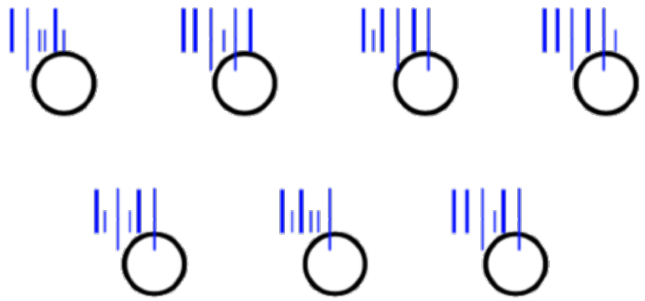


What do we need?

Which graph does data live on?



Extending CNNs to graph signals



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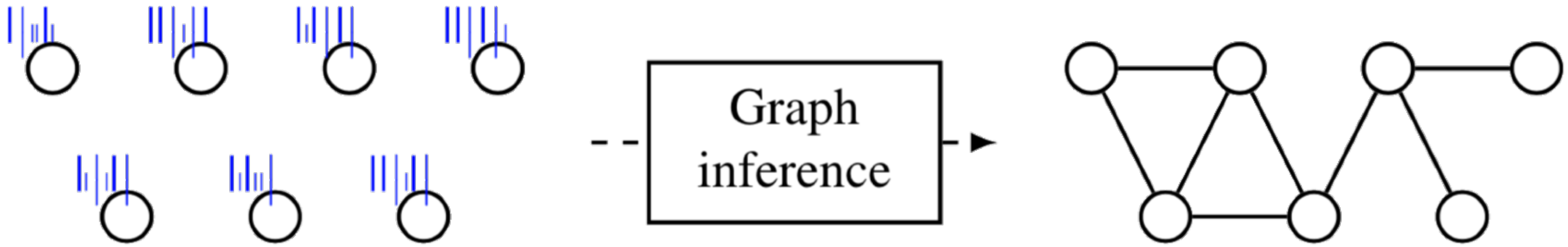

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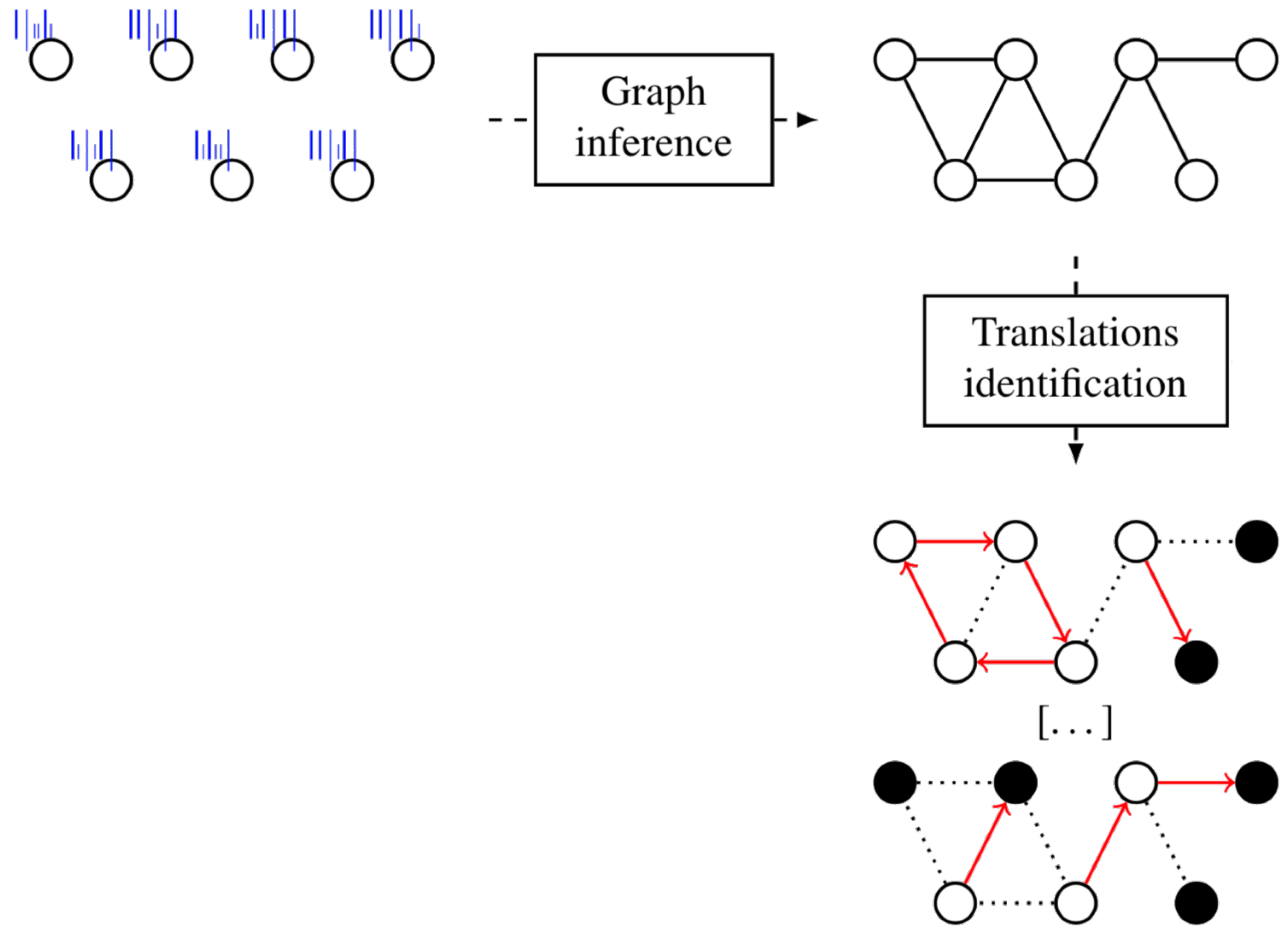

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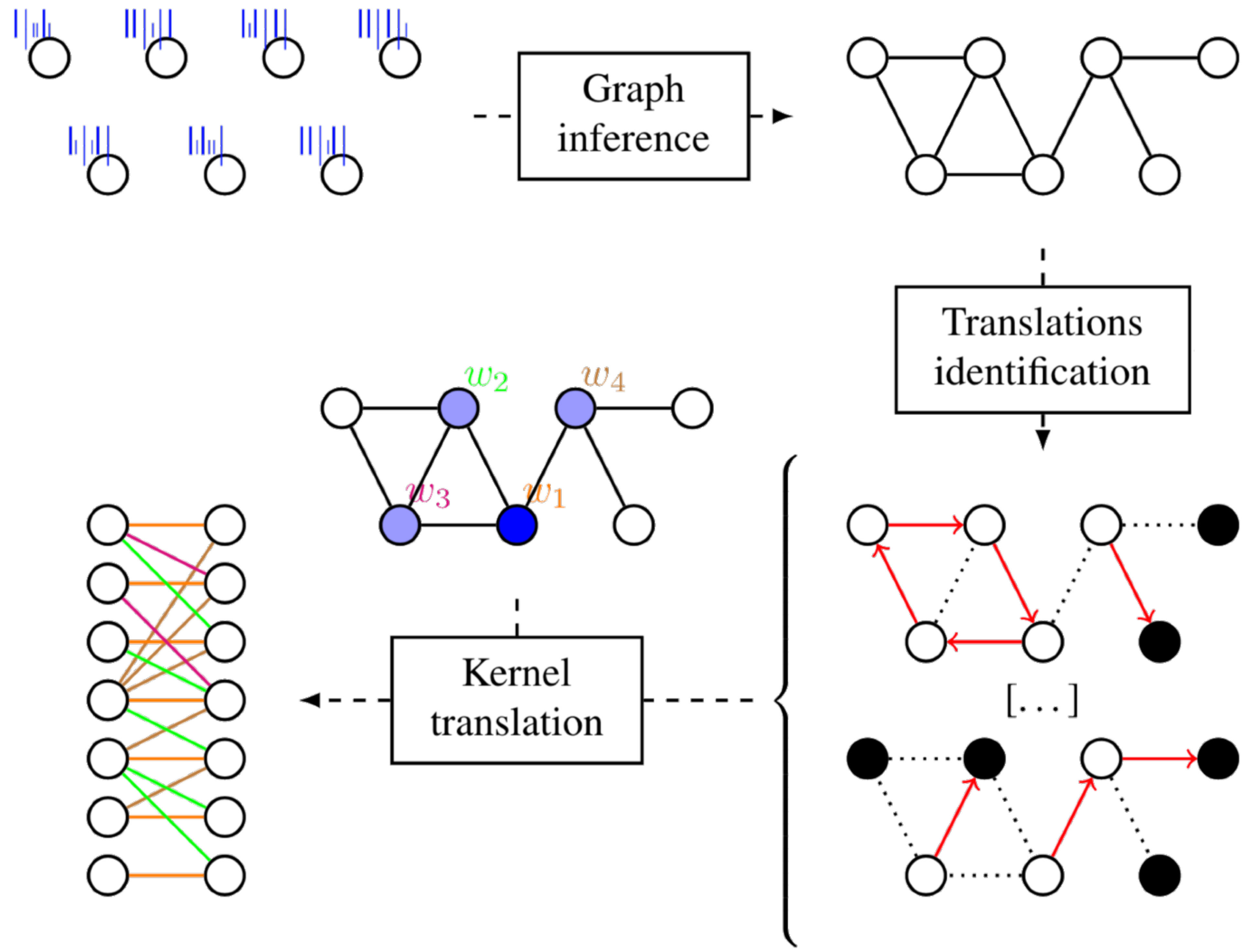

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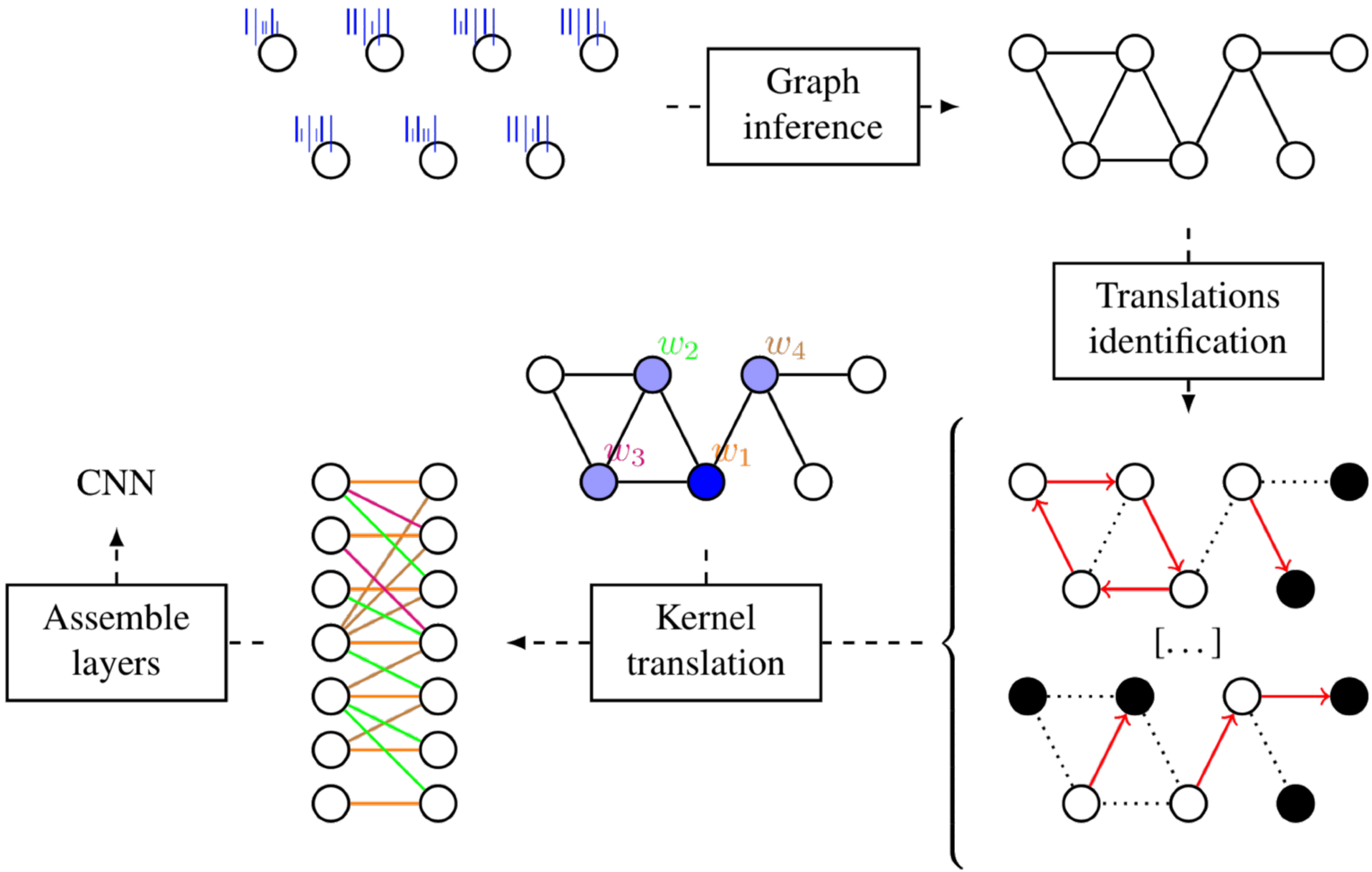

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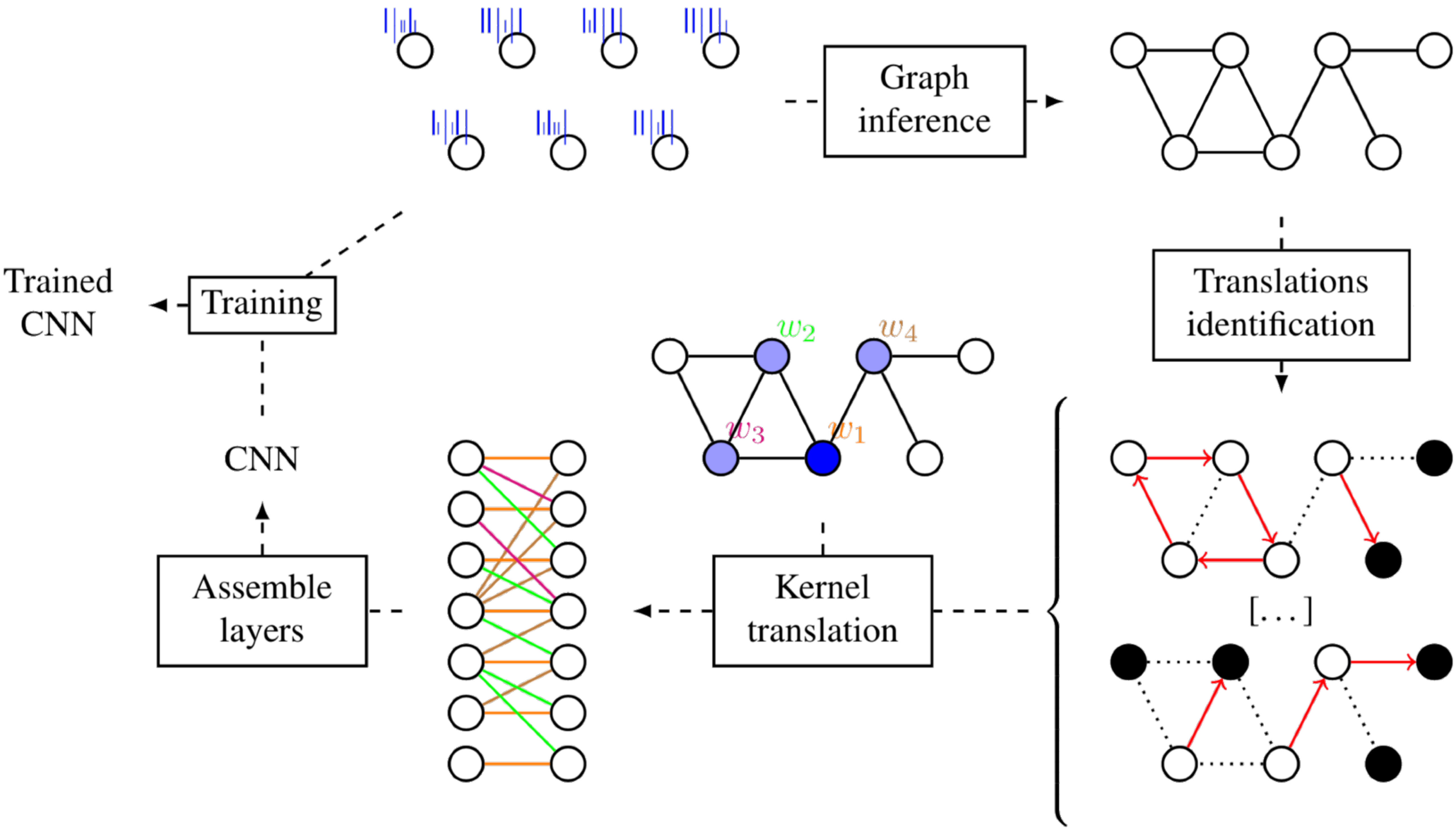
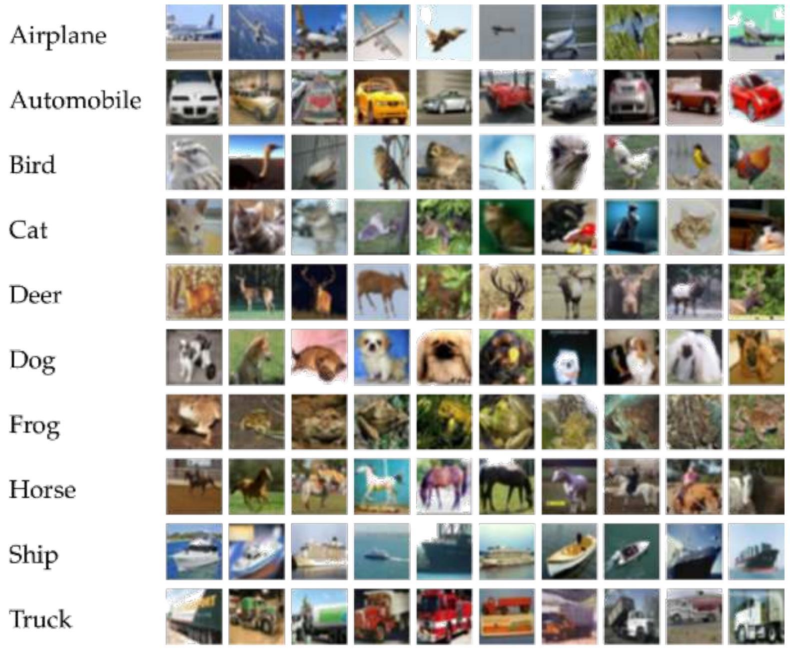
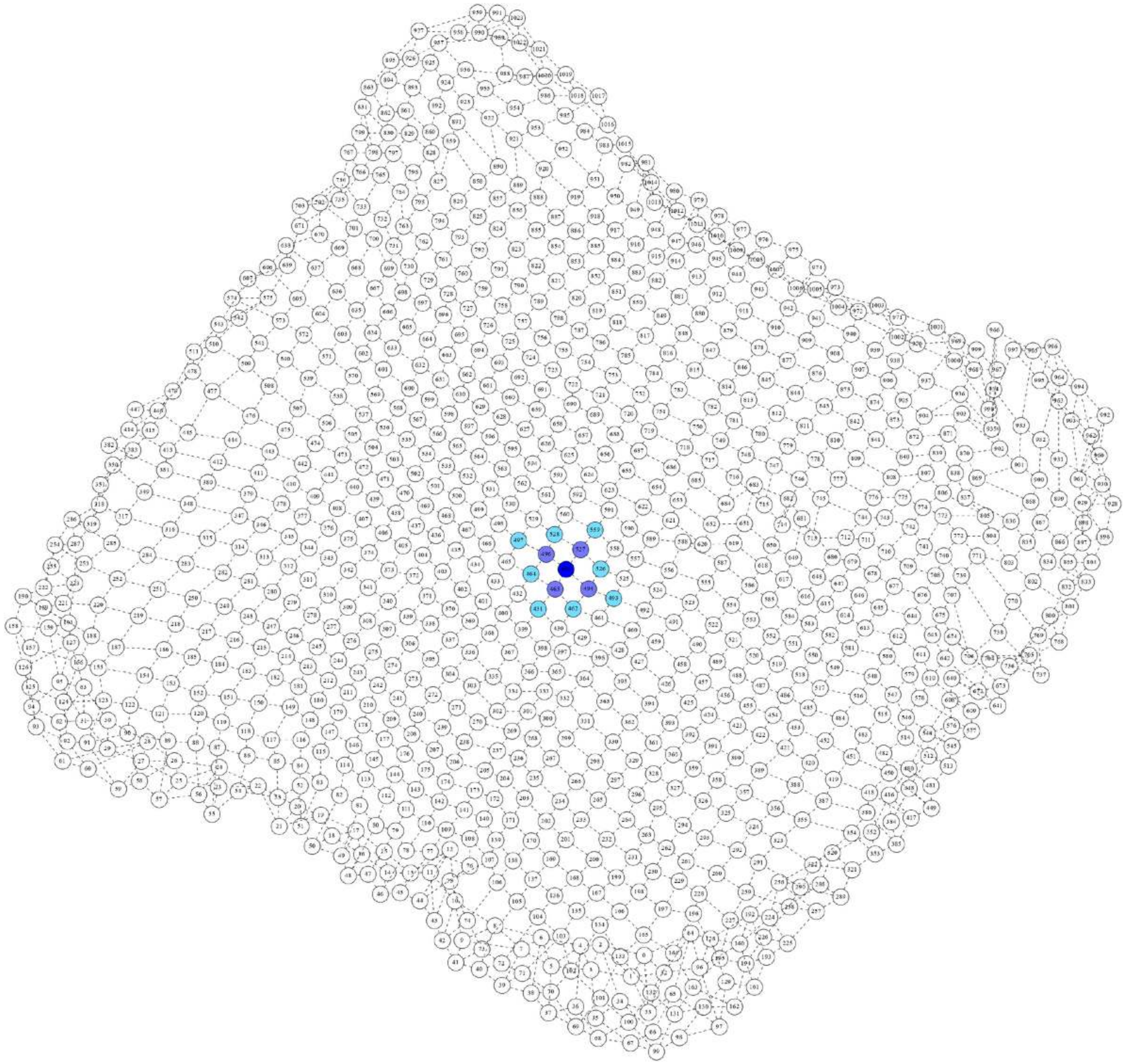


Results





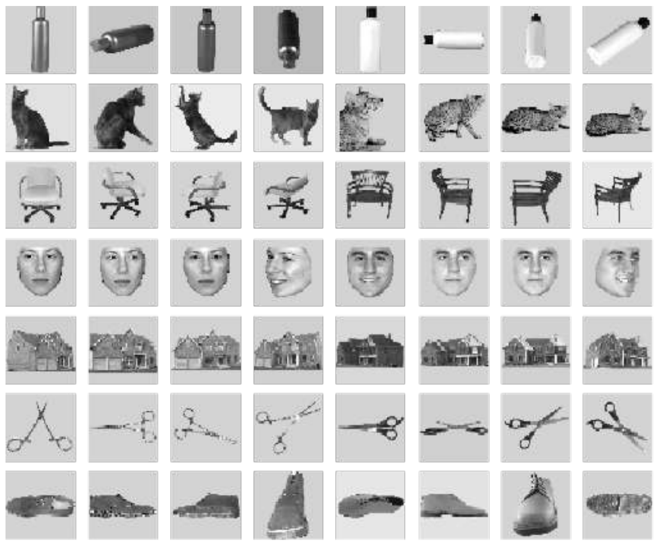
No knowledge on topology [1]: 72.7%

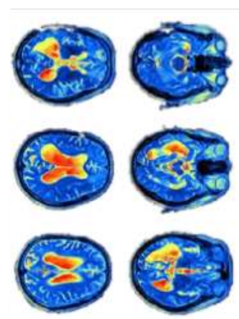
Introduced method: 85.52%

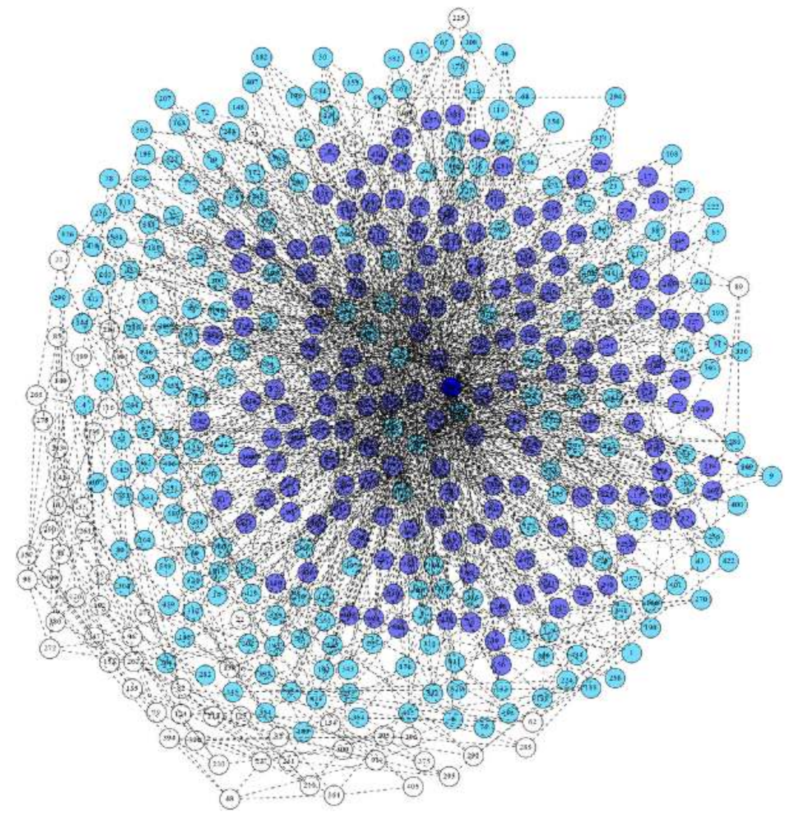
Same architecture with 3x3 kernels: 85.98%

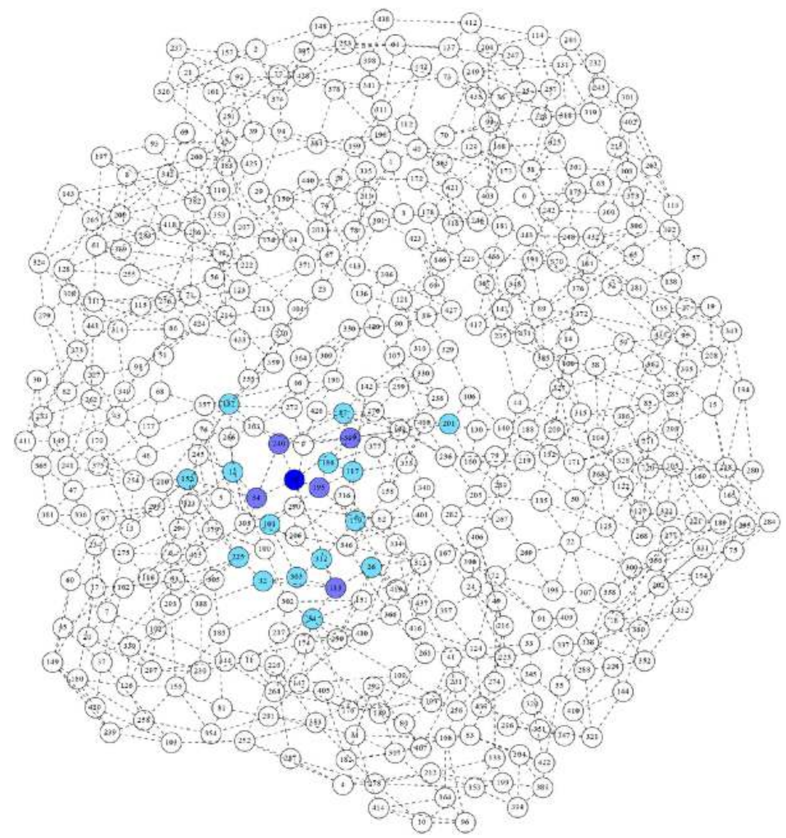
[1] Cheng et al. - Deep Haar scattering networks (2016)

Results









	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Subject 6	Average
Our method with $\mathcal{G}_{geo}$	56%	52%	47%	53%	42%	49%	49.9%
$\text{Def}_5(\mathcal{G}_{geo})$	60%	**56%**	**51**%	54%	**45%**	51%	52.8%
$\text{Def}_{10}(\mathcal{G}_{geo})$	61%	55%	50%	55%	42%	51%	52.3%
$\text{Def}_{20}(\mathcal{G}_{geo})$	**62%**	54%	**51%**	**56%**	43%	**53%**	**53.2%**
$\text{Def}_5(\mathcal{G}^*_K)$	61%	53%	50%	53%	43%	49%	51.5%
$\text{Def}_{10}(\mathcal{G}^*_K)$	59%	54%	50%	54%	**45%**	51%	52.2%
$\text{Def}_{20}(\mathcal{G}^*_K)$	61%	51%	49%	54%	42%	49%	51%
MLP	57%	51%	**51%**	51%	42%	49%	50.2%
SVM	50%	42%	35%	32%	31%	40%	38.2%
LR ($C = 1$, l_1)	51%	48%	44%	34%	42%	42%	43.6%
LR ($C = 50$, l_1)	47%	47%	38%	28%	34%	40%	38.9%
LR ($C = 1$, l_2)	49%	47%	41%	34%	31%	38%	39.7%
Ridge	35%	34%	28%	22%	23%	31%	28.8%

Ongoing work on this

$$\forall v_1, v_2 \in \mathcal{V} : (\psi(v_1) = \psi(v_2) \neq \perp) \Rightarrow (v_1 = v_2)$$

$$\forall v \in \mathcal{V} : (\{v, \psi(v)\} \in \mathcal{E}) \vee (\psi(v) = \perp)$$

Too strict
NP-complete

$$\forall v_1, v_2 \in \mathcal{V} : (\{v_1, v_2\} \in \mathcal{E} \Leftrightarrow \{\psi(v_1), \psi(v_2)\} \in \mathcal{E}) \\ \vee (\psi(v_1) = \perp) \vee (\psi(v_2) = \perp)$$

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$$\begin{aligned} s(\widetilde{\psi}) &= \alpha\ loss(\widetilde{\psi}) \\ &+ \beta\ \left|\left\{v \in \mathcal{V}_1 : \left\{v, \widetilde{\psi}(v)\right\} \notin \mathcal{E} \wedge \widetilde{\psi}(v) \neq \perp\right\}\right| \\ &+ \gamma \sum_{\substack{\{v_1, v_2\} \in \binom{\mathcal{V}_1}{2} \\ \widetilde{\psi}(v_1) \neq \perp \\ \widetilde{\psi}(v_2) \neq \perp}} \left| d(v_1, v_2) - d\left(\widetilde{\psi}(v_1), \widetilde{\psi}(v_2)\right) \right| \end{aligned}$$

Ongoing work on this

Algorithm 1: $v_{src}\_to\_v_{tgt}$ $(\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle, \mathcal{V}_1, v_{src}, v_{tgt})$

foreach $v \in \mathcal{V}$ **do**
- $visited[v]$:= **false**;
- $predecessors[v]$:= **NULL**;

$Q := \{\langle 0, v_{src}, \textbf{NULL}, \textbf{NULL} \rangle\}$;

while $Q \neq \emptyset$ **do**
- $total\_score_{v_1}, v_1, pred, \widetilde{\psi}_{pred \to v_1} := \underline{extract\_min}(Q)$;
- **if not** $visited[v_1]$ **then**
  - $visited[v_1]$:= **true**;
  - $predecessors[v_1] := \langle pred, \widetilde{\psi}_{pred \to v_1} \rangle$;
  - **if** $v_1 = v_{tgt}$ **then**
    - **break**;
  - **if** $\widetilde{\psi}_{pred \to v_1} \neq$ **NULL then**
    - $\mathcal{V}_1 := \left\{ v \in \mathcal{V} \,\middle|\, \widetilde{\psi}_{pred \to v_1}(v) \neq \bot \right\}$;
  - $\mathcal{V}_2 := \underline{choose\_subset}(\mathcal{V})$;
  - **foreach** $v_2 \in \mathcal{V}_2 \setminus \{v_1\}$ **do**
    - **if not** $visited[v_2]$ **then**
      - $\widetilde{\psi}_{v_1 \to v_2}, score_{\widetilde{\psi}_{v_1 \to v_2}} := \underline{minimize\_s}(v_1, v_2, \mathcal{G}, \mathcal{V}_1, \mathcal{V}_2)$;
      - $total\_score_{v_2} := total\_score_{v_1} + score_{\widetilde{\psi}_{v_1 \to v_2}}$;
      - $Q := Q \cup \left\{ \langle total\_score_{v_2}, v_2, v_1, \widetilde{\psi}_{v_1 \to v_2} \rangle \right\}$;

return $predecessors$;

From here

To here

Target of approx. trans.

NP-complete Heuristics!

Ongoing work on this

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            break;
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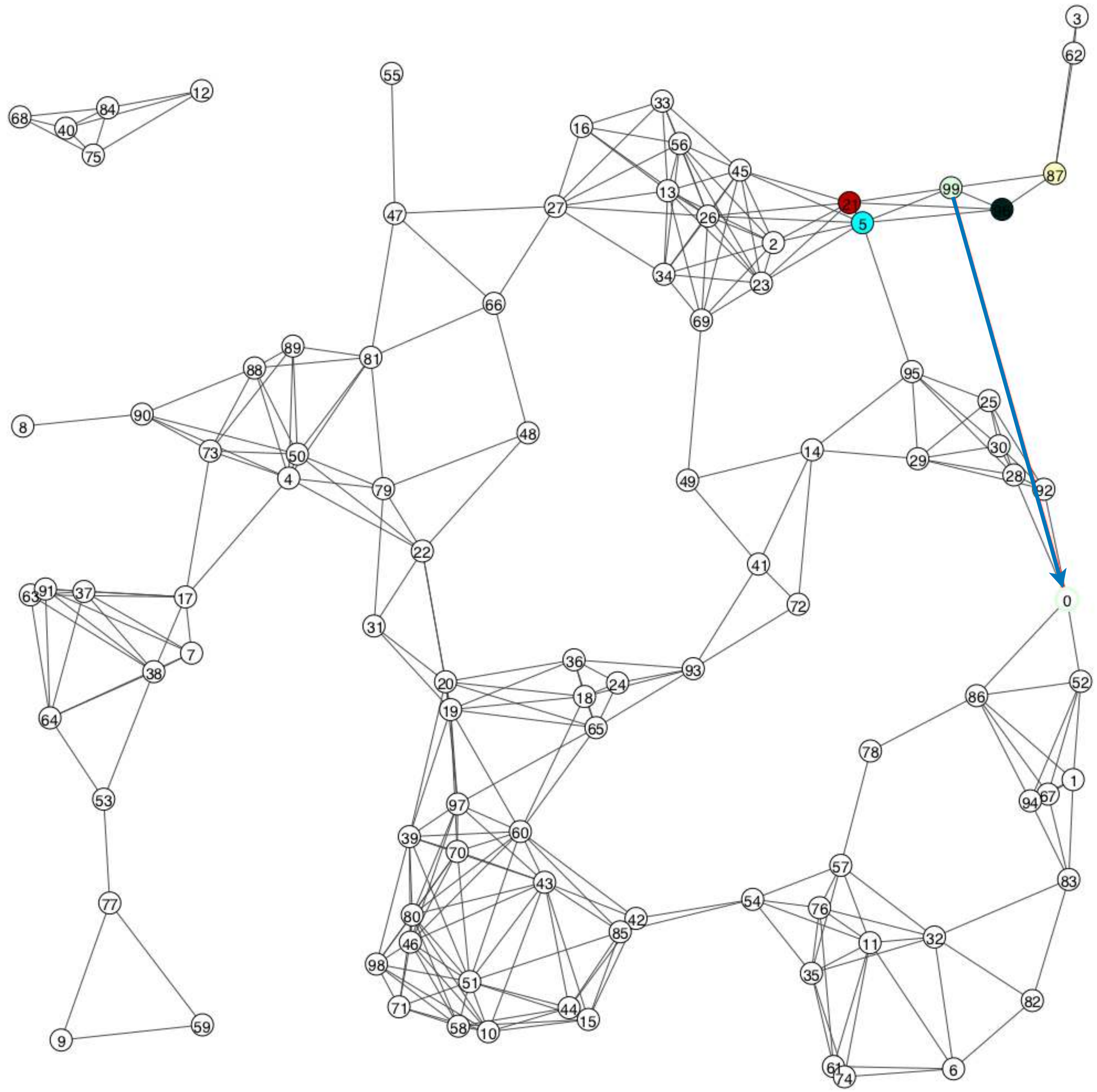
                $total\_score_{v_2} :=$ $total\_score_{v_1} + score_{\widetilde{\psi}_{v_1 \rightarrow v_2}}$;
                $Q :=$ $Q \cup \left\{ \langle total\_score_{v_2}, v_2, v_1, \widetilde{\psi}_{v_1 \rightarrow v_2} \rangle \right\}$;
return $predecessors$;

Algorithm 2: $minimize\_s$ $(v_1, v_2, \mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle, \mathcal{V}_1, \mathcal{V}_2)$

foreach $v \in \mathcal{V}_1$ **do**
    $\widetilde{\psi}_{v_1 \rightarrow v_2}(v) :=$ **NULL**;
$\widetilde{\psi}_{v_1 \rightarrow v_2}(v_1) := v_2$;
$score_{\widetilde{\psi}_{v_1 \rightarrow v_2}} := 0$;
for $i = 1$ **to** $\left\lceil \frac{|\mathcal{V}_1|}{K} \right\rceil$ **do**
    $current\_best_{\widetilde{\psi}} :=$ **NULL**;
    $current\_best\_score := \infty$;
    $\mathcal{V}_1^- := \mathcal{V}_1 \setminus \left\{ v \in \mathcal{V}_1 \;\middle|\; best_{\widetilde{\psi}_{v_1 \rightarrow v_2}}(v) \neq \mathbf{NULL} \right\}$;
    $\mathcal{V}_2^- := \mathcal{V}_2 \setminus \left\{ best_{\widetilde{\psi}_{v_1 \rightarrow v_2}}(v), \forall v \in \mathcal{V}_1 \right\} \cup \underbrace{\{\bot, \ldots, \bot\}}_{K\ times}$;
    foreach $(v_{11}, \ldots, v_{1K}) \in \underline{permutations}(\mathcal{V}_1^-, K)$ **do**
        foreach $(v_{21}, \ldots, v_{2K}) \in$ $\underline{unique\_permutations}(\mathcal{V}_2^-, K)$ **do**
            $\widetilde{\psi} := \widetilde{\psi}_{v_1 \rightarrow v_2}$;
            for $j = 1$ **to** K **do**
                $\widetilde{\psi}[v_{1j}] := v_{2j}$;
            $score := \underline{s}(\widetilde{\psi})$;
            if $score < current\_best\_score$ **then**
                $current\_best\_score := score$;
                $current\_best_{\widetilde{\psi}} := \widetilde{\psi}$;
    $\widetilde{\psi}_{v_1 \rightarrow v_2} := current\_best_{\widetilde{\psi}}$;
    $score_{\widetilde{\psi}_{v_1 \rightarrow v_2}} := current\_best\_score$;
return $\widetilde{\psi}_{v_1 \rightarrow v_2}, score_{\widetilde{\psi}_{v_1 \rightarrow v_2}}$;

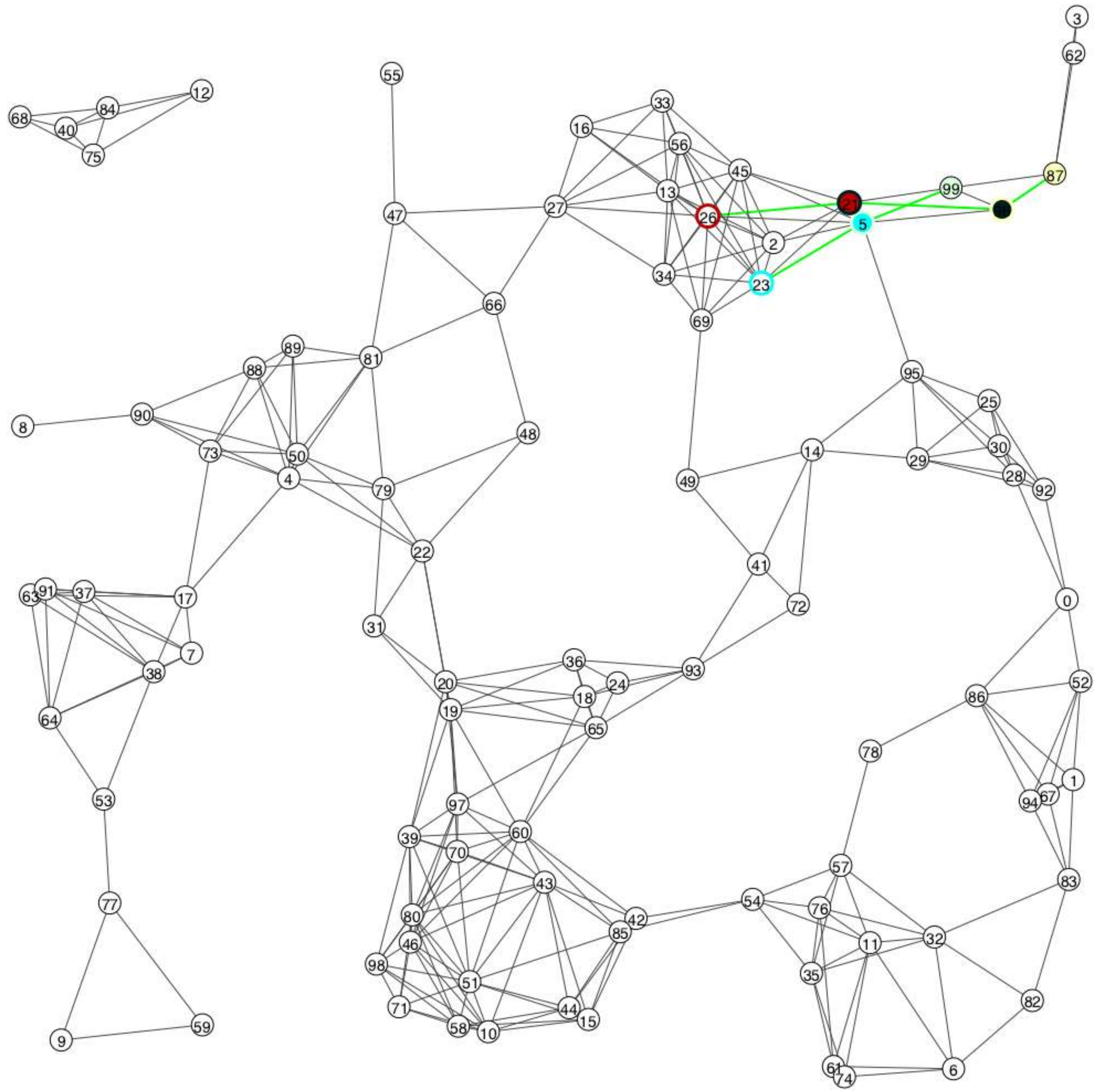
Controls complexity

Results — Random geometric graph (1)



Objective

Results — Random geometric graph (1)



$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss = 40%* *~~SNP~~ = 0%*

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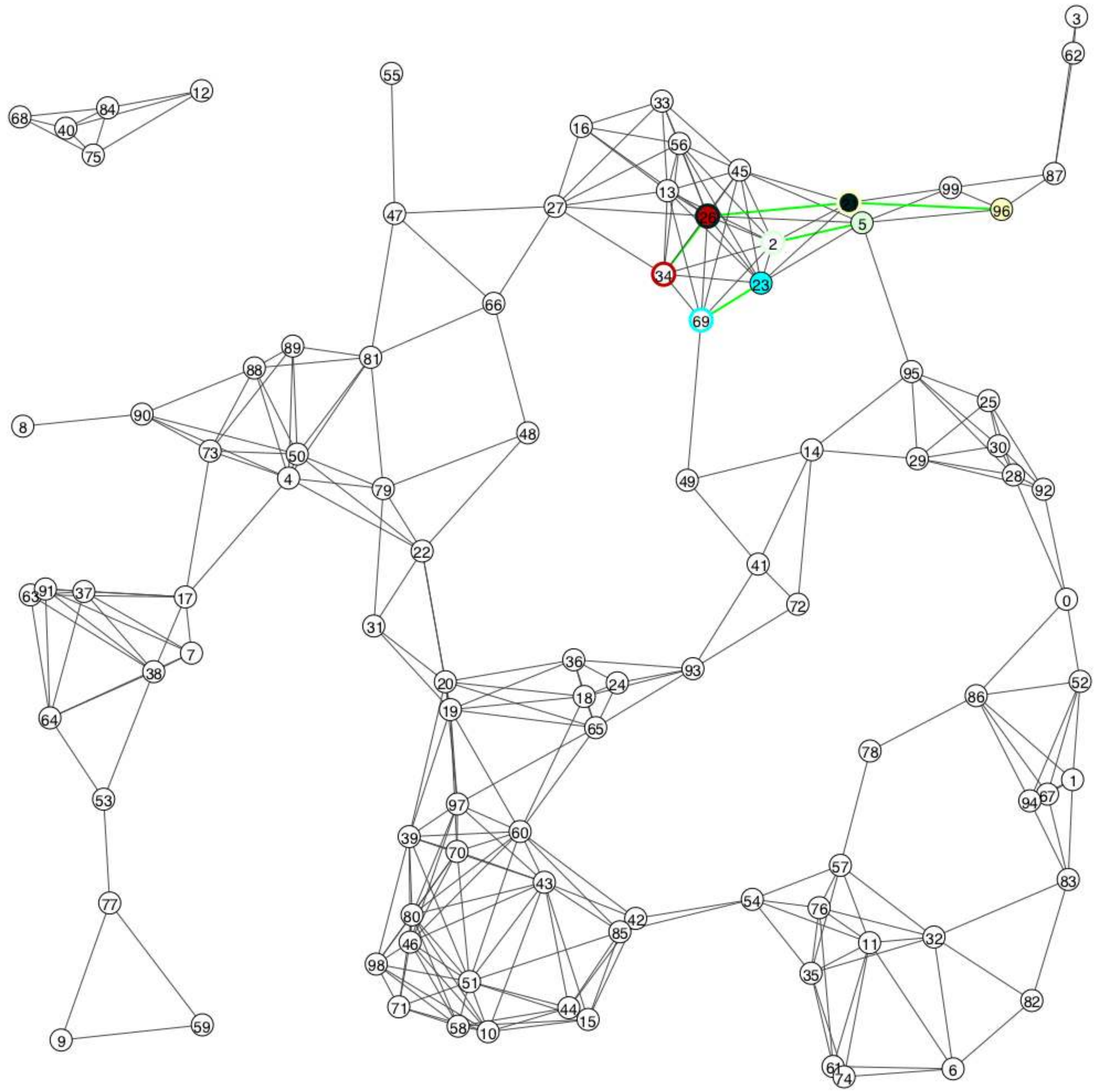

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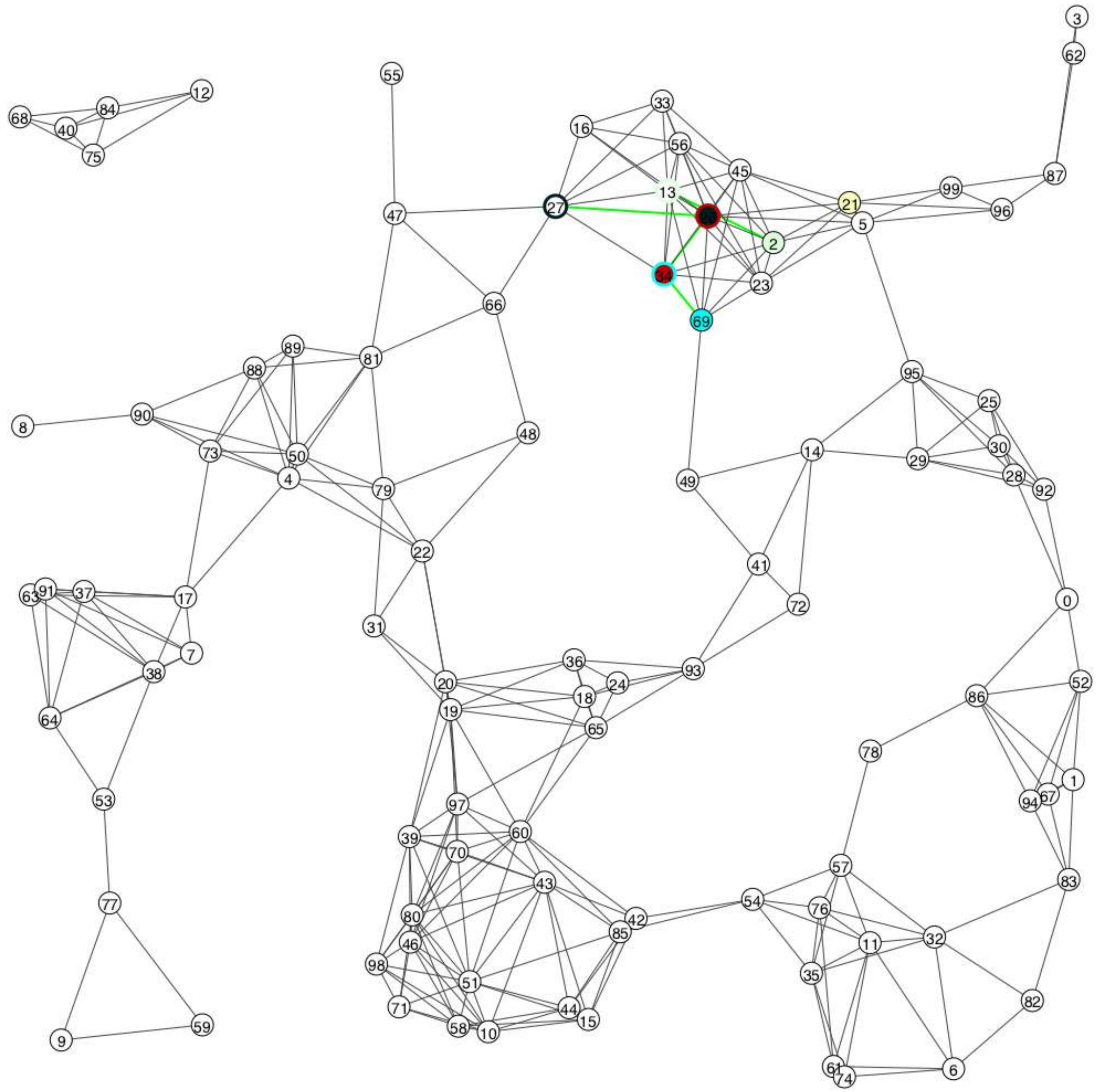

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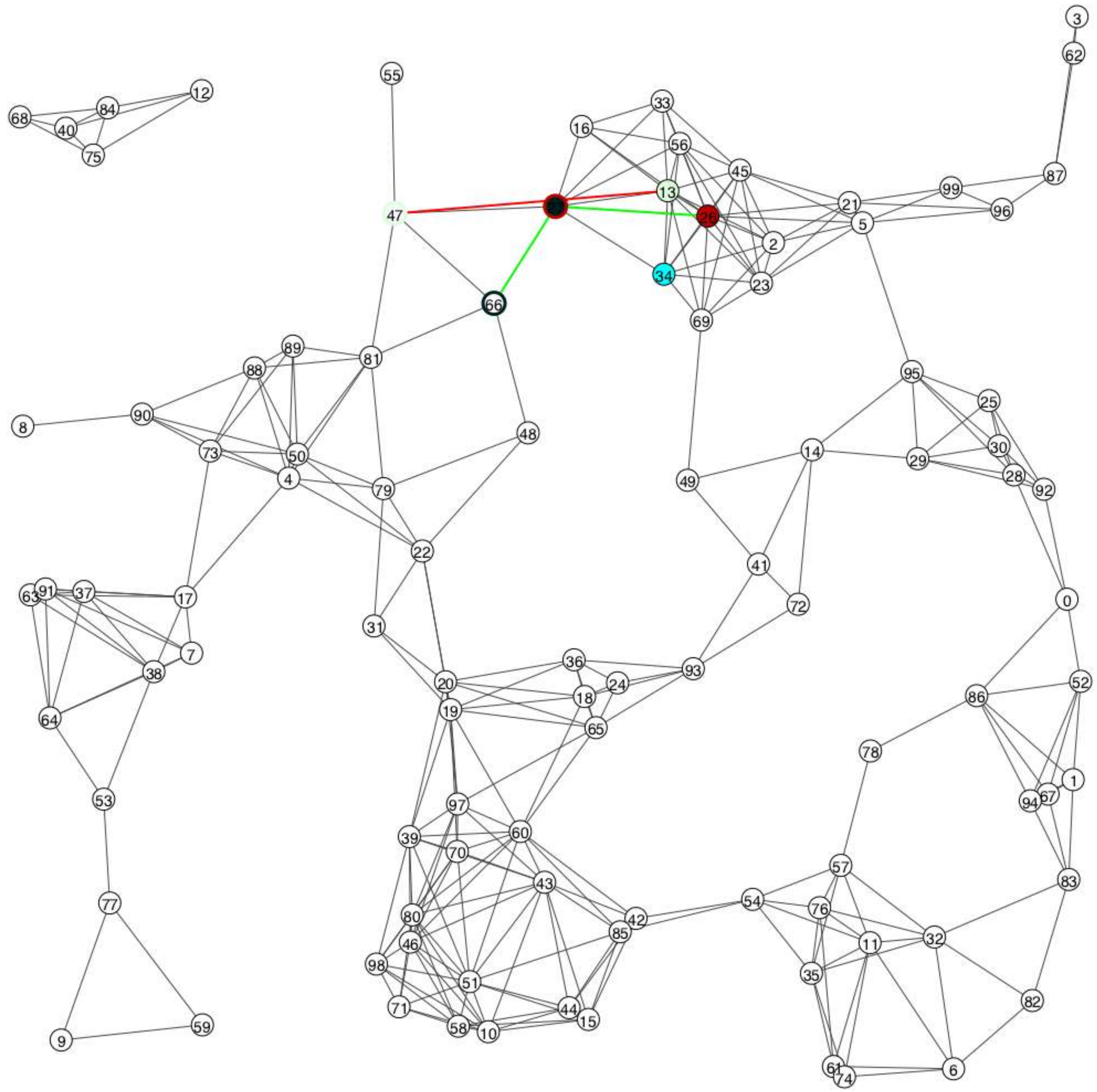

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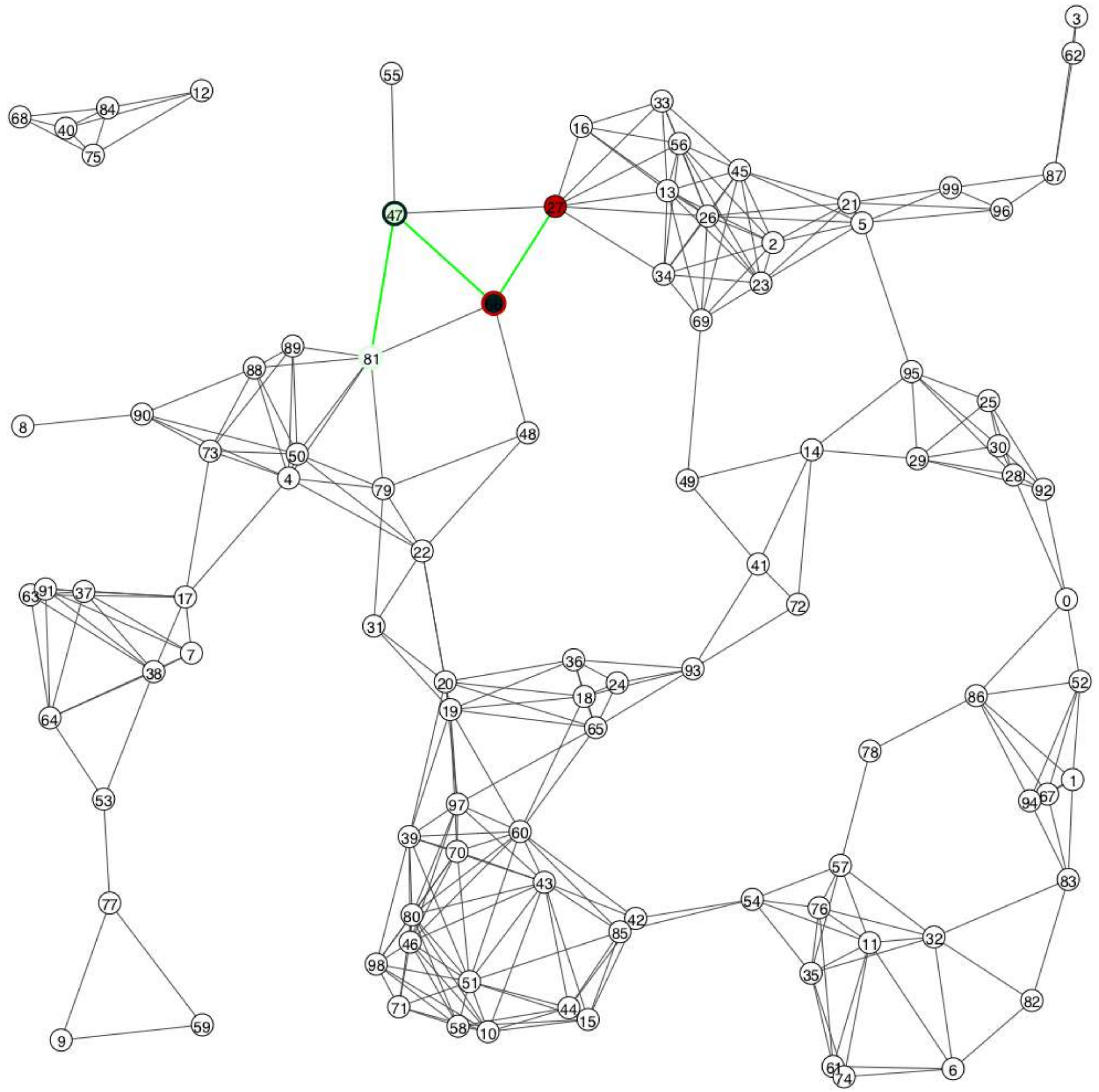

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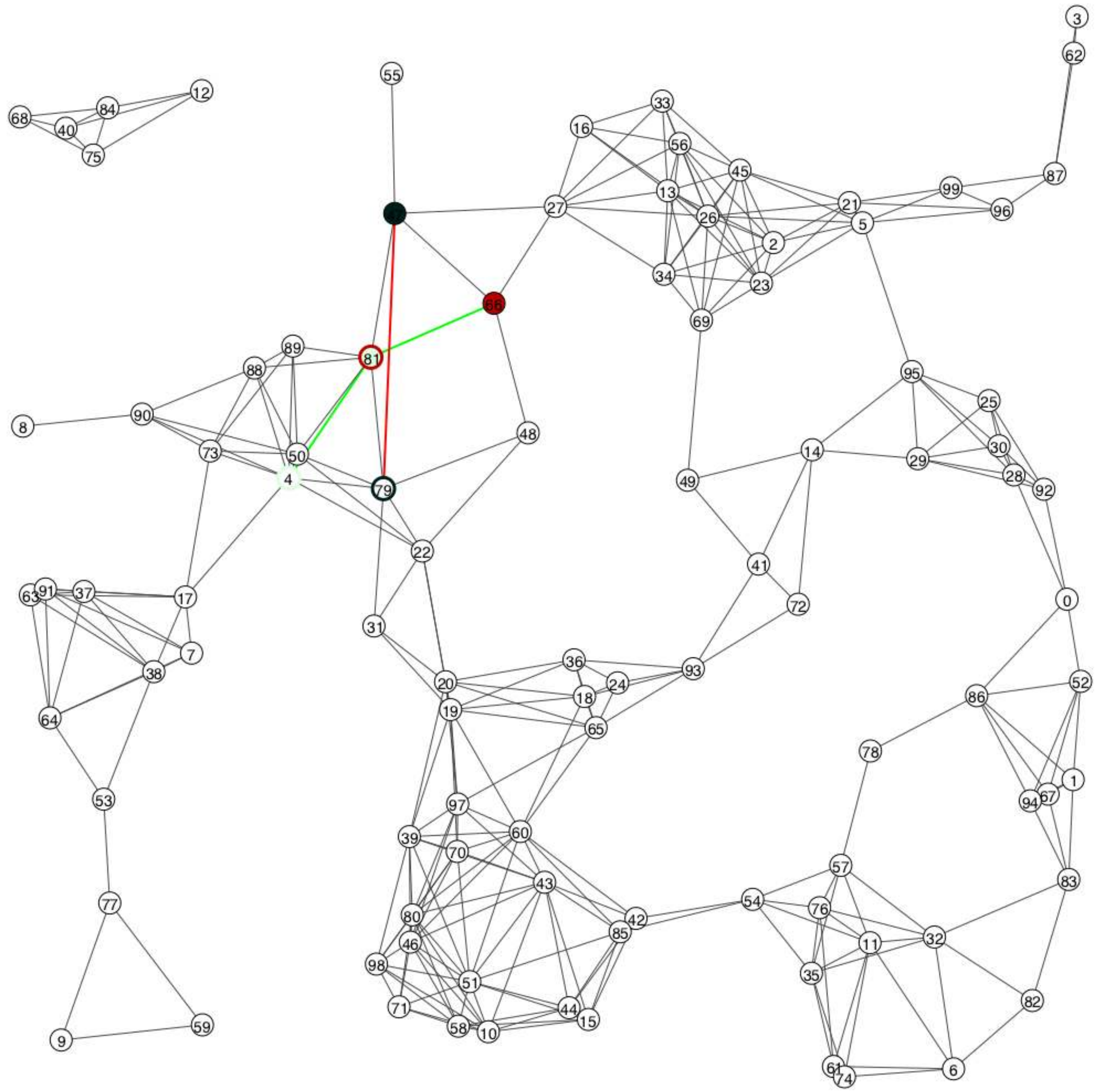

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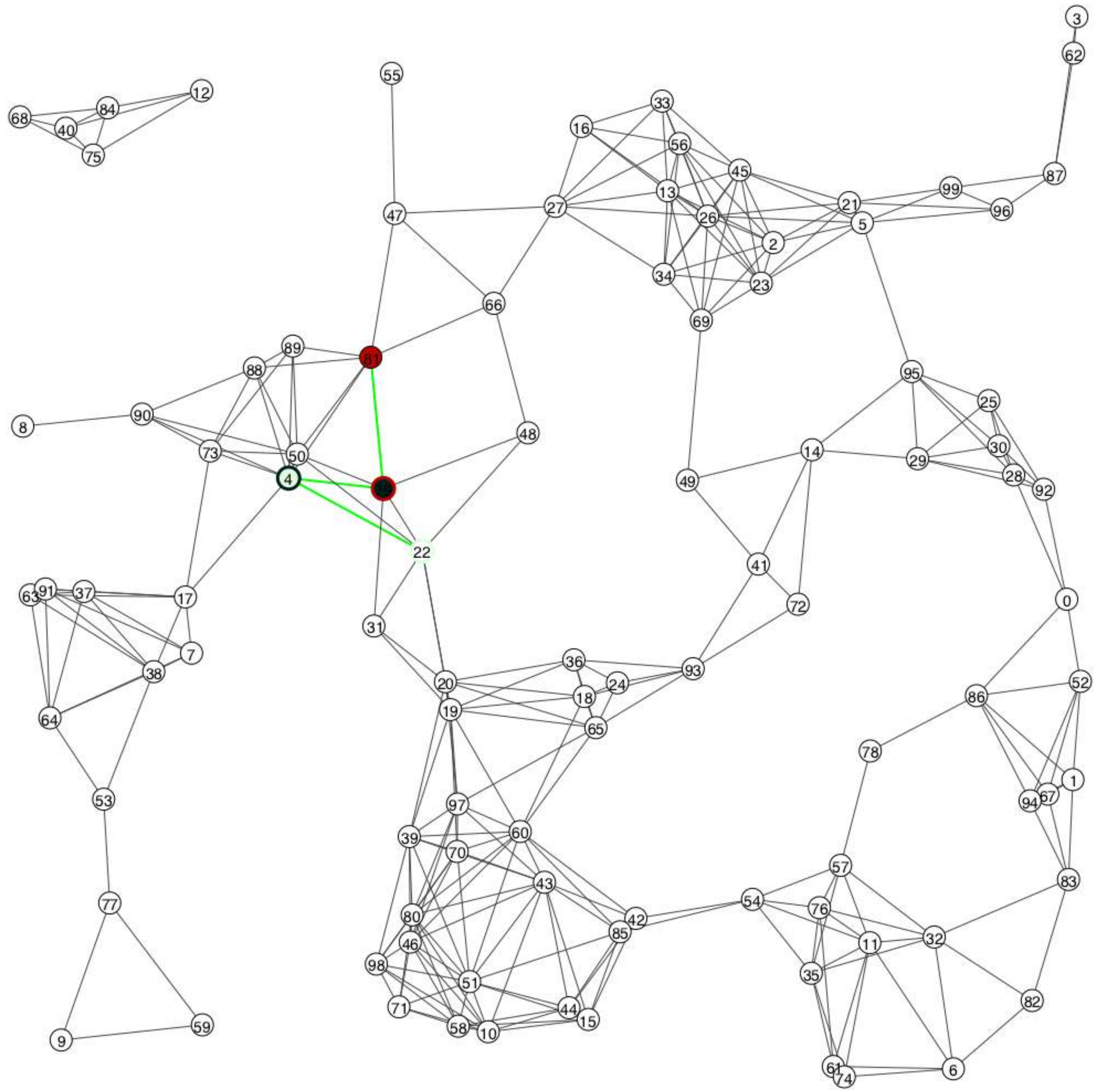

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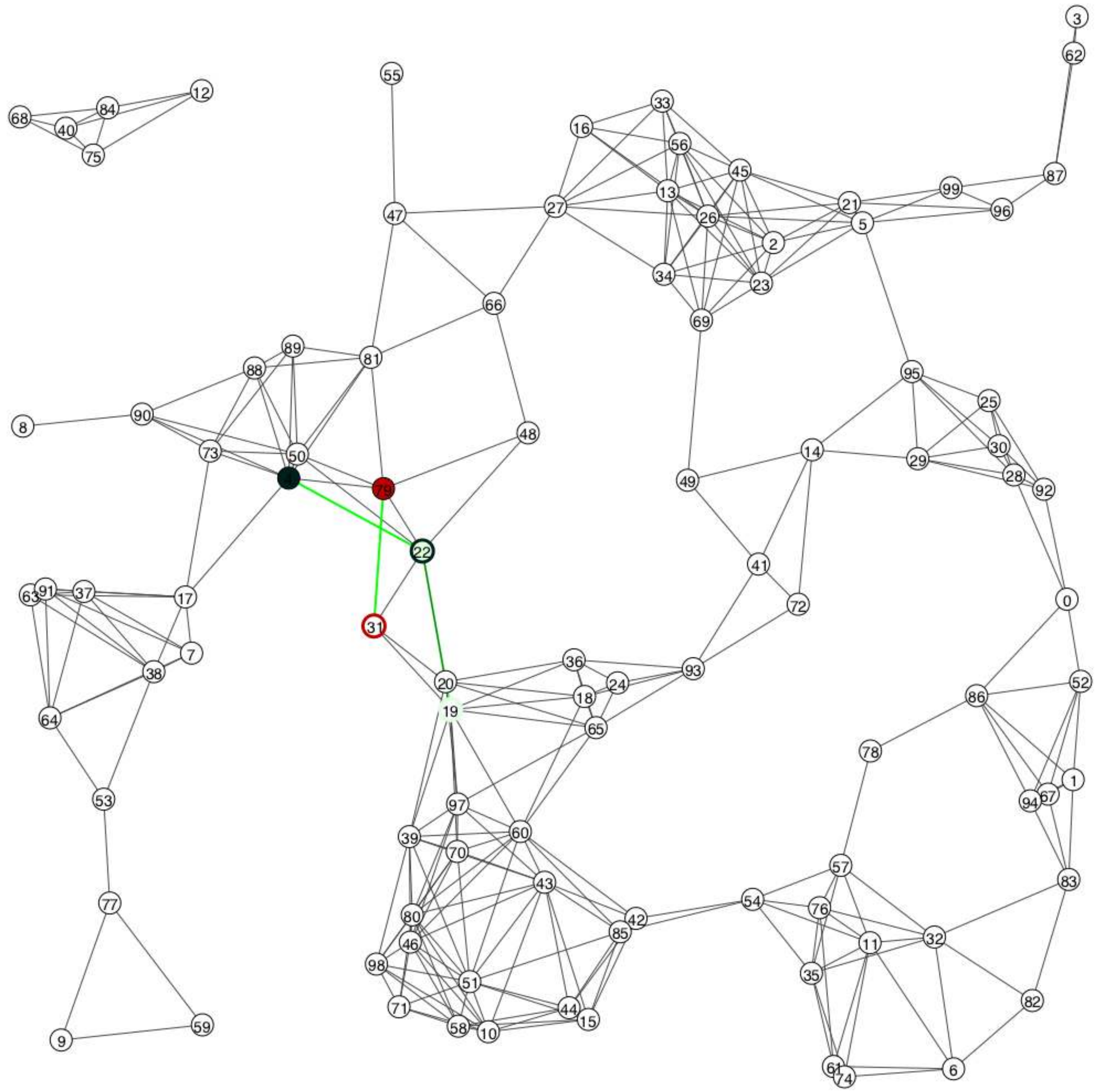
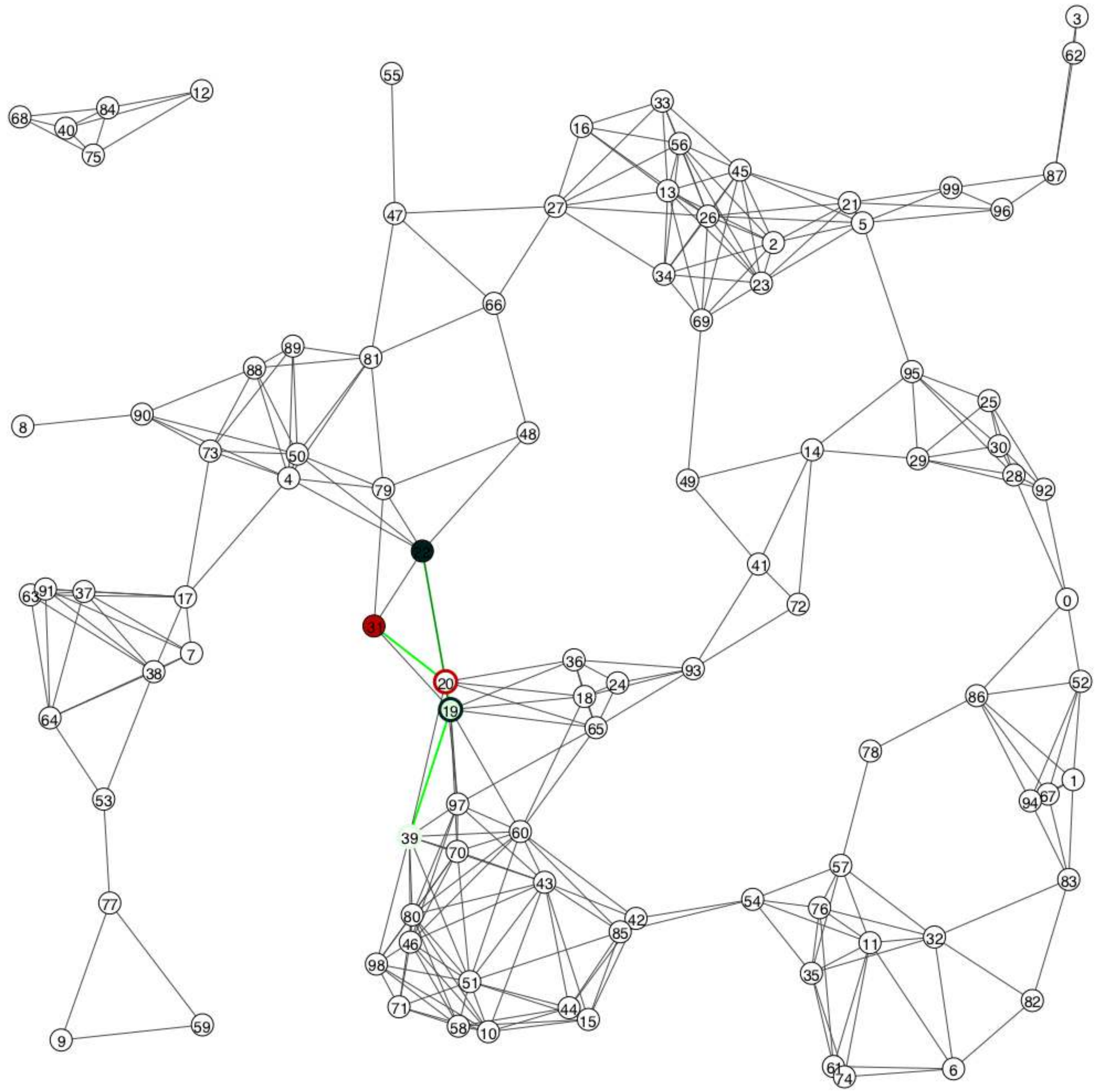


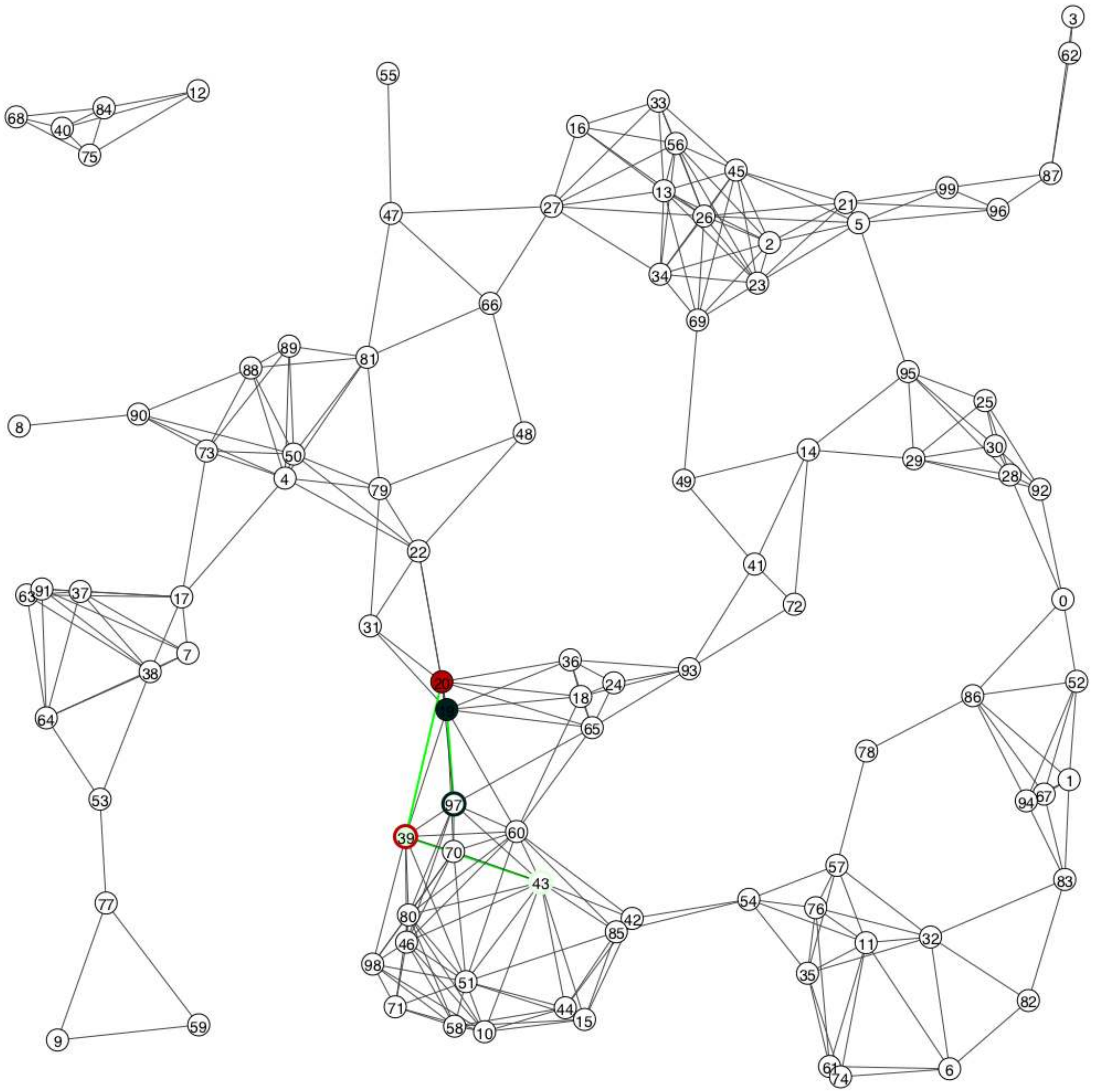
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss = 40%* *~~SNP~~ = 0%*

Results — Random geometric graph (1)



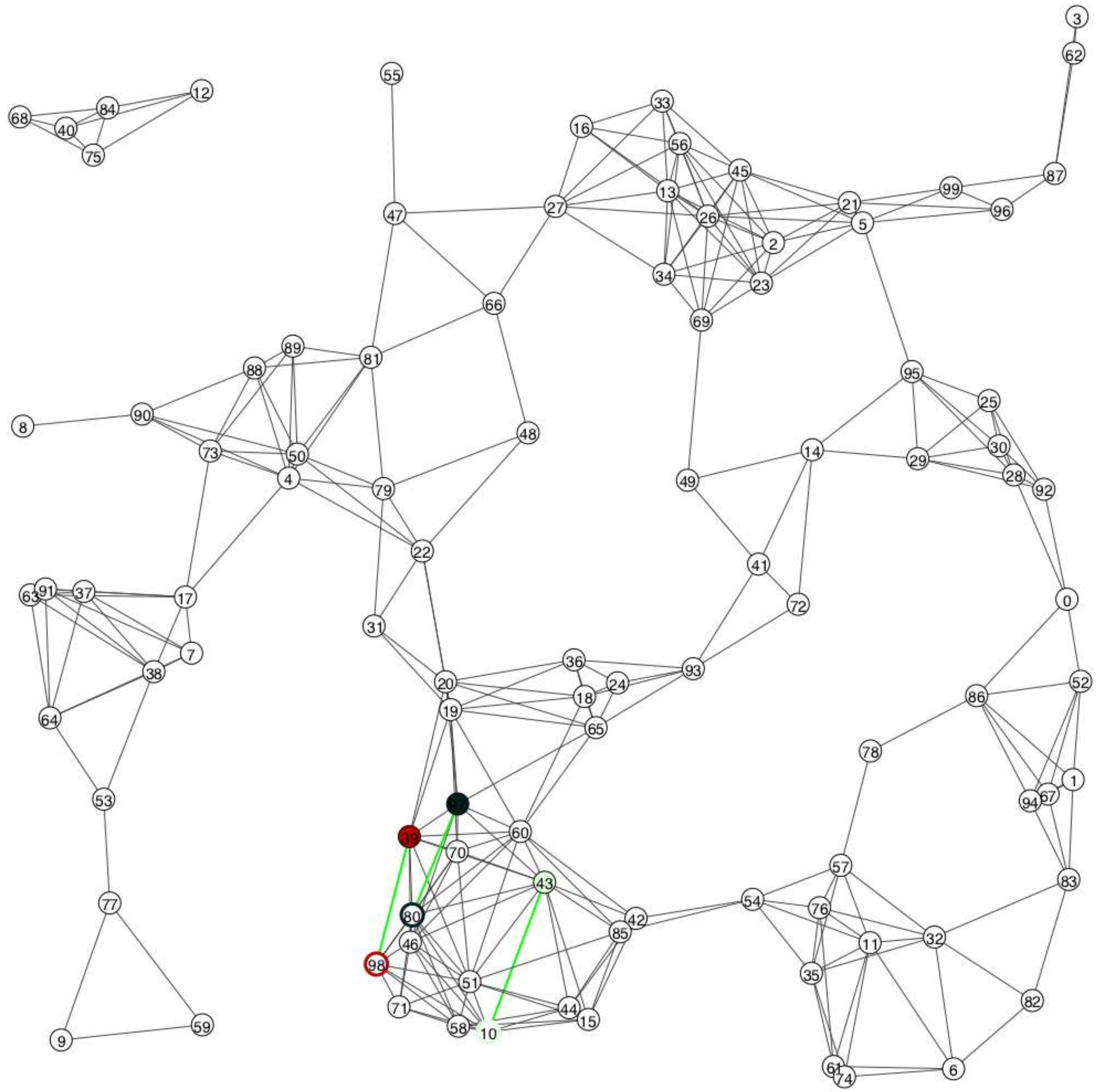
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss* = 40% ~~*SNP*~~ = 0%

Results — Random geometric graph (1)



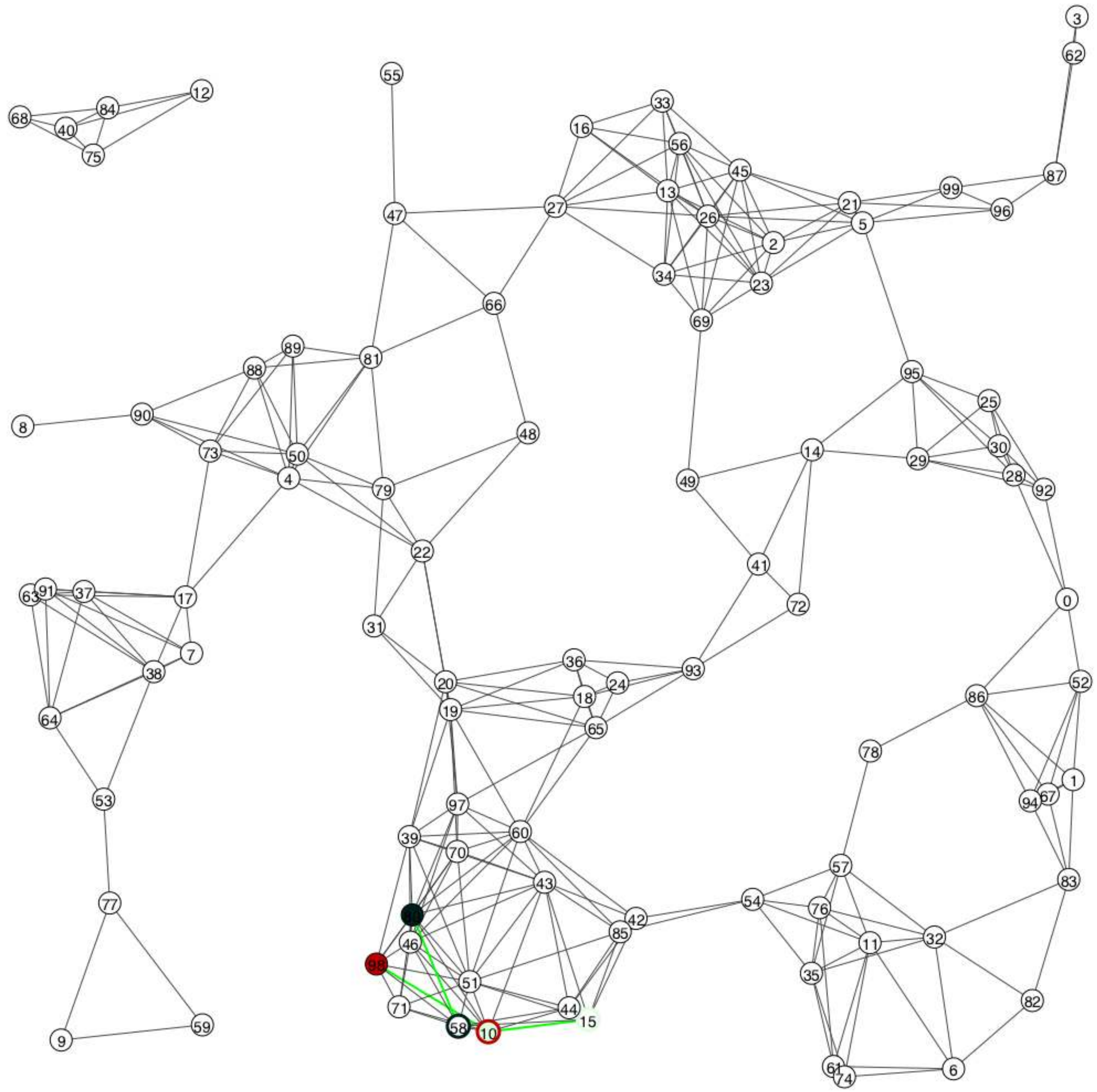
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss = 40%* *~~SNP~~ = 0%*

Results — Random geometric graph (1)



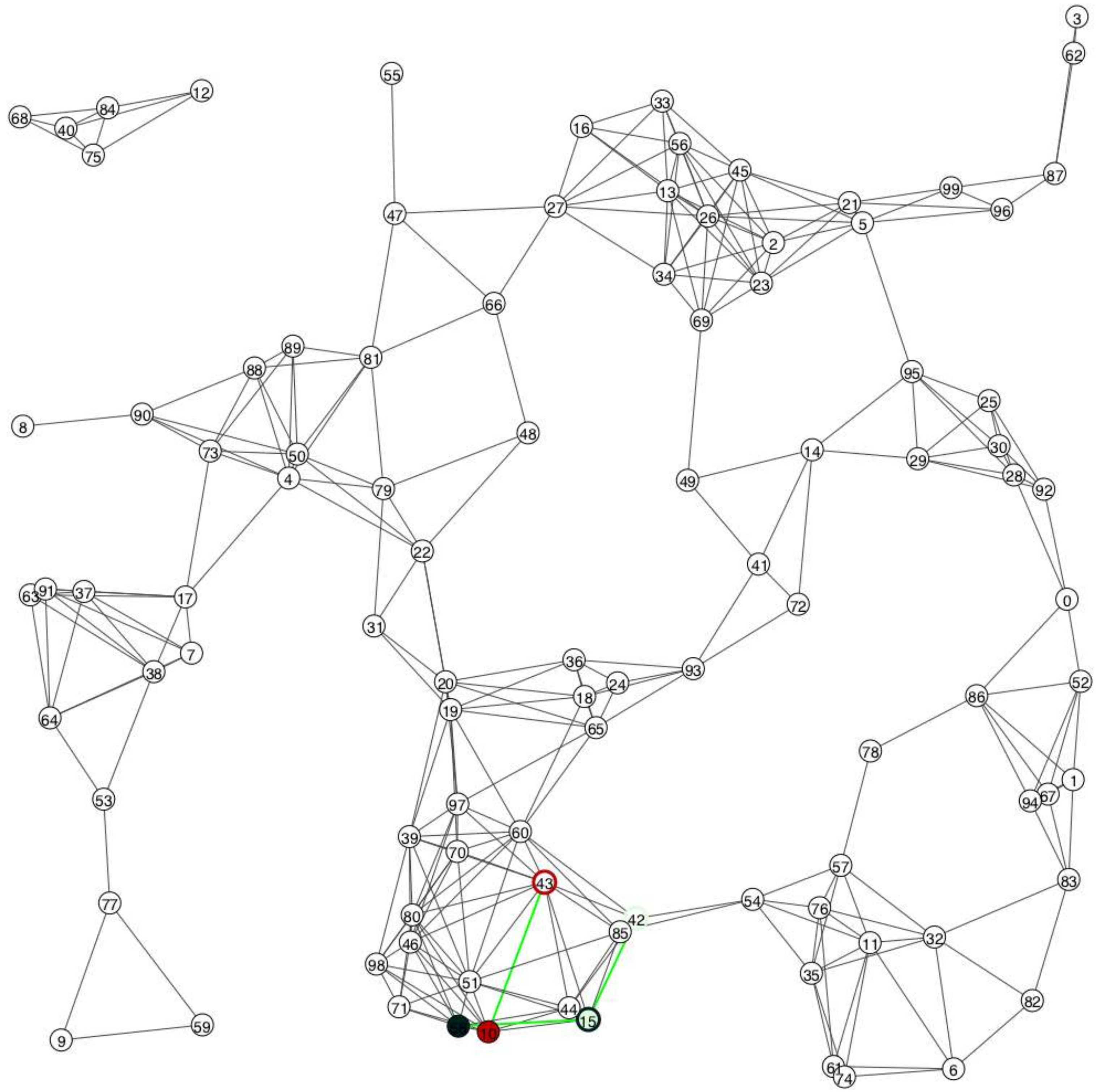
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ → *loss = 40%* $\cancel{SNP}$ = 0%

Results — Random geometric graph (1)



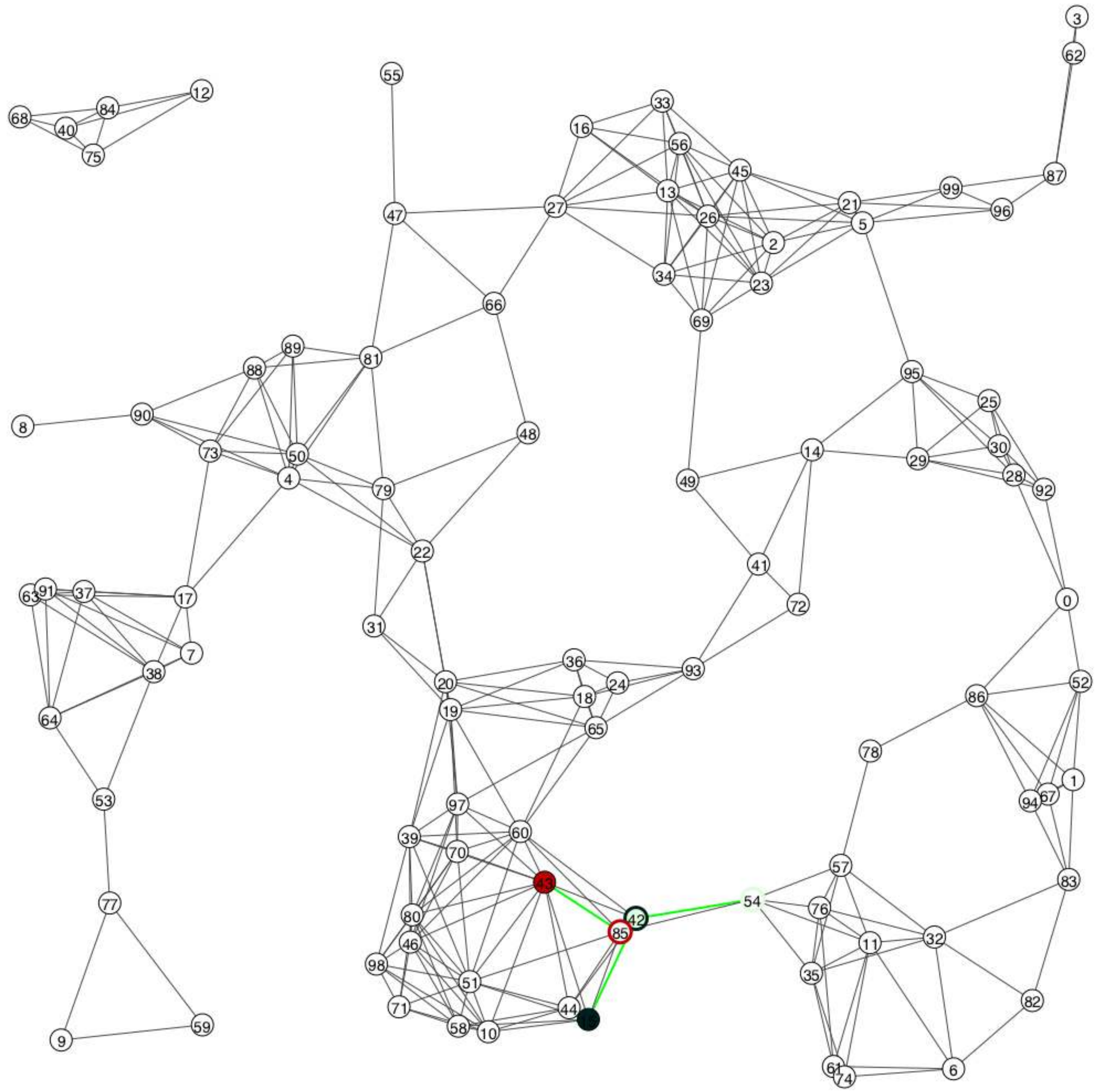
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ $loss = 40\%$ $\cancel{SNP} = 0\%$

Results — Random geometric graph (1)



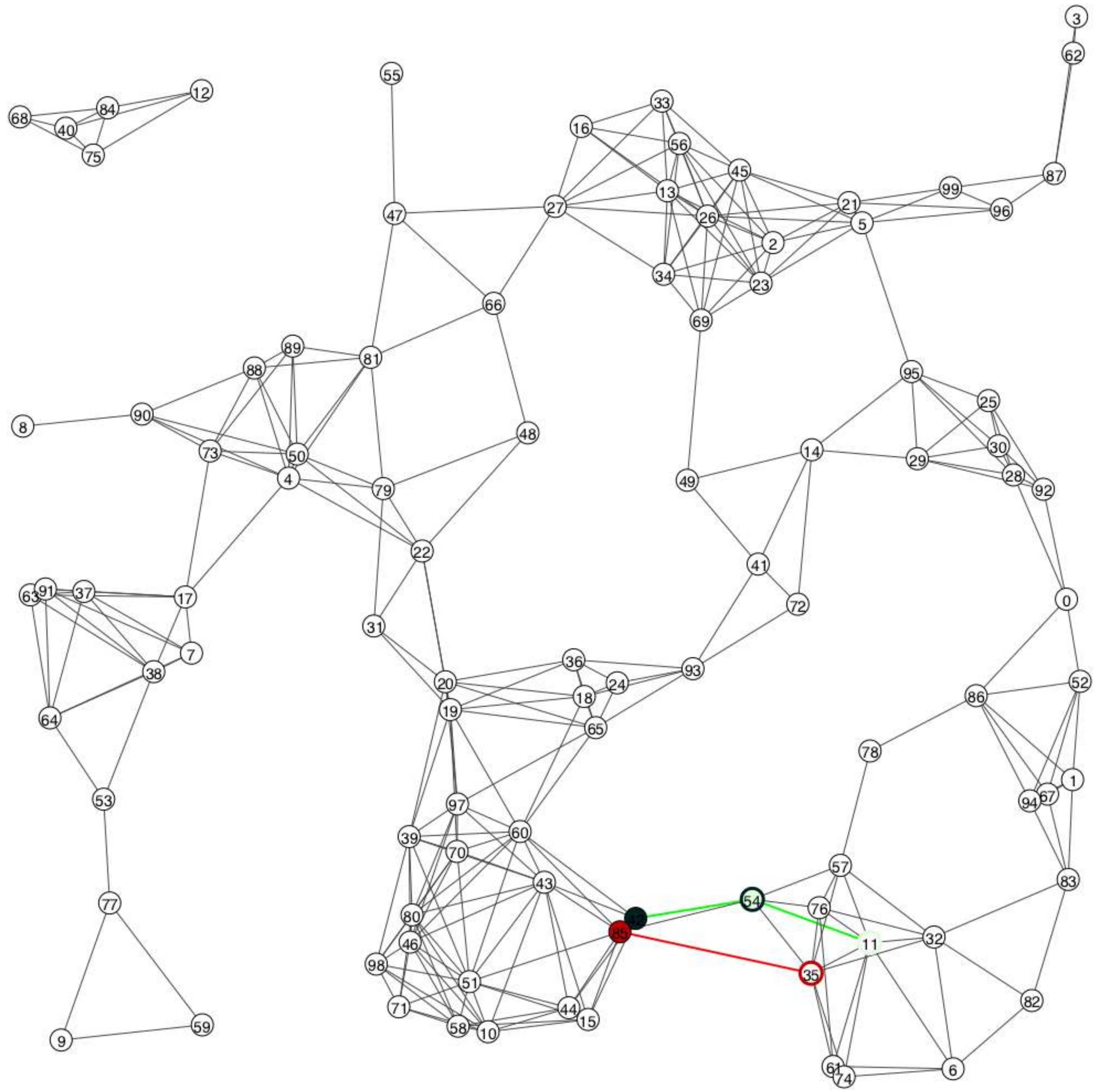
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss = 40%* *~~SNP~~ = 0%*

Results — Random geometric graph (1)



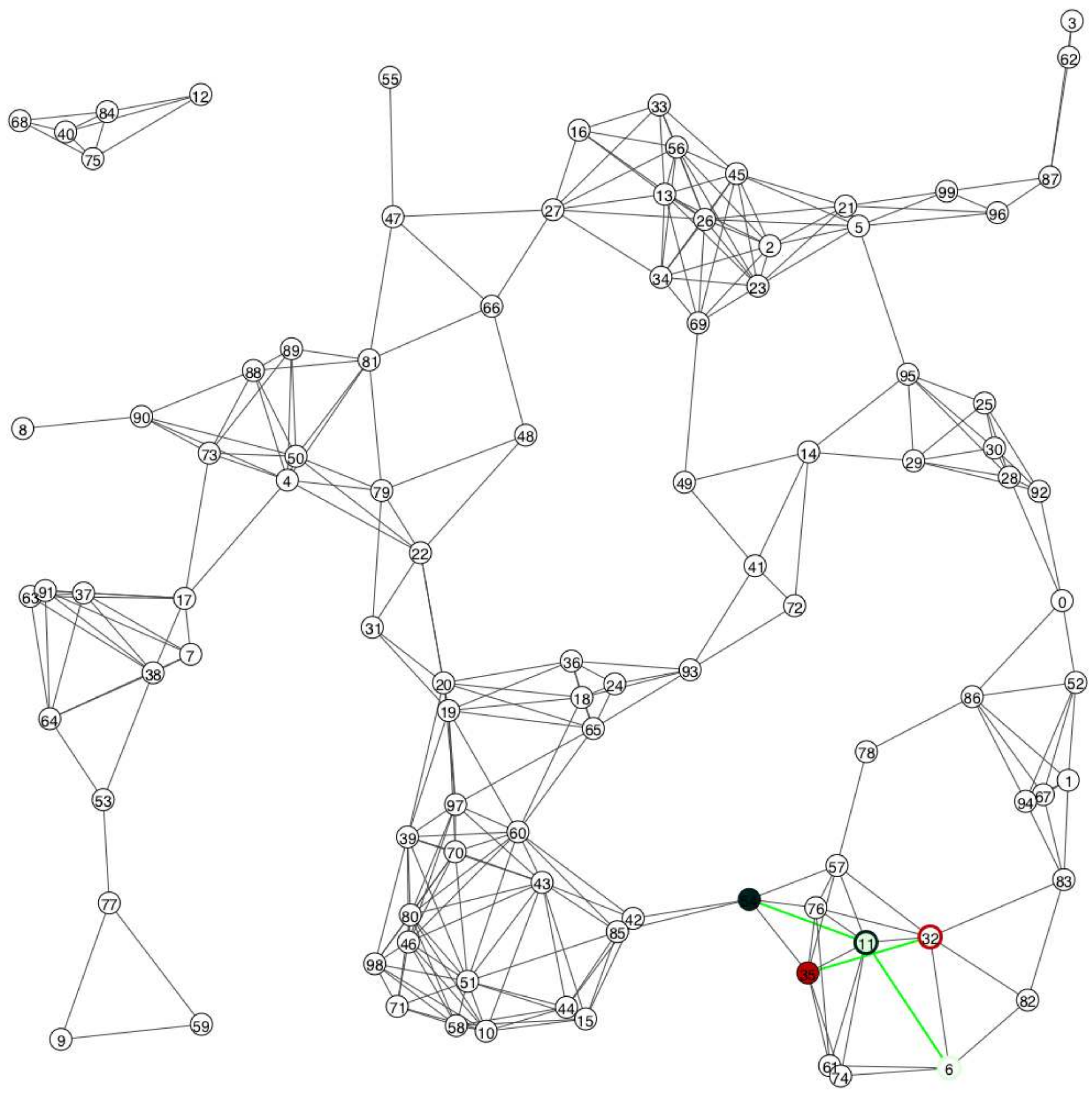
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss* = 40% ~~*SNP*~~ = 0%

Results — Random geometric graph (1)



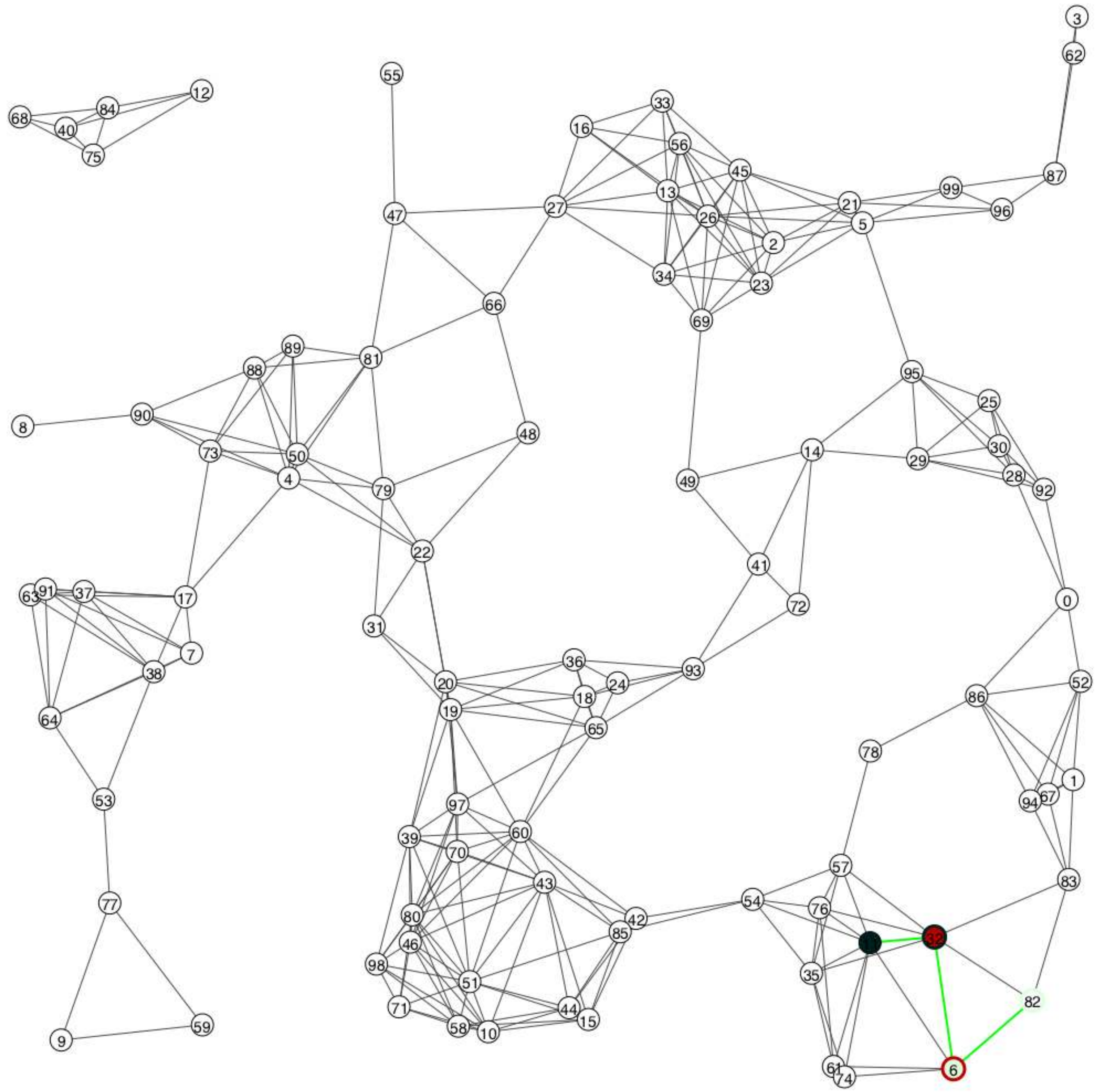
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ $loss = 40\%$ $\cancel{SNP} = 0\%$

Results — Random geometric graph (1)



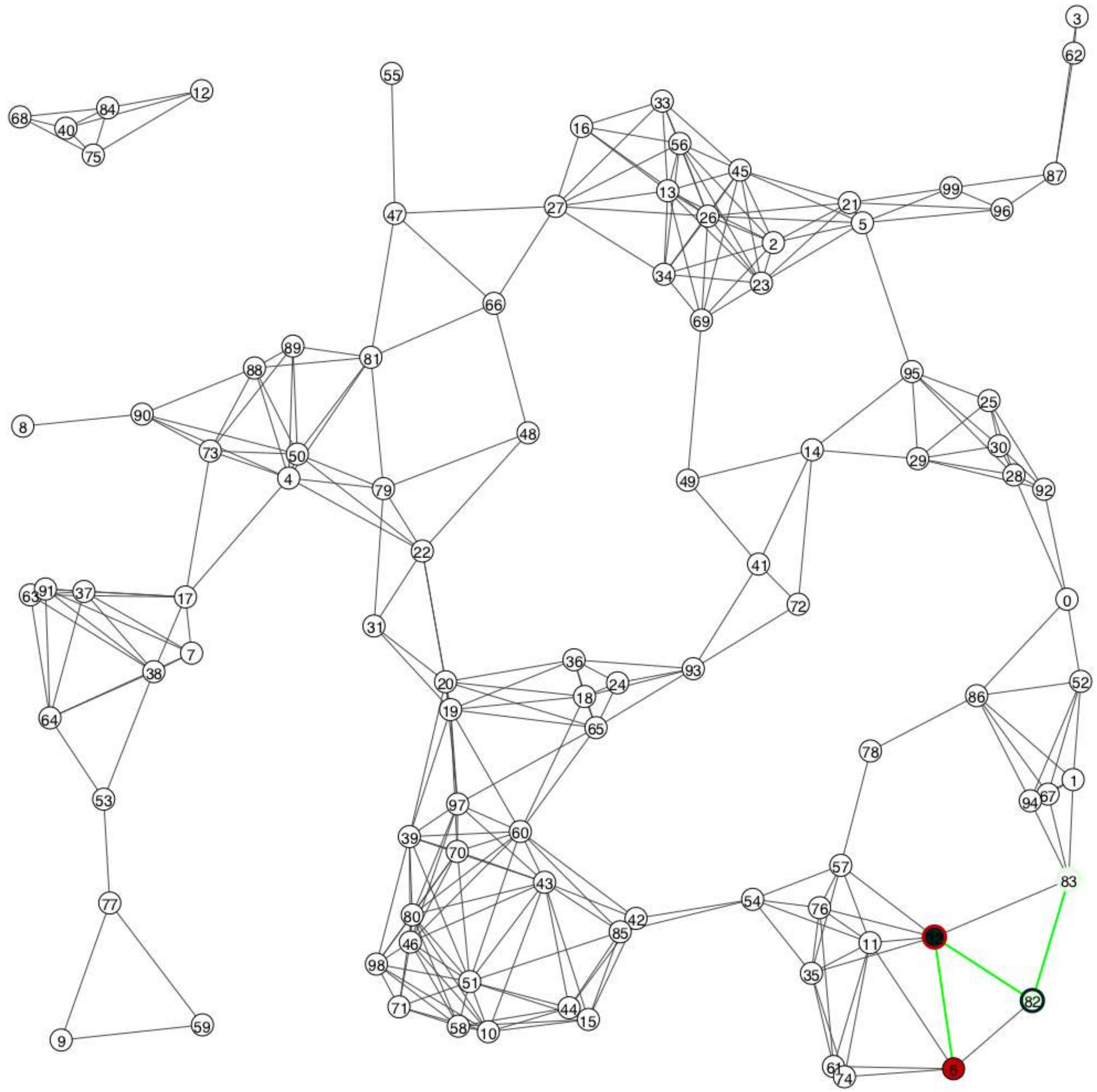
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss = 40%* *~~SNP~~ = 0%*

Results — Random geometric graph (1)



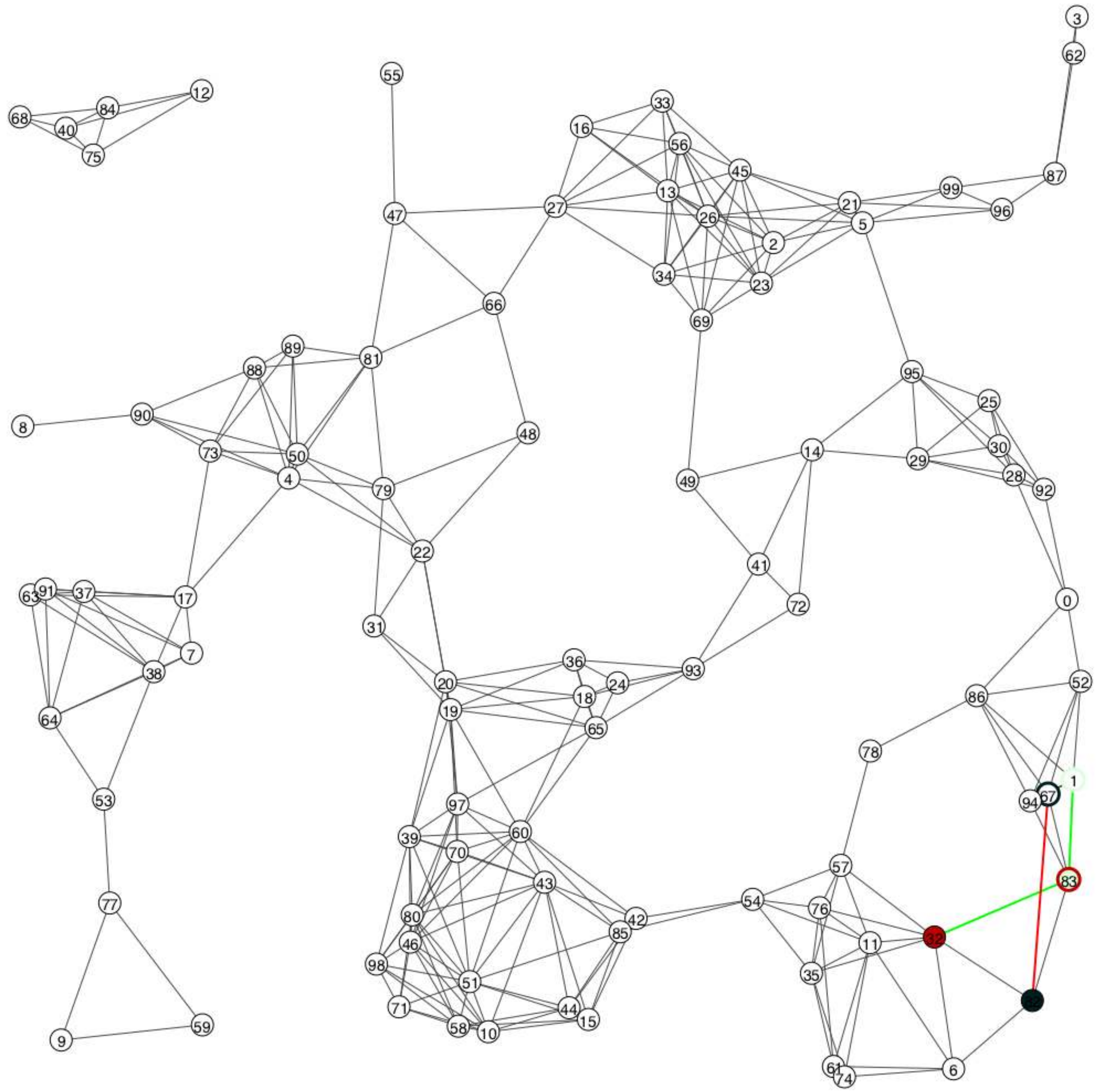
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss = 40%* *~~SNP~~ = 0%*

Results — Random geometric graph (1)



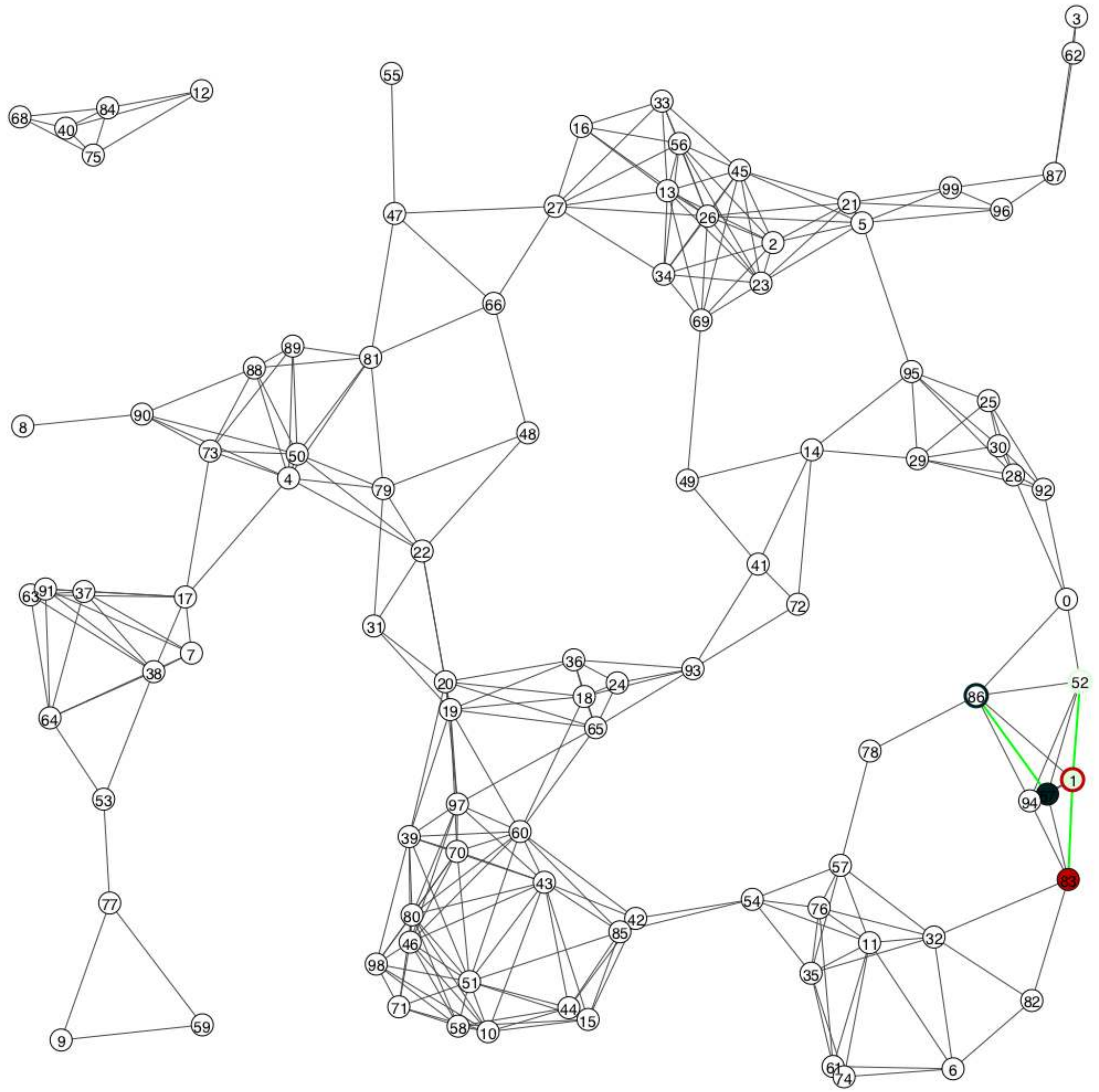
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss* = 40% ~~*SNP*~~ = 0%

Results — Random geometric graph (1)



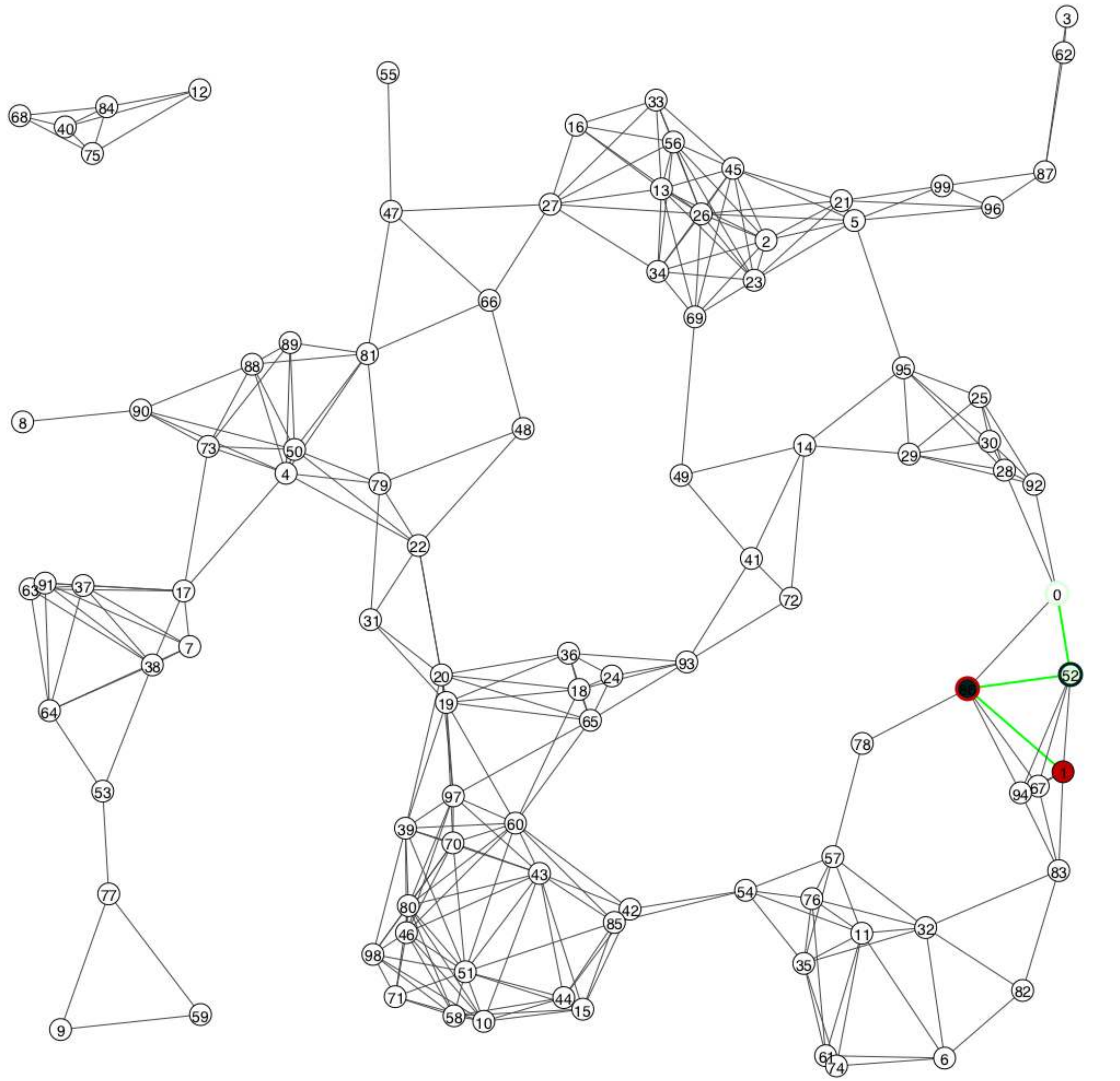
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss* = 40% ~~*SNP*~~ = 0%

Results — Random geometric graph (1)



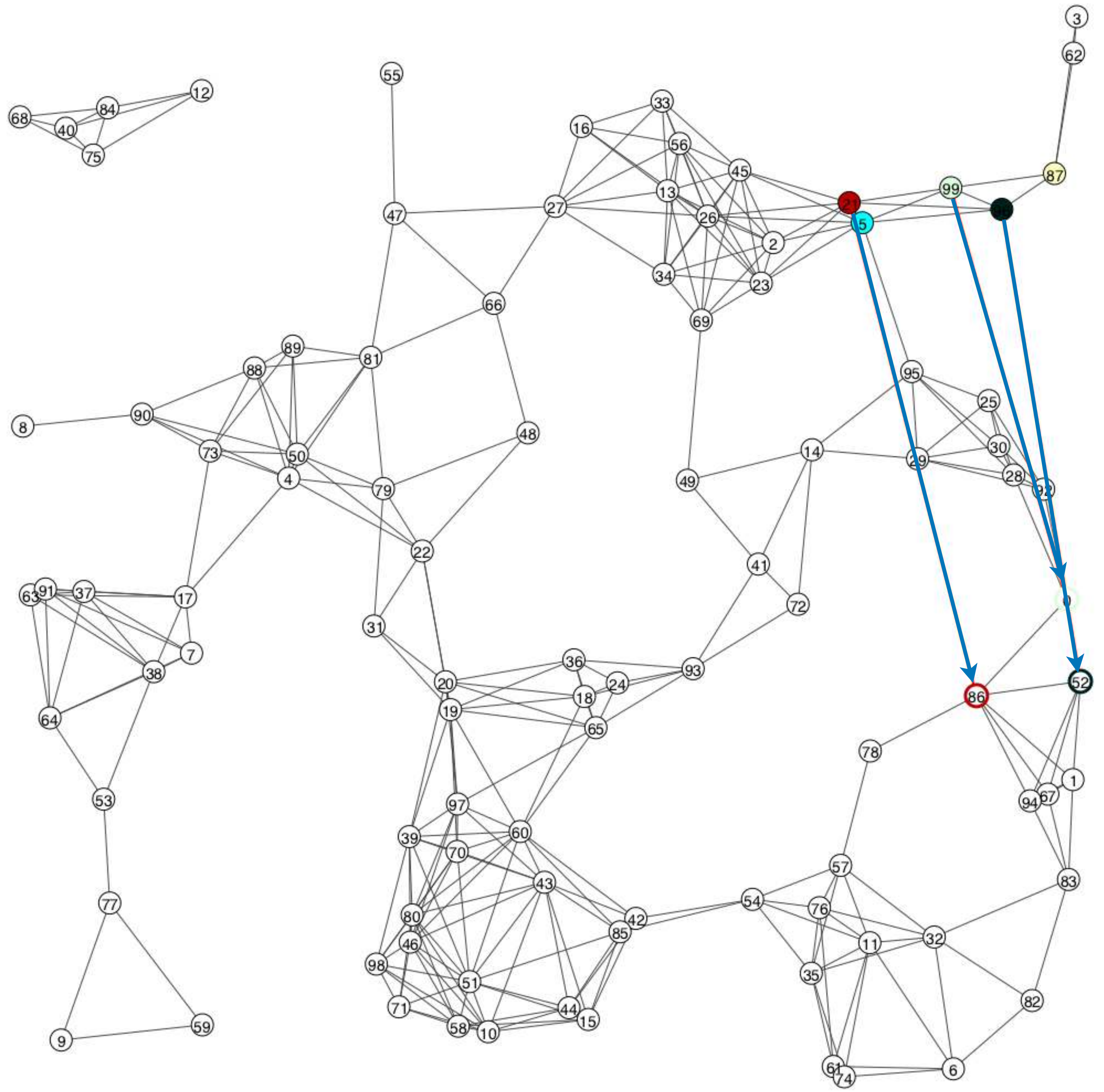
K $\in$ {1, 2, 3, ∞} $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔   *loss* = 40%   ~~*SNP*~~ = 0%

Results — Random geometric graph (1)



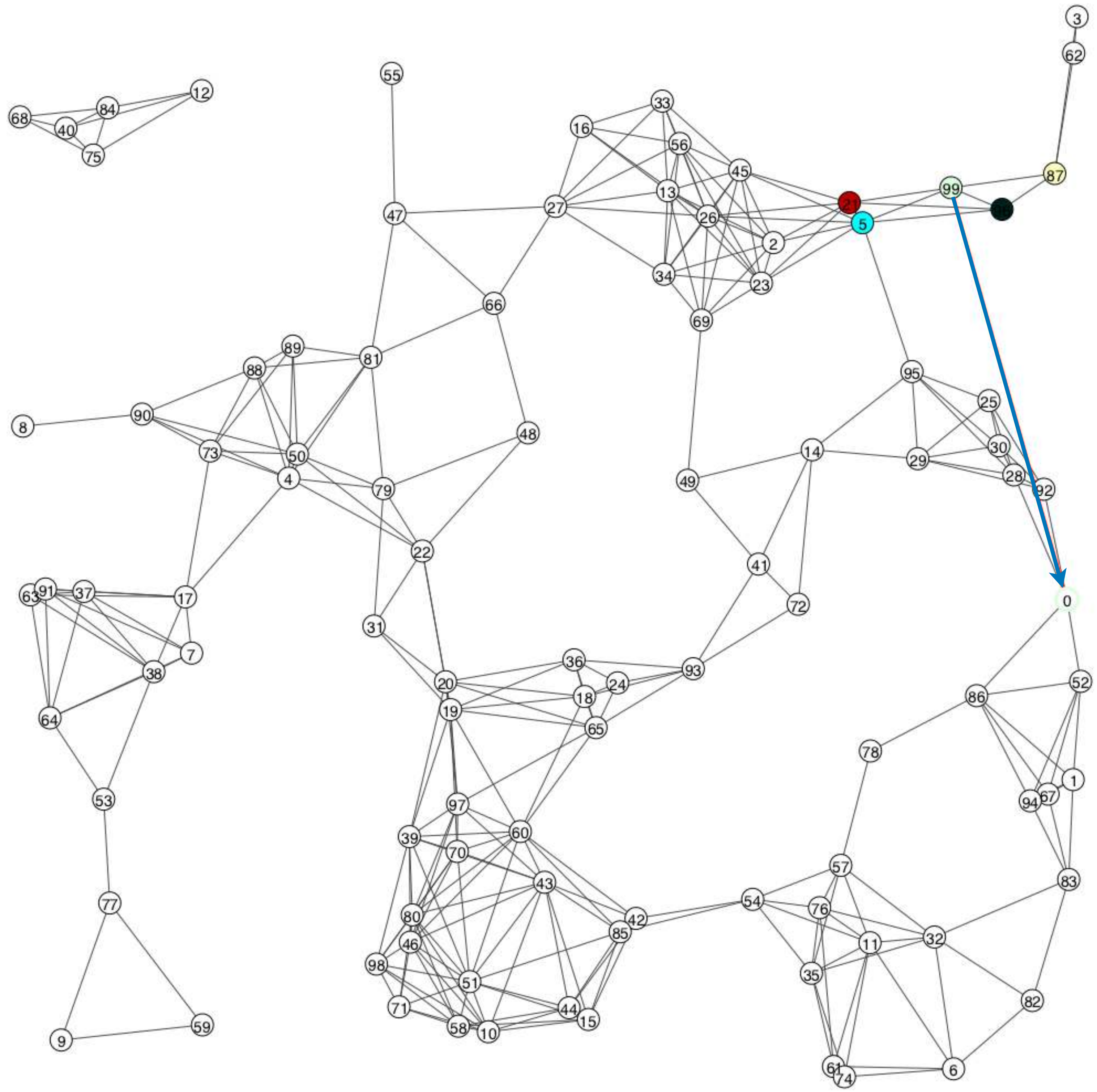
$K \in \{1, 2, 3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 20$ ➔ *loss = 40%* $\cancel{SNP} = 0\%$

Results — Random geometric graph (1)



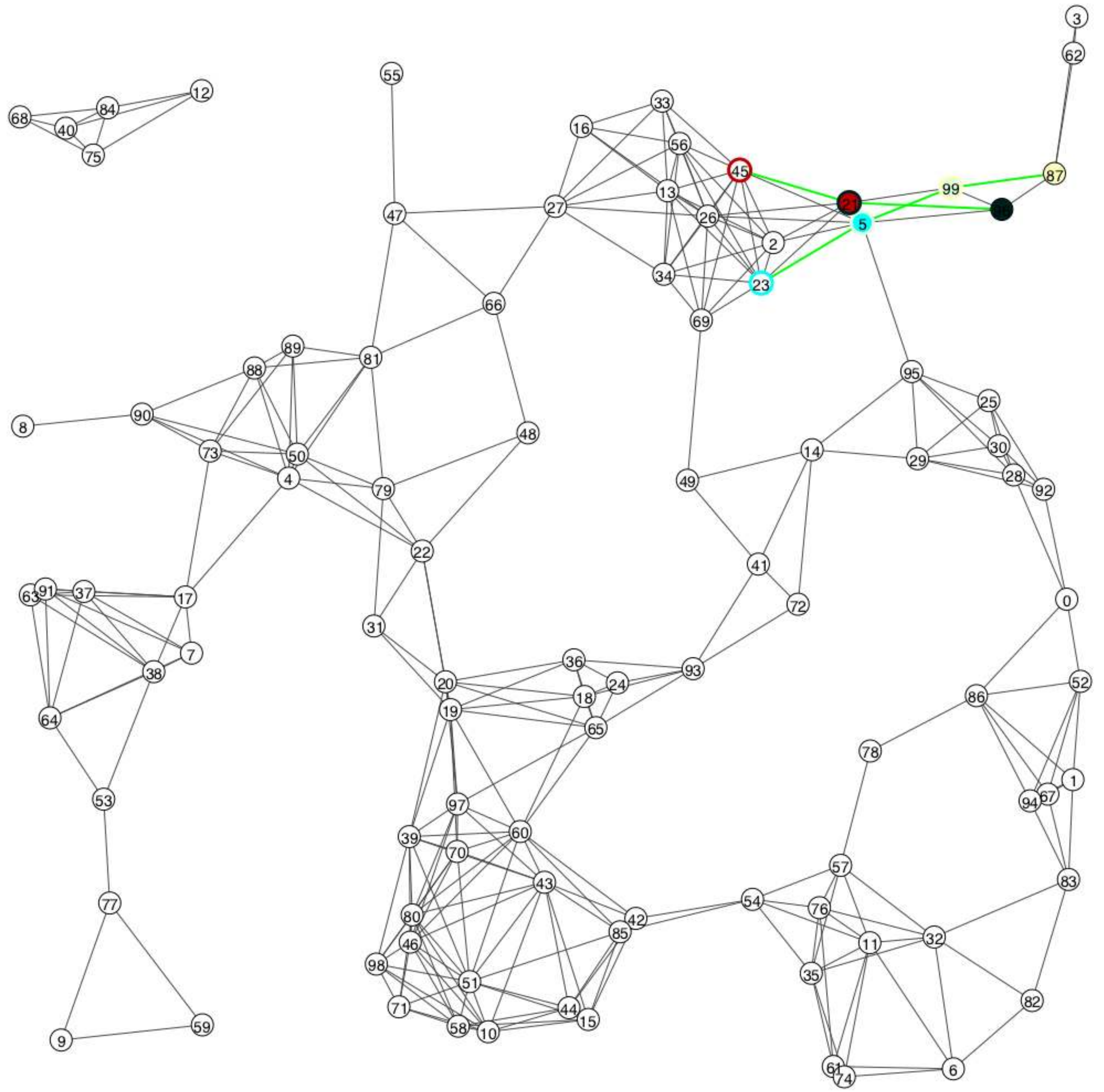
Composition

Results — Random geometric graph (1)



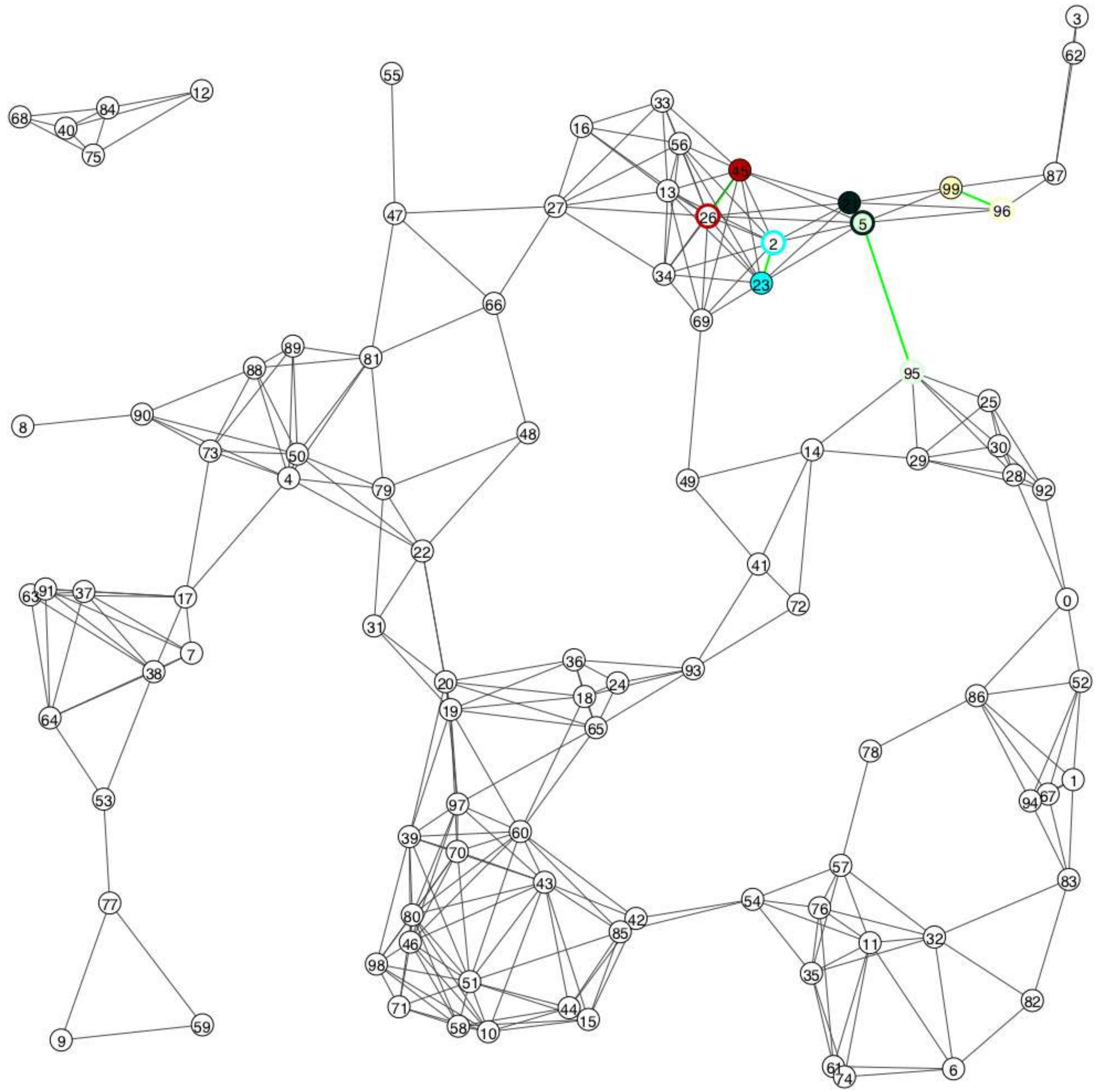
Objective

Results — Random geometric graph (1)



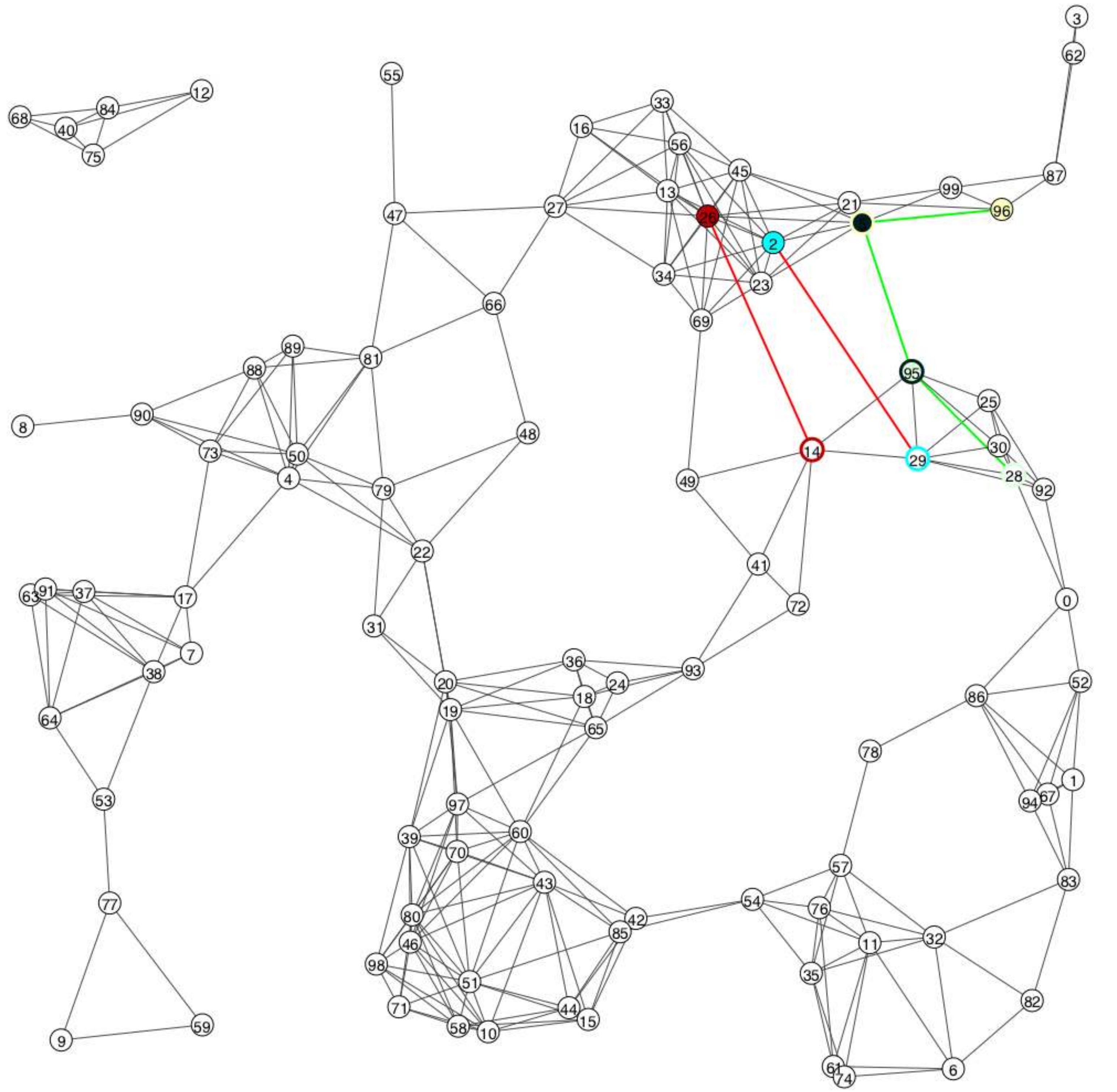
$K \in \{3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 1$ ➔ *loss = 0%* ~~*SNP*~~ *= 30%*

Results — Random geometric graph (1)



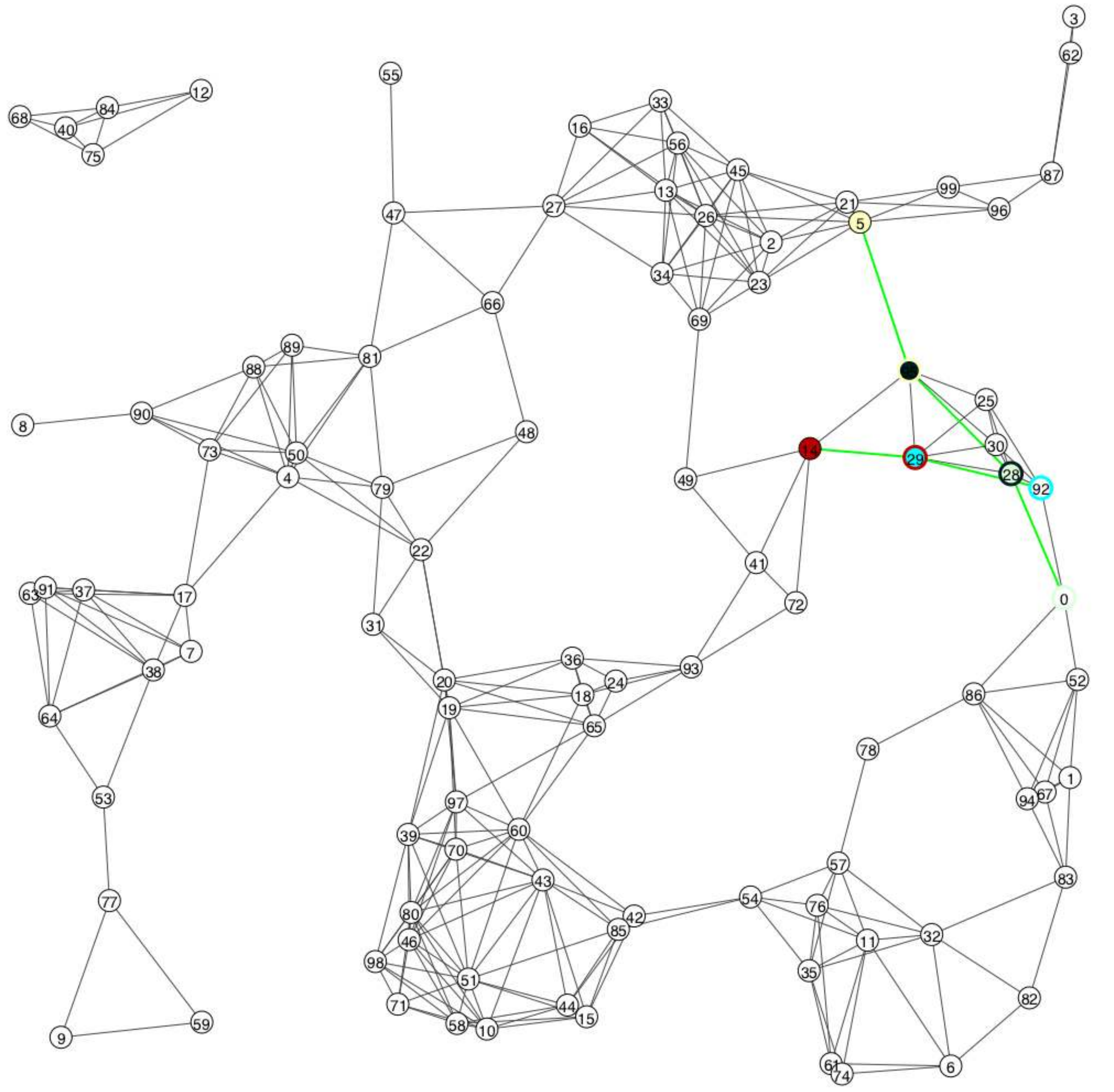
$K \in \{3, \infty\}$ $\quad \alpha = 5$ $\quad \beta = 1$ $\quad \gamma = 1$ $\quad \rightarrow$ $\quad$ *loss* = 0% $\quad$ *~~SNP~~* = 30%

Results — Random geometric graph (1)



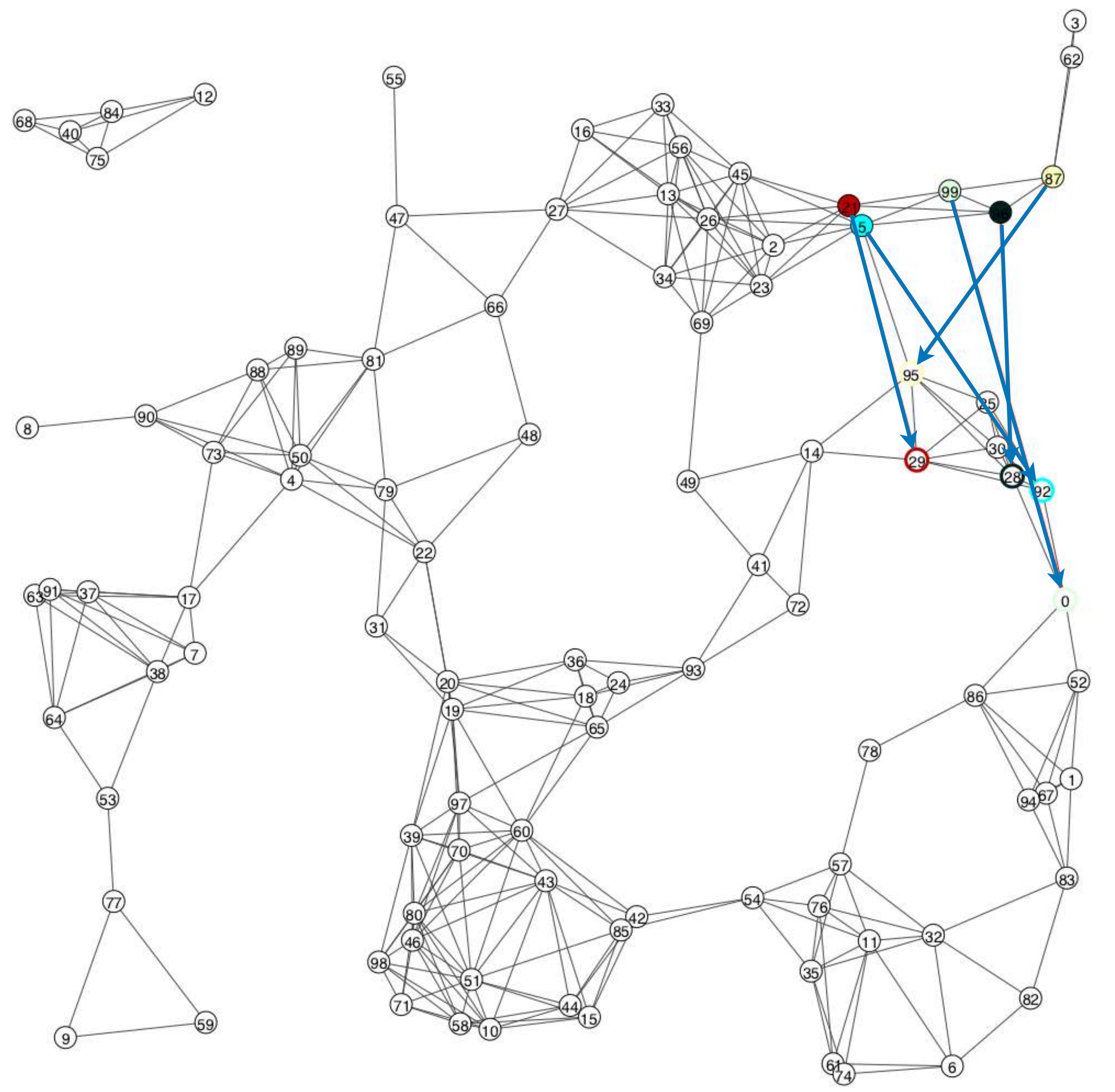
$K \in \{3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 1$ ➔ *loss = 0%* $\cancel{SNP}$ *= 30%*

Results — Random geometric graph (1)



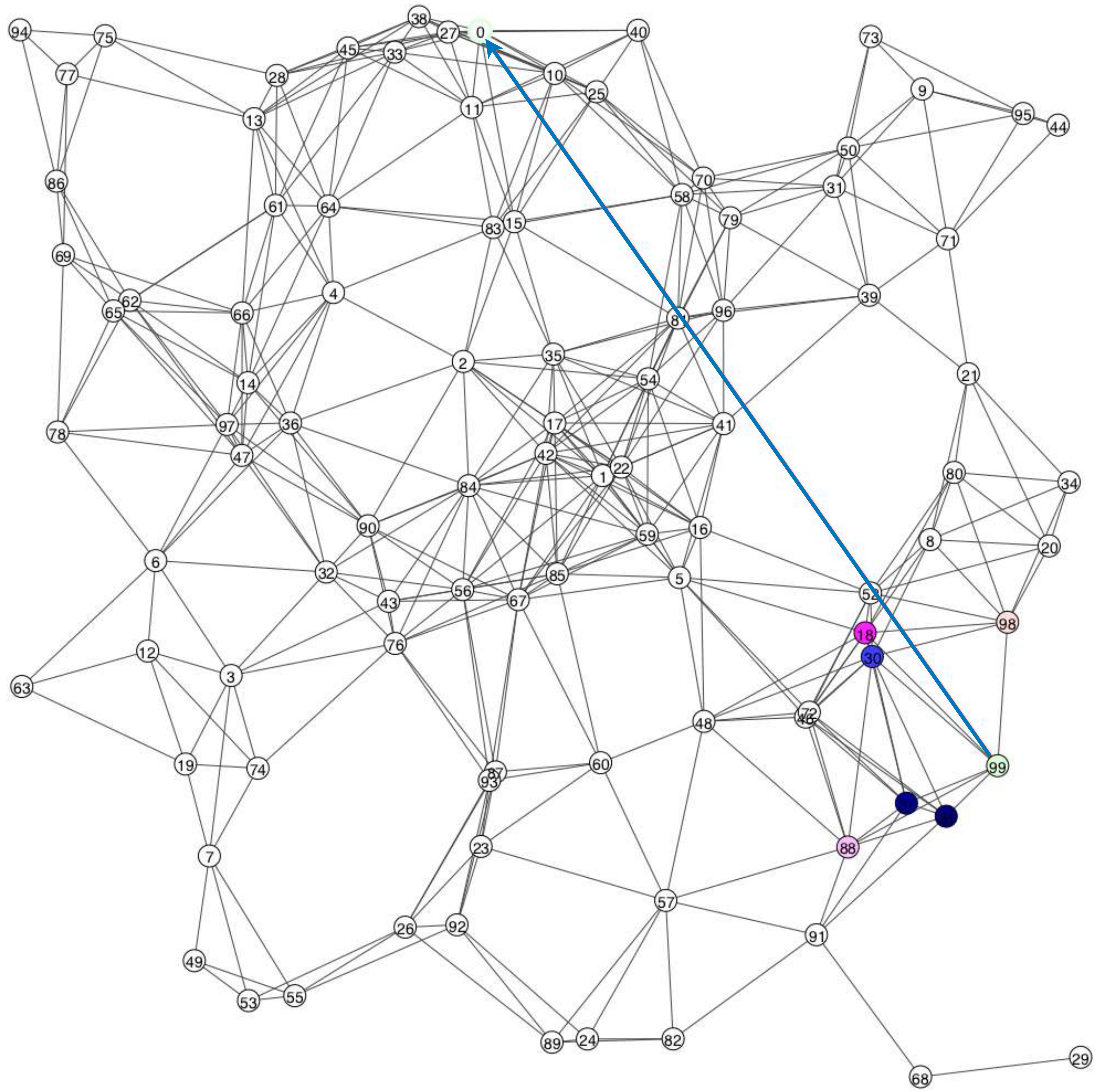
$K \in \{3, \infty\}$ $\alpha = 5$ $\beta = 1$ $\gamma = 1$ $\rightarrow$ $loss = 0\%$ $\cancel{SNP} = 30\%$

Results — Random geometric graph (1)



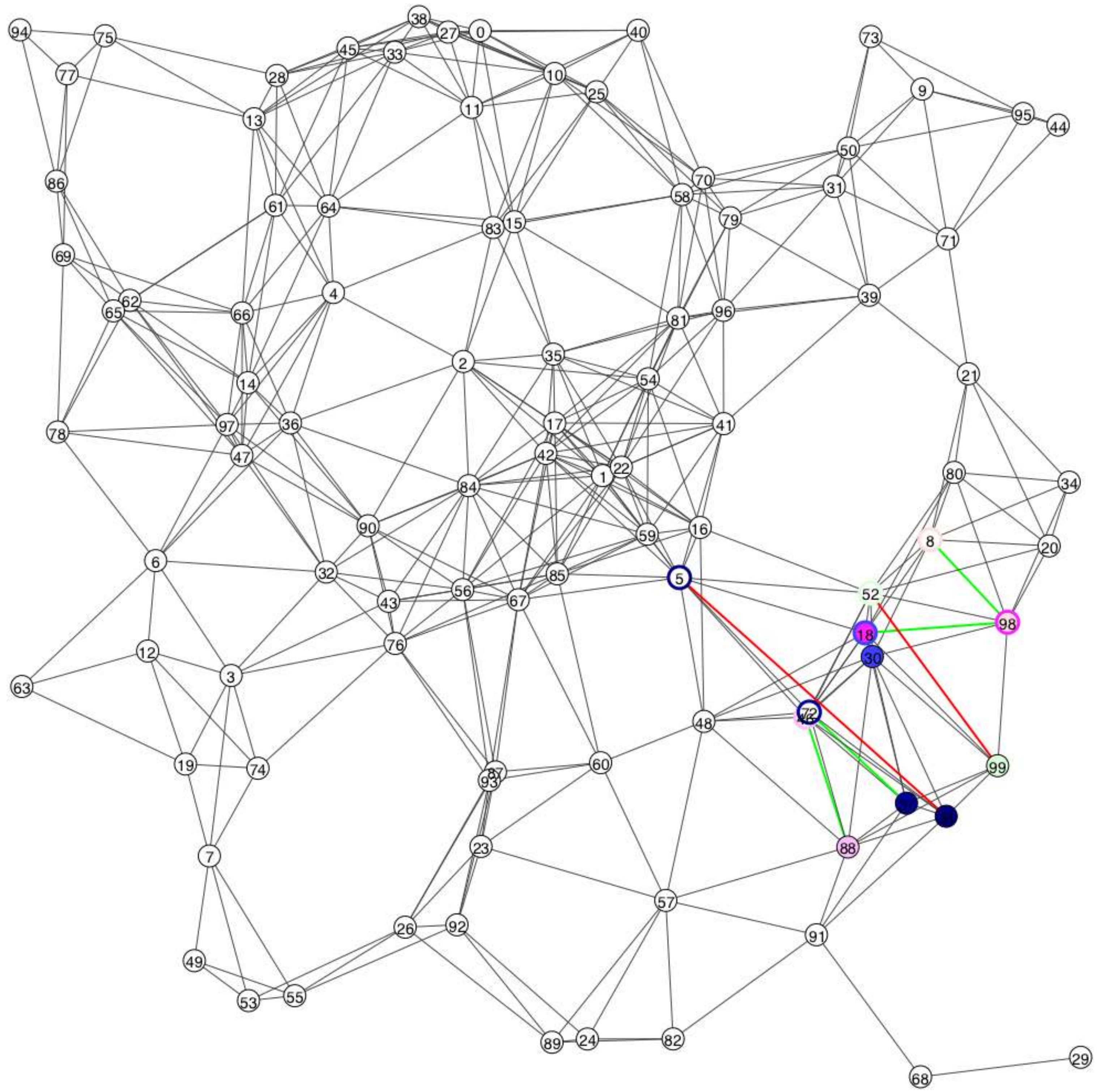
Composition

Results — Random geometric graph (2)



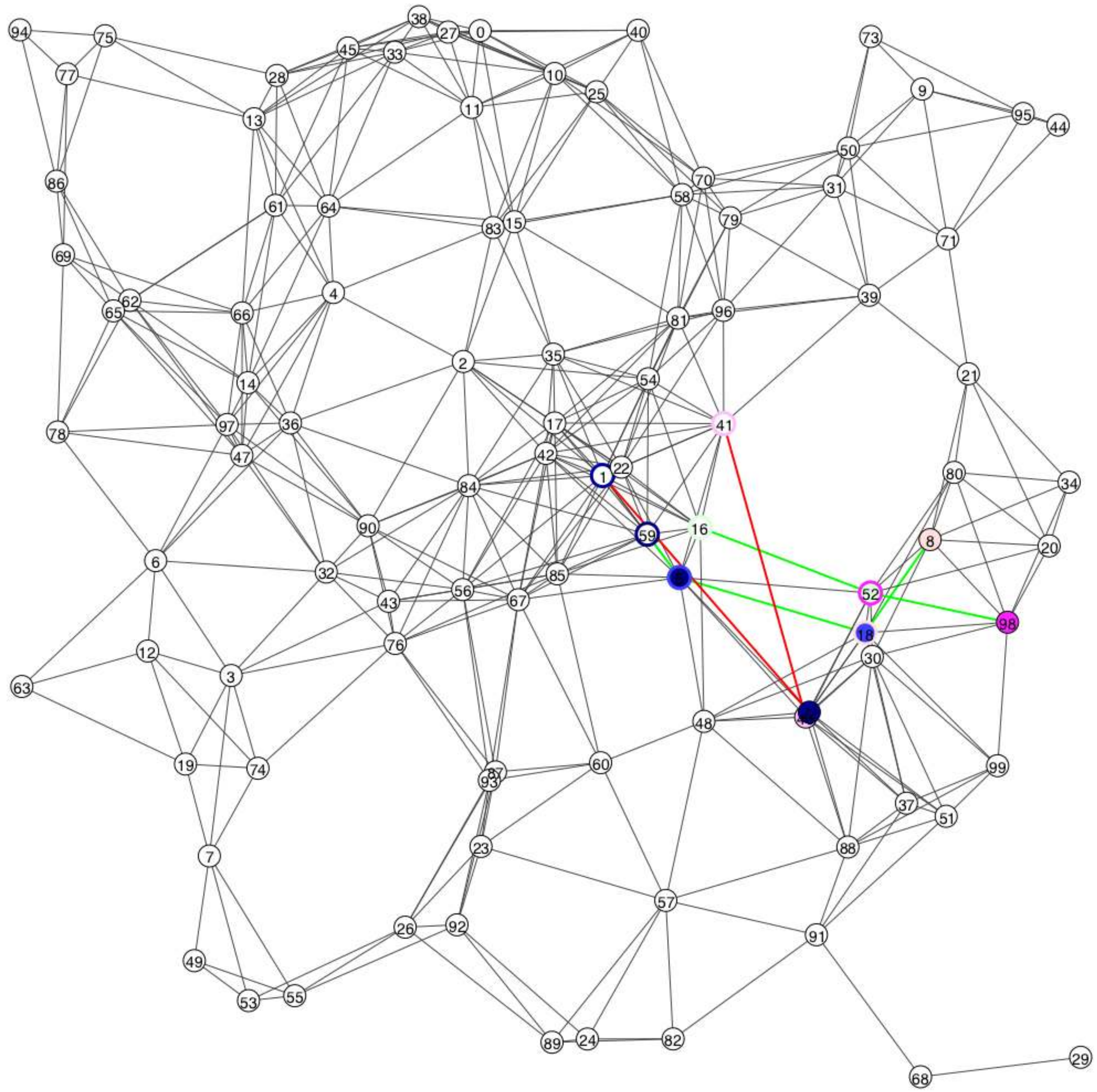
Objective

Results — Random geometric graph (2)



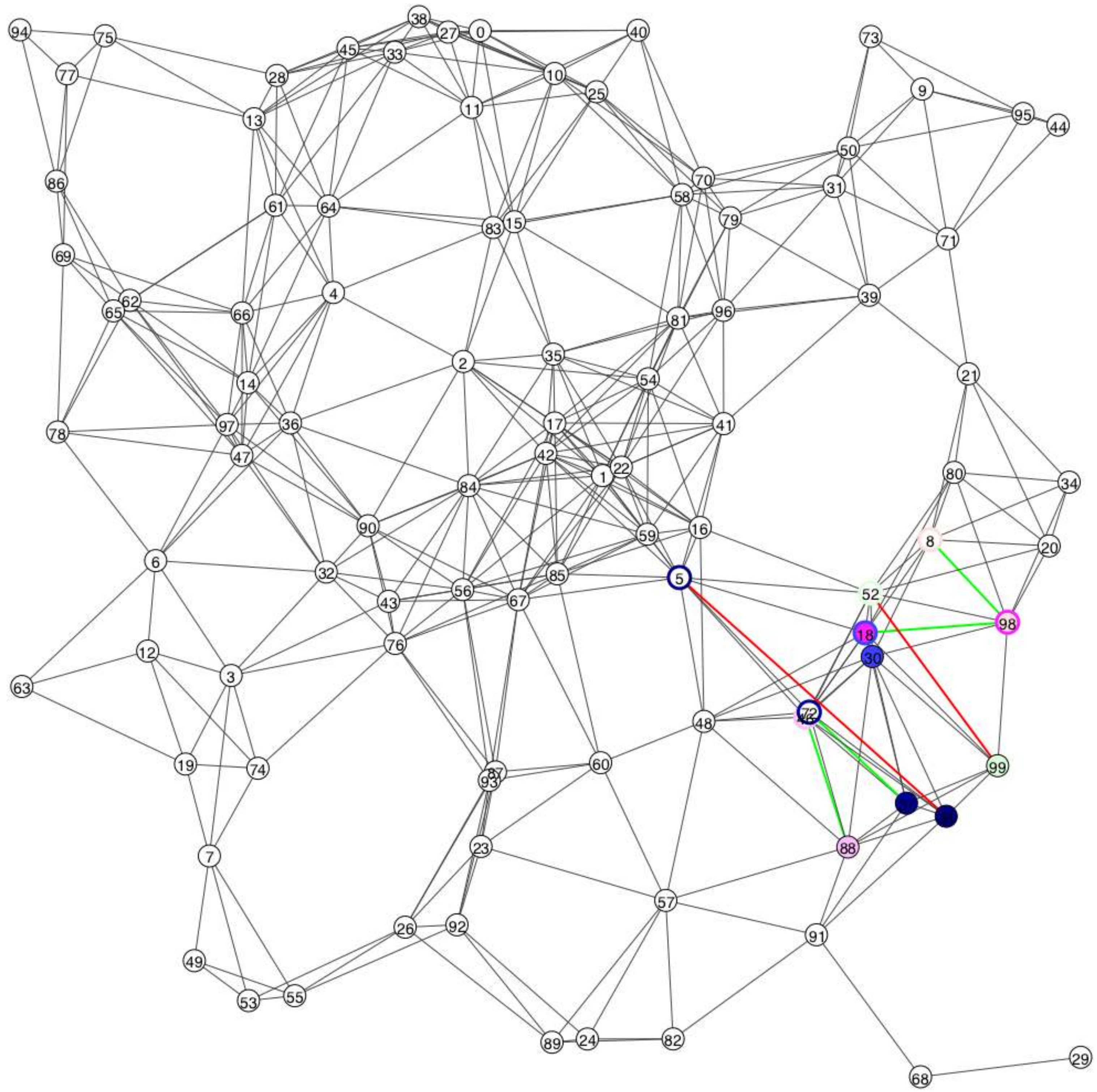
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ ➔ loss = 0% ~~SNP~~ = 14.29%

Results — Random geometric graph (2)



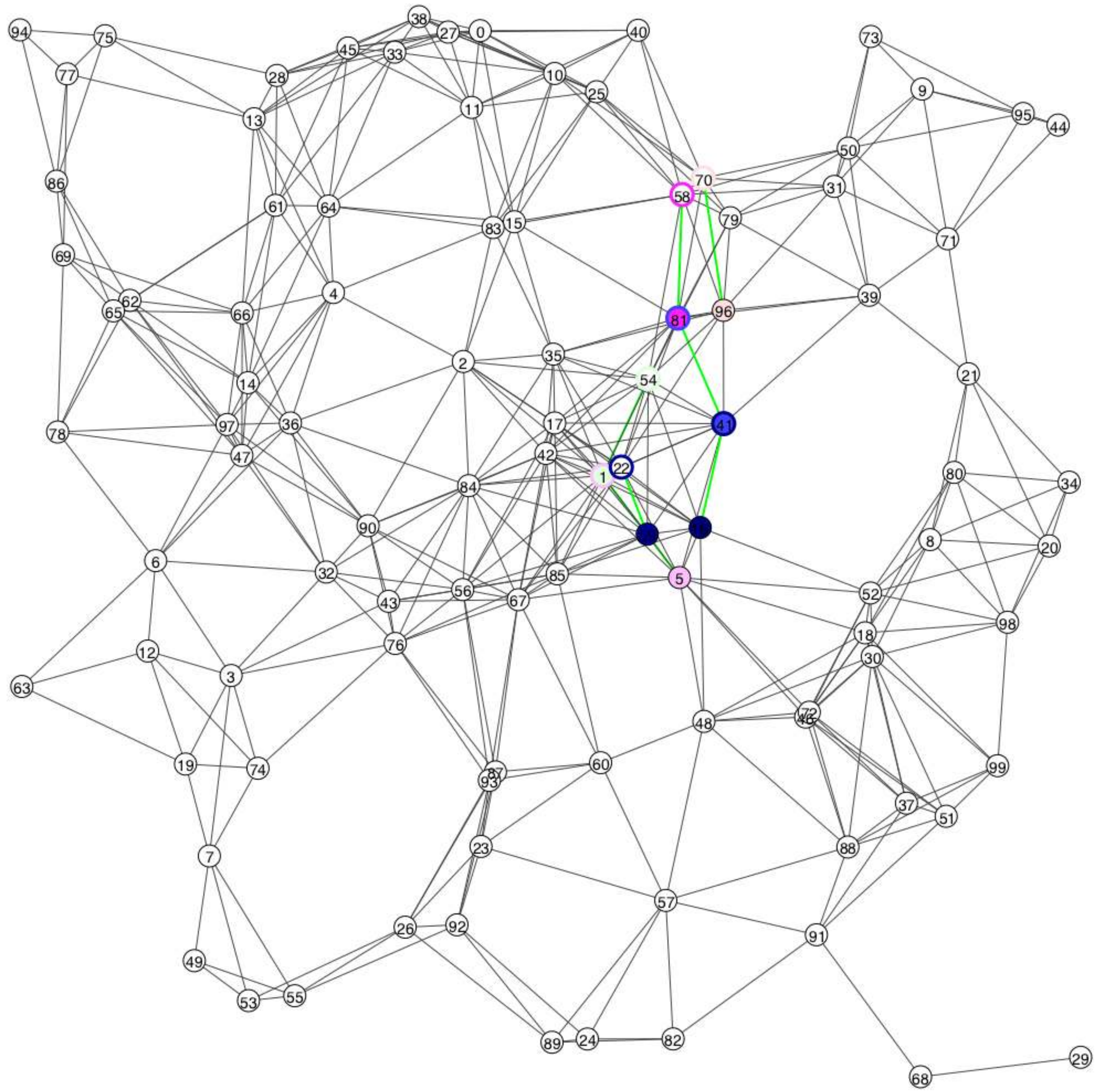
$K = 1 \qquad \alpha = 20 \qquad \beta = 1 \qquad \gamma = 5 \qquad \rightarrow \qquad loss = 0\% \qquad \cancel{SNP} = 14.29\%$

Results — Random geometric graph (2)



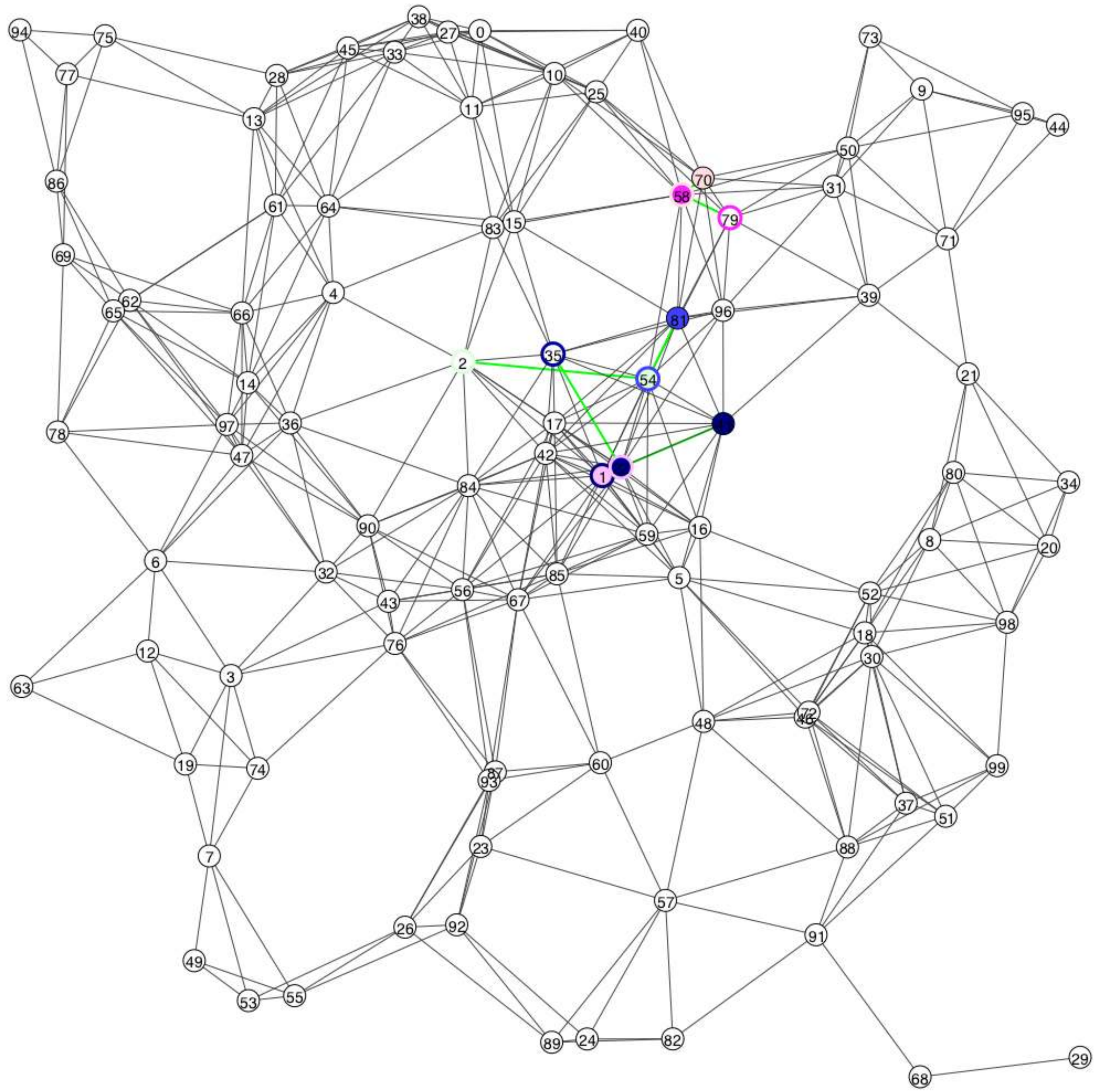
K = *1* *α* = *20* *β* = *1* *γ* = *5* ➔ *loss* = *0%* *~~SNP~~* = *14.29%*

Results — Random geometric graph (2)



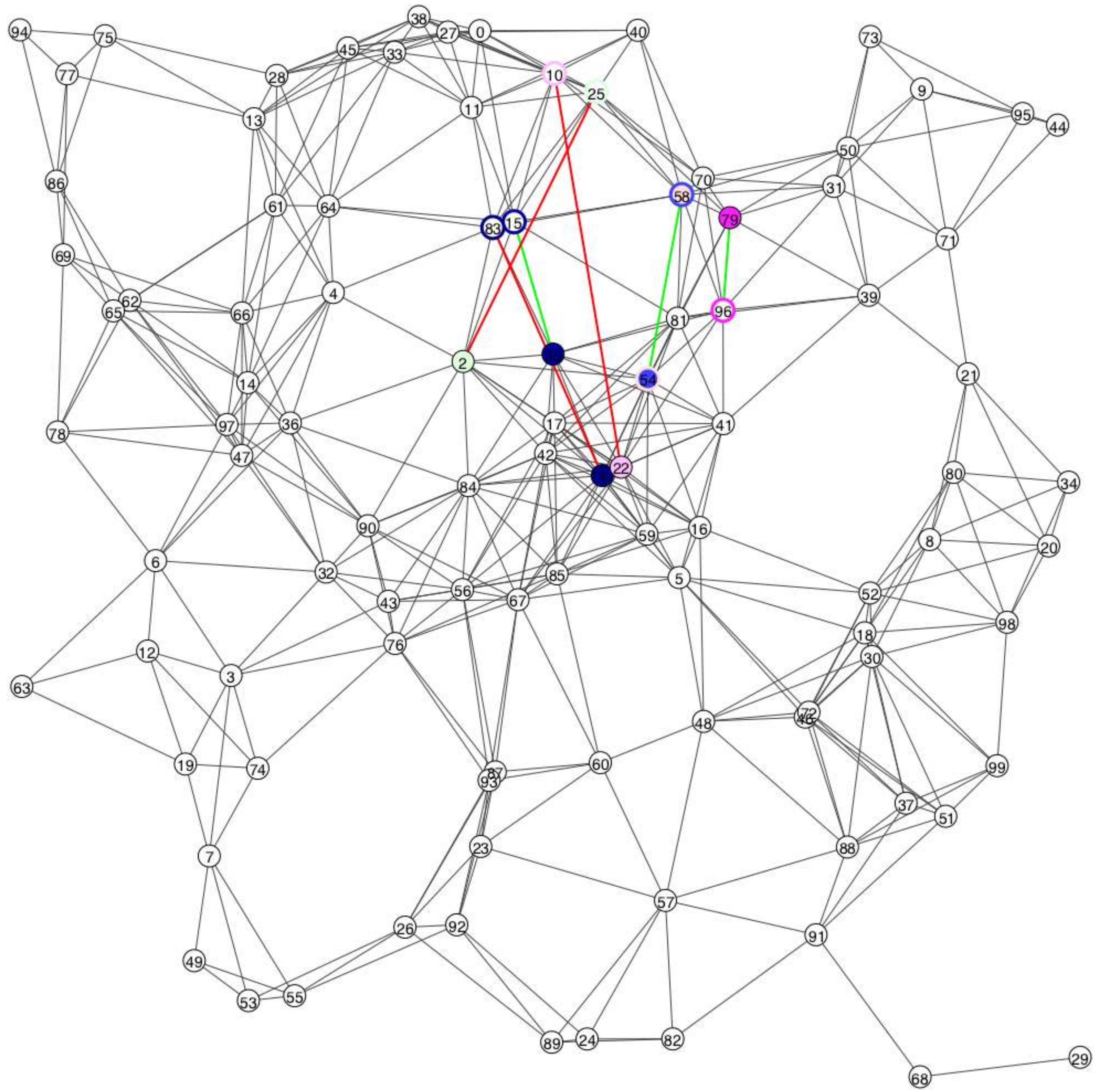
K = 1 $\quad$ *α = 20* $\quad$ *β = 1* $\quad$ *γ = 5* $\quad$ ➔ $\quad$ *loss = 0%* $\quad$ *~~SNP~~ = 14.29%*

Results — Random geometric graph (2)



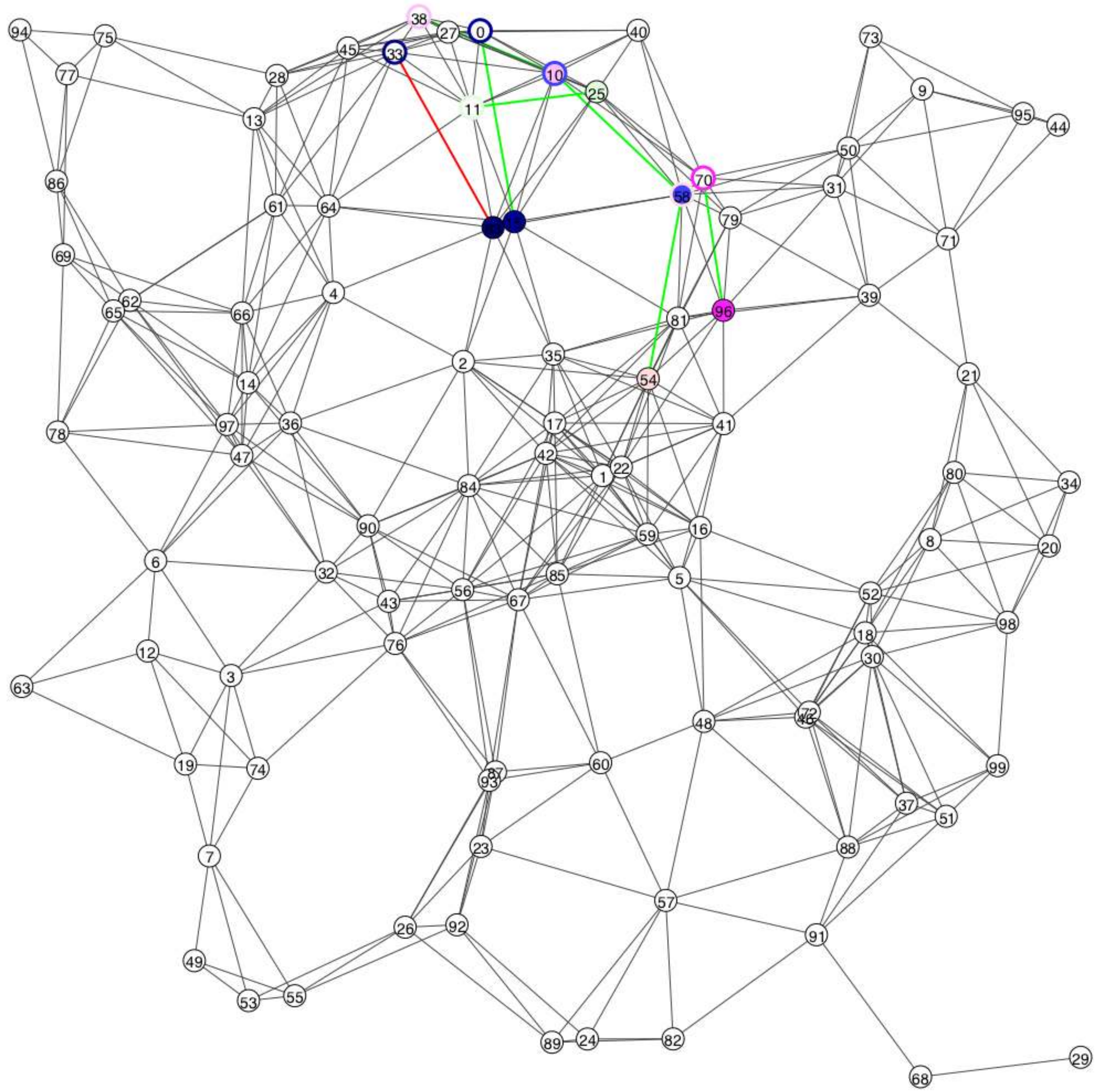
$K = 1$ $\alpha = 20$ $\beta = 1$ $\gamma = 5$ ➔ $loss = 0\%$ $\cancel{SNP} = 14.29\%$

Results — Random geometric graph (2)



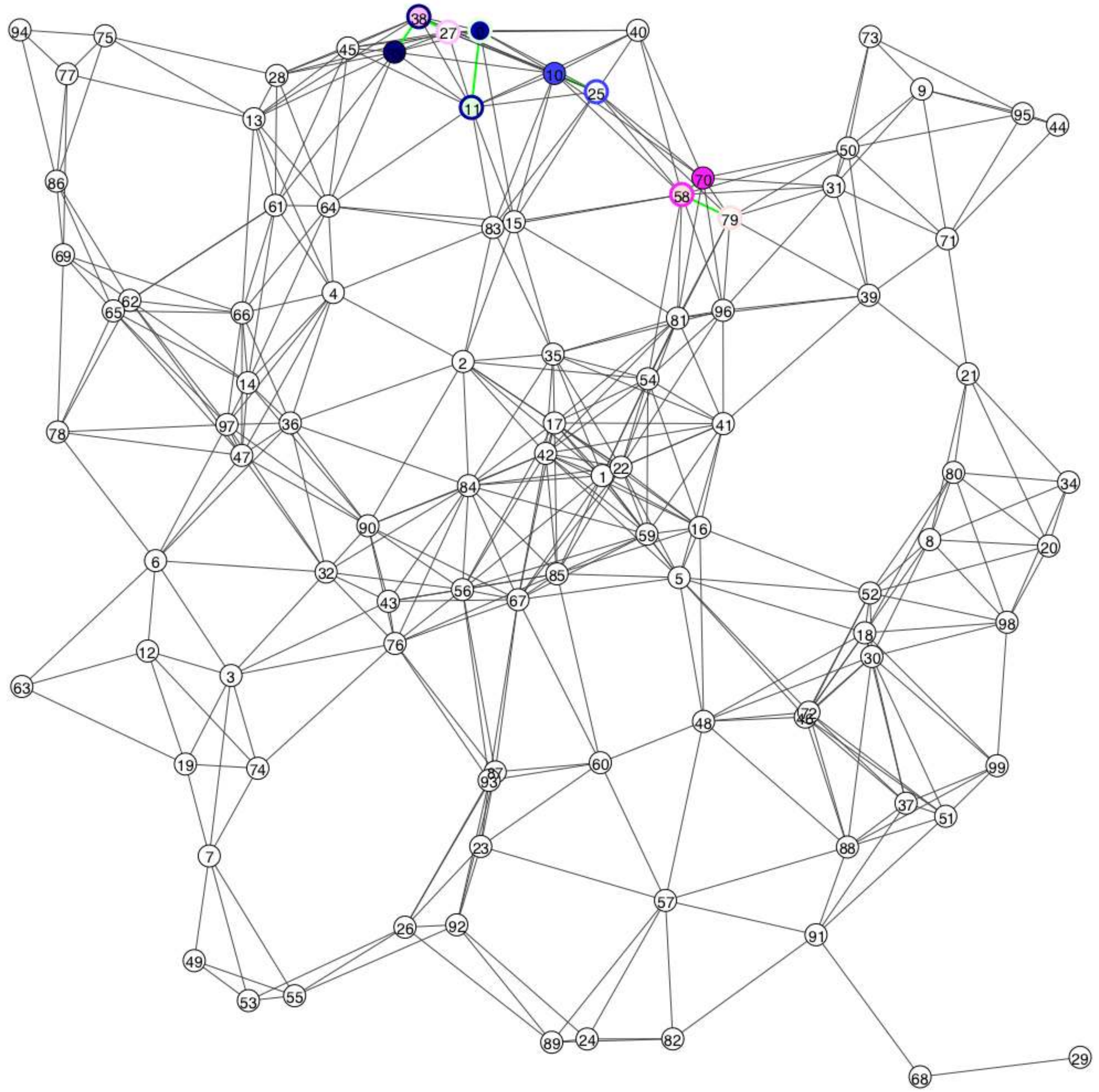
K = 1 $\alpha = 20$ $\beta = 1$ $\gamma = 5$ ➔   *loss* = 0%   ~~*SNP*~~ = 14.29%

Results — Random geometric graph (2)



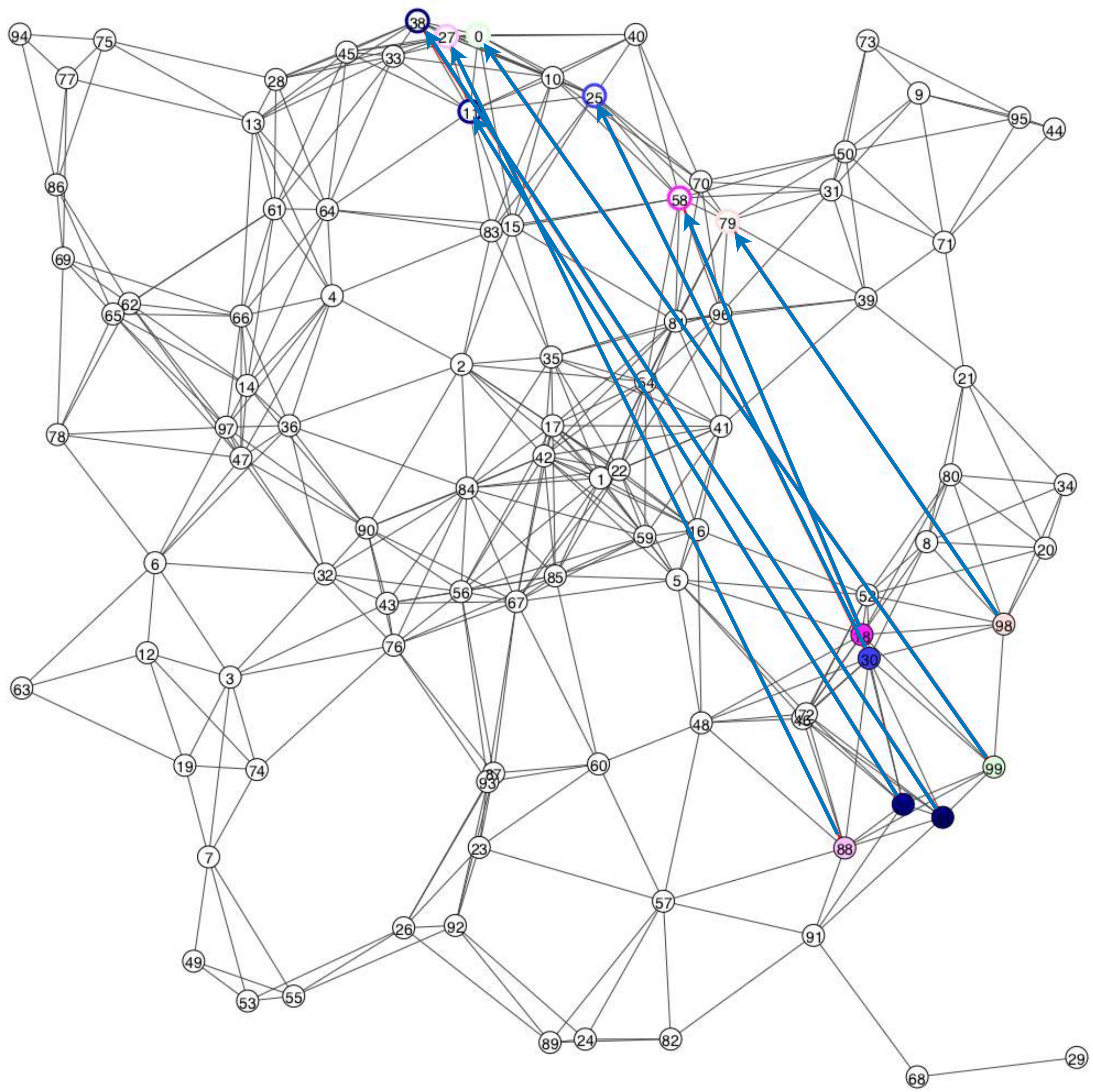
K = *1* $\quad$ $\alpha = 20$ $\quad$ $\beta = 1$ $\quad$ $\gamma = 5$ $\quad$ ➔ $\quad$ *loss = 0%* $\quad$ *~~SNP~~ = 14.29%*

Results — Random geometric graph (2)



K = 1 *α = 20* *β = 1* *γ = 5* ➔ *loss = 0%* ~~*SNP*~~ *= 14.29%*

Results — Random geometric graph (2)

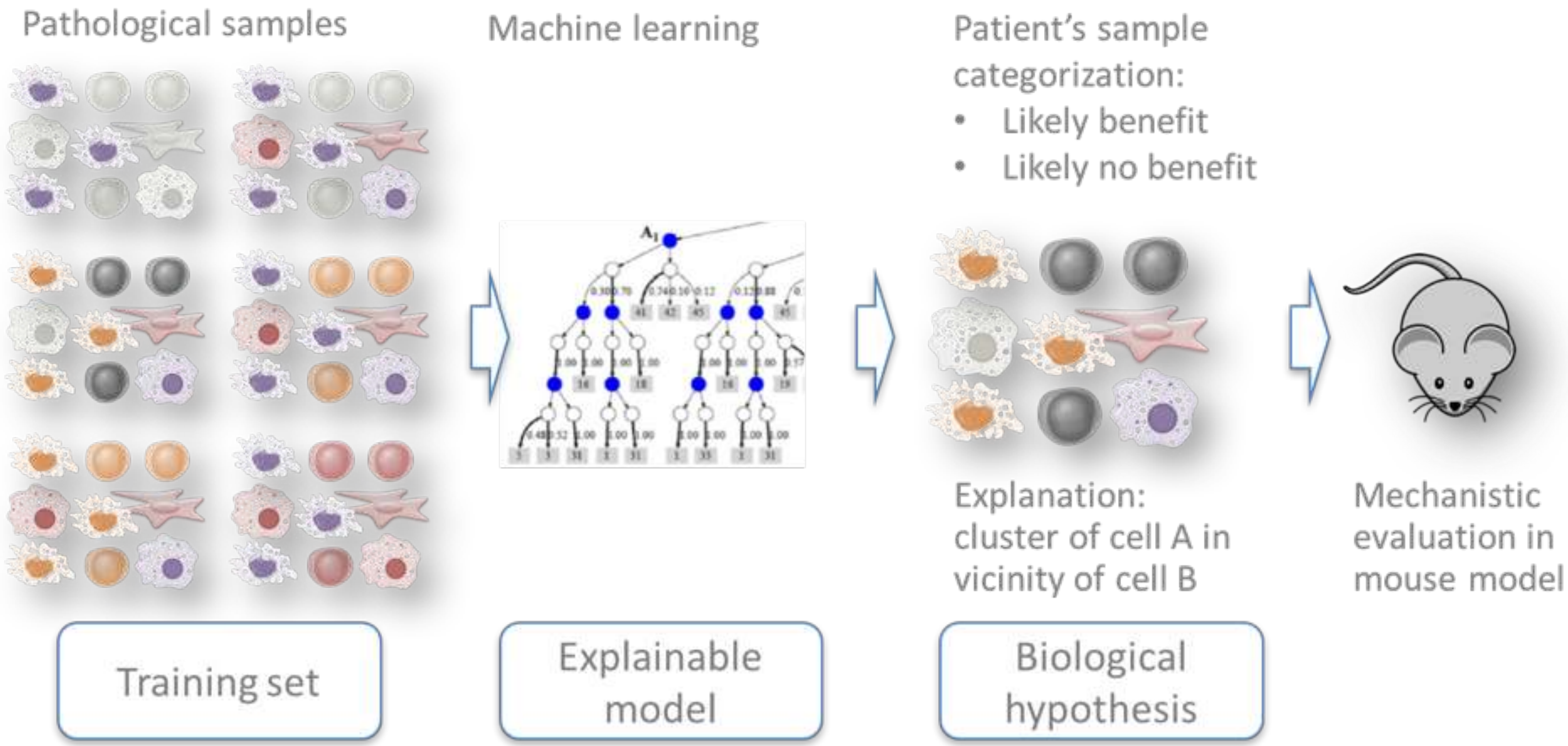


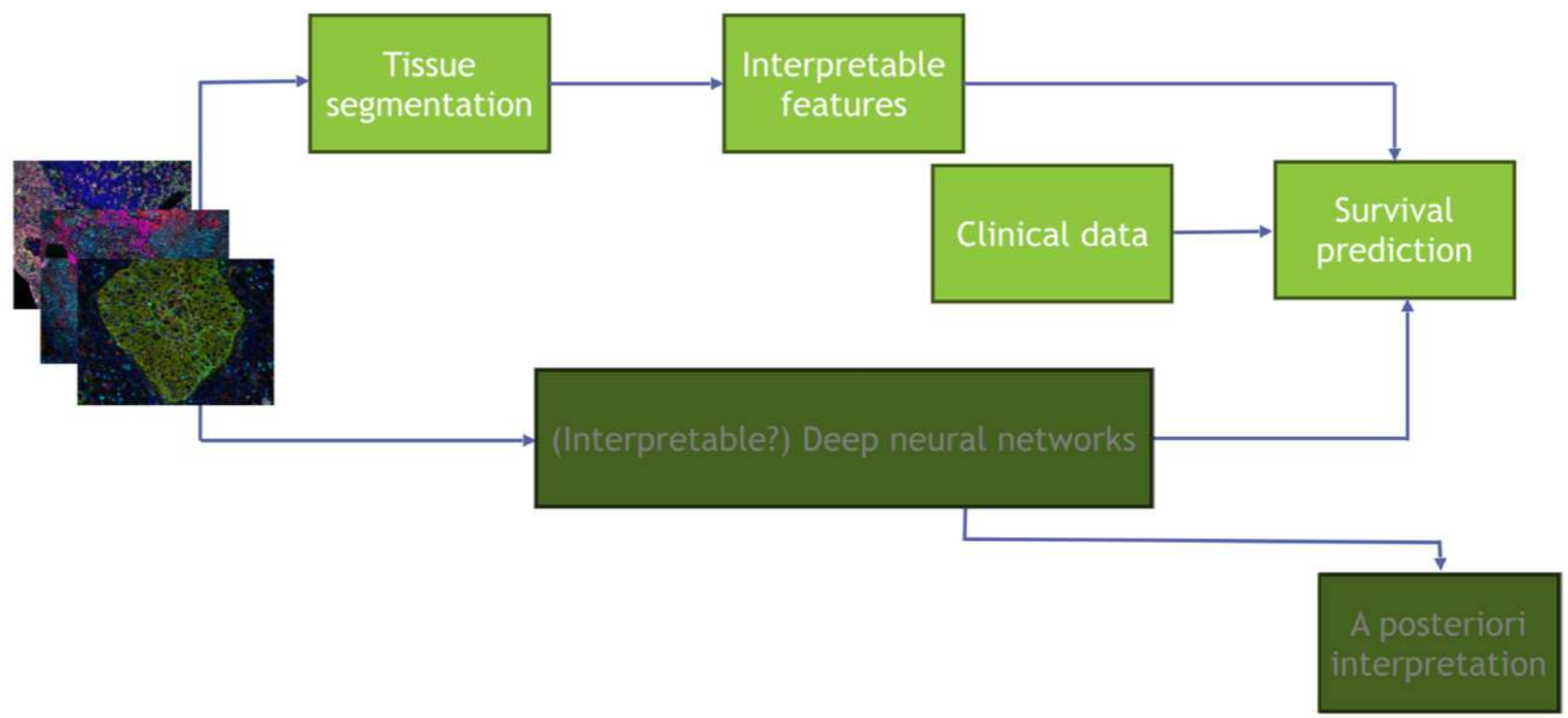
Composition

Outline

- GSP for AI, *a.k.a.* "What I did during my Ph.D. and a bit after"
- **GSP and personalized healthcare**
- GSP and brain signals

Why personalized oncology?





Datasets are becoming larger

SKCM

Experimental Strategy	Cases (n=470)	Files (n=12 724)
Diagnostic Slide	433	475
Tissue Slide	469	475

BRCA

Experimental Strategy	Cases (n=1 098)	Files (n=31 511)
Diagnostic Slide	1 062	1 133
Tissue Slide	1 093	1 978

COAD

Experimental Strategy	Cases (n=461)	Files (n=14 270)
Diagnostic Slide	451	459
Tissue Slide	460	983

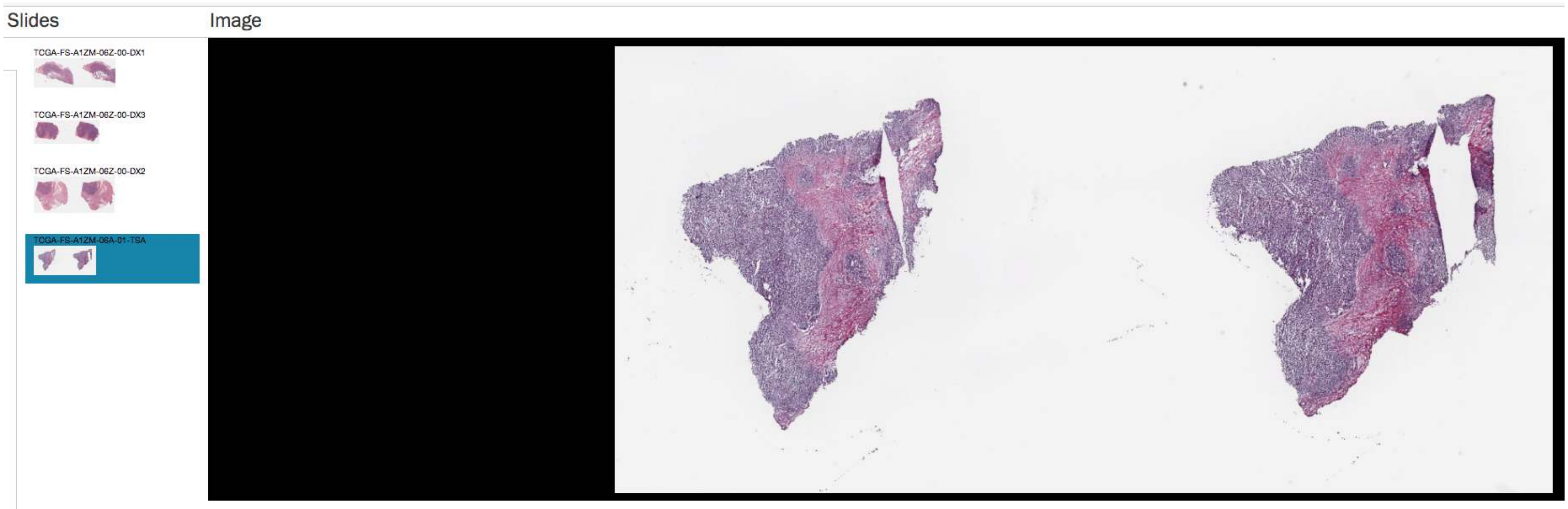
LUAD

Experimental Strategy	Cases (n=585)	Files (n=17 051)
Diagnostic Slide	478	541
Tissue Slide	514	1 067

PRAD

Experimental Strategy	Cases (n=500)	Files (n=14 287)
Diagnostic Slide	403	449
Tissue Slide	490	723

Datasets are becoming larger



File_id	8b36b58e-1f2d-4419-98b9-eabfd60e76cb
Submitter_id	TCGA-FS-A1ZM-06A-01-TSA
Percent_lymphocyte_infiltration	--
Percent_tumor_nuclei	75
Percent_monocyte_infiltration	--
Percent_normal_cells	--
Percent_stromal_cells	30
Percent_eosinophil_infiltration	--
Slide_id	b045f429-078b-4431-8691-3a9604b818ce
Percent_neutrophil_infiltration	--
Section_location	TOP
Percent_granulocyte_infiltration	--
Percent_necrosis	--
Percent_inflam_infiltration	--
Number_proliferating_cells	--
Percent_tumor_cells	70

UUID	8b74dba2-0f4c-5e4e-8de1-62397a19dcbd
Classification Of Tumor	not reported
Alcohol Intensity	--
Age At Diagnosis	74 years 182 days
Days To Birth	-27210
Days To Death	--
Days To Last Follow Up	3080
Days To Last Known Disease Status	--
Days To Recurrence	--
Last Known Disease Status	not reported
Morphology	8720/3
Primary Diagnosis	C77.9
Prior Malignancy	not reported
Progression Or Recurrence	not reported
Site Of Resection Of Biopsy	C77.9
Tissue Or Organ Of Origin	C77.9
Tumor Grade	not reported
Tumor Stage	stage iii
Vital Status	alive

Datasets are becoming larger

UUID	8b74dba2-0f4c-5e4e-8de1-62397a19dcbd
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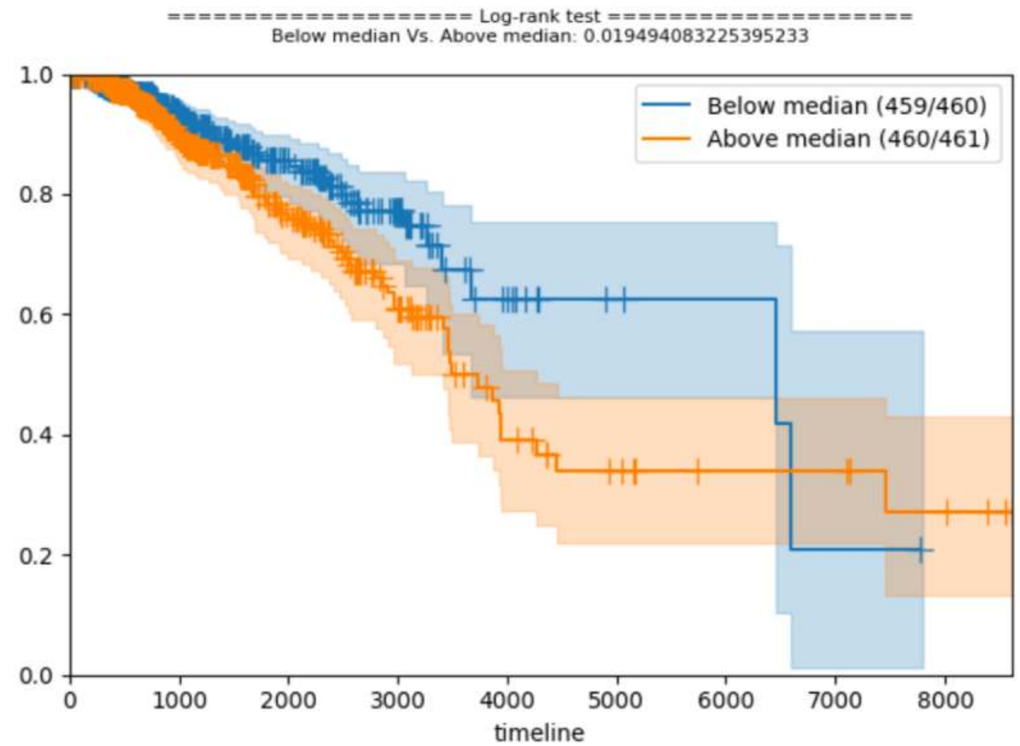
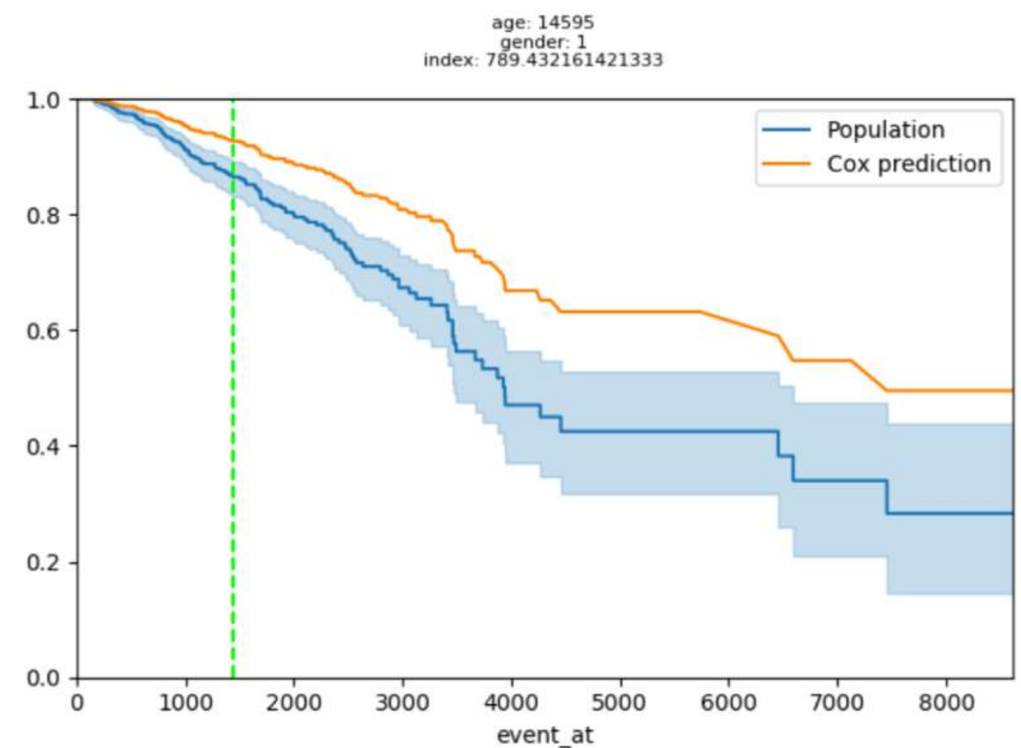
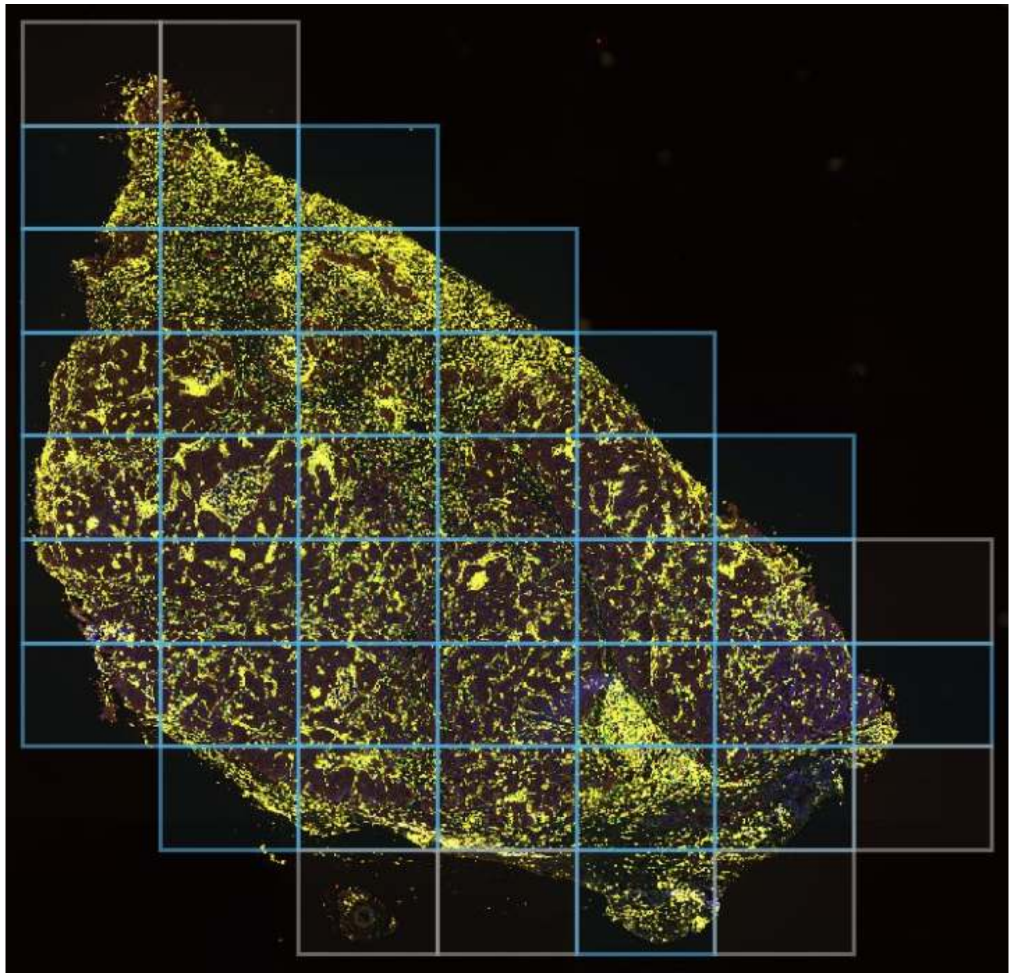
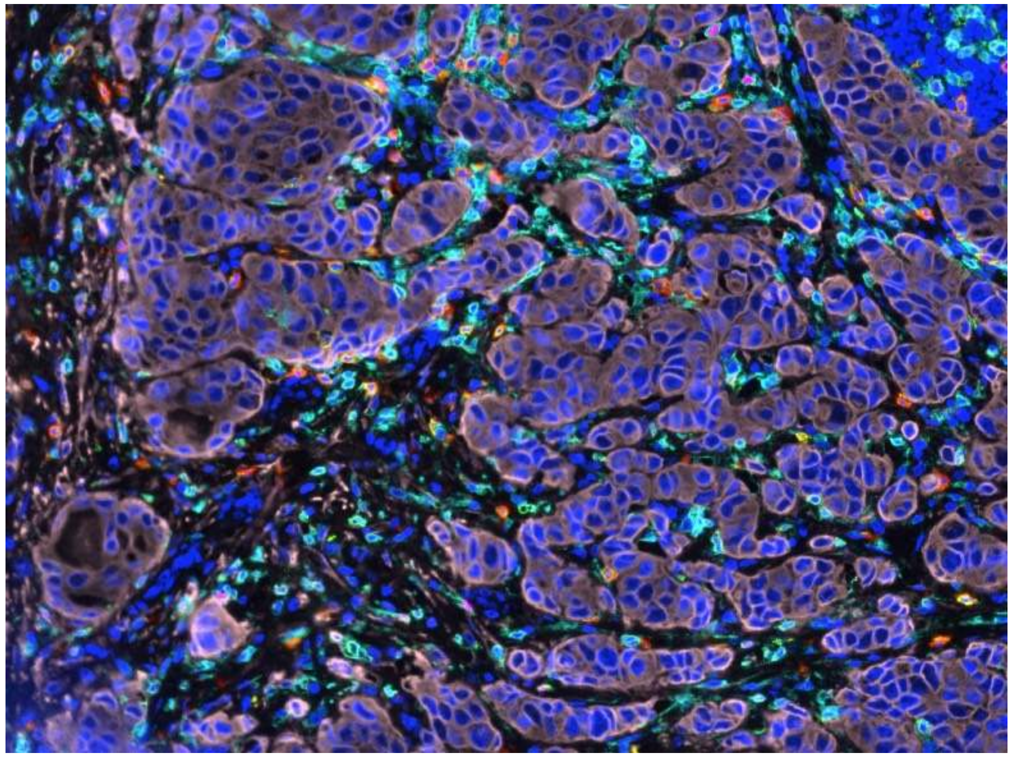
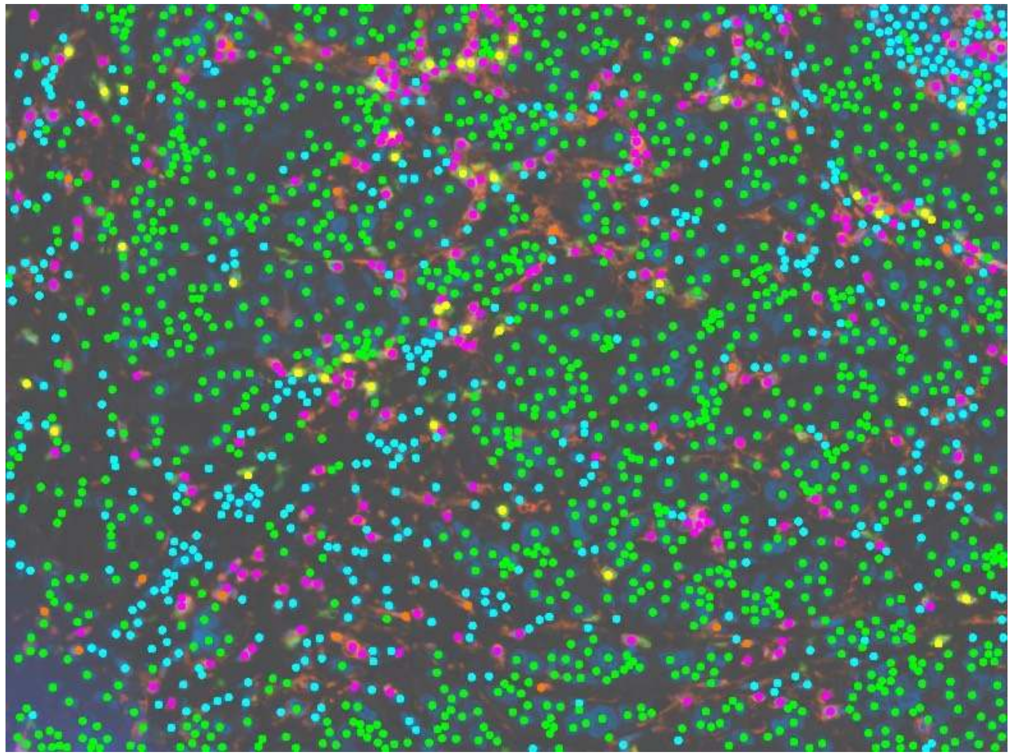




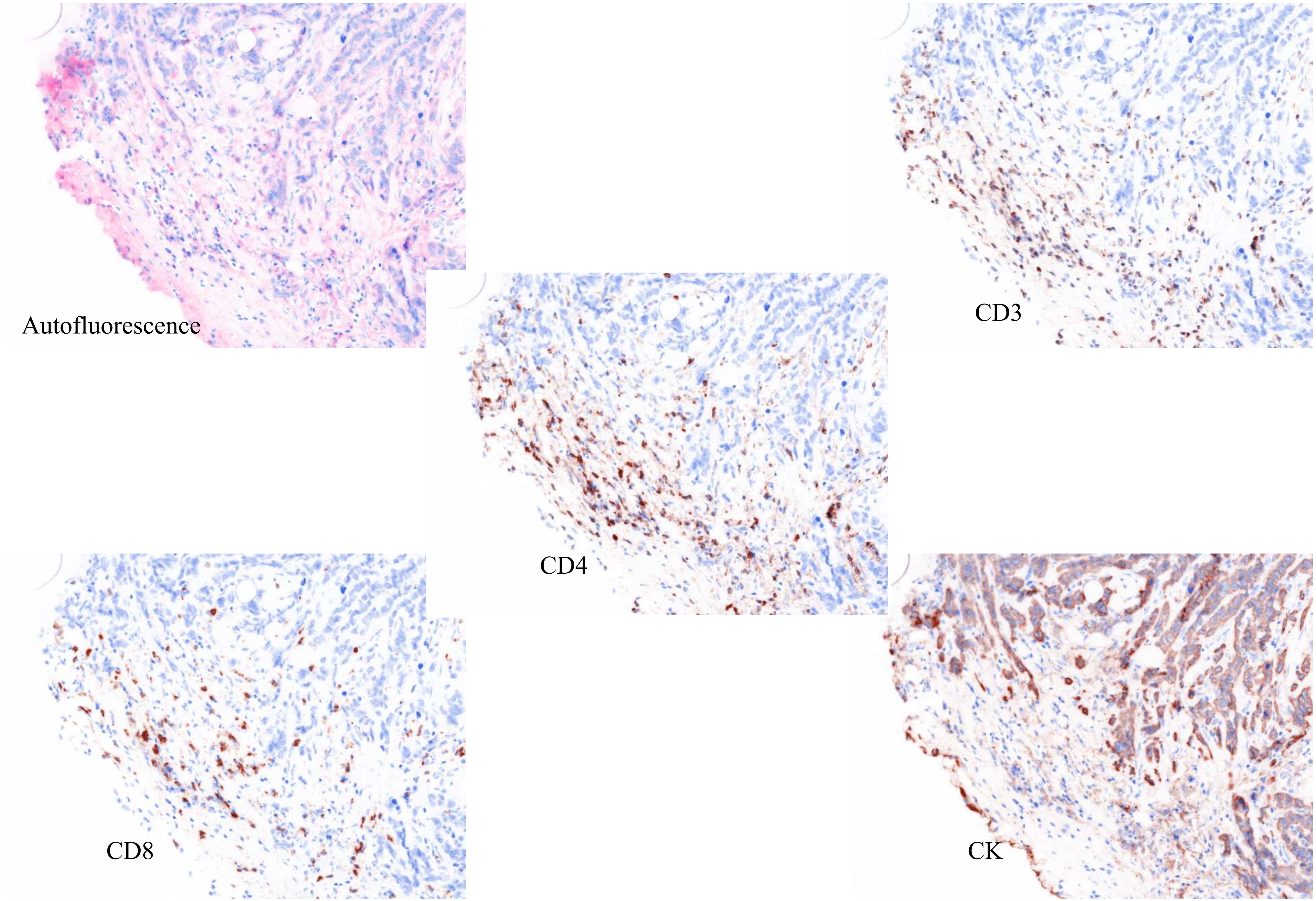
Techniques are improving



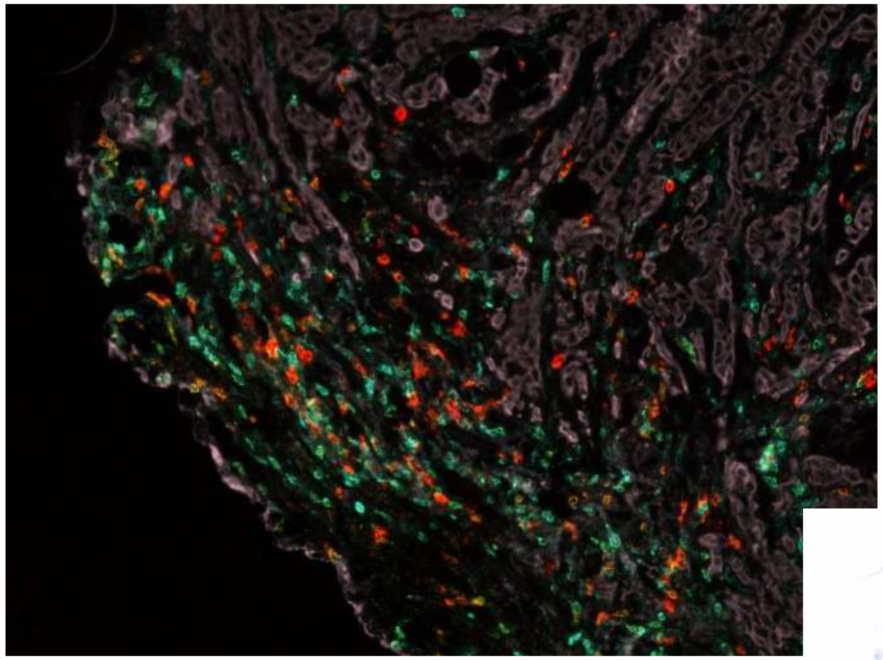




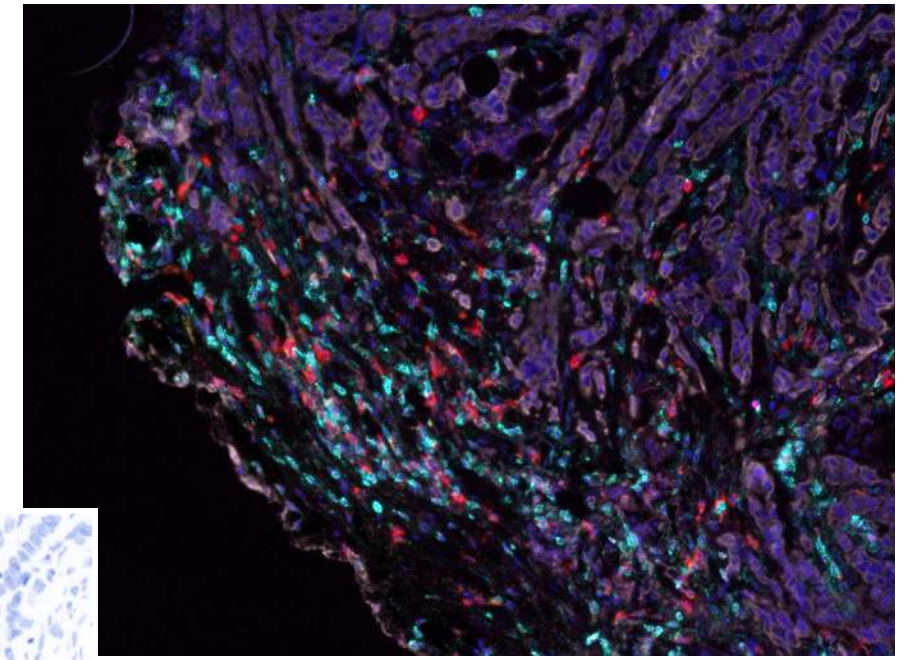
Techniques are improving



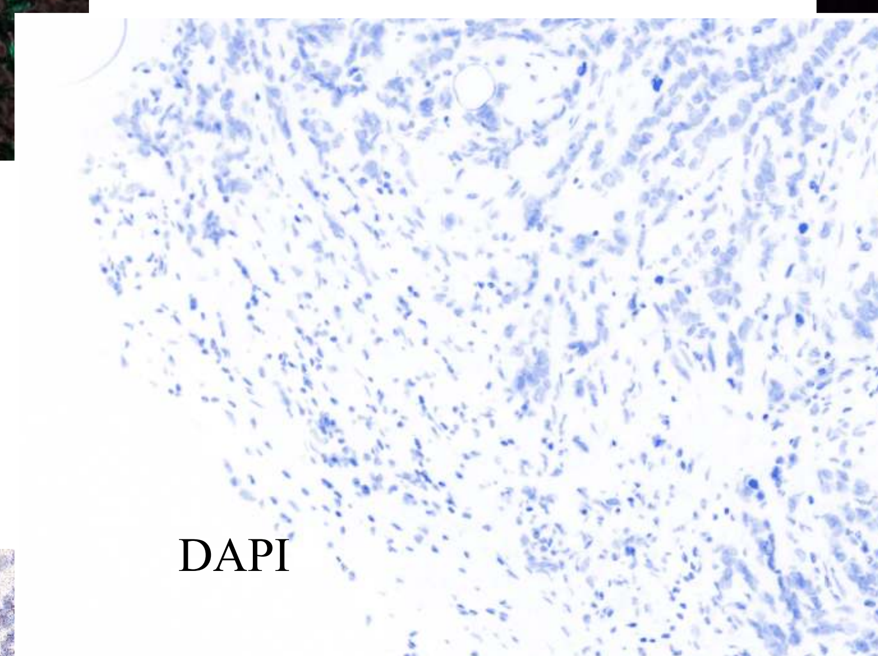
Techniques are improving



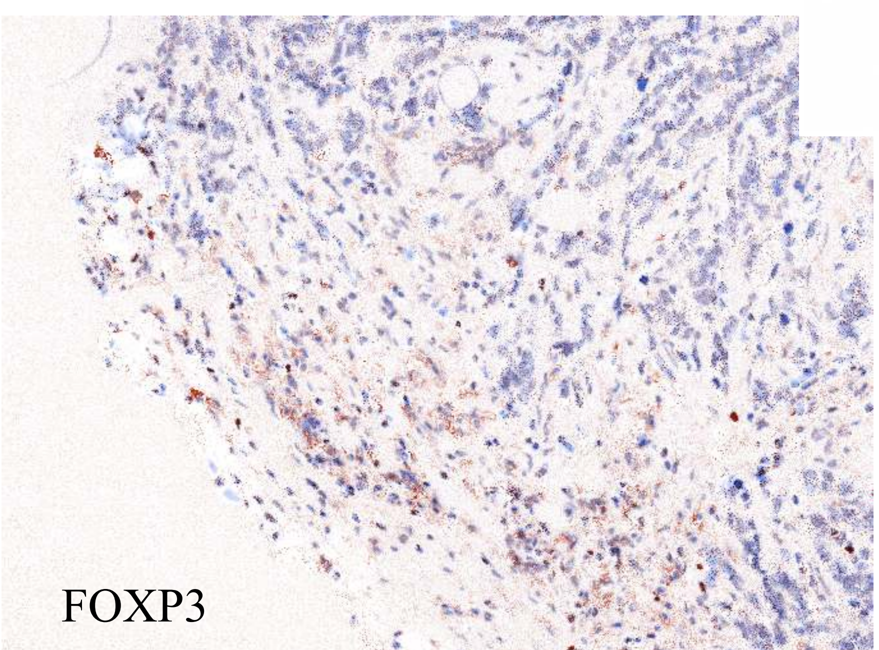
Composite



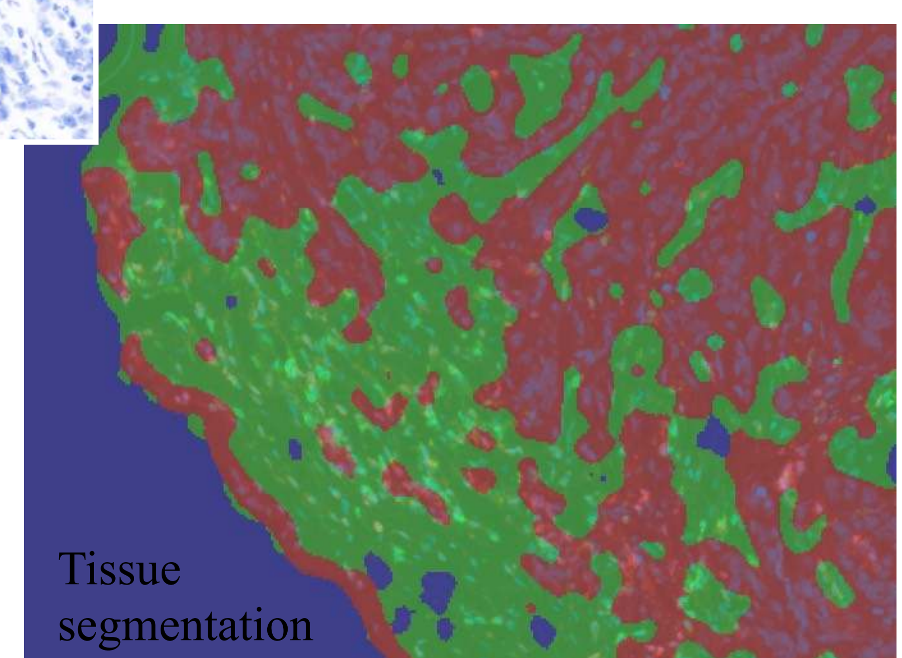
Composite



DAPI

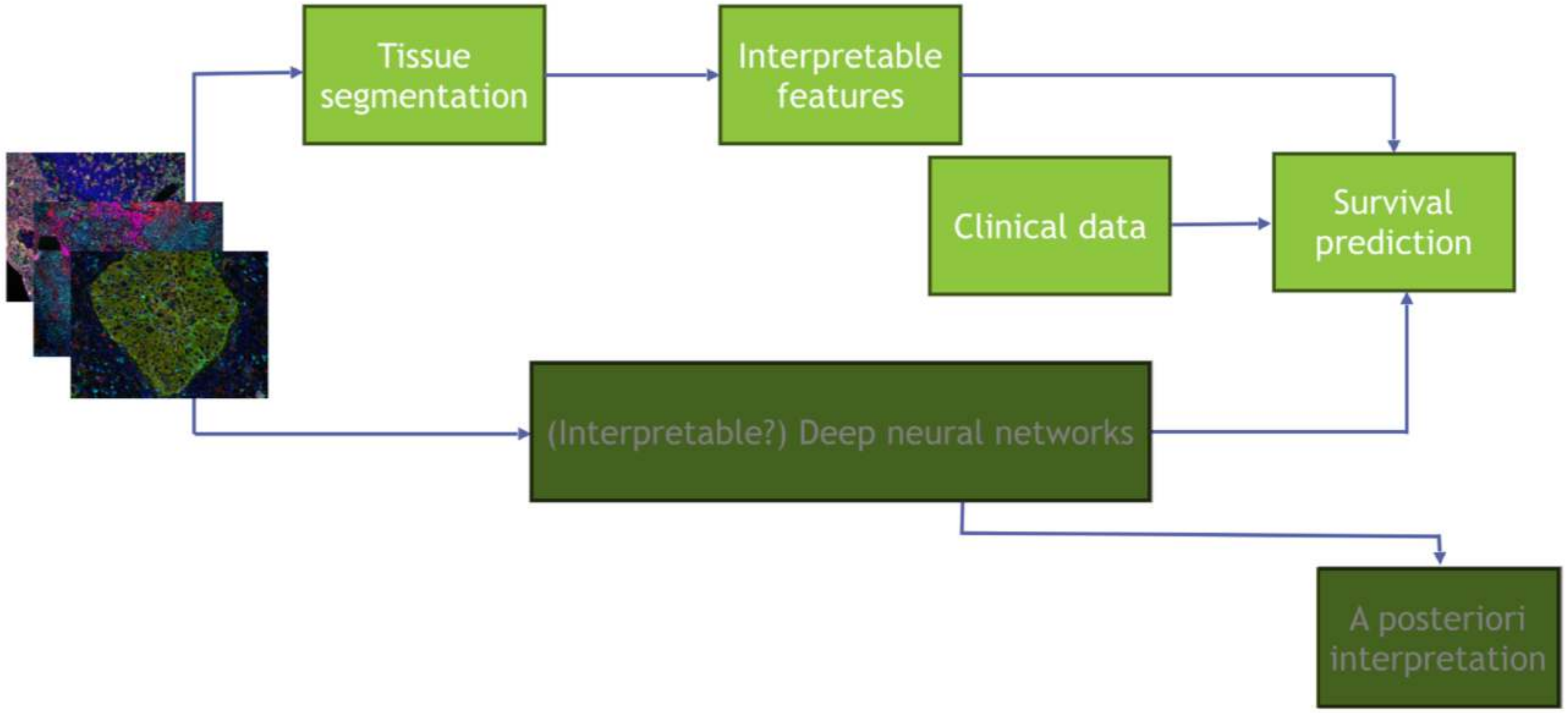


FOXP3

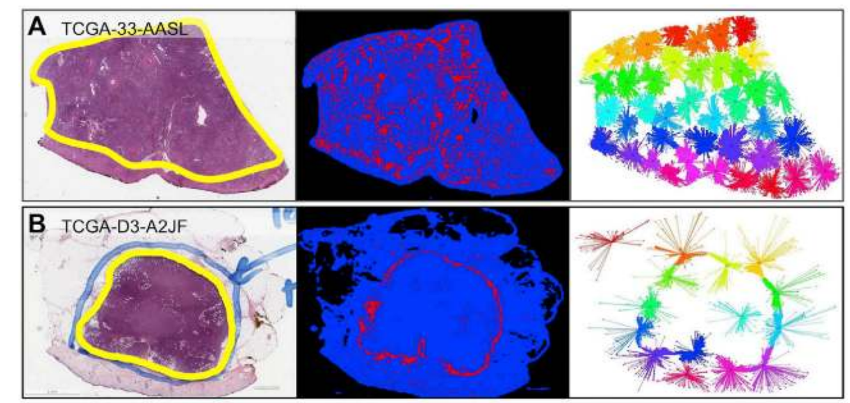


Tissue
segmentation

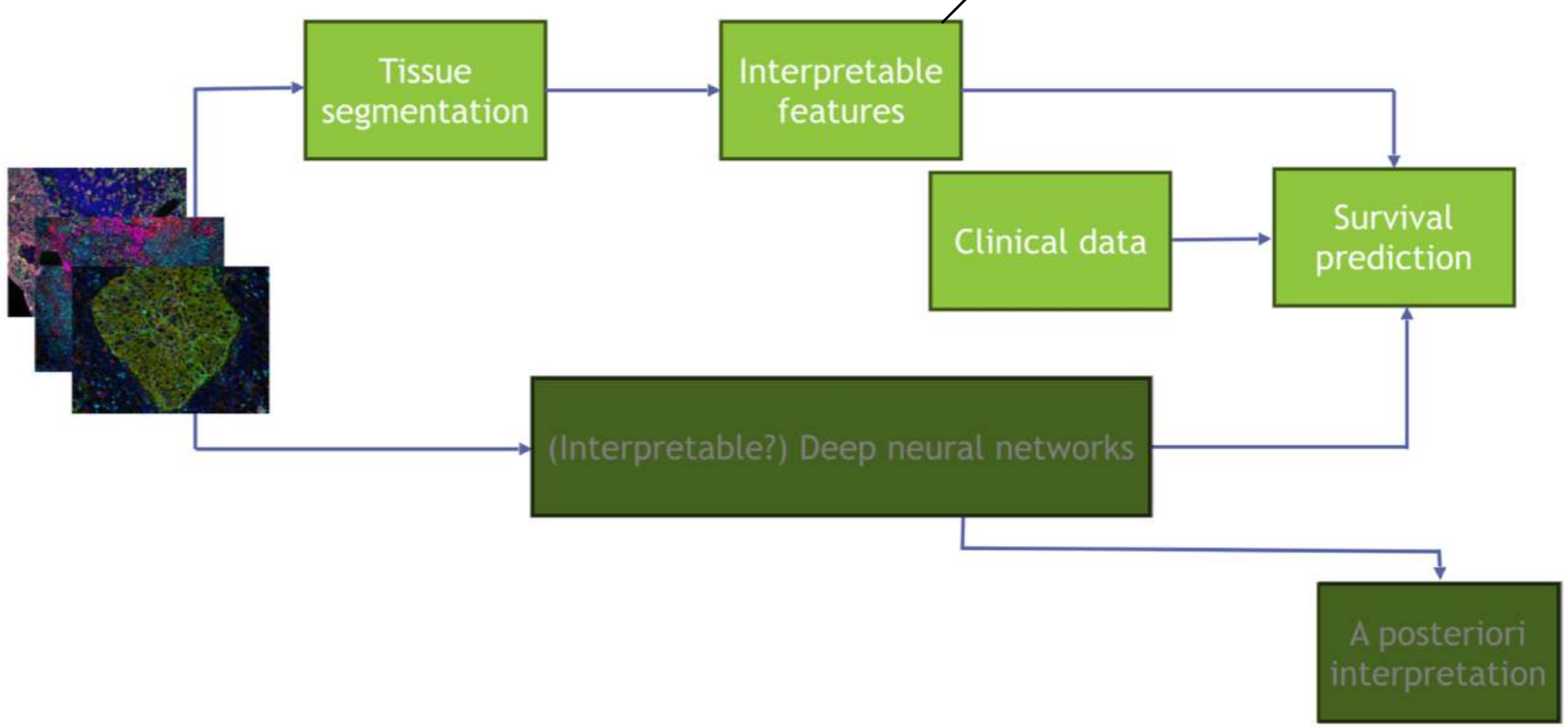
What is the link with GSP?



What is the link with GSP?

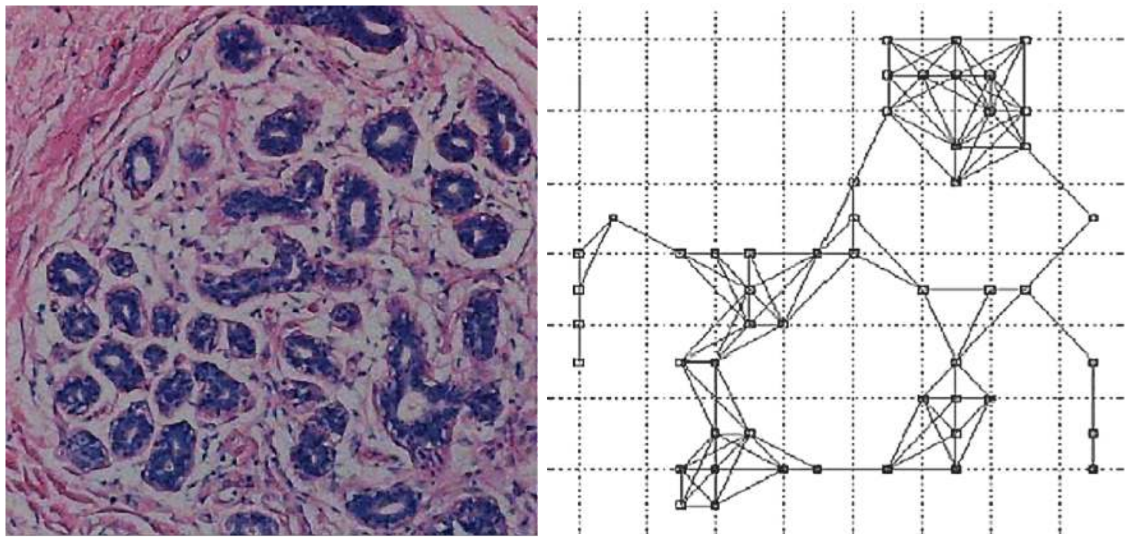


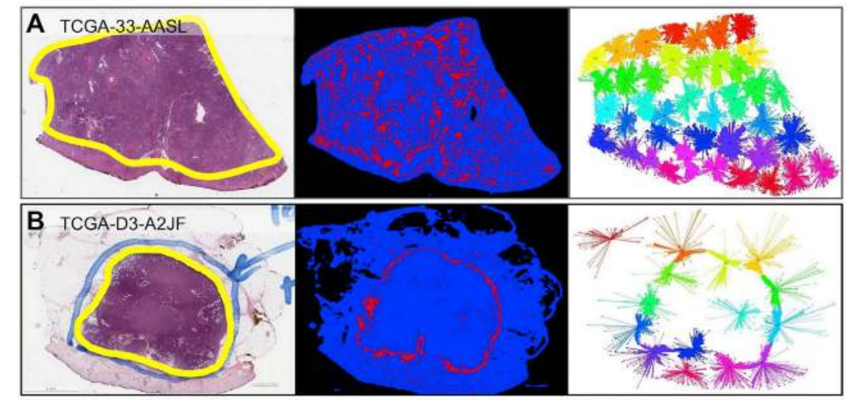
Mostly statistical [1]



[1] Saltz et al. - Spatial Organization and Molecular Correlation of Tumor-Infiltrating Lymphocytes Using Deep Learning on Pathology Images (2018)
[2] Yener - Cell-Graphs: Image-Driven Modeling of StructureFunction Relationship (2017)

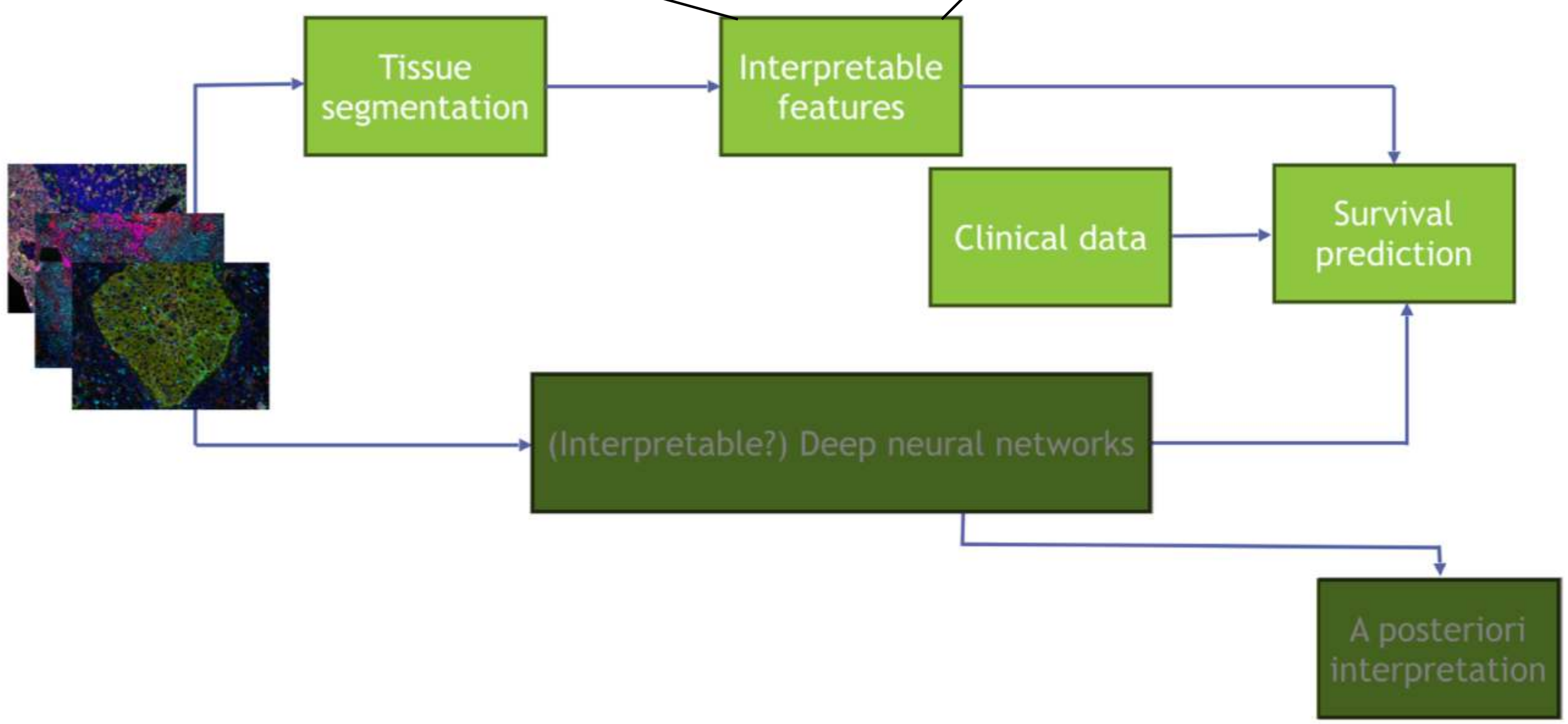
What is the link with GSP?





Graph features [2]

Mostly statistical [1]



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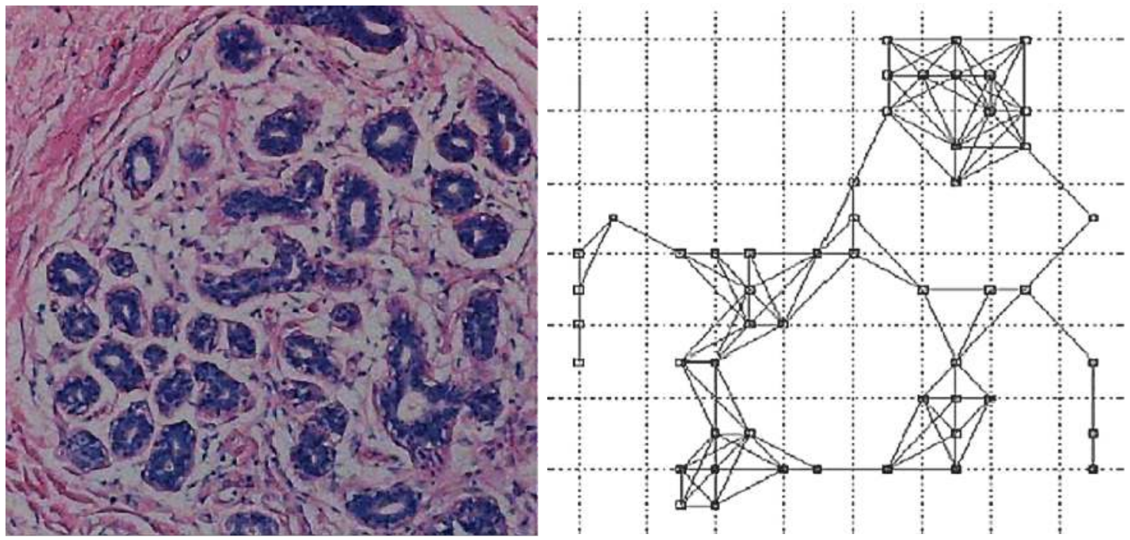
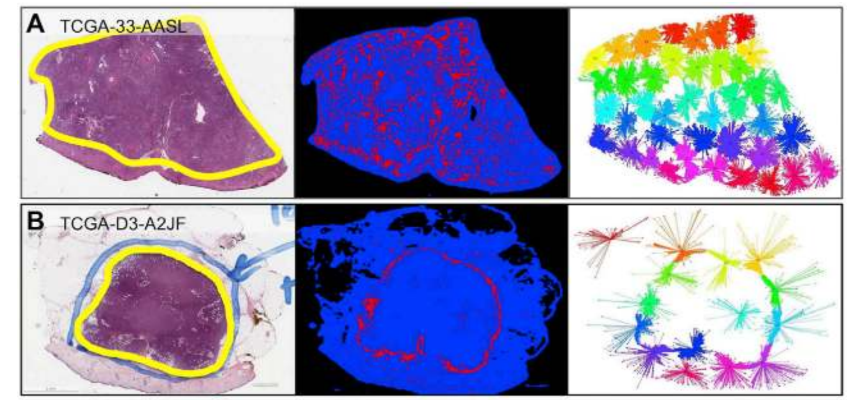
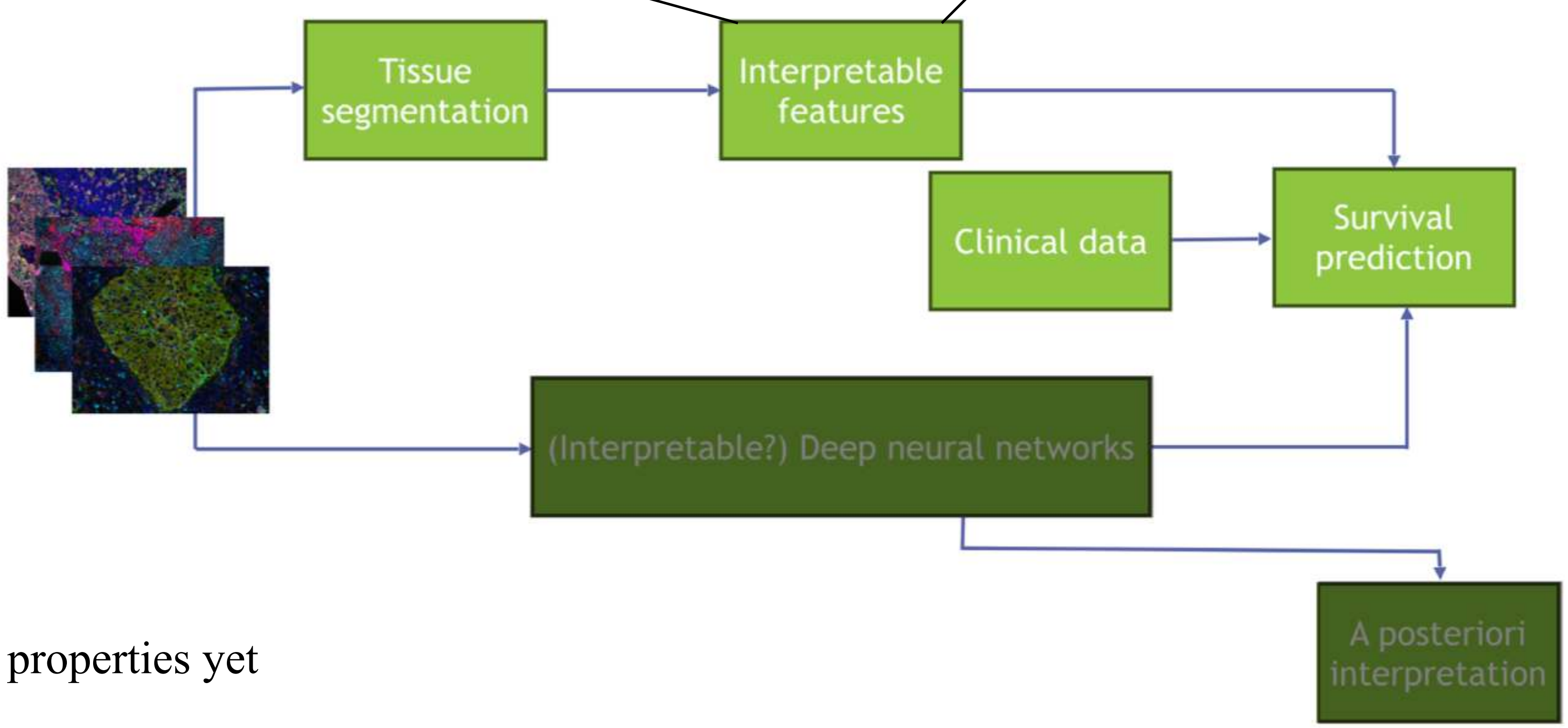
What is the link with GSP?





Graph features [2]

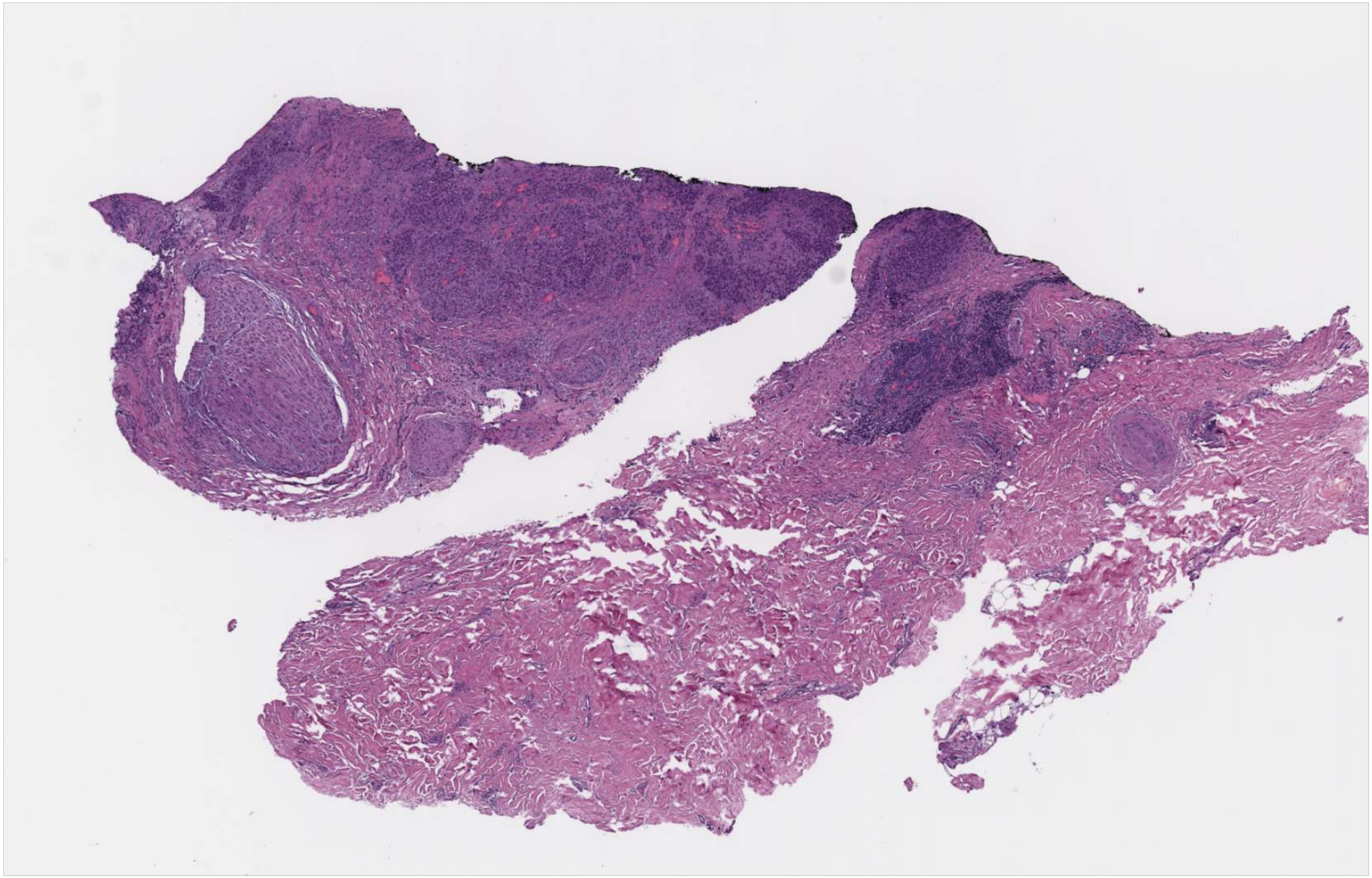
Mostly statistical [1]



No work uses signal properties yet

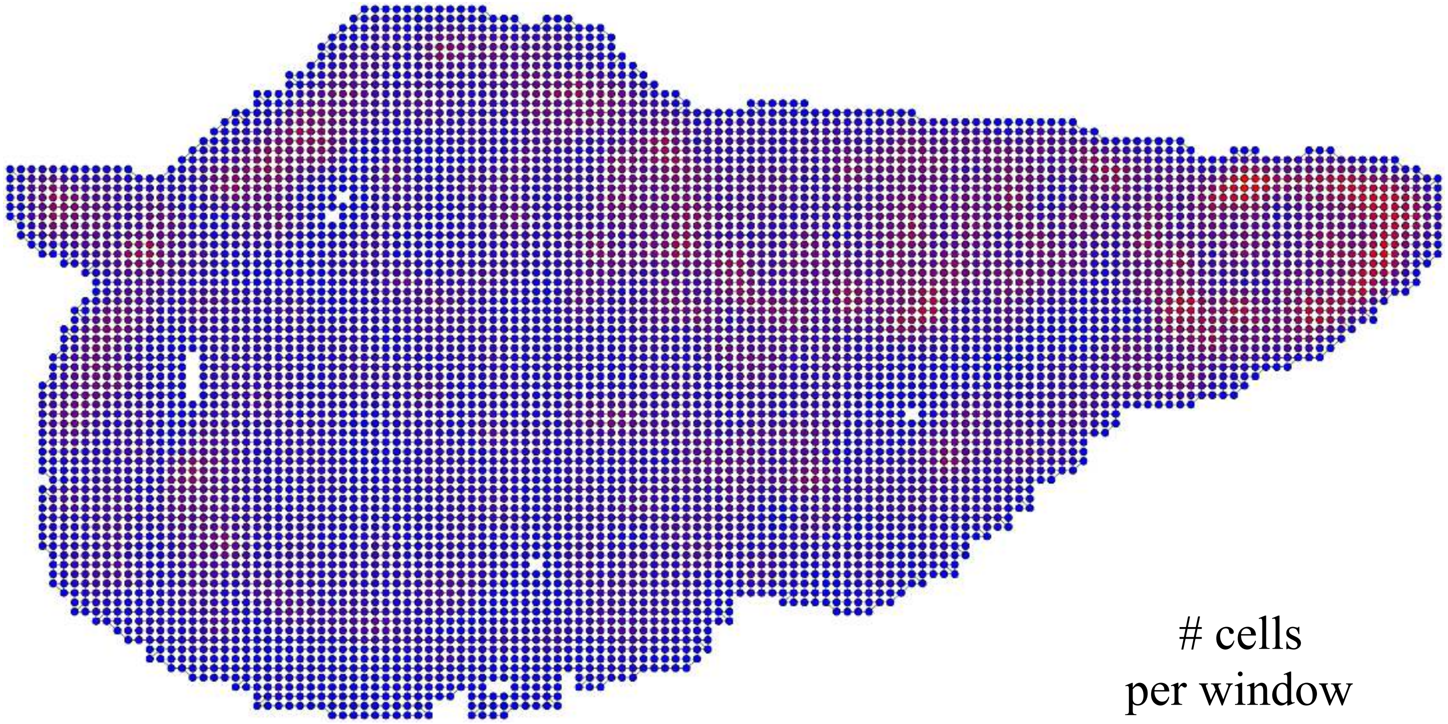
[1] Saltz et al. - Spatial Organization and Molecular Correlation of Tumor-Infiltrating Lymphocytes Using Deep Learning on Pathology Images (2018)
[2] Yener - Cell-Graphs: Image-Driven Modeling of StructureFunction Relationship (2017)

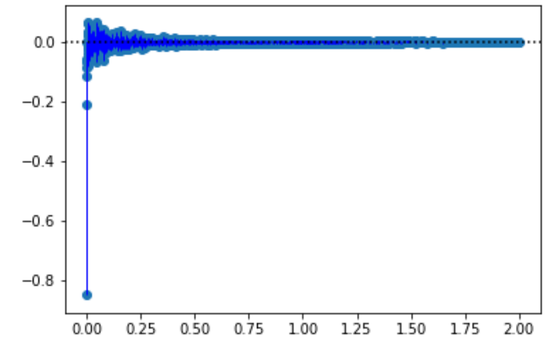
Our approach



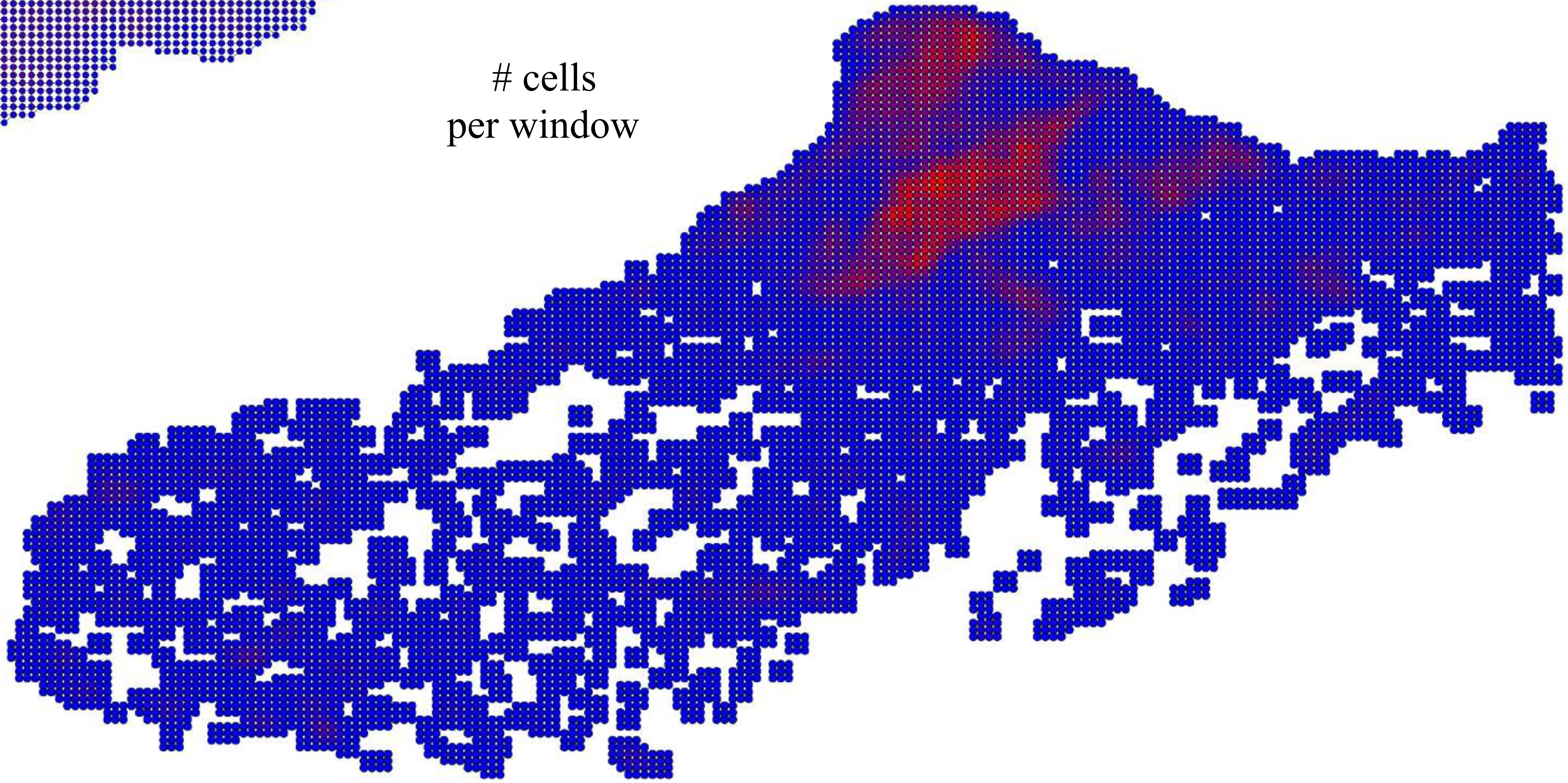
Our approach

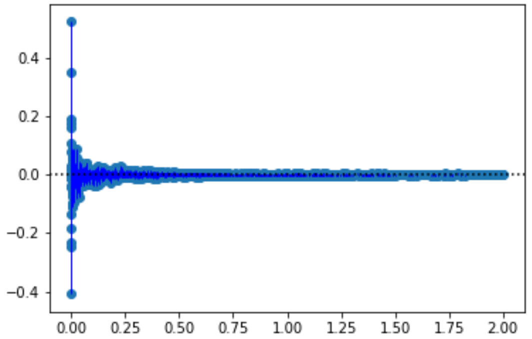
Our approach



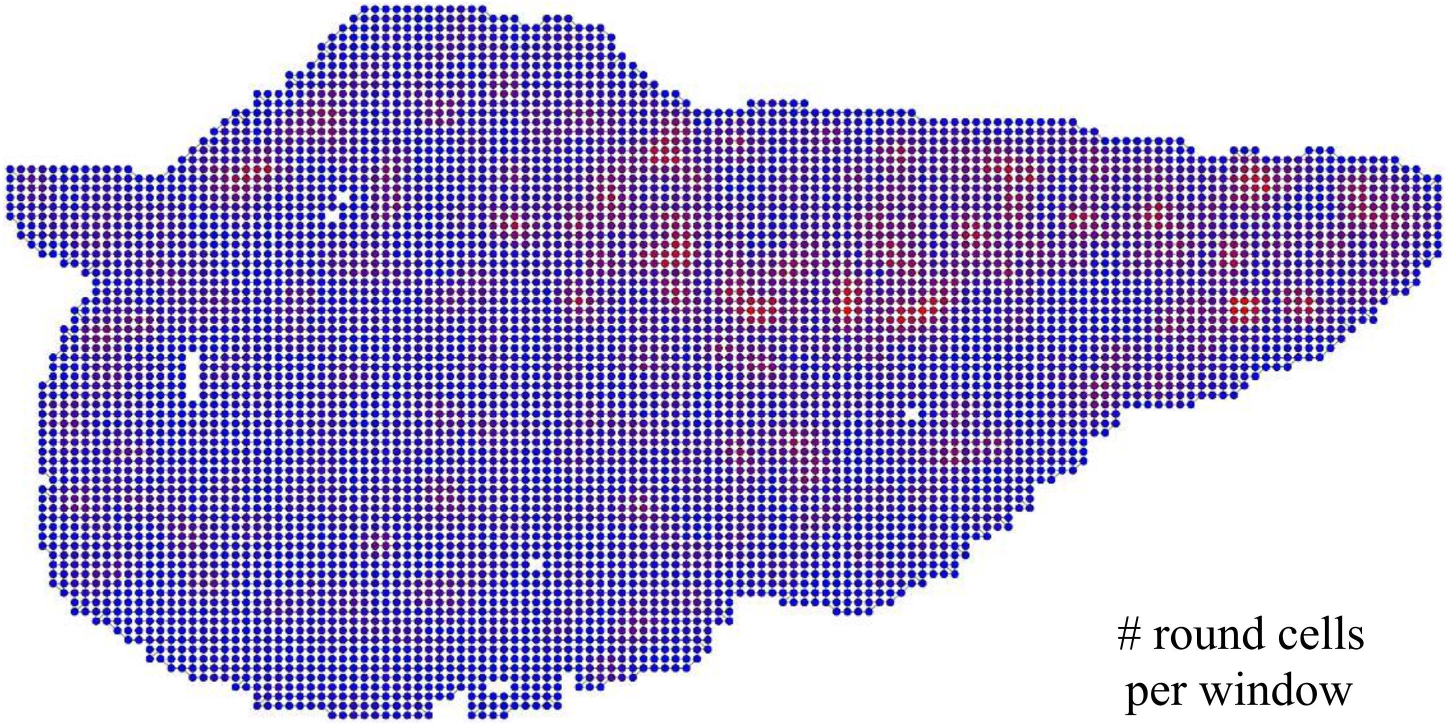


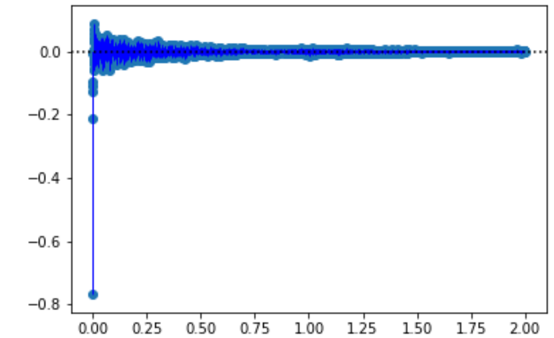
cells
per window



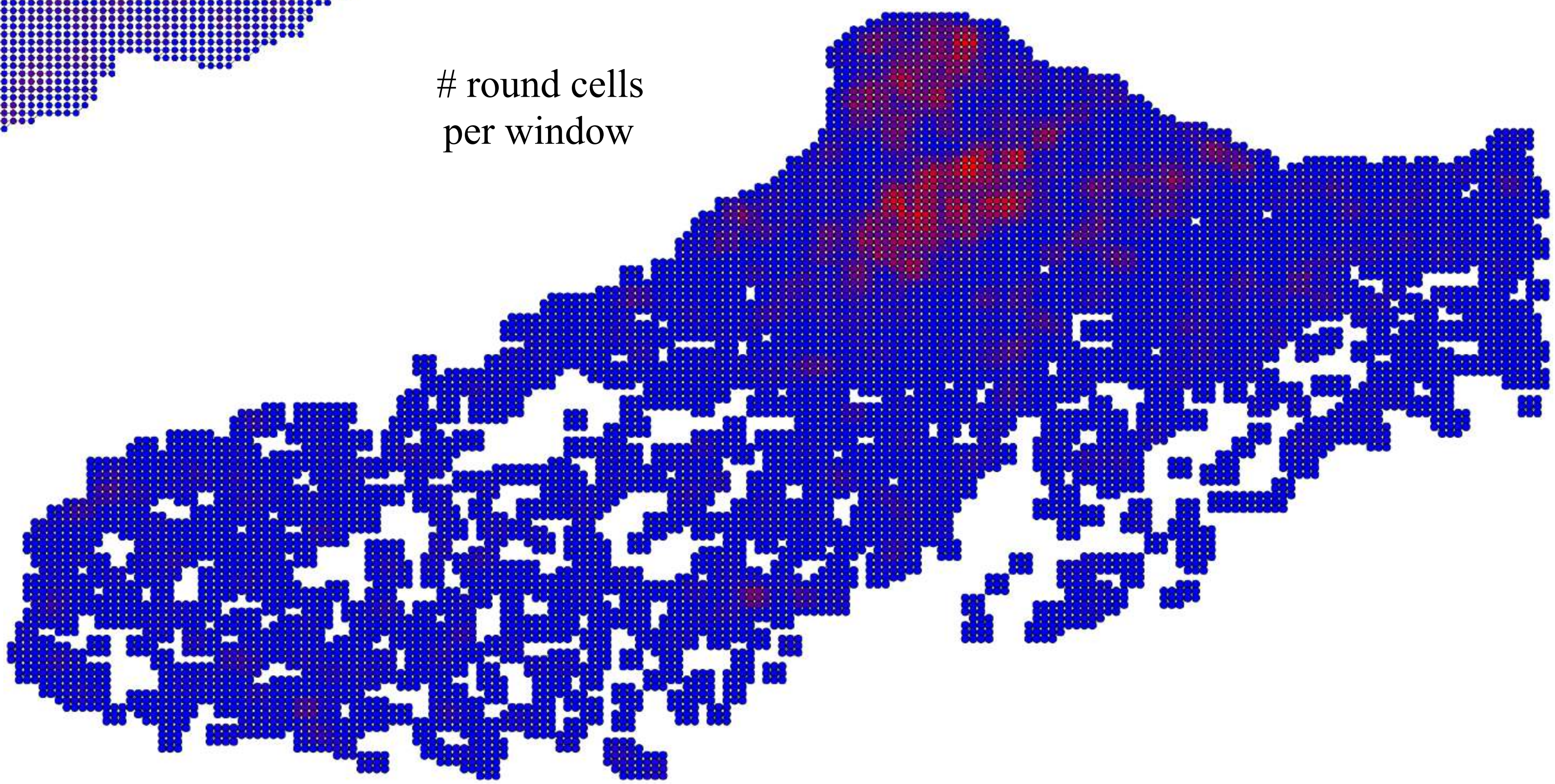


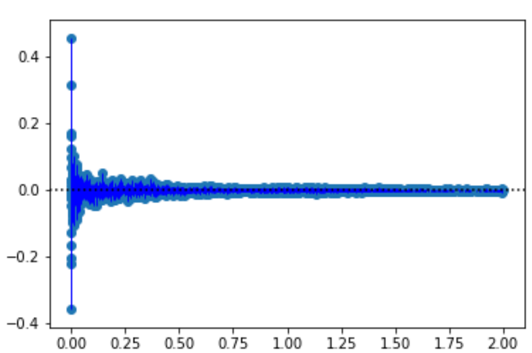
Our approach



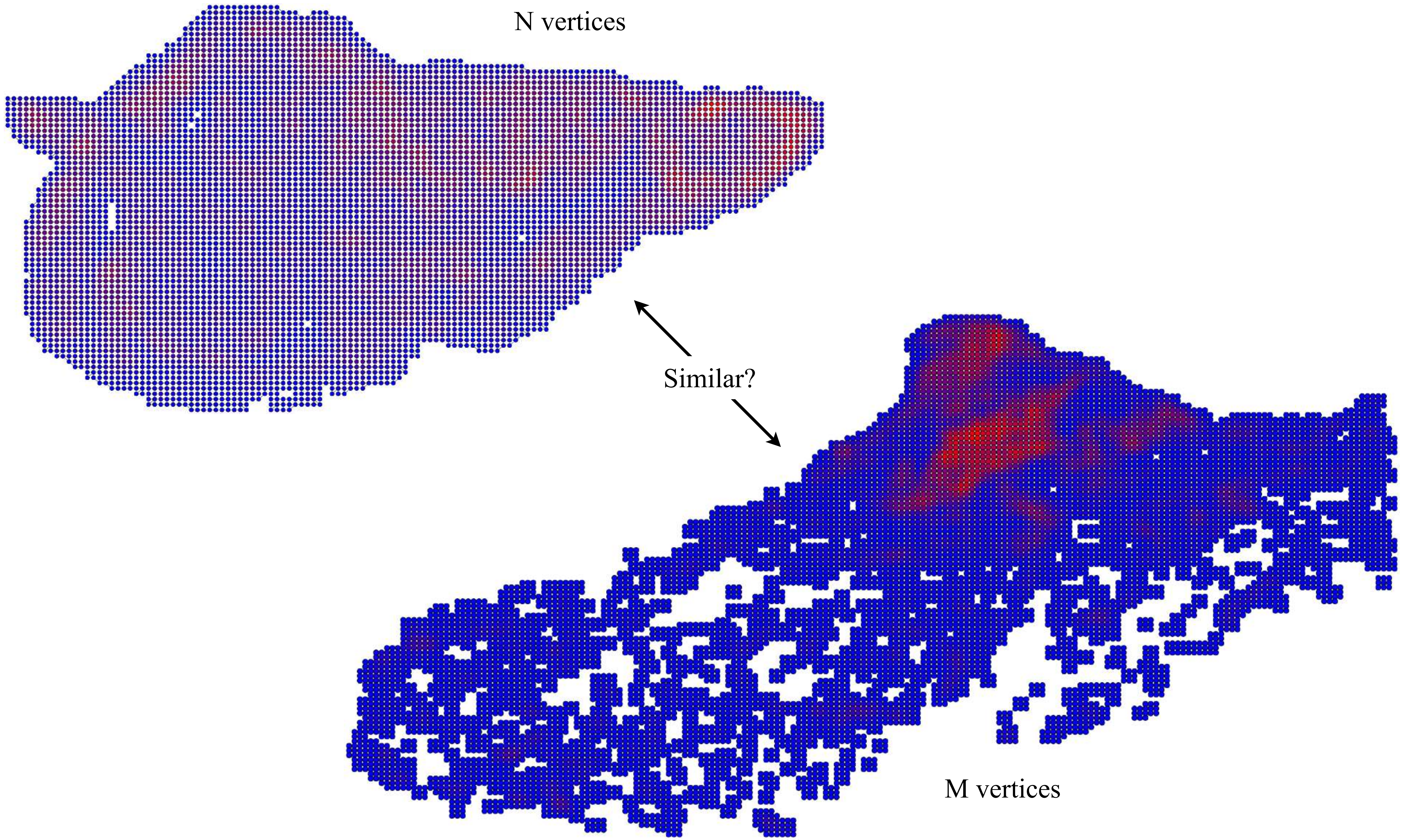


round cells
per window





An interesting GSP problem



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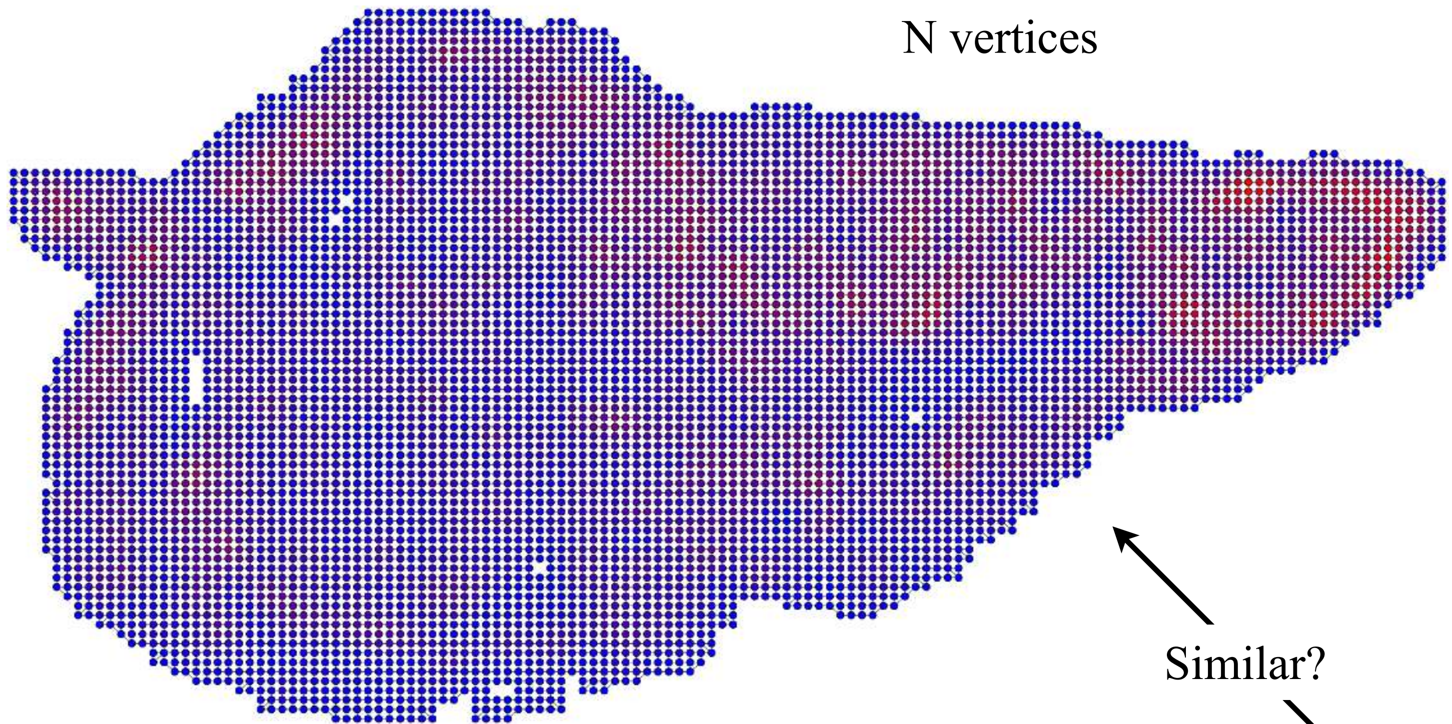
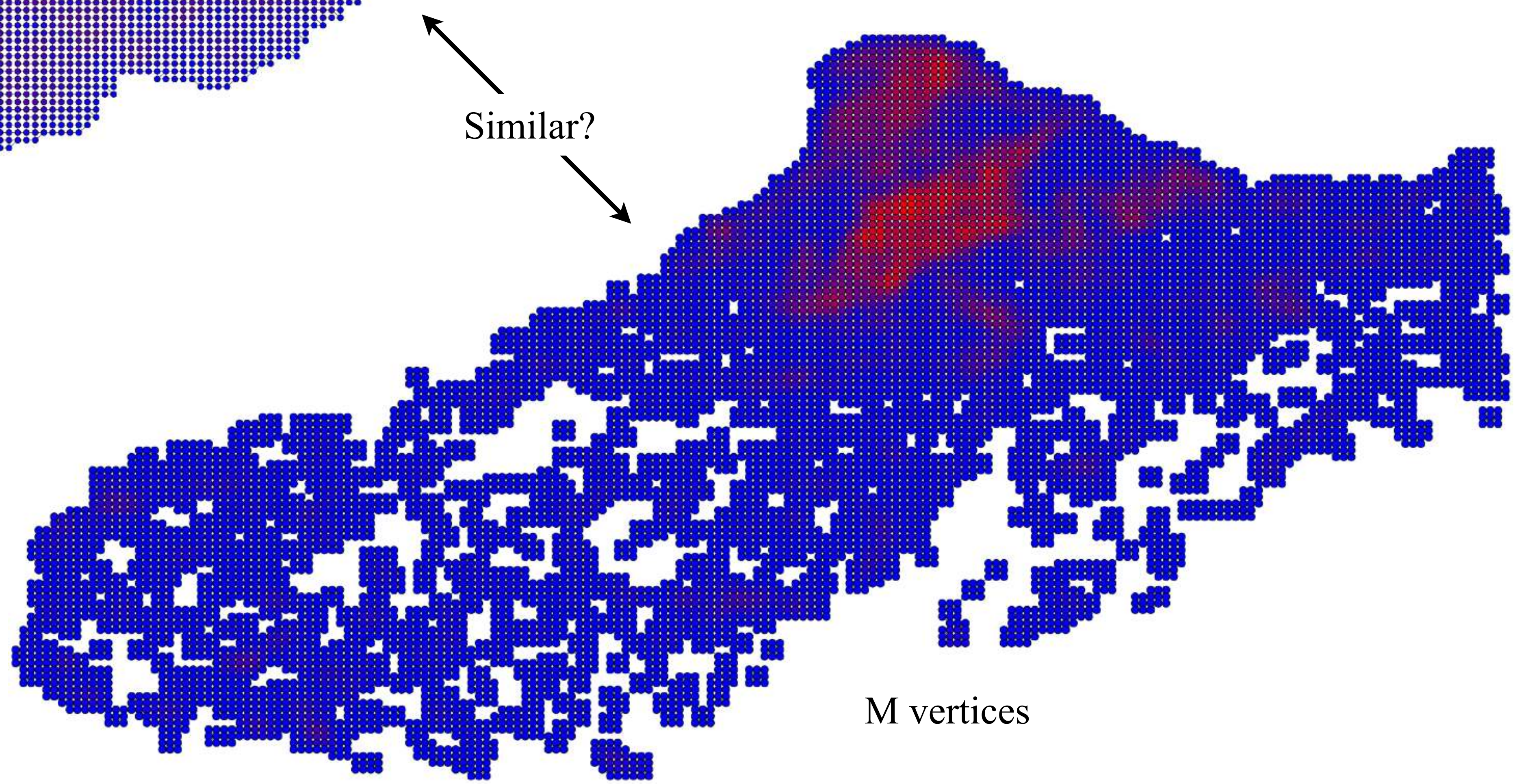


<u>Solution 1:</u> features

<u>Solution 2:</u> common representation



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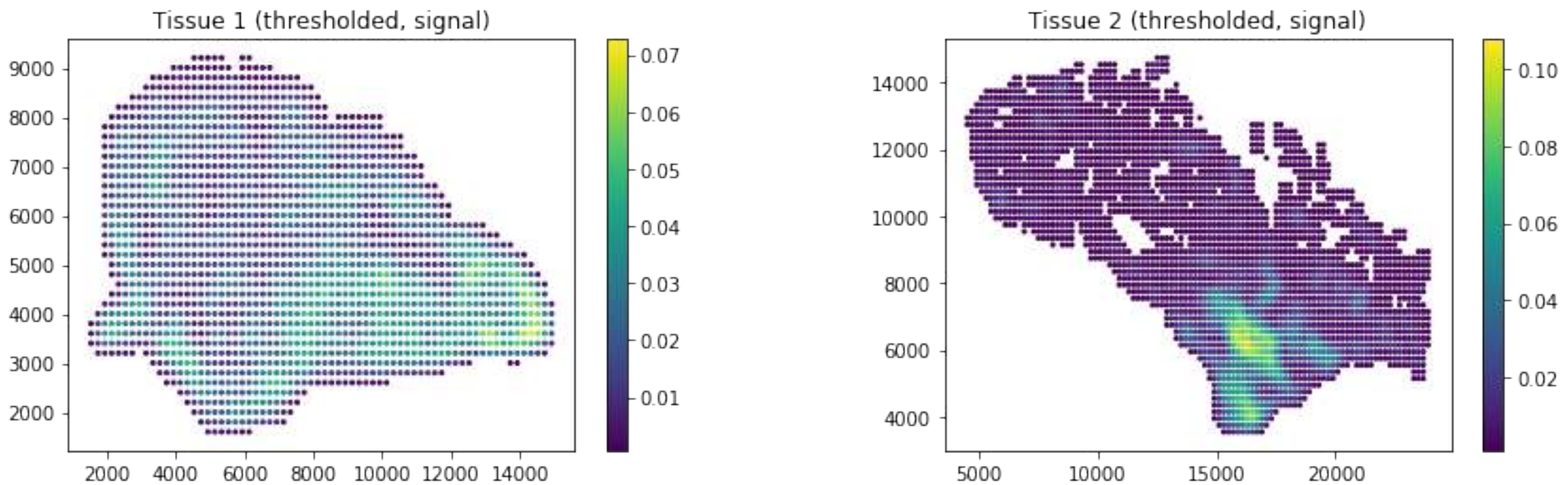
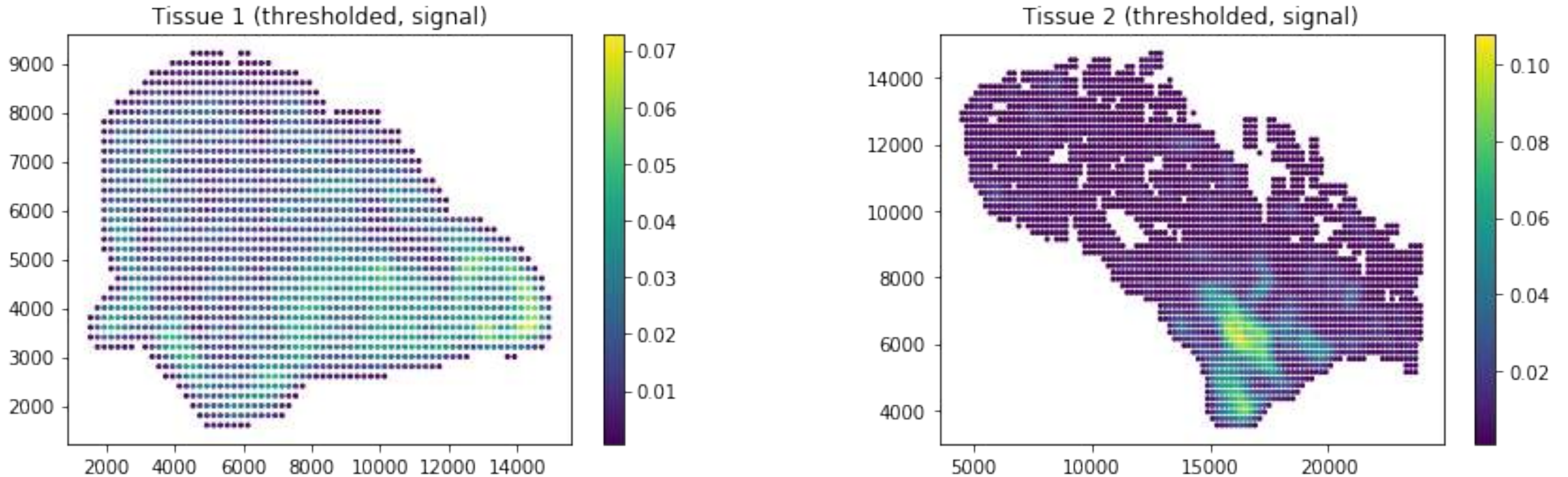


Learned with smoothness prior

An interesting GSP problem



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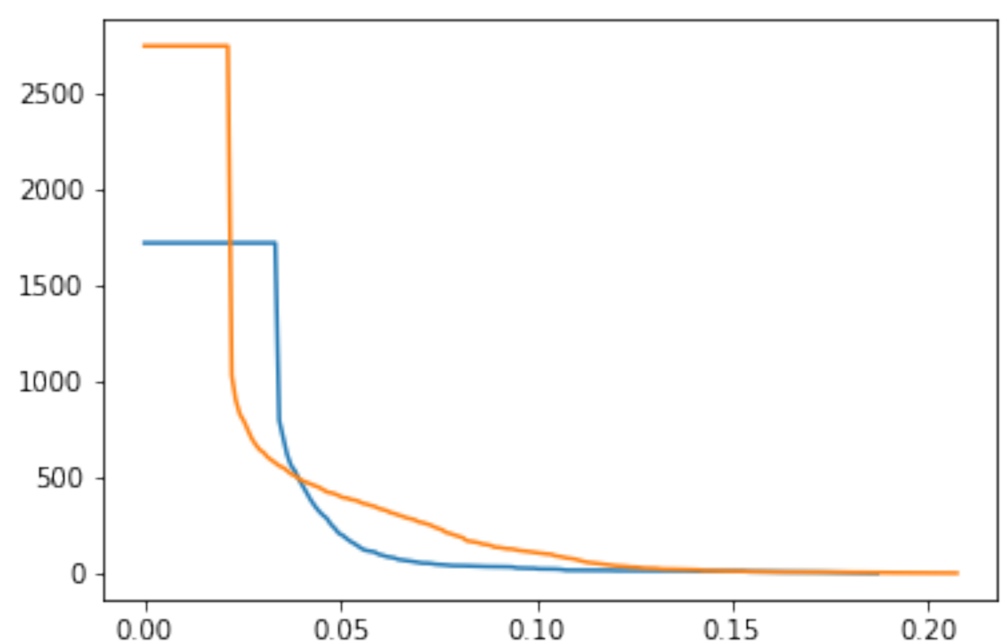

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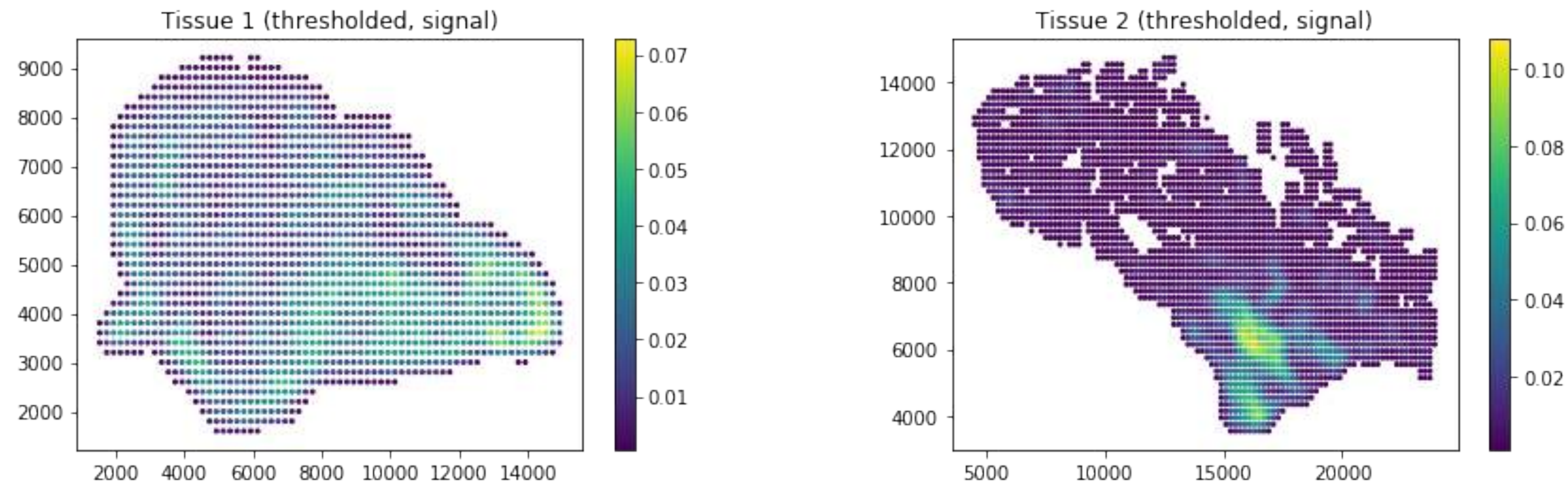
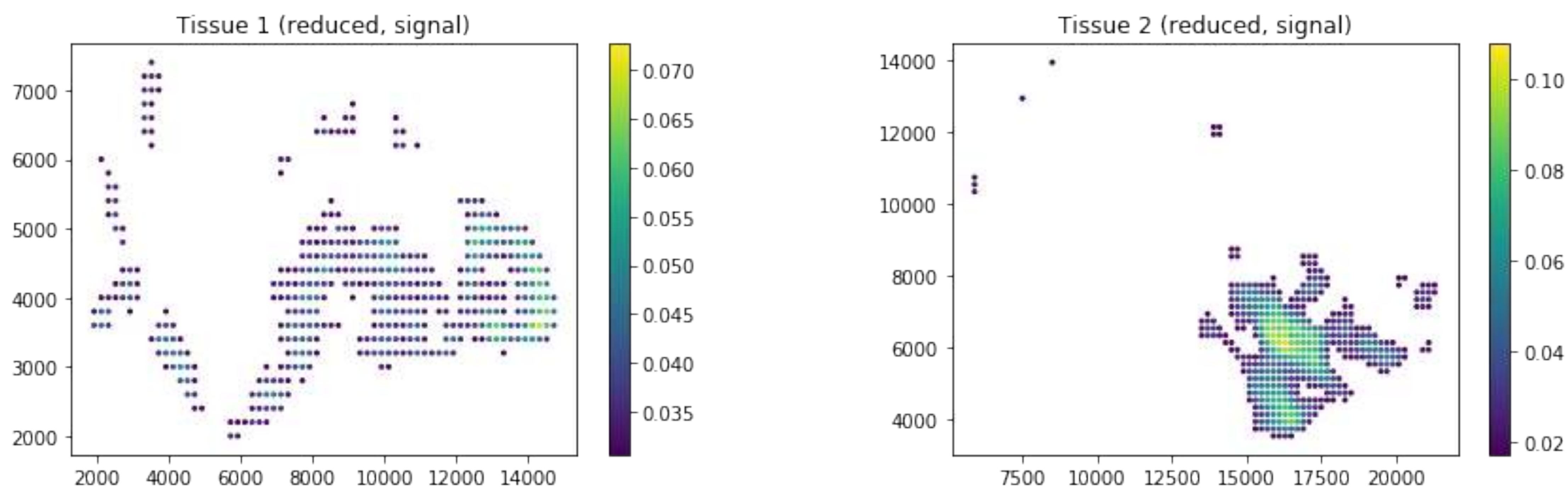
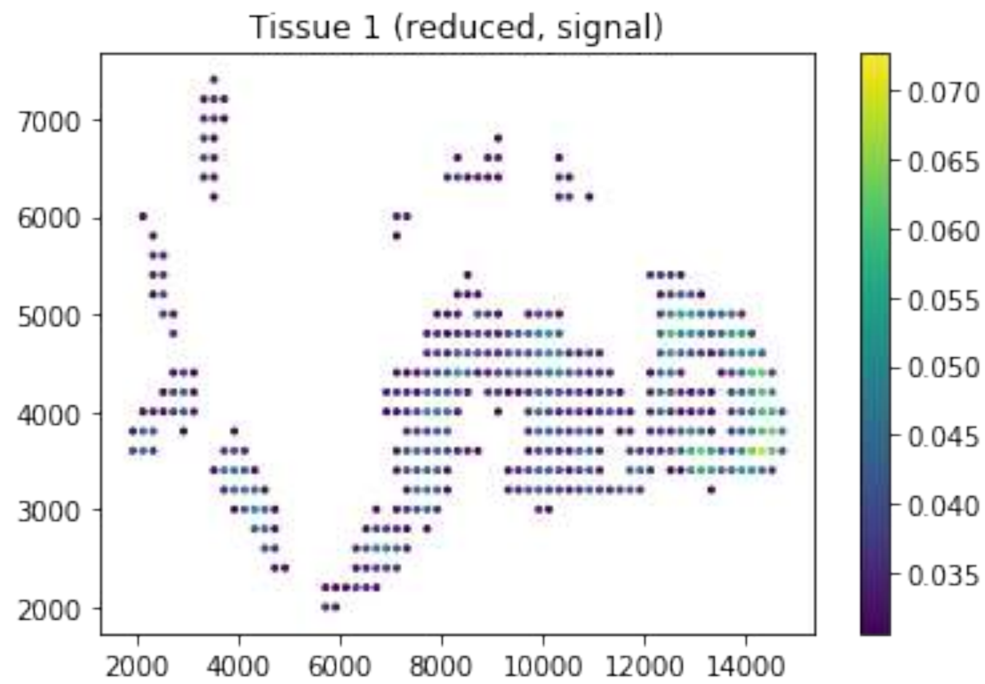


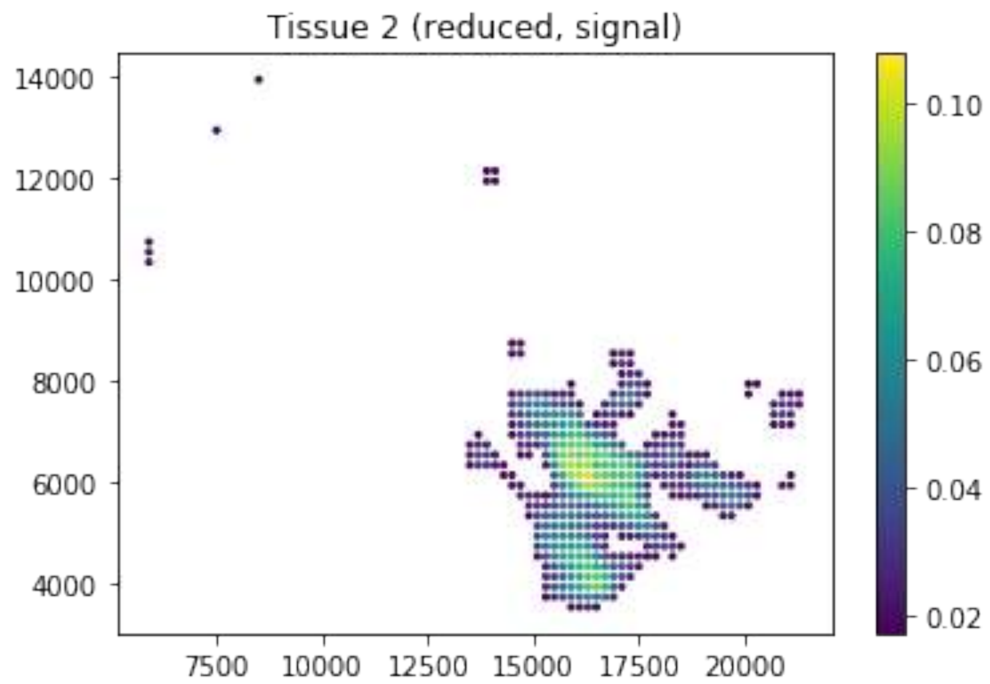
Learned with smoothness prior



Slightly different vertex count

An interesting GSP problem



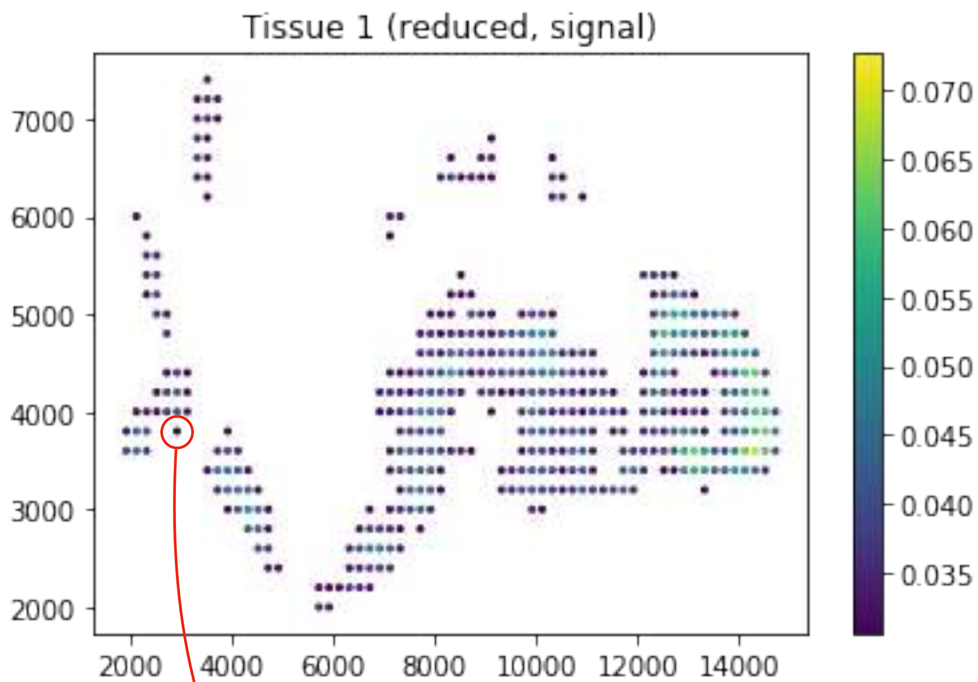


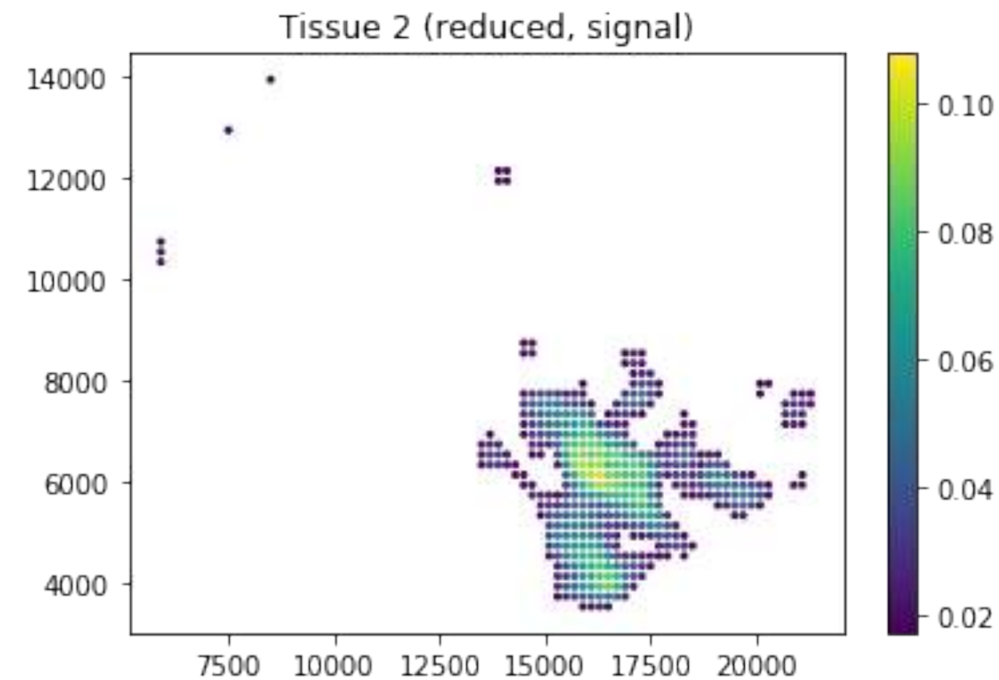
$$d_{\text{geo}}(\mathbf{A}_1, \mathbf{A}_2) = \sqrt{\sum_{1 \leq i \leq N} |\log \lambda_i|^2} \quad [1]$$

Laplacians Generalized eigenvalues

[1] Laguna et al. - Graph similarity based on graph Fourier distances (2018)

An interesting GSP problem





Disconnect the minimizer

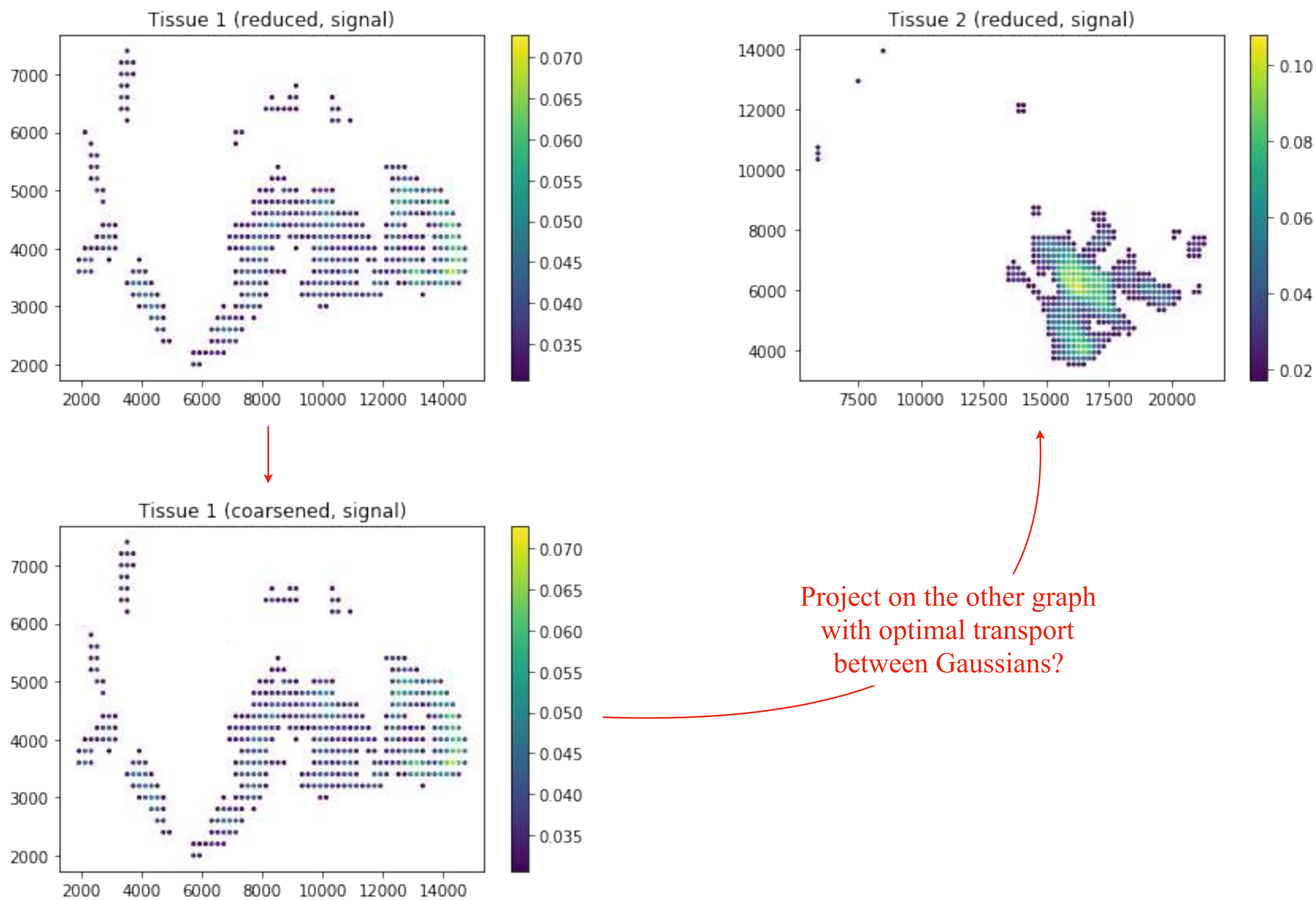
$$d_{\text{geo}}(\mathbf{A}_1, \mathbf{A}_2) = \sqrt{\sum_{1 \leq i \leq N} |\log \lambda_i|^2} \quad [1]$$

Laplacians

Generalized eigenvalues

[1] Laguna et al. - Graph similarity based on graph Fourier distances (2018)

An interesting GSP problem

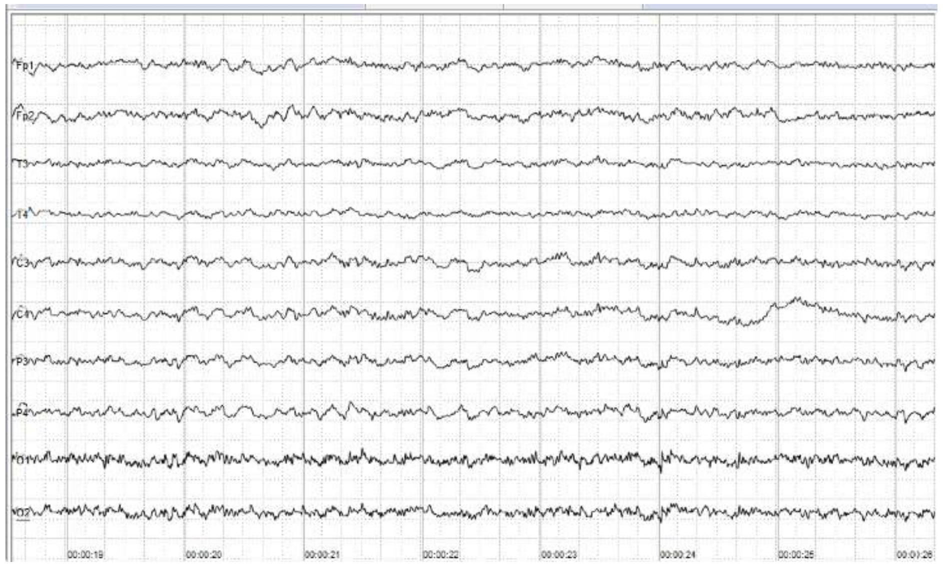


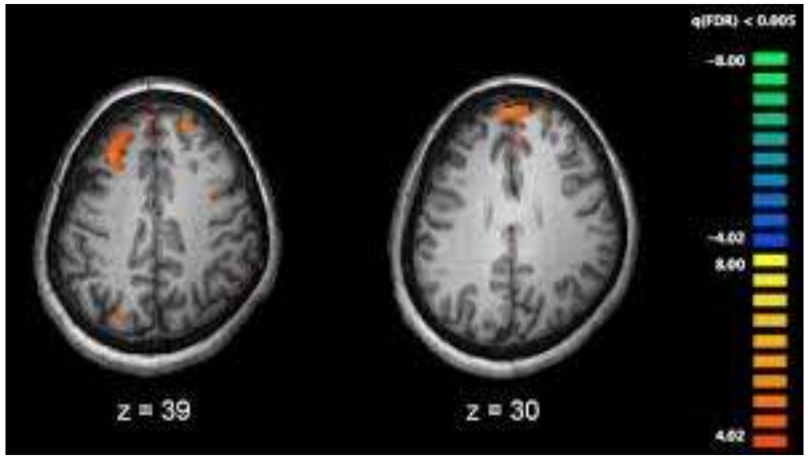
Project on the other graph
with optimal transport
between Gaussians?

Outline

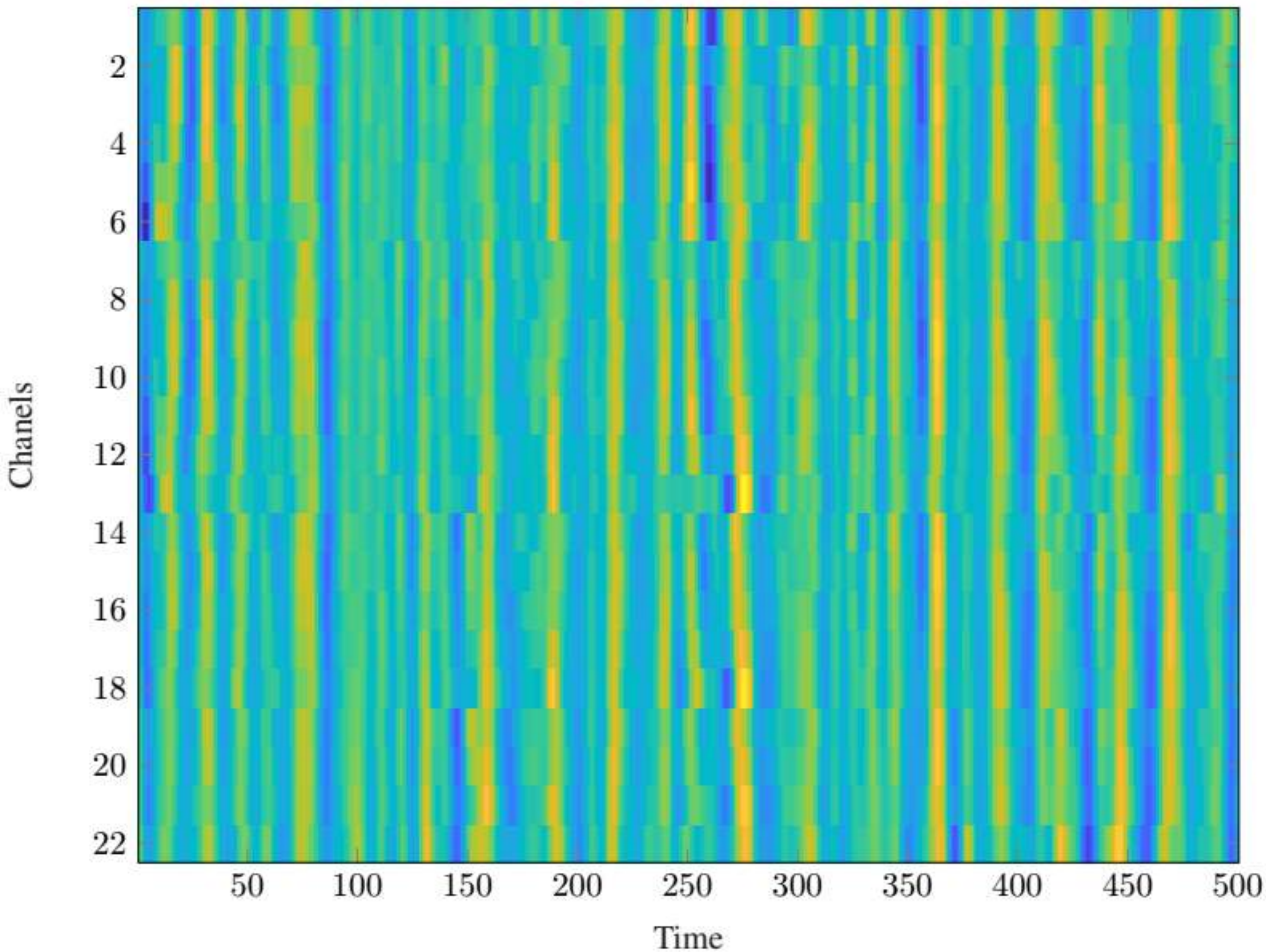
- GSP for AI, *a.k.a.* "What I did during my Ph.D. and a bit after"
- GSP and personalized healthcare
- **GSP and brain signals**

EEG and fMRI measure brain activity

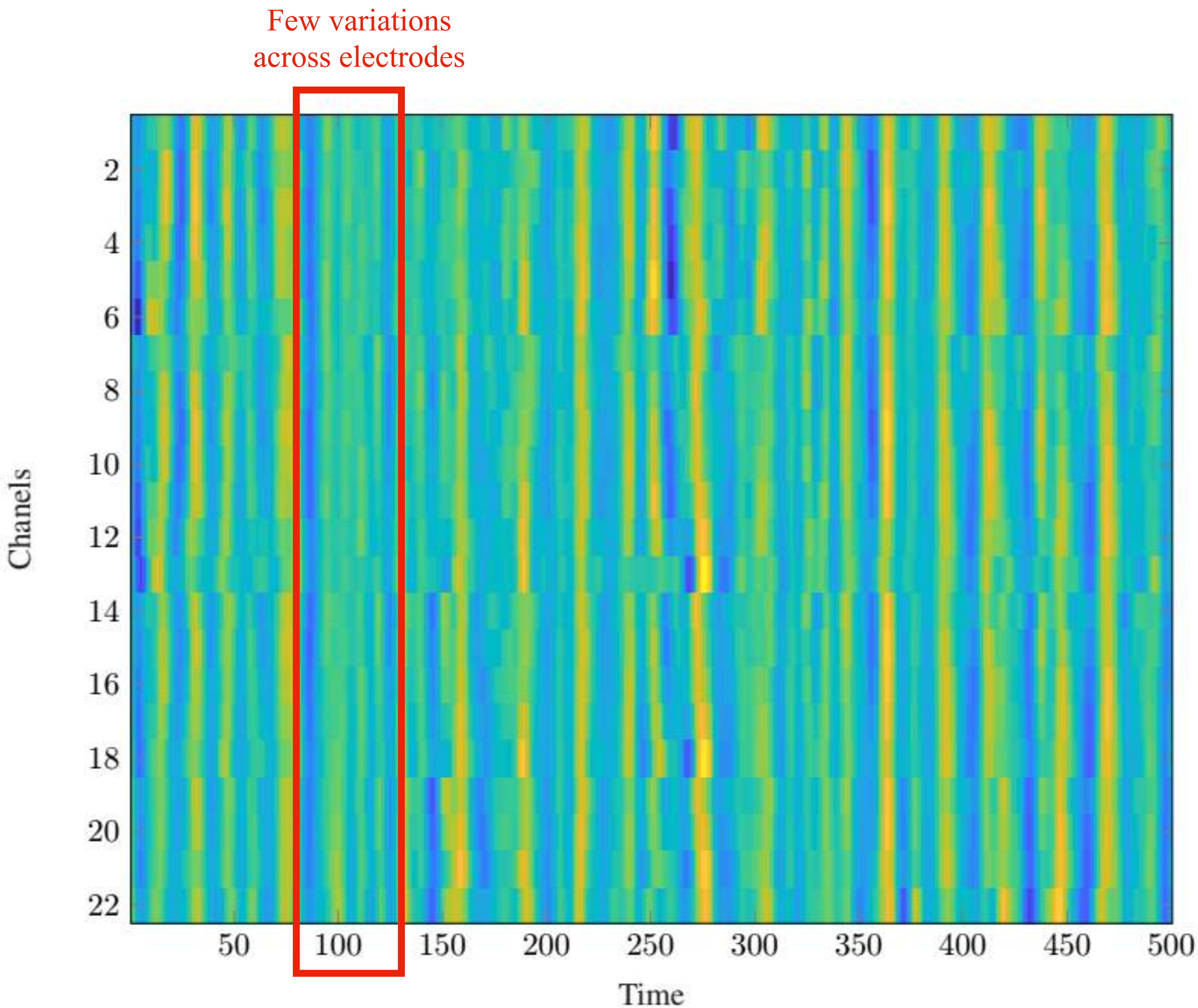




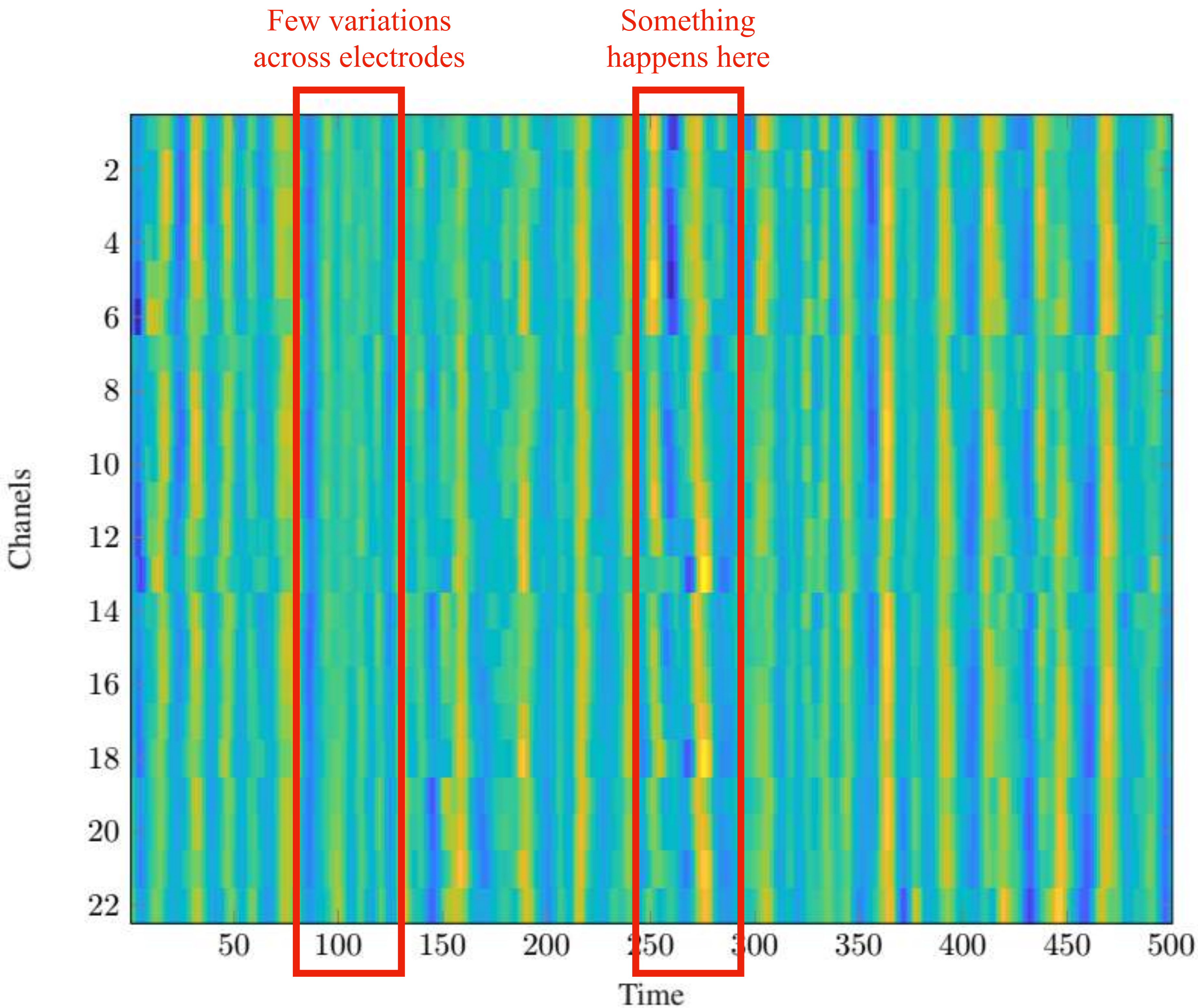
Task seen as non-stationary behavior



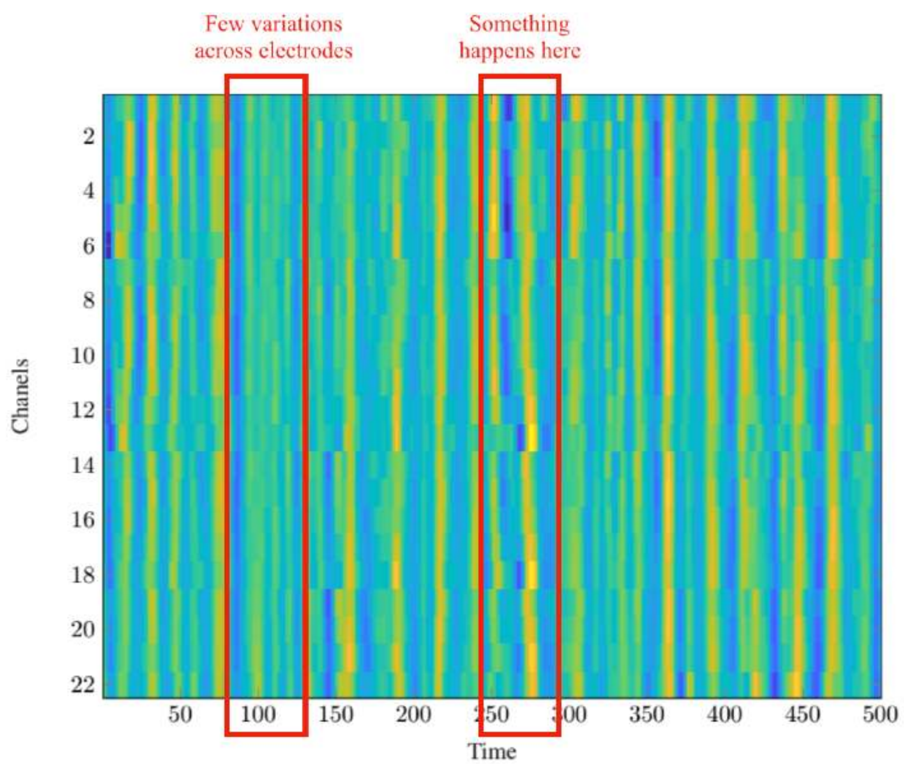
Task seen as non-stationary behavior



Task seen as non-stationary behavior



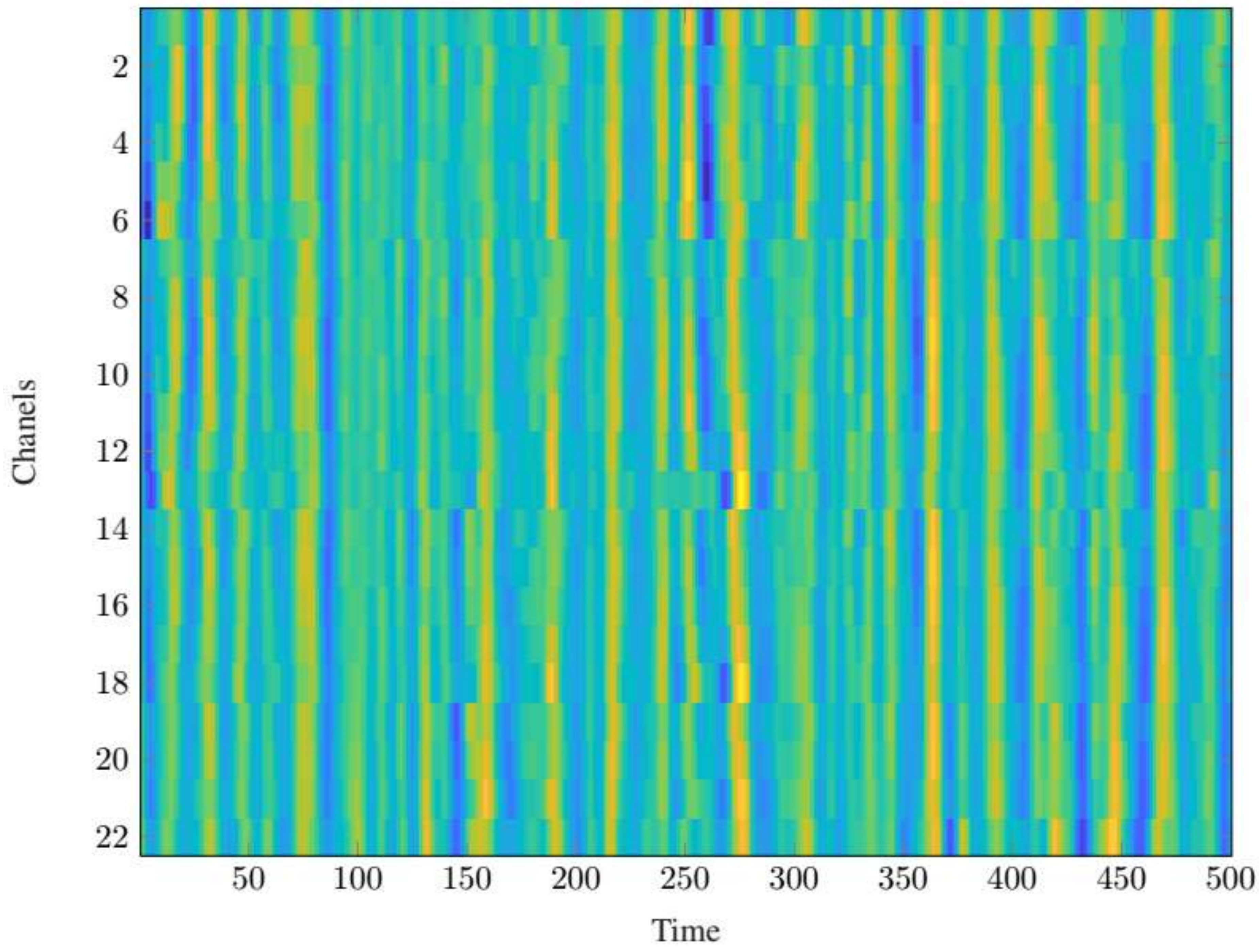
How do we measure stationarity?



$$
\begin{aligned}
\widetilde{\boldsymbol{\Sigma}} &= \sum_{i=1}^{M} \mathbf{X}(i)\mathbf{X}(i)^\top \\
&= \sum_{k \in \mathbf{k}} \sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i)=k}} \mathbf{T}^k \mathbf{Y}(i)\mathbf{Y}(i)^\top {\mathbf{T}^k}^\top \\
&= \sum_{k \in \mathbf{k}} \mathbf{T}^k \left(\sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i)=k}} \mathbf{Y}(i)\mathbf{Y}(i)^\top \right) {\mathbf{T}^k}^\top \\
\boldsymbol{\Sigma} &= \sum_{k \in \mathbf{k}} \mathbf{T}^k \mathbb{E}_{\mathbf{Y}} \left[\sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i)=k}} \mathbf{Y}(i)\mathbf{Y}(i)^\top \right] {\mathbf{T}^k}^\top \\
&= \sum_{k \in \mathbf{k}} \left| \{ i, \mathbf{k}(i) = k \} \right| \mathbf{T}^{2k} \ ,
\end{aligned}
$$

The covariance matrix of signals that are stationary on a graph shares the same eigenvectors as the graph itself

How do we measure stationarity?



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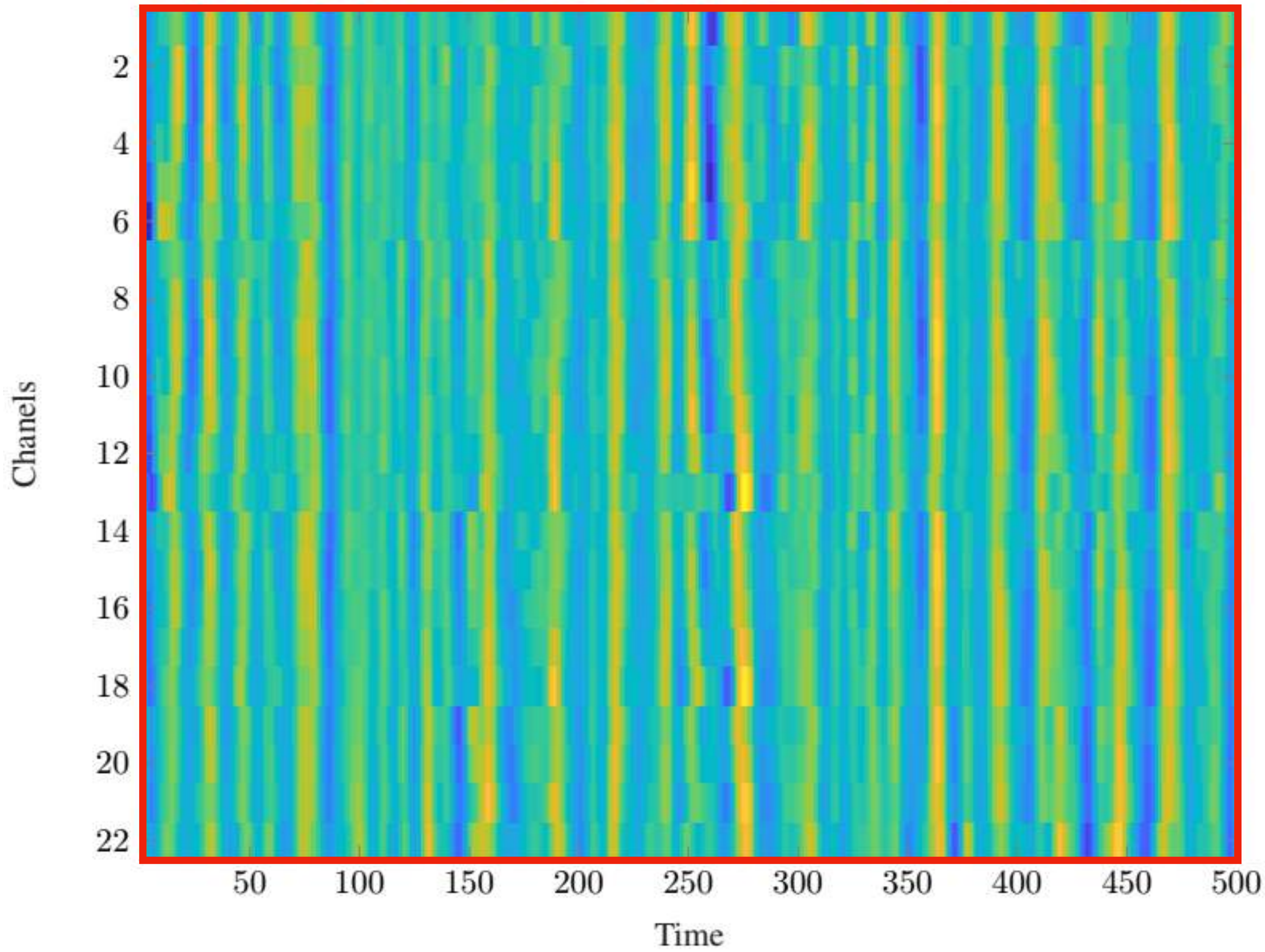
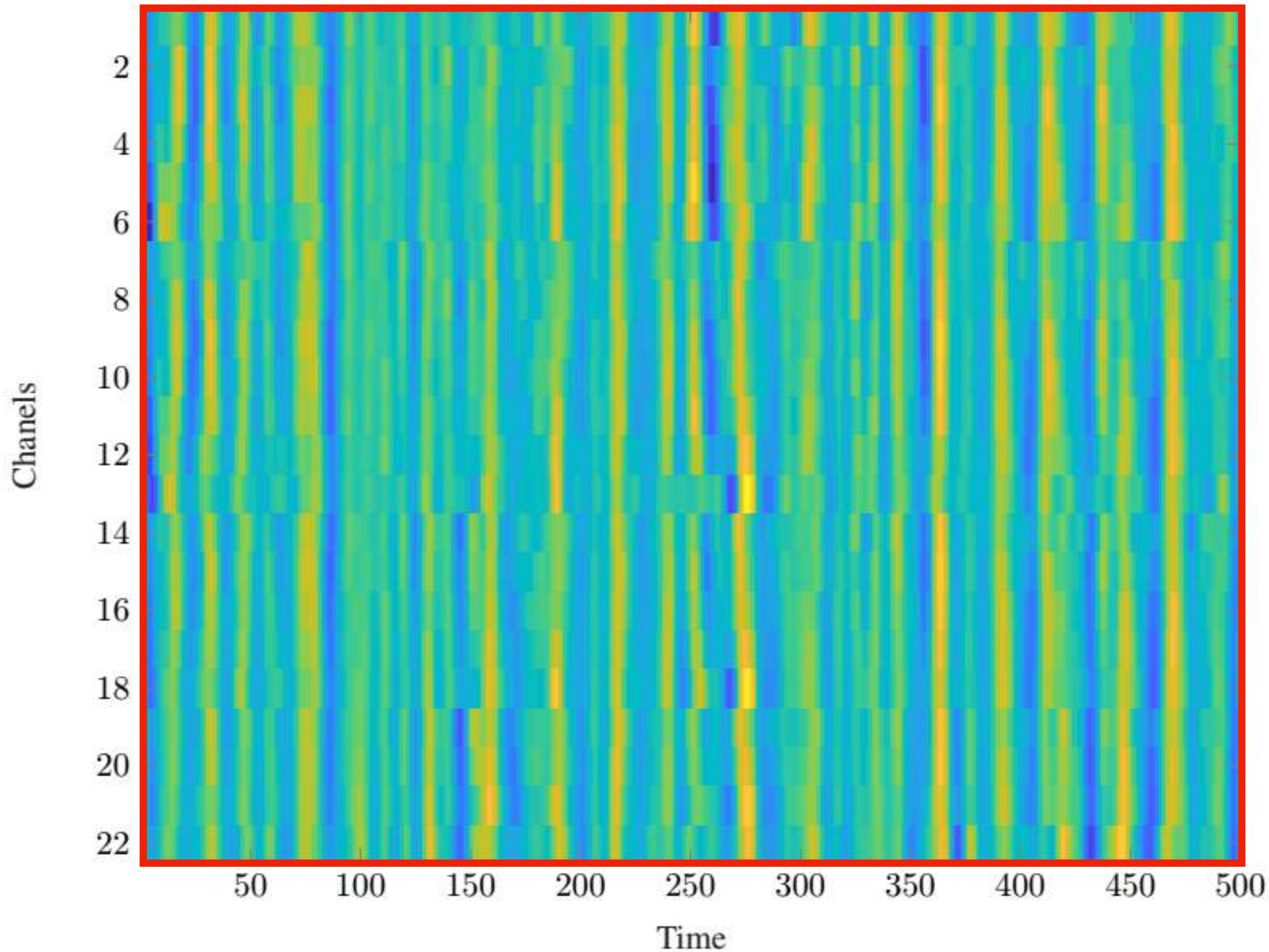


Σ

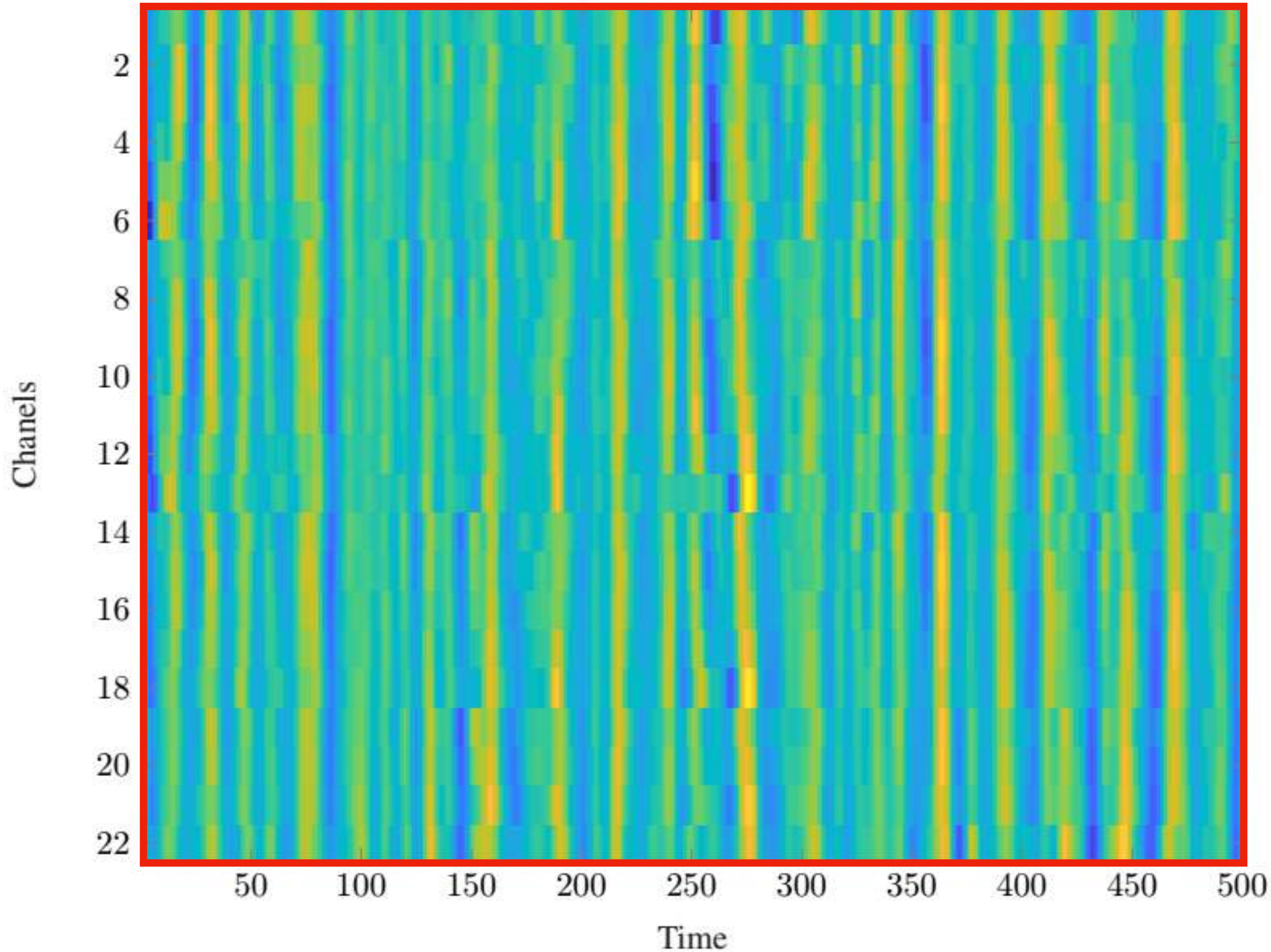
How do we measure stationarity?



Σ

Σ_1

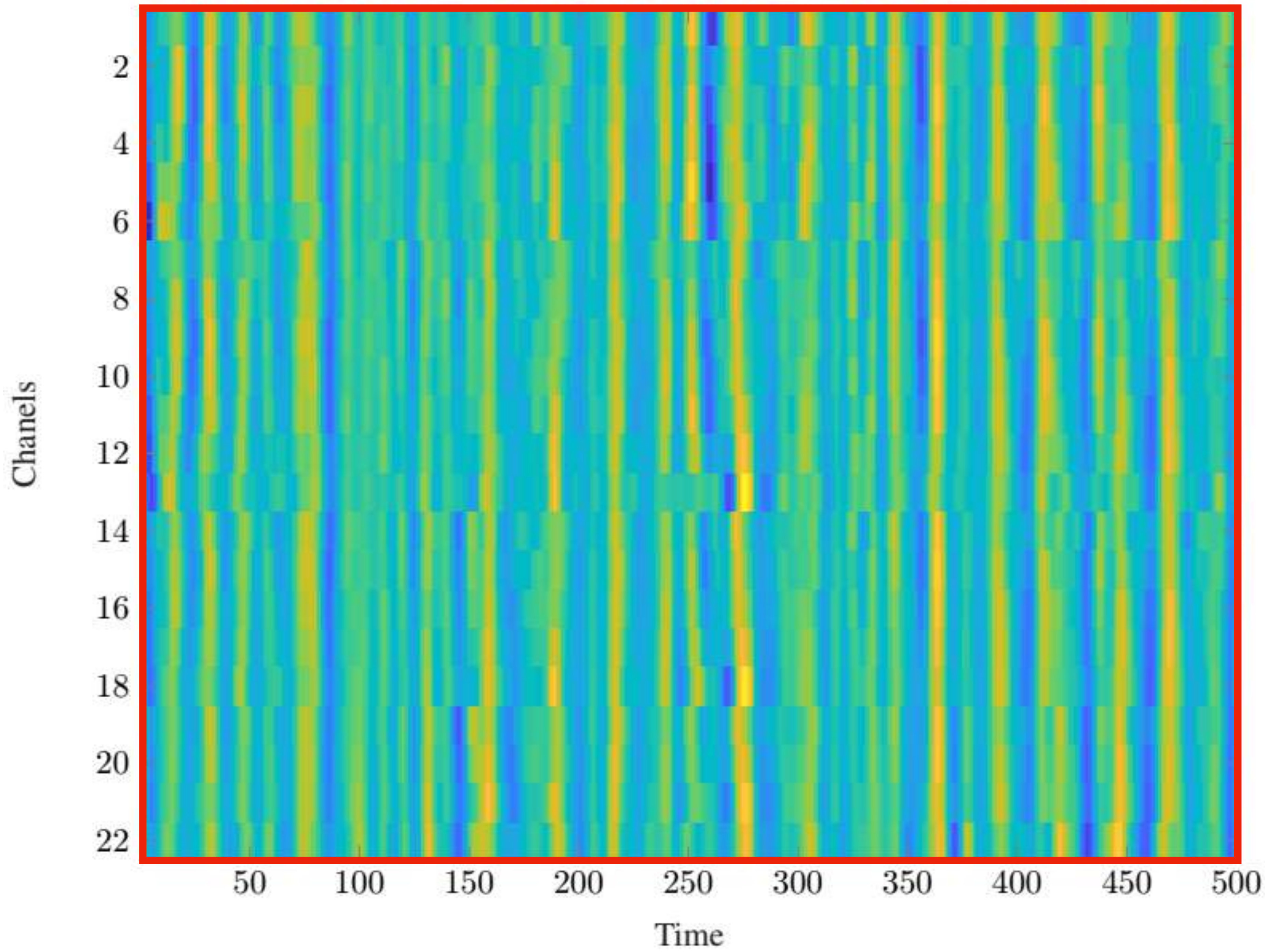
How do we measure stationarity?



$\boldsymbol{\Sigma}$

$\boldsymbol{\Sigma}_1 \quad \boldsymbol{\Sigma}_2$

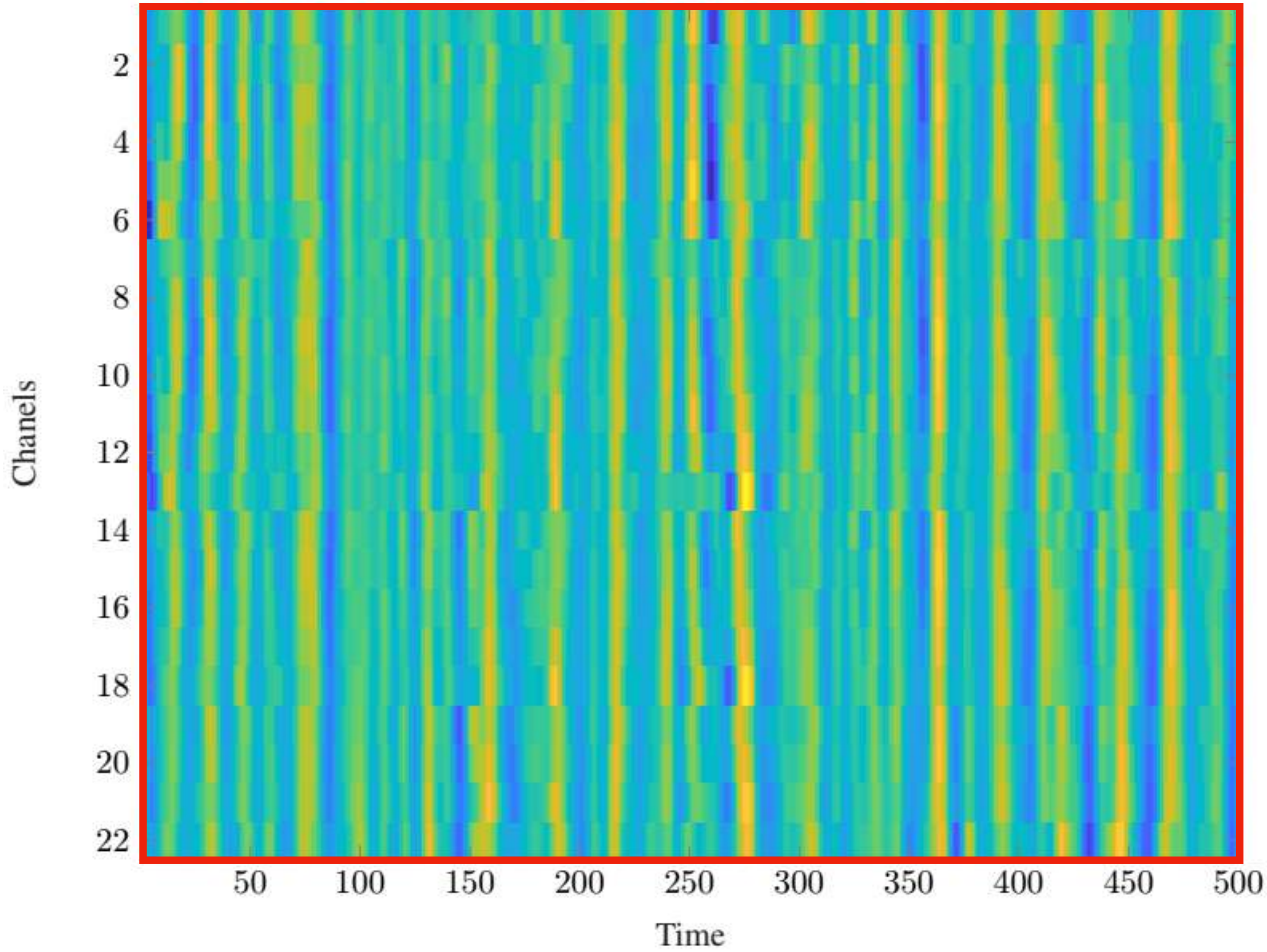
How do we measure stationarity?



$\boldsymbol{\Sigma}$

$\boldsymbol{\Sigma}_1 \quad \boldsymbol{\Sigma}_2 \quad \boldsymbol{\Sigma}_3$

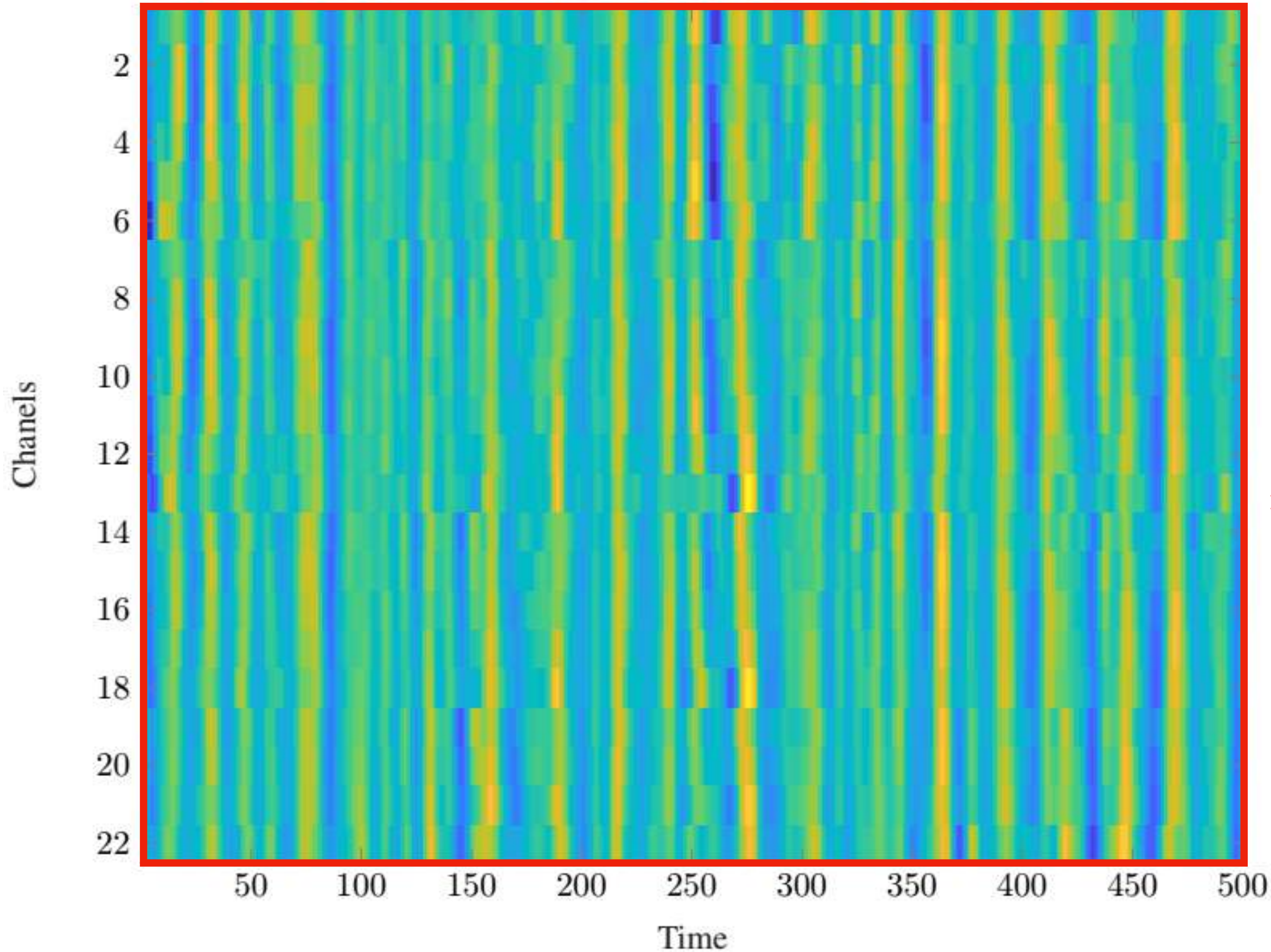
How do we measure stationarity?



$\boldsymbol{\Sigma}$

$\boldsymbol{\Sigma}_1$ $\boldsymbol{\Sigma}_2$ $\boldsymbol{\Sigma}_3$ $\cdots$ $\boldsymbol{\Sigma}_M$

How do we measure stationarity?



$$\boldsymbol{\Sigma} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$$

$\boldsymbol{\Sigma}_1 \quad \boldsymbol{\Sigma}_2 \quad \boldsymbol{\Sigma}_3 \quad \cdots \quad \boldsymbol{\Sigma}_M$

$$= \mathbf{V}_i\boldsymbol{\Lambda}_i\mathbf{V}_i^\top$$

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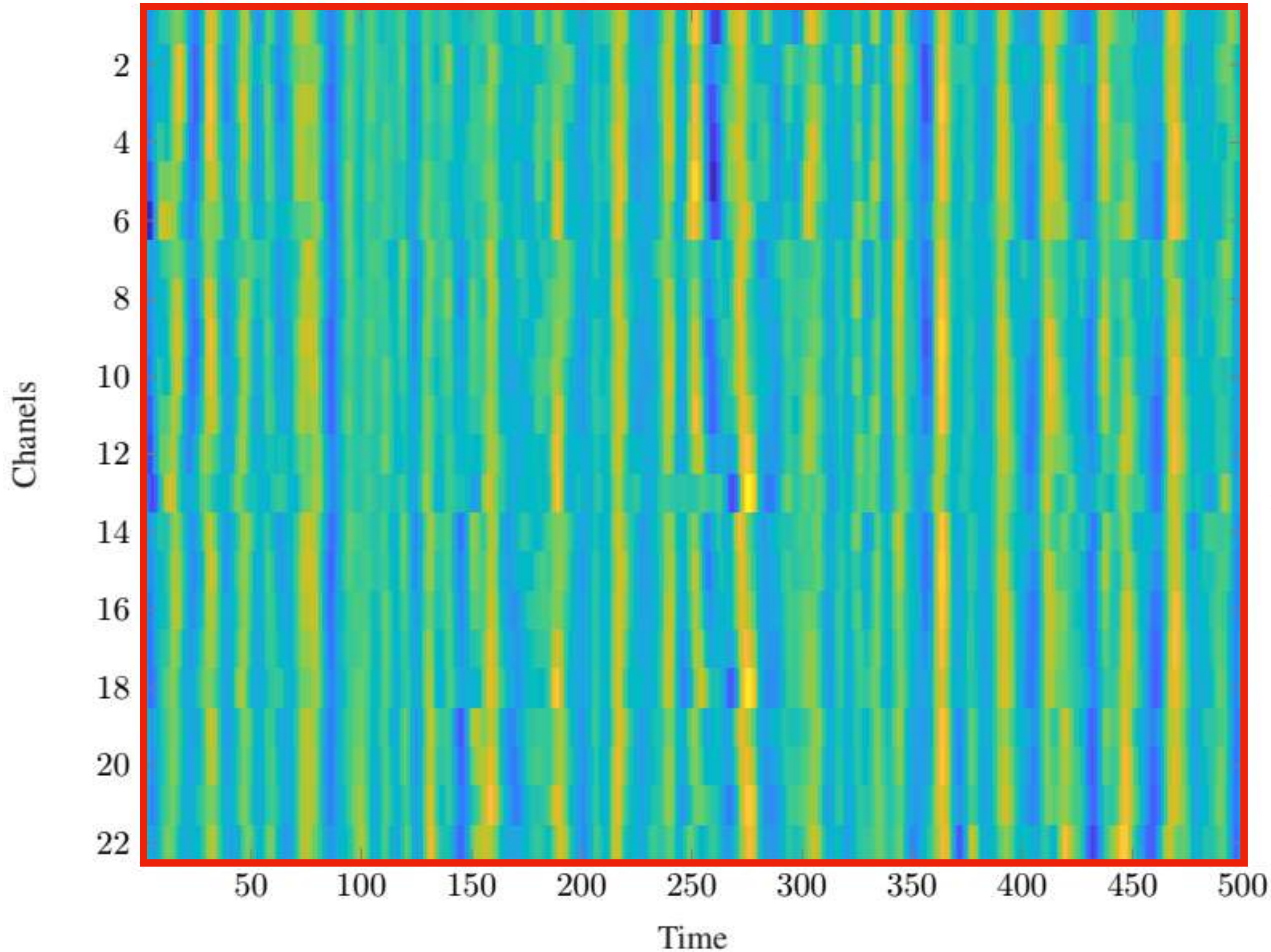

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$$\boldsymbol{\Sigma}_1 \quad \boldsymbol{\Sigma}_2 \quad \boldsymbol{\Sigma}_3 \quad \cdots \quad \boldsymbol{\Sigma}_M$$

$$= \mathbf{V}_i\boldsymbol{\Lambda}_i\mathbf{V}_i^\top = \mathbf{V}\mathbf{V}^\top\mathbf{V}_i\boldsymbol{\Lambda}_i\mathbf{V}_i^\top\mathbf{V}\mathbf{V}^\top$$

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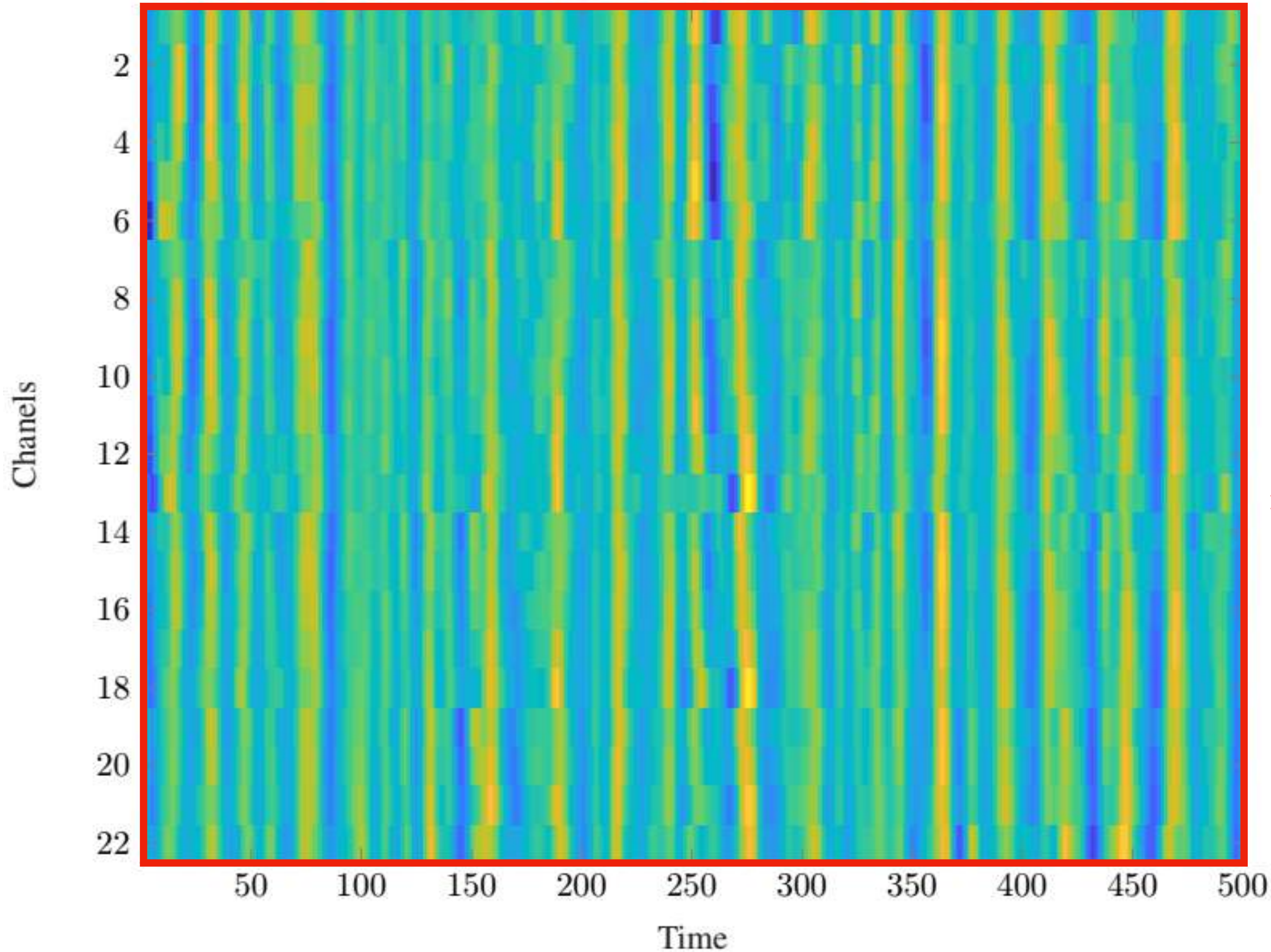

$$\boldsymbol{\Sigma} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$$

$$\boldsymbol{\Sigma}_1 \quad \boldsymbol{\Sigma}_2 \quad \boldsymbol{\Sigma}_3 \quad \cdots \quad \boldsymbol{\Sigma}_M$$

$$= \mathbf{V}_i\boldsymbol{\Lambda}_i\mathbf{V}_i^\top = \mathbf{V}\mathbf{V}^\top\mathbf{V}_i\boldsymbol{\Lambda}_i\mathbf{V}_i^\top\mathbf{V}\mathbf{V}^\top = \mathbf{V}\mathbf{A}\mathbf{V}^\top$$

How do we measure stationarity?

$$\mathbf{\Sigma}_i = \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^\top = \color{red}{\mathbf{V}\mathbf{V}^\top} \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^\top \color{red}{\mathbf{V}\mathbf{V}^\top} = \color{red}{\mathbf{V}} \mathbf{A} \color{red}{\mathbf{V}^\top}$$

How do we measure stationarity?

$$\mathbf{\Sigma}_i = \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^\top = \mathbf{V} \mathbf{V}^\top \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^\top \mathbf{V} \mathbf{V}^\top = \mathbf{V} \mathbf{A} \mathbf{V}^\top$$

$\mathbf{A}$ is diagonal iff
same bases

How do we measure stationarity?

$$\boldsymbol{\Sigma}_i = \mathbf{V}_i \boldsymbol{\Lambda}_i \mathbf{V}_i^\top = \mathbf{V}\mathbf{V}^\top \mathbf{V}_i \boldsymbol{\Lambda}_i \mathbf{V}_i^\top \mathbf{V}\mathbf{V}^\top = \mathbf{V}\mathbf{A}\mathbf{V}^\top$$

A is diagonal iff
same bases

$$\Gamma(\boldsymbol{\Sigma}_i) = \|\mathbf{A}_i \; diag(\mathbf{A}_i)^{-1}\|_F - \sqrt{N}$$

How do we measure stationarity?

$$\mathbf{\Sigma}_i = \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^\top = \mathbf{V} \mathbf{V}^\top \mathbf{V}_i \mathbf{\Lambda}_i \mathbf{V}_i^\top \mathbf{V} \mathbf{V}^\top = \mathbf{V} \mathbf{A} \mathbf{V}^\top$$

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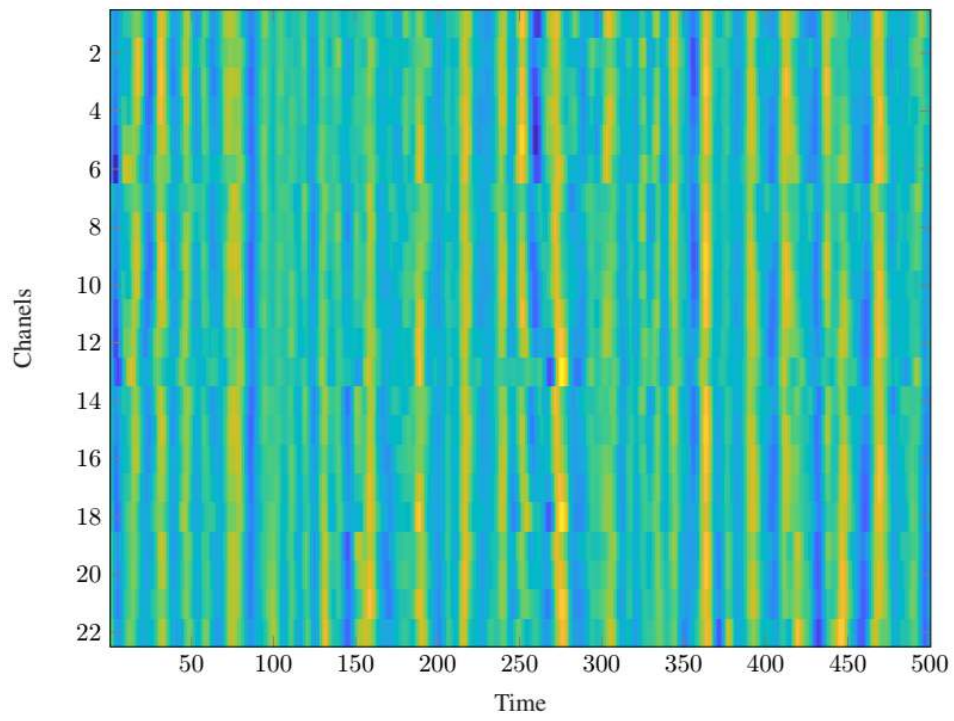

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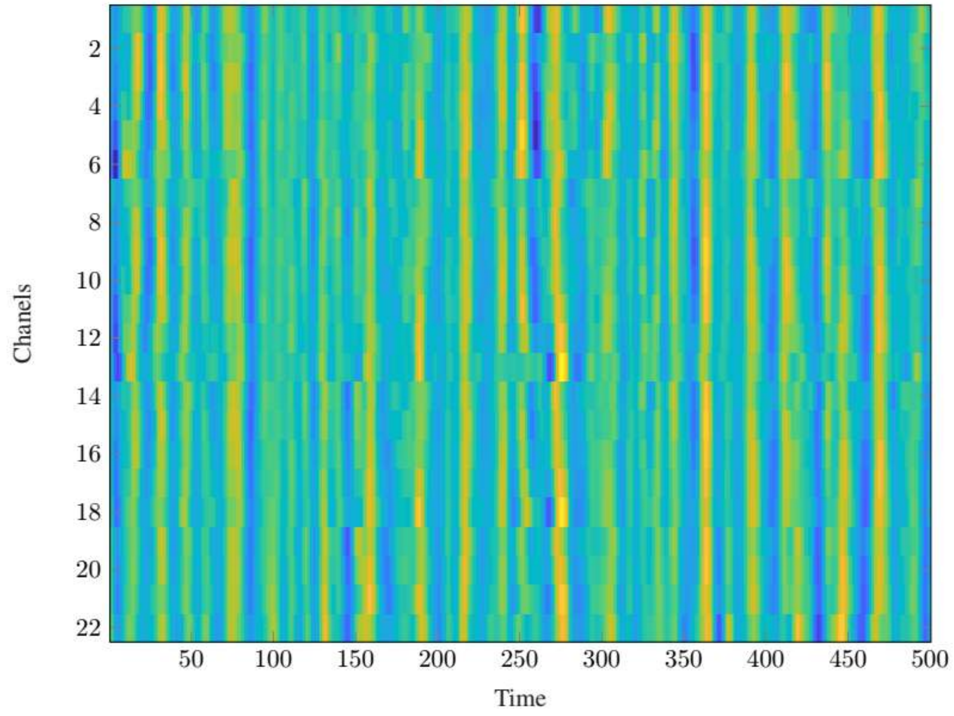

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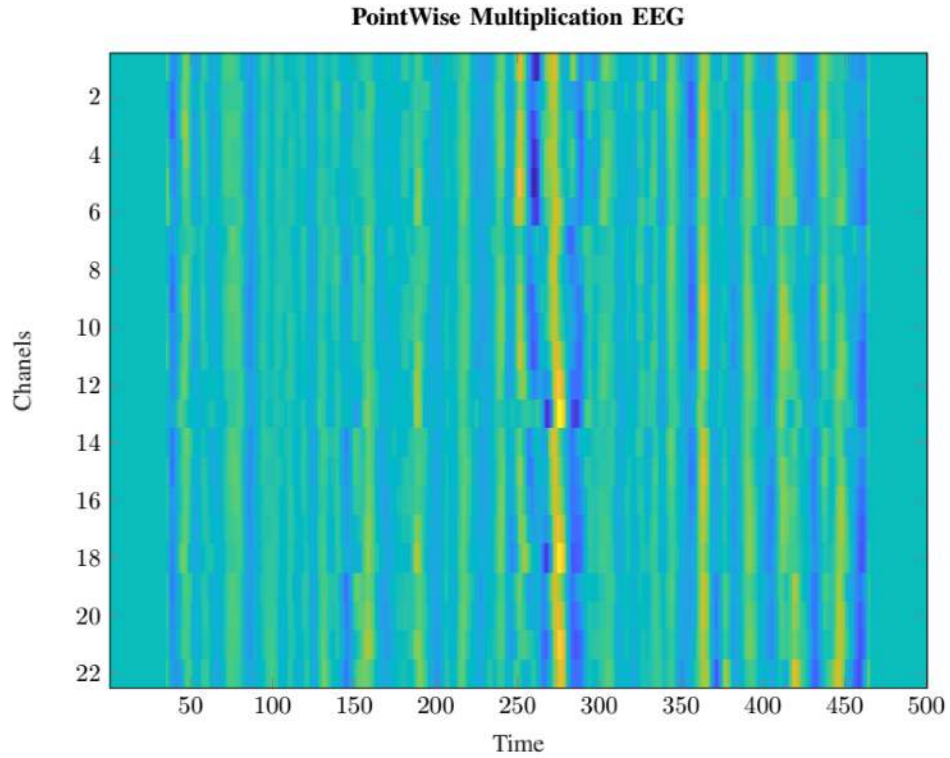

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A is diagonal iff
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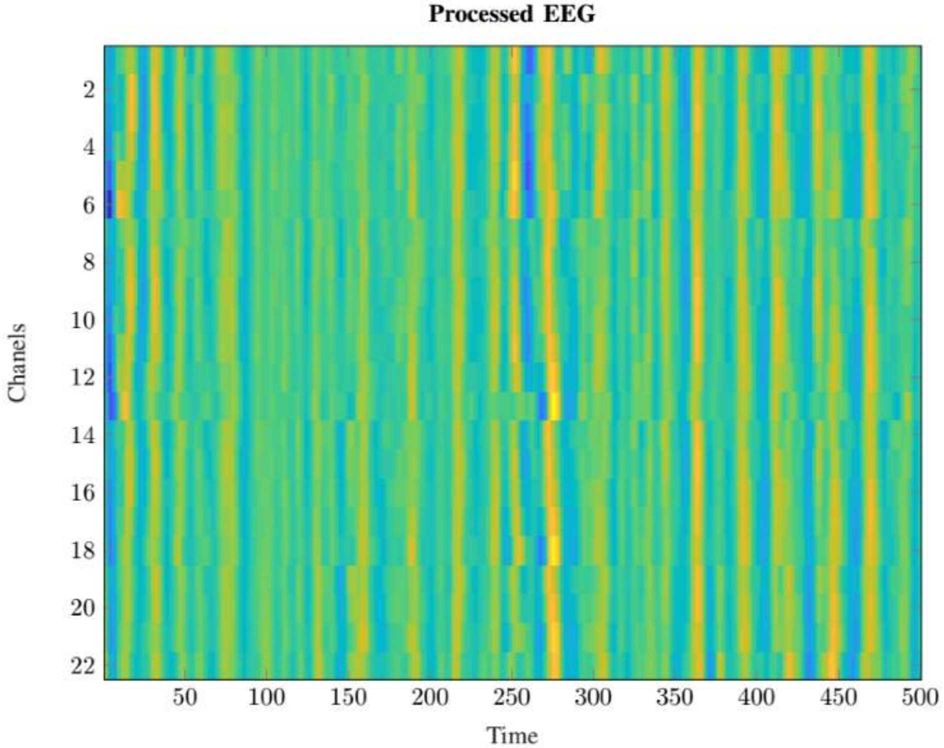
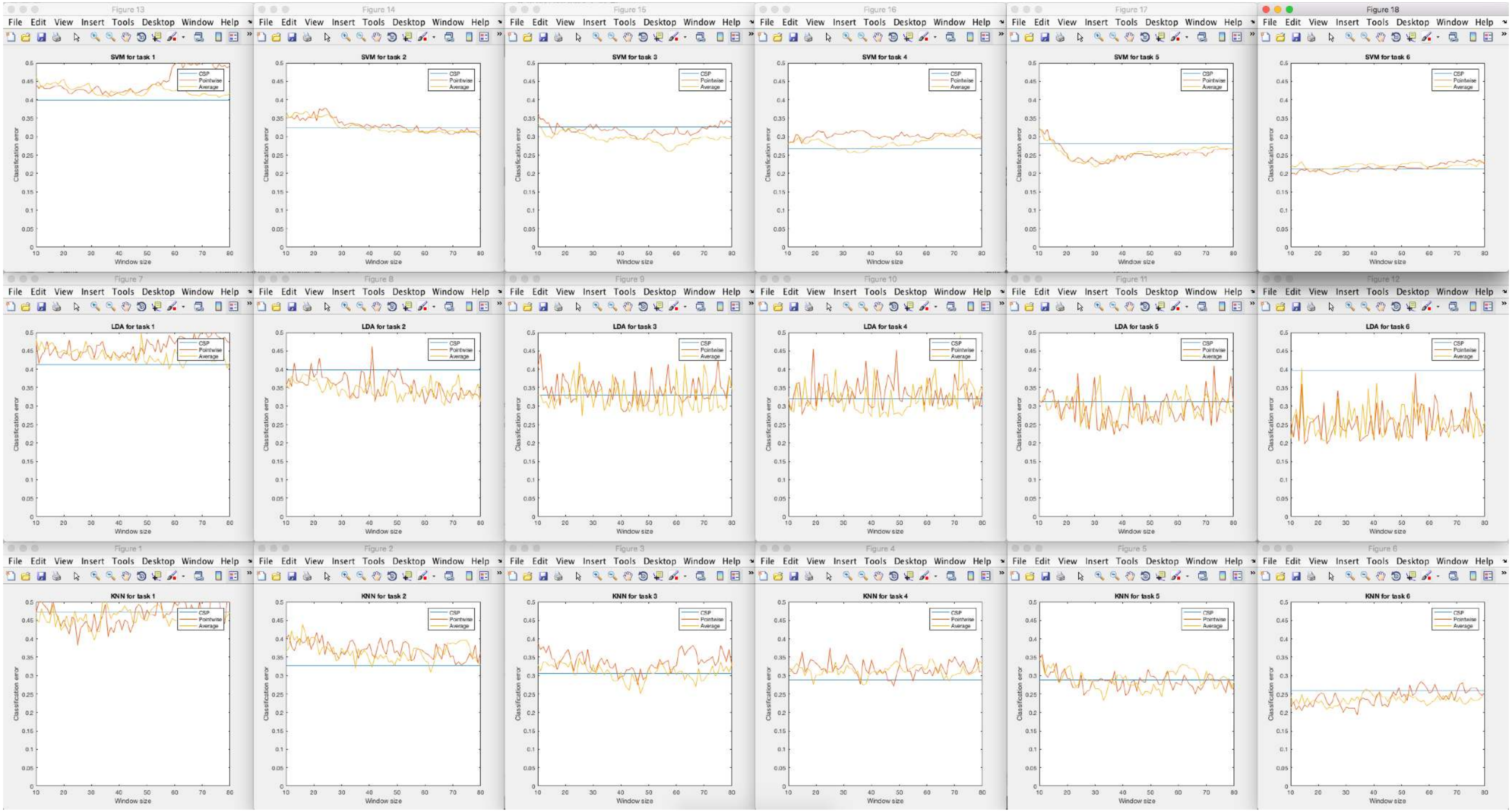


Preliminary results (classification)



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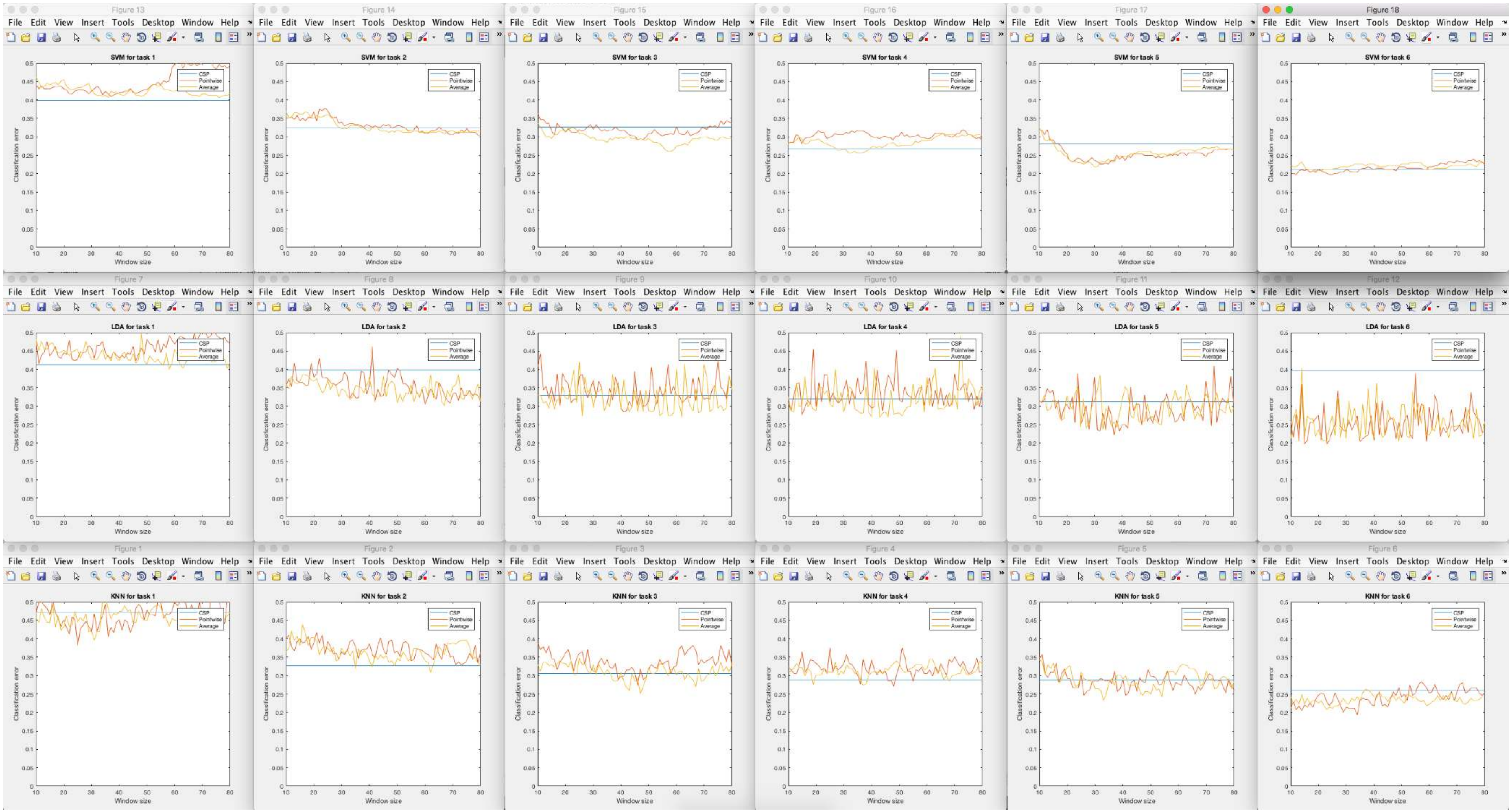
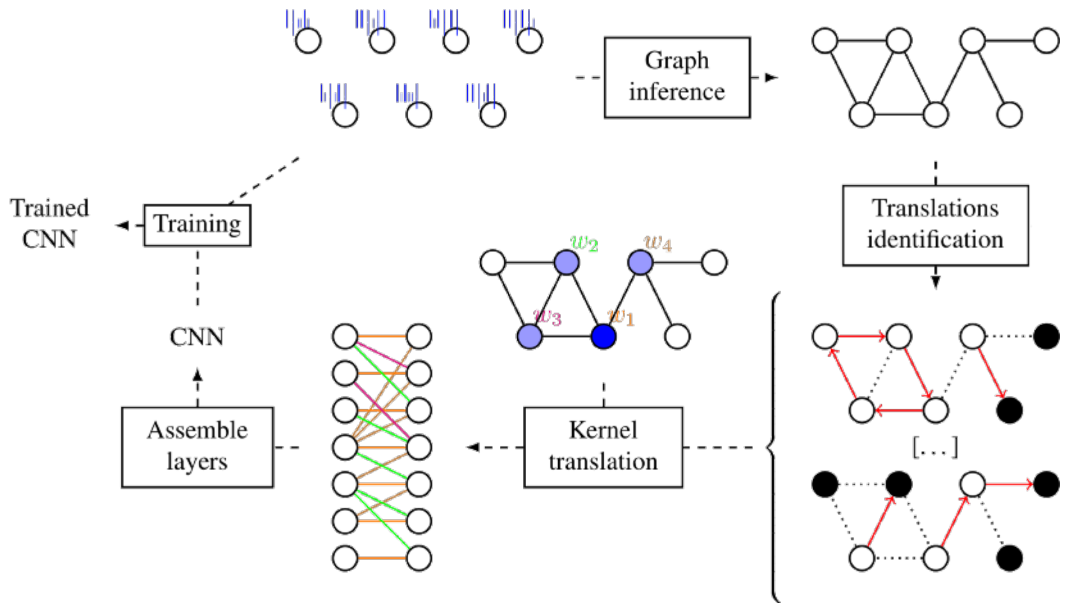


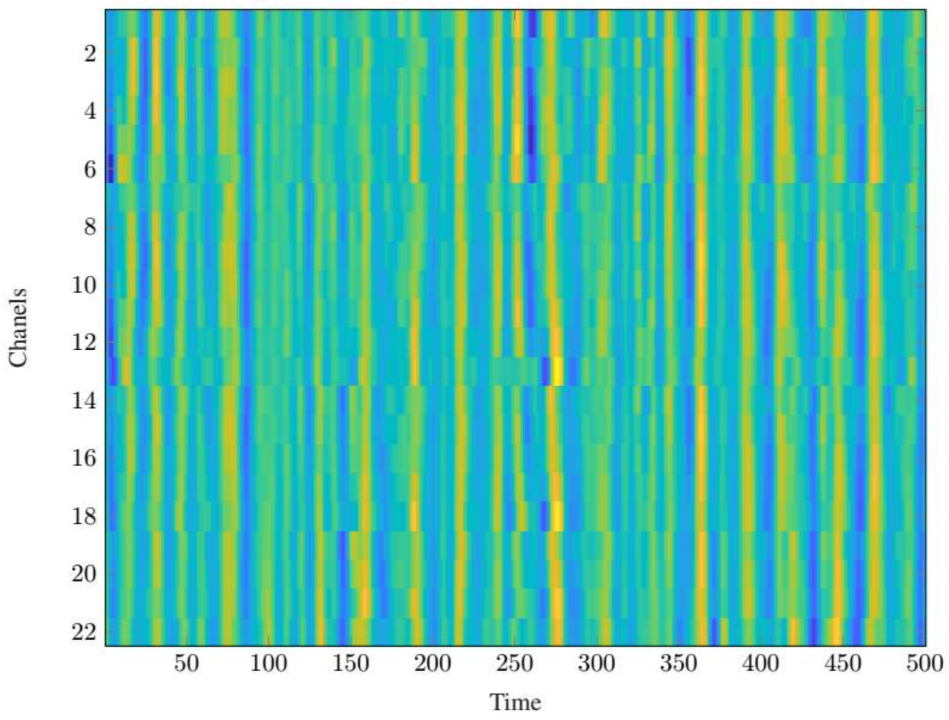
Room for improvement (cov. estimation, resting state, one window per trial…)

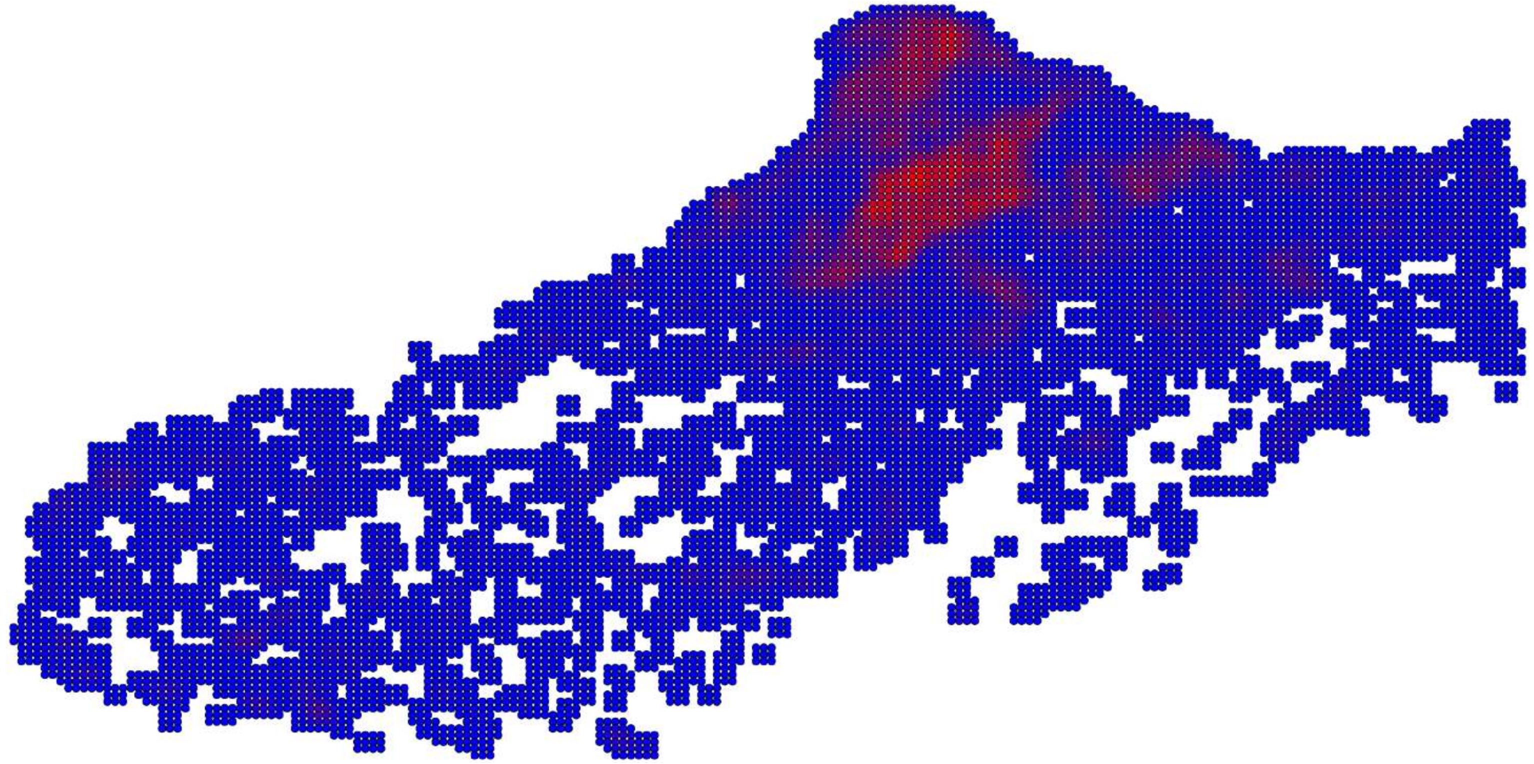
Outline

Conclusions

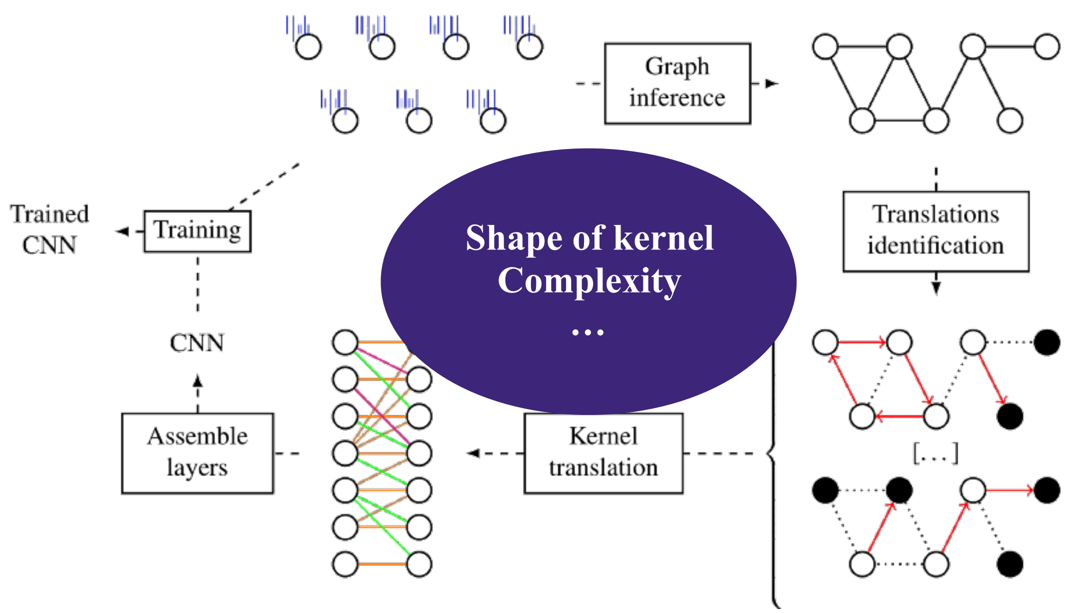
Those are only a few examples

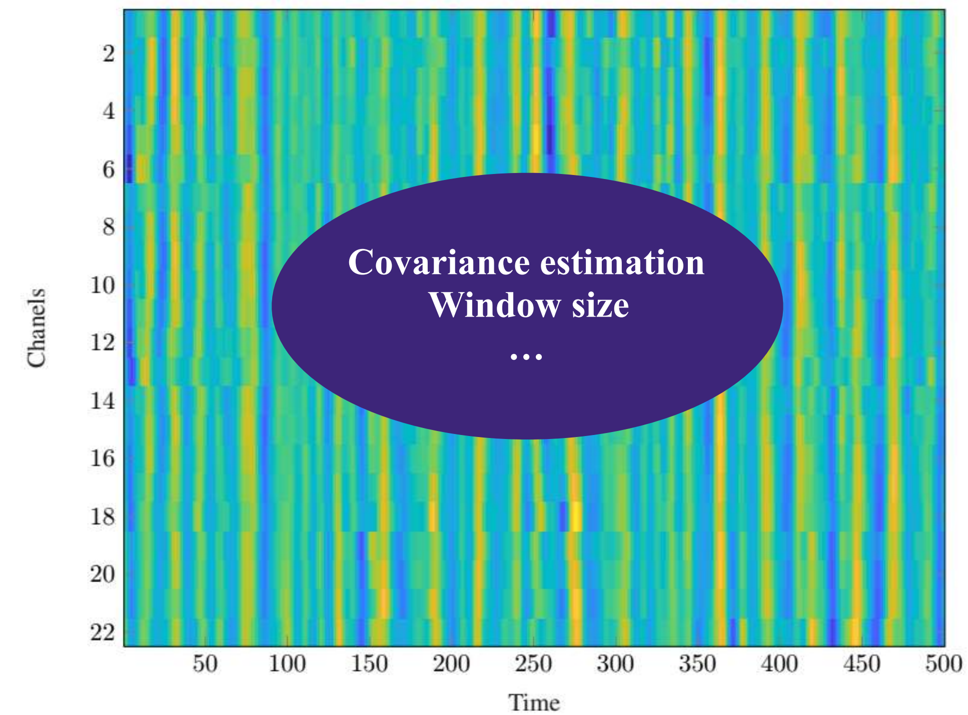


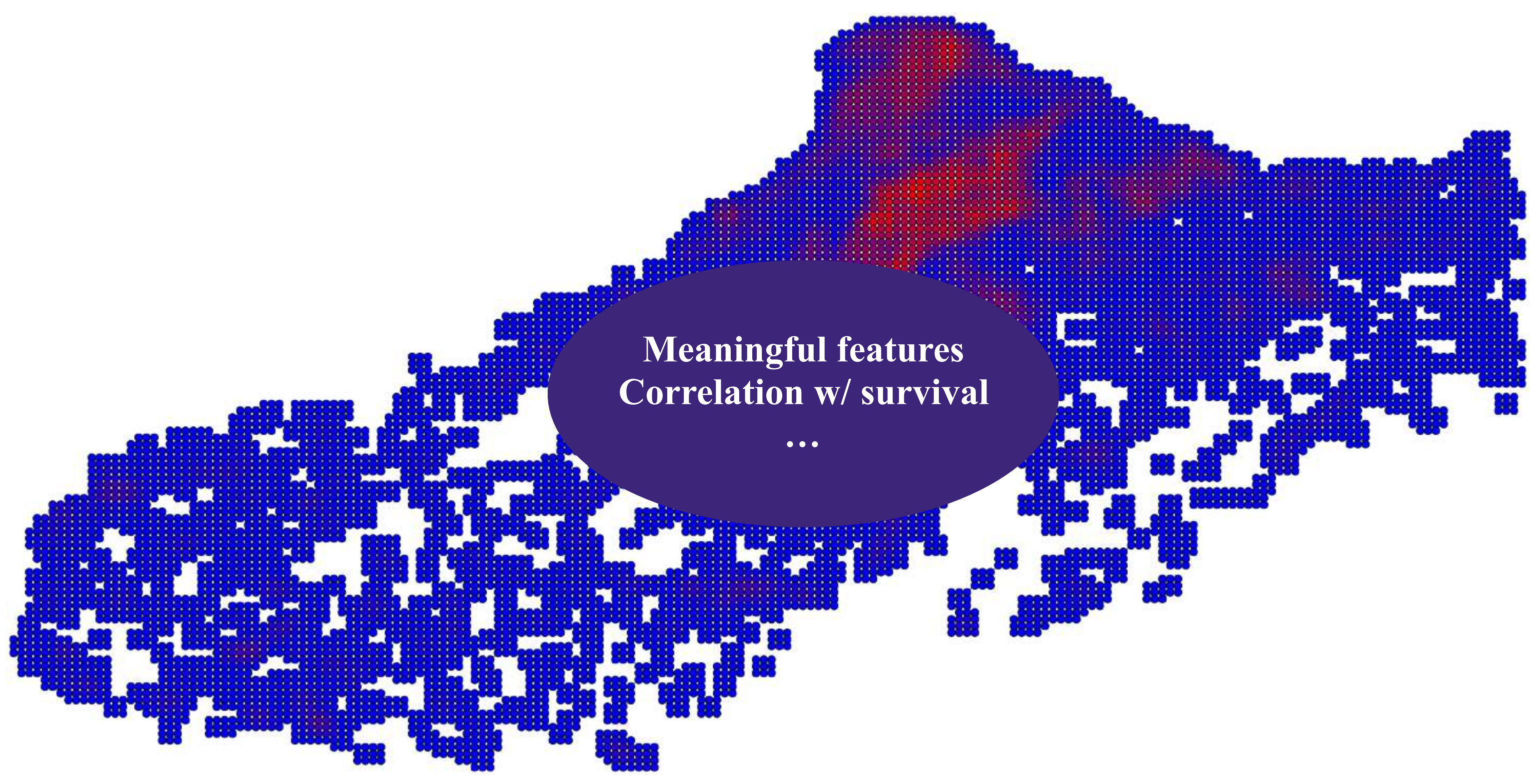




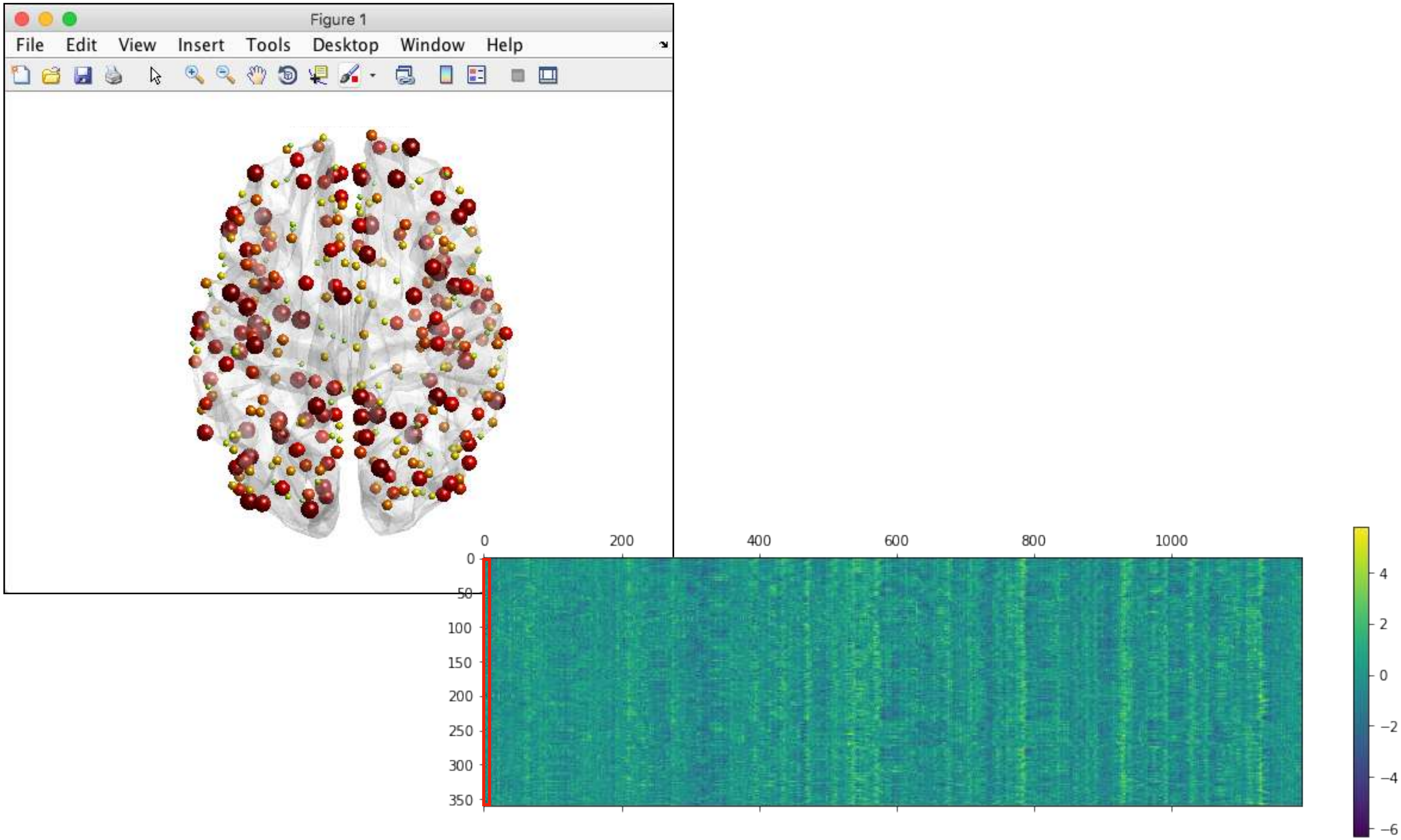
A glimpse on future work





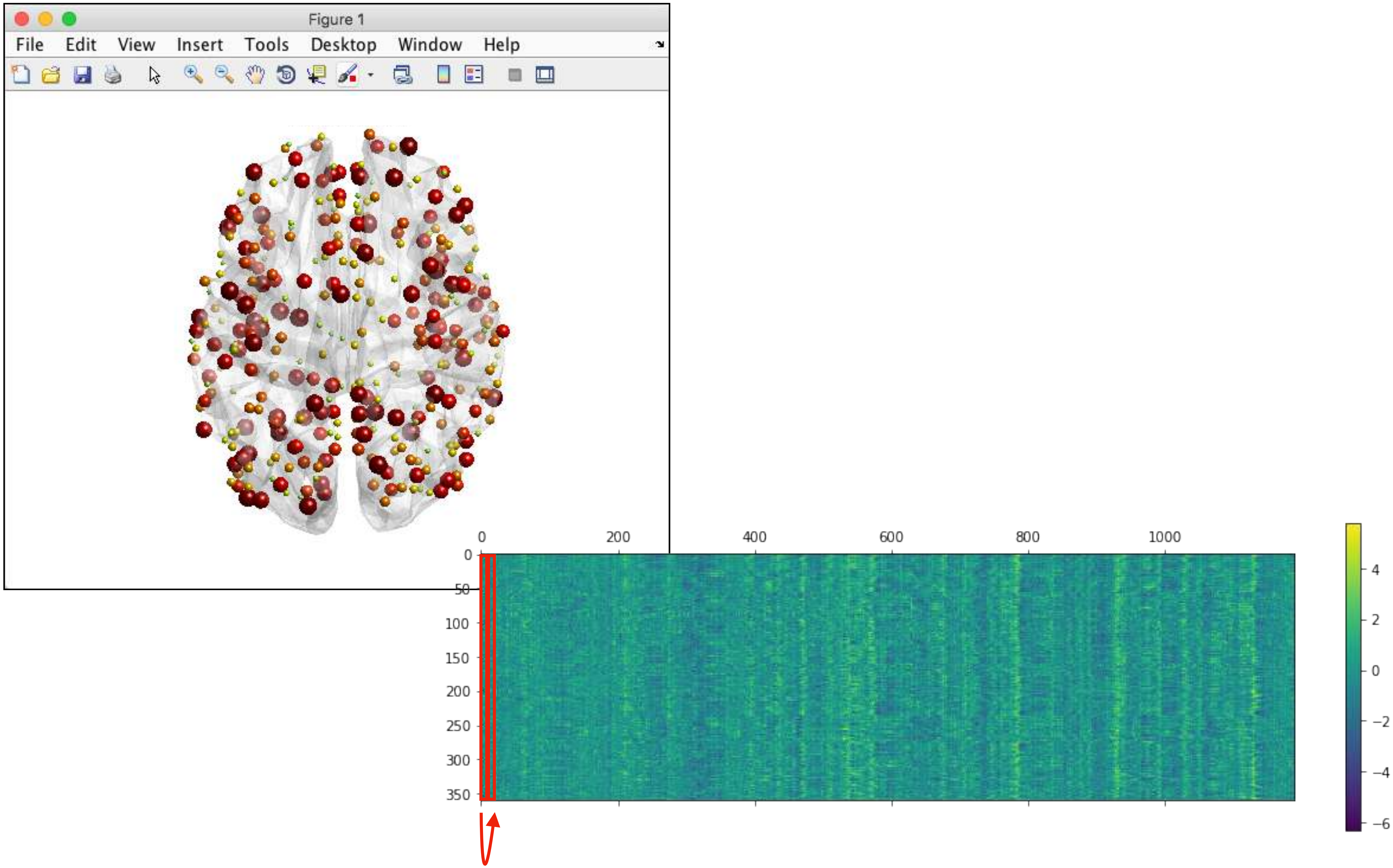


A glimpse on future work



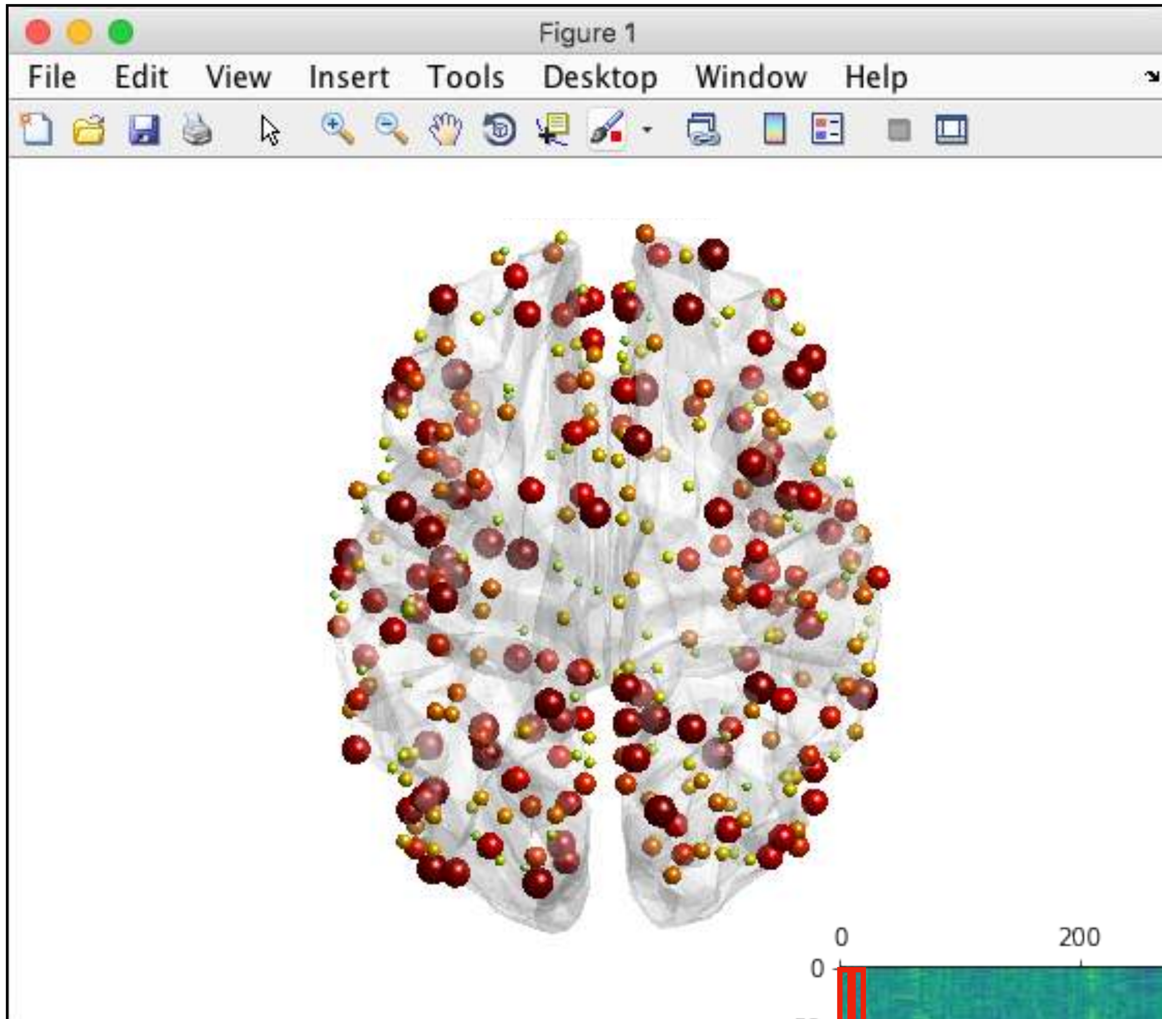
How to go
from this

A glimpse on future work

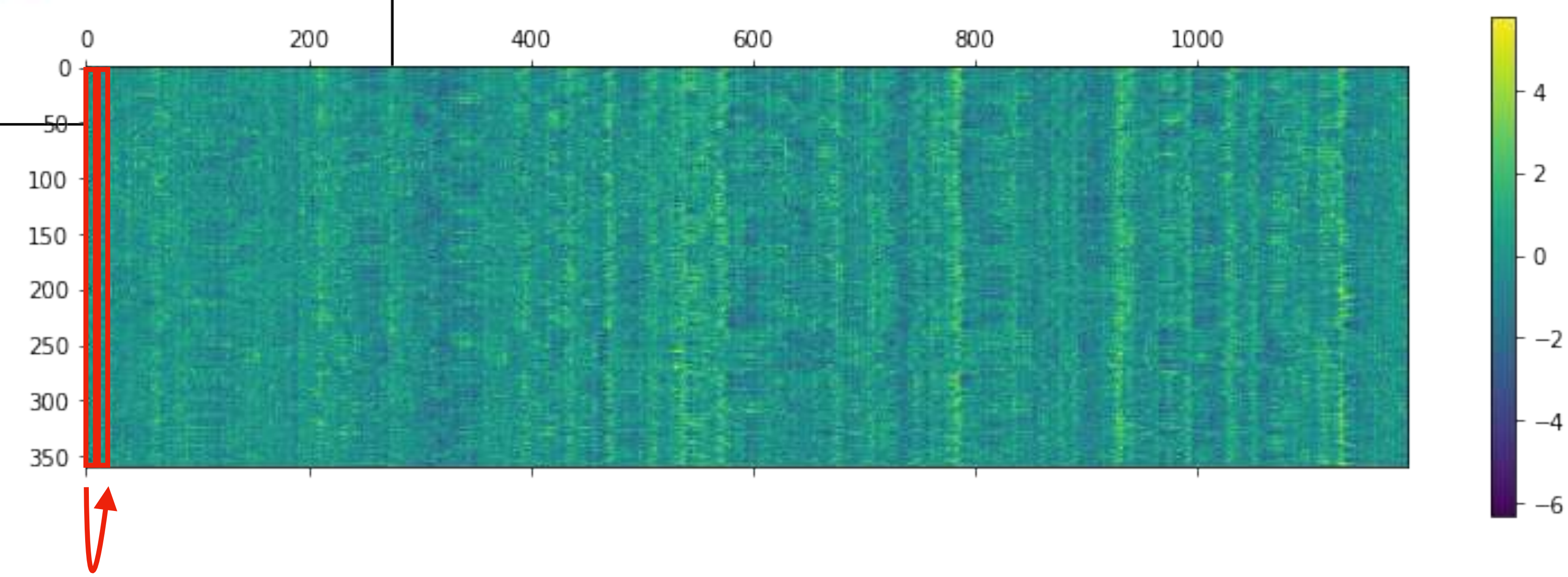


To this

A glimpse on future work



Using graph-constrained
optimal transport on the connectome



To this

A few applications of GSP to machine learning and medicine

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bastien.pasdeloup@epfl.ch

ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE